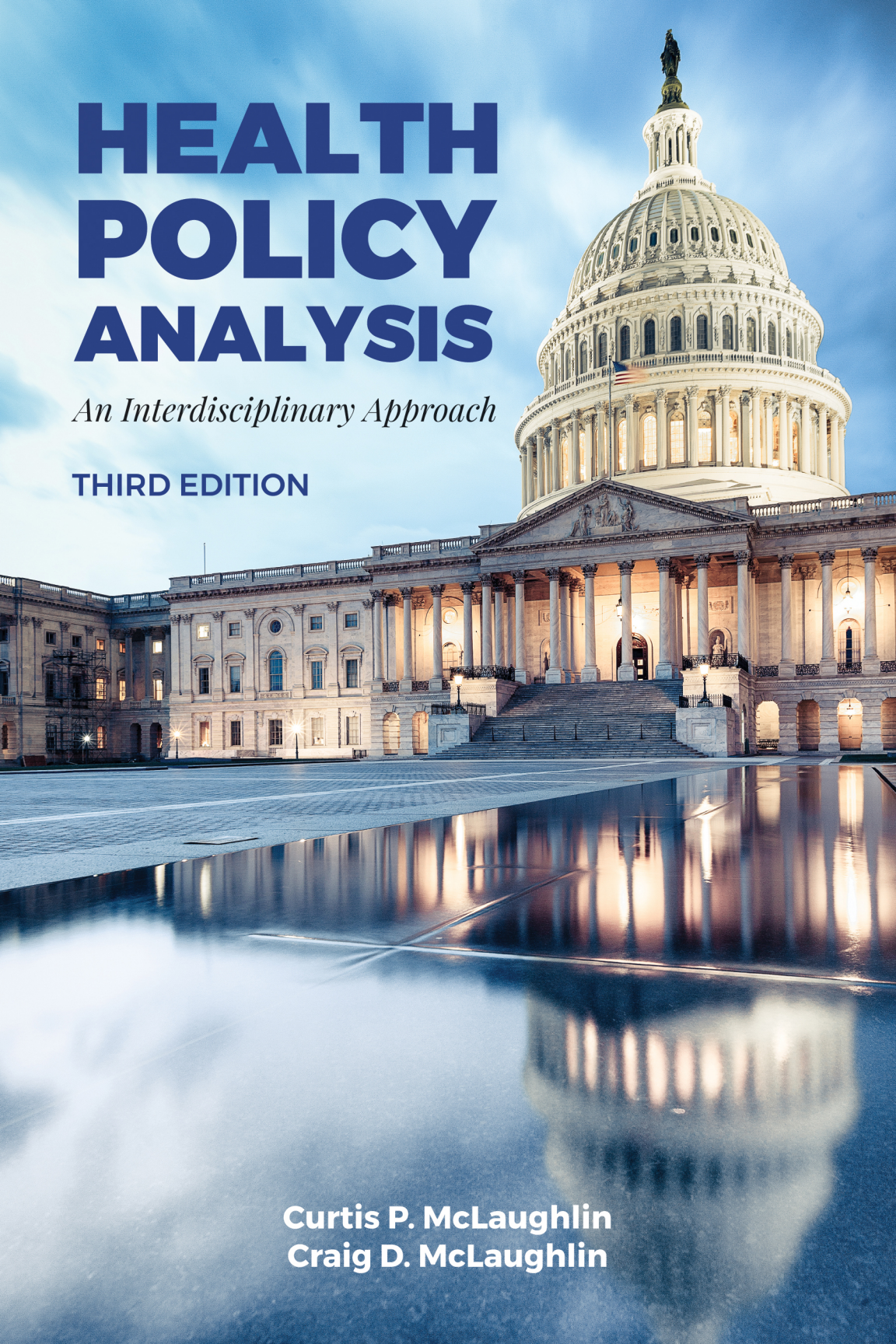




HEALTH POLICY ANALYSIS

An Interdisciplinary Approach

THIRD EDITION

Curtis P. McLaughlin
Craig D. McLaughlin

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*In memory of Barbara Nettles-Carlson,
RN, FNP, MPH—wife, mother, stepmother,
trailblazer, educator, and
dedicated health professional.*



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Preface

► The Policy Analysis Process and Health Professionals

This text is about the process of developing health policy relevant to the United States. We have included the perspectives of a number of disciplines and professions. Because our country has many actors but no coherent, integrated, systematic health policy at the federal level, even after the passage of the Patient Protection and Affordable Care Act (ACA), we have drawn heavily on our personal experiences and backgrounds, which include economics, political science, management, communications, and public health. We have also drawn on the experiences of other countries. Although the federal government has taken on a greater role with the passage of the ACA, states and even smaller jurisdictions will continue to play a major role in health planning. Values, economics, and health risks may vary among them, which suggests a need for independence in planning and execution. Canada's experience with a broad policy and specific health systems for each province has seemed to work as well, or better than, a centralized bureaucracy might have. Even the health services of a number of European countries have tended toward more decentralization as time has passed.

This text is organized into three parts: "The Context," "The Policy Analysis Process," and "The Professional as a Participant." We have anticipated that this text will be used to review health system issues and policy planning for health in a variety of graduate professional programs. We have not assumed zero knowledge of the U.S. health system, but we have not anticipated that the reader will have a great deal of background about how and why the U.S. health system developed as it did, nor about the efforts that took place in the past to reform it. Therefore, Part I, "The Context," explores current issues with the system (Chapters 1 and 2) and the history of how that system has evolved (Chapter 3). Chapter 4 challenges readers to ask about where we want to be, and Chapter 5 reviews policy alternatives that seem to have strong support for getting from where we are to where we might want to be. Some of these are reflected in the ACA, while others are not. These chapters do not purport to be "value free," but this text is different from most books on health policy because it does not attempt to push a single solution set. Studying the present is important for research and understanding, but the educational purpose of this text, and presumably of any course in which it is assigned, is to prepare students to meet whatever new, and perhaps unforeseen, challenges that develop in the future.

Part II, "The Policy Analysis Process," develops a set of tools for future use. Chapter 6 deals with identification and definition of the issues to be studied. Chapter 7 introduces some of the concepts of technology assessment applicable

to health care. Chapter 8 adds more concepts of technology assessment related to evidence-based clinical innovation and management. Chapter 9 reviews the political processes that influence planning in various settings, especially the public-sector health arena. Chapter 10 presents the accepted methods of economic and financial analysis that determine the economic viability of healthcare plans. Chapter 11 addresses the ethical and other value considerations that must enter into the health policy process. In our deeply divided country, value issues are important. They crop up in just about every context and influence the outcome of most analyses. We have put this chapter after the other three process chapters to try to offset the tendency of many less sophisticated students to start with the qualitative and never get to the rewarding, but demanding, work of including the quantitative. Part II ends with Chapter 12, which focuses on implementation. Policies and plans must take into account the capacities of organizations and societies to implement them. At the same time, how the policy-making process proceeds becomes a part of the context within which the implementation will take place. Yes, there is a problem of circularity here, but that is real life.

Part III, “The Professional as Participant,” deals with the roles, skills, and leadership that health professionals can bring to the policy-making process in their local and national communities. It also acknowledges that one has to act out of a personal set of values and point of view, while at the same time preserving one’s flexibility to make incremental progress if that is all that can be achieved. Chapter 13 suggests that there are important roles for healthcare professionals in the change process. It also discusses the skills that health professionals need to acquire if they are going to be accepted into the process and work effectively on its tasks, either from the inside or the outside. Chapter 14 provides summary material and concluding material for the text.



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Mr. McLaughlin received his BA with honors in chemistry from Wesleyan University and his MBA with distinction and his DBA from Harvard Business School. While there, he studied and then taught in the Harvard interdisciplinary program in healthcare economics and management. At the Business School in Chapel Hill, he developed management programs for health professionals and directed the Operations Management Area and the Doctoral Program. He has served as a consultant to the World Health Organization and a number of businesses and organizations.

Craig D. McLaughlin, MJ, retired as the executive director of the Washington State Board of Health, a position he held when he coauthored the first edition of this text. He joined the board as senior health policy manager in 2001 and served as executive director from 2004 through 2011. Immediately prior, he served as director of college relations and adjunct faculty for The Evergreen State College. As a newspaper editor and freelance journalist for more than a decade, Mr. McLaughlin wrote and edited articles on a broad range of health issues. He has served as a communications consultant to foundations and as a management consultant to media organizations. He continues to serve as a public health policy consultant as well as a freelance journalist and motivational speaker.

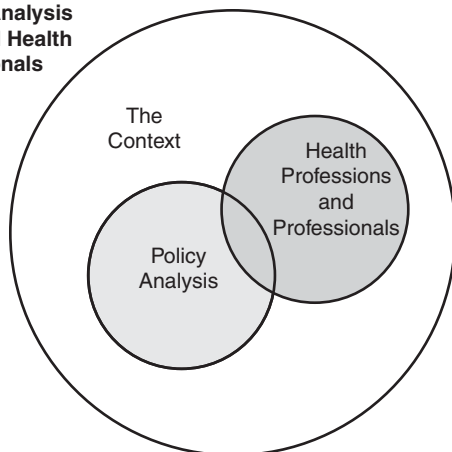
Mr. McLaughlin earned his BA in biology from Wesleyan University and his masters in journalism from the University of California at Berkeley. He also completed all coursework toward an MPA with a concentration in health administration at the University of New Mexico.



PART I

The Context

**The Policy Analysis
Process and Health
Professionals**



Although this book is designed to be valuable to anyone engaged in health policy development, its primary purpose is to enable current and future health professionals to understand and then participate in the health policy process. The figure above shows policy analysis and the work of the health professions taking place within the context of the healthcare system. The first section of this book develops that context. It begins with an explanation of what health policy analysts think about and do (Chapter 1). This is followed by a discussion of the current status of the U.S. healthcare system (Chapter 2) and a review of factors that influenced its development as the decentralized system we have today (Chapter 3). The case accompanying Chapter 3 provides a chance to look at the experiences of other countries and develop some hypotheses about how these countries achieved their current status. Chapter 4 reviews the many and varied objectives for the U.S. healthcare system being expressed by various policy participants. Chapter 5 presents many of the policy choices being suggested. One educational outcome you should try to achieve is to understand these positions, their underlying assumptions, and their strengths and weaknesses.

These chapters provide both the context and vocabulary for moving on to the second part of this book, which outlines available tools for rational policy analysis—one of the circles within a circle in the diagram. The third part of this book looks at the role of the health professions and professionals and, in particular, how they can and should participate in policy analysis.



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CHAPTER 1

Introduction

President after president has pushed for an overhaul to our healthcare system and remedies for its access problems. Only Lyndon Johnson and Barack Obama shepherded through dramatic changes. Attempts by Truman, Eisenhower, Nixon, and Clinton were less successful. In the more recent past, the rapid growth of healthcare costs has expanded the policy debate. So has growing recognition of medical errors and other quality problems. In the meantime, policy makers struggle with a highly fragmented system and a divided body politic. At the same time, the rest of the developed world has advanced, used, and institutionalized increasingly sophisticated approaches to policy analysis. Such efforts have supported these countries in doing a better job delivering quality care at less cost.

More recently, the United States has faced continuous near-miss attempts to replace the Affordable Care Act (ACA), as well as tax proposals, executive orders, and administrative policy revisions that undermined the act. Uncertainty has dominated the industry and added to costs. This ongoing debate centered on contending definitions of affordability, adequate coverage, consumer choice, and acceptable wealth transfer mechanisms.

► The Many Actors

Policy decisions are made at multiple levels of U.S. society:

- National government
- State and local governments
- Healthcare institutions
- Provider professionals
- Payer organizations (employers and insurers)
- Employers (meeting the mandate)
- Individuals (consumers)

BOX 1-2 through **BOX 1-7** distributed throughout this chapter provide samples of health policy questions faced in each of these domains. Like most tables and lists in this text, they are meant to be illustrative, not exhaustive.

In such a decentralized environment, government may take a hands-on approach, treating health care as a public good, as it does transportation and education, or a hands-off approach, favoring market-driven outcomes. Therefore,

government's stance and specific policies may swing dramatically as political power shifts. For example, during the 2016 presidential campaign, one side vowed to repeal the ACA if it gained complete control of the political process, undoing a major accomplishment of the Obama administration. Wide swings in public attitudes are not unknown. The 1988 Medicare Catastrophic Coverage Act had a favorable rating with the public when passed, but was repealed in November 1989 as the public, especially the wealthier elderly, learned more about how they would have to pay for it.

This chapter describes what healthcare policy is, how the policy analysis process works, and the different roles health professionals can play in setting and implementing health policy over time. The role of a policy analyst is described quite completely in the excerpt from the U.S. Office of Personnel Management Operating Manual displayed in **BOX 1-1**.

BOX 1-1 Excerpts from the Office of Personnel Management Qualification Standards for General Schedule Positions—Policy Analysis Positions

- Knowledge of a pertinent professional subject-matter field(s). Typically there is a direct, even critical, relationship between the possession of subject-matter expertise and successful performance of analytical assignments.
- Knowledge of economic theories including micro-economics and the effect of proposed policies on production costs and prices, wages, resource allocations, or consumer behavior; and/or macro-economics and the effect of proposed policies on income and employment, investment, interest rates, and price level.
- Knowledge of public policy issues related to a subject-matter field.
- Knowledge of the executive/legislative decision-making process.
- Knowledge of pertinent research and analytical methodology and ability to apply such techniques to policy issues, such as:
 - Qualitative techniques, such as performing extensive inquiry into a wide variety of significant issues, problems, or proposals; determining data sources and relevance of findings and synthesizing information; evaluating tentative study findings and drawing logical conclusions; and identifying omissions, questionable assumptions, or inadequate data in the analytical work of others.
 - Quantitative methods, such as cost benefit analysis, design of computer simulation models and statistical analysis including survey methods and regression analysis.
- Knowledge of the programs or organizations and activities to assess the political and institutional environment in which decisions are made and implemented.
- Skill in dealing with decision makers and their immediate staffs. Skill in interacting with other specialists and experts in the same or related fields.
- Ability to exercise judgment in all phases of analysis, ranging from sorting out the most important problems when dealing with voluminous amounts of information to ensure that the many facets of a policy issue are explored, to sifting evidence and developing feasible options or alternative proposals and anticipating policy consequences.
- Skill in effectively communicating highly complex technical material or highly complex issues that may have controversial findings, or both, using language appropriate to specialists and/or nonspecialists, facilitating the formulation of a decision.
- Skill in written communication to organize ideas and present findings in a logical manner with supporting, as well as adverse, criteria for specific issues, and to prepare material complicated by short deadlines and limited information.

- Skill in effective oral communication techniques to explain, justify, or discuss a variety of public issues requiring a logical presentation of appropriate facts and information or analysis.
- Ability to work effectively under the pressure of tight time frames and rigid deadlines.

Reproduced from www.opm.gov/policy-data-oversight/classification-qualifications/general-schedule-qualification-standards/specialty-areas/policy-analysis-positions/; accessed 10/13/17. For more detail see Section IV-A (pp. 33-34) of the Operational Manual for Qualification Standards for General Schedule Positions.

BOX 1-2 Illustrative Health Policy Issues at the U.S. Federal Level

- How should otherwise healthy people be motivated to participate in health insurance programs, thus lowering the average premium?
- What population groups should receive subsidized coverage from tax revenues?
- Because the Constitution does not include the topic of health care as a federal responsibility, how should the federal government participate in supporting health care for all?
- How should the federal government support quality improvement efforts if state boards are not effectively addressing medical error rates?
- The cost of malpractice insurance in some states threatens the supply of providers in some specialties and appears to raise the cost of care, so what is the role of the federal government in avoiding the negative effects of malpractice lawsuits?
- Progress in information technology implementation in health care has lagged behind most other information-intensive service sectors. Are the provisions of the Health Information Technology for Economic and Clinical Health (HITECH) Act sufficient to overcome this problem?
- What services should be covered under Medicare? Medicaid?
- How many health professionals in a subspecialty are sufficient? Armed with the right answer, what should we be doing about any shortages? About any surpluses?

► Health Care: What Is It?

The terms *health* and *health care* are used loosely in the United States. Before exploring the question of meaning, however, a note on style: Most U.S. style manuals have called for health care to be open (two words) as a noun, but hyphenated as an adjective, such as when referring to the *health-care* sector. Writers in the United Kingdom have tended to use health care as a noun and healthcare as an adjective. The style is evolving, however—more and more United States-based publications are using healthcare as an adjective, and healthcare has begun to appear regularly as a noun. This book uses health care as a noun, but healthcare as an adjective.

On the issue of meaning, what people in the United States often mean by health is an absence of notable ailments. The World Health Organization (2005), however, defines health as “a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity.”

Similarly, when people utter the phrase *the healthcare system*, they often are talking about the system for financing and delivering personal medical services—what

some refer to as illness care and we will refer to often as the medical care system. The system that promotes health and wellness is actually far more complex. Other health-related systems include public health, mental health, and oral health. Moreover, much of our health is the result of *social determinants*, such as housing, education, social capital, our natural environment, and the way we construct our built environment, all shaped by decisions made outside the healthcare system.

Thinking about health in terms of population outcomes can dramatically shift the way problems are defined and addressed. One example is identifying the leading causes of death. Using a disease model, the leading killers are ailments such as heart disease, cancer, stroke, injury, and lung disease, but McGinnis and Foege (1993), using a population-based, prevention-oriented perspective, identified the “real causes of death” as behaviors such as tobacco use, improper diet, lack of physical activity, and alcohol misuse. They argued that 88% of what we spend on health nationally pays for access to medical care, but in terms of influence on health status, medical care accounts for a mere 10%. This view attributes 50% of our health status to our behaviors, 20% to genetics, and 20% to environmental factors. Yet only 4% of health spending has been going to promote healthy behaviors and 8% to all other nonmedical health-related activities (Robert Wood Johnson Foundation, 2000). Since the mid-1960s, public health spending as a percentage of overall spending on health care has fluctuated between 1% and 1.5% (Frist, 2002), and yet 25 years of the 30-year increase in life expectancy between 1900 and 1995 can be attributed to public health interventions.

This text focuses mostly on access, cost, and quality issues related to personal medical services. That is because our primary intended audience is healthcare professionals (people who work primarily within the medical care system), and it is also due to the simple fact that the United States is wrestling with so many current policy issues related to medical care access, cost, and quality. Keep that intentional bias in mind. Stop occasionally to think about how a big-picture view of health might change the way problems and solutions are identified.

BOX 1-3 Illustrative Health Policy Issues at State and Local Levels

- What services should be provided and to whom under Medicaid options and waivers?
- How should the professional licensure be conducted so as to encourage quality of care, adequate access, and appropriate competition?
- How should the public university system decide how many professionals to train to ensure adequate access to all sections of the state? To all target groups?
- How aggressive should our state be in implementing and supporting health insurance exchanges?
- What should be the roles of the state insurance regulations and oversight boards in ensuring access to care for the general public and for special populations?
- Should the curative healthcare system, the mental health system, and public health clinics be merged as healthcare access becomes universal?
- What are intended and unintended consequences of sex education policies on health and health services?
- How do we undertake healthcare emergency planning for responses to floods, earthquakes, pandemics, and terrorism? What is the relationship between the state systems (public health and military) and local first responders?

► Health Policy: What Is It?

Most of us are clear on what health policy is about in general terms. Simply stated, it addresses questions such as:

- How well are we doing delivering health care?
- How did we get here?
- Where do we want to be?
- What other alternatives are available here and throughout the world?
- What is likely to work in the future given our political process?
- What roles should health professionals and ordinary citizens play in this process?
- How can we become better prepared for such roles?

We cannot expect any representative cross section of participants to agree on the answers to all of these questions because their interests often conflict. One goal of this text is to encourage development of an objective, managerial approach to decision-making—one that uses precise definitions of terms and relationships and carefully considers the key issues (and walks in the shoes of key actors). Readers should come away with a set of tools for interpreting and analyzing events, situations, and alternatives—tools that add to the skills already developed through professional training and experience.

► The Policy Analysis Process

The policy analysis process usually involves the following activities:

- **Problem identification.** Why do we need to evaluate and possibly change the way we do things? What kinds of actions are people asking for? What are the drivers that require that scarce resources be devoted to this policy area? What is the intended output? What is the expected result?
- **Process definition.** What is the current situation? Why are current results unsatisfactory to some? What is being done about it? Who are the current actors, and what are their roles? Are people framing the issue effectively? What are reasonable expectations for results over a relevant time horizon?

BOX 1-4 Illustrative Health Policy Issues for Healthcare Institutions

- How much charitable (uncompensated) care should we provide beyond that which is mandated?
- What should be our health information technology strategy?
- Should we undertake joint planning for future services with our local health department?
- How should we go about increasing the proportion of the local population who volunteer as local organ donors?
- Can we rationalize the services provided by local providers, reducing duplication and waste, and still avoid charges of anticompetitive practices?
- What should we be doing to become an effective learning organization?

- **Process analysis.** What is happening in practice? How are outputs and outcomes measured? What are interested parties recommending? What are the resource inputs? Are they appropriate? Are the outputs distributed fairly? Process analysis can be approached using a consistent set of steps:
 - Map out the existing processes that yield the outputs and outcomes of concern in as much detail as necessary to be operational.
 - Generate a list of solution strategies and narrow it to viable alternatives.
 - Map out the best processes for the more promising alternatives.
 - Ask where, how, and when new technologies might change each process within the relevant time horizon.
 - Determine the resource requirements of the most promising alternatives and then cost them.
 - Calculate other process parameters, such as lives saved, hospital days avoided, or persons served.
- **Qualitative analysis.** Identify and assess the nonquantitative issues related to valuation of benefits, quality, equity, and distribution and perceived fairness of outcomes.
- **Evaluation and choice.** Take steps to evaluate the options and make a choice:
 - Weigh the evidence, quantitative and qualitative, and review the conclusions to evaluate for:
 - Technical feasibility (medical evidence and operational effectiveness)
 - Political feasibility
 - Economic viability
 - Status with value-laden issues
 - Choose a preferred policy.
 - Prepare to report your findings and conclusions.
- **Implementation strategy.** How do we gain public, professional, and consumer support for change and backing for the most appropriate alternative(s)?

BOX 1-5 Illustrative Health Policy Issues for Provider Professionals

- Should I accept Medicaid patients?
- What services should I provide in addition to those normally provided by my specialty?
- Should I accept an invitation to join the local consortium for accountable care organizations (ACO)?
- What should I do to help the local populace understand the risks of potential pandemics without arousing unnecessary concerns?
- What positions should I encourage my local, state, and national professional organizations to take on current health policy issues?
- Should I volunteer to serve on local or state committees assessing and advocating on health policy issues? Should I seek or accept a leadership role? How do I prepare for that possibility?
- Should I make my information systems meet current “meaningful use” standards and take the subsidy or forget about it until I’m forced to convert?
- Should I enter (or stay in) private practice, or should I join a large group with ties to a dominant delivery network (hospital, health maintenance organization [HMO], ACO, pharmacy chain, etc.)?

How do we ensure early implementer and consumer buy-in? How do we mediate conflicting interests?

- **Implementation planning.** What steps do we need to take to ensure the successful implementation of the chosen alternative? How will we evaluate the level of improvement?
- **Feedback on policy processes.** Have we been making the right choices? If not, why not? What might we do to enable better policy choices in the future?

► Professionals and the Policy Process

An unusual aspect of health care in the United States is the low level of influence that health professionals have on policy formulation. All too often health professionals refer to what policy makers are doing to them, not on what they are doing to contribute to policy processes. Prepared professional leadership is extremely important if policies are going to be accepted and effectively implemented.

One reason U.S. healthcare professionals have been relatively uninvolved in policy making has been the very high opportunity cost of time devoted to policy matters. Most countries have a Ministry of Health that oversees the national health system. Their professionals compete for higher administrative posts that offer better salaries, and especially better locations. Most key positions below the political level in the ministry are held by health professionals, and the directors of most divisions, departments, and institutions are physicians. At one time, U.S. health department directors were all expected to be MDs, and so were many hospital administrators. During and after World War II, when physicians were in short supply, other administrators were called on to run those institutions and new cadres of administrators were trained in the nation's schools of public health, public administration, and business administration. Rapidly rising physician incomes, especially after the introduction of Medicare and Medicaid in 1965, increased the demand for physician services, but not the supply. Providing

BOX 1-6 Illustrative Health Policy Issues for Payer Organizations (Employers and Insurers)

- What kinds of options should I offer as health benefits? Given that employees need choices, should I offer high-deductible insurance policies to go with medical savings accounts?
- How much money and effort should we allocate to prevention? What about the argument that people change plans so often that our investment in prevention won't pay off?
- We have a lot of data on health care utilization. Should I mine that data and suggest choices of procedures? Providers? Lifestyle changes?
- Ethically, how much should we know about our employees' (the insureds') lifestyles that may affect future healthcare costs, and how should we use that knowledge?
- Now that healthcare benefits are mandated for most employers, how do we balance competing for the right labor force, avoiding or not avoiding the tax penalties for those employees not covered, and keeping premium costs under control?
- Should we participate in the new insurance exchanges? If so, what should be offered?

care paid so much better than administration that few physicians sought training in health administration. Educational institutions and health agencies again expanded their training programs for health administrators without requiring health profession credentials. As managed care has begun to constrain provider income and consolidation has increased administrator incomes, professionals have taken a stronger interest in managerial training programs. This interest has been reinforced by provider dissatisfaction with the changes in professional autonomy and working conditions under managed care. Professionals are waking up to the opportunities to participate.

► **National Systems Differ but Parallels Exist**

Every country's healthcare system is unique, a result of culture, history, and happenstance. Yet the issues policy makers face can parallel each other. Many developed countries are struggling with the burden of their social programs, including health care. Even in countries that have long had national health services, there have been many efforts to decentralize them, to make them more responsive to local needs, to tap into tax revenues available at the regional and local levels, and to expand private insurance alternatives. Medical care systems in the United Kingdom and Scandinavia provide examples of this. No other developed country, however, spends as much per capita or as a percentage of the national income (gross domestic product) as the United States, and many of them have better health outcomes across their population. The results achieved in the United States should be better.

It is important to understand that both developed and less developed countries have taken different routes to more or less successful healthcare systems, leading, in turn, to differences in costs and outcomes. Their results have been achieved over decades of adaptation to the cultures and institutions of those countries and may or may not be models for the United States.

All countries are aiming at targets that shift as their populations age, as new technologies become available, and as new diseases and environmental threats emerge. Although health care is not officially a right in the United States, all levels of government and the body politic have been concerned about the proportion of the population forced either to forgo care or to seek some form of public assistance. The provisions of the ACA went a long way toward meeting those needs. Early attempts to repeal and replace the act ran afoul of the public's desire to not increase the number of uninsured.

BOX 1-7 Illustrative Health Policy Issues for Individuals

- Should I purchase health insurance if my employer does not pay for it, or should I pay the tax penalty?
- What should I do about my increasing weight and high blood pressure?
- When I retire, how much should I plan to rely on Medicare to cover my healthcare costs as I continue to age?
- Certain medical specialties are not available in my area. My county government wants to issue tax-exempt bonds to finance a new doctors' office wing on the county hospital site. Should I support the referendum on the bonds?
- My daughter is 24 years old and waiting tables at the Pizza Palace. The company's health benefits are minimal. Should I keep her on my health insurance policy until she turns 26?

► Key Policy Categories

The major policy categories in the first column of **TABLE 1-1** relate to quality of care, availability of resources, payment and funding, motivation of patients and providers, volume and price of services, competition, and cost drivers. (McLaughlin, 2014). The other columns represent major disciplines needed by policy analysis teams. Any significant policy analysis is likely to need data and other contributions from experts in medicine, economics (including finance), political science, and services management (including behavioral and operational skills). The Xs identify the major roles of each discipline in the analysis of that particular issue.

► Overarching Medico-Social Issues

In addition to these specific policy categories, a number of overarching social issues need to be kept in mind. They include:

- Ongoing relationships between health insurance and employment
- Employment status, compensation, and autonomy of healthcare professionals
- Equity in access to services, including affordability, adequacy of coverage, consumer choices available
- Fairness in intergenerational transfers and acceptability of wealth transfer mechanisms
- Professional versus institutional responsibilities for process development and improvement

Relationships Between Healthcare Financing and Employment

Increased international competition for jobs has highlighted the high costs of U.S. health care and the impact of concentrating those costs onto large employers who purchase health insurance for employees and retirees. The proportion of workers receiving health insurance coverage at their place of employment has decreased in recent years. Employers have sought to control costs through the use of managed care organizations. Believing this effort has reaped the bulk of its potential savings, employers now are shifting more of the financial burden to workers by requiring higher individual premiums, reducing subsidies for dependents' coverage, relying more on independent contractors, or eliminating benefits. This has forced the nation to wrestle with the question of whether health insurance coverage should be dependent on employer decisions. The response in the ACA was that the employer had to contribute but the employee was to make the decision about which plan option to purchase with the combination of employer, government, and personal funds.

Small employers, the ones most likely to drop their health benefits, were exempt from the requirements of the ACA. There was also concern about whether the penalties in the ACA would be sufficient to change employer behaviors significantly (Wilensky, 2012).

TABLE 1-1 Matrix of Major Policy Categories Versus Major Skills Disciplines				
Major Policy Categories	Major Skills Disciplines			
	Medical	Economic/ Financial	Political	Operational/ Managerial
Quality				
Access		X	X	X
Technical management	X	X		X
Interpersonal relationships	X			X
Continuity of care	X		X	
Measurement and reporting	X			X
Resource Availability				
Personnel	X	X	X	
Technology				
Evidence-based medicine	X	X		X
Process rationalization	X	X		X
Information systems	X	X		X
Payment				
Insurance/allocation of risk		X	X	
Motivating Patients and Payers				
Consumer-oriented care	X	X	X	X
Mandated payments		X	X	
Price transparency			X	X
Motivating Providers				
Volume				
Fee-for-service		X	X	X
Capitation/vouchers		X	X	
Bundling	X	X	X	X
Budgets/salaries				
Pay-for-performance	X	X	X	X
Price competition				
Antitrust		X	X	
Labor substitution	X	X	X	X
Increased buyer power		X	X	
Cost-efficiency and Effectiveness				
Malpractice			X	X
Fraud and abuse			X	X
Cost-reduction measures	X	X	X	X
Organizational learning	X		X	

Employment Status, Compensation, and Autonomy of Health Professionals

For many years, physicians and pharmacists were independent business people. Hospitals employed some specialists (e.g., radiologists, pathologists, anesthesiologists), often under profit-sharing agreements, but medical practice acts in many states prohibited the use of employed physicians. Movement toward managed care and the consolidation and industrialization of the healthcare industry, however, prompted states to change these laws and prompted more and more organizations to buy practices and to serve customers that had previously turned to independent practices and pharmacies. The ability of large organizations to buy and sell goods and services at deep discounts forced more and more small provider groups to sell out. Increasingly, healthcare professionals were employed by large organizations and experienced conflicts over their professional independence and autonomy. This has led to patient concerns about providers' disinterestedness, a concern that tends to weaken the status of the health professions.

Equity in Access to Services

The Centers for Disease Control and Prevention (CDC) issued a set of targets for *Healthy People 2020*, a federal strategic plan for improving health status and reducing health disparities. Reducing health disparities involves easing the disproportionate burden of disease, disability, and death among a population or group. Disparities can result from cultural factors, behaviors, social determinants (such as low socioeconomic status), and not receiving quality or culturally and linguistically appropriate care when it is accessed. The problem of health disparities is not unique to the United States. The equivalent term used in much of the rest of the world, and increasingly in the United States, is *health equity*. **BOX 1-8** shows some of the base-lines and the targets displayed in the CDC's *Healthy People 2020 Objectives* (2010).

Fairness in Intergenerational Transfers

Recent debates about the national debt have focused on entitlement reform. Recommended reforms include raising the starting age for full Medicare benefits. Some have objected to the use of the word *entitlement* to refer to Medicare. They prefer the term *earned benefit*. For someone who has been paying into Medicare and Social Security for 20 years but is still relatively young, this represents a loss of expected return on the investment and an intergenerational transfer to the elderly who are already on Medicare. How might we deal with this fairness issue?

The individual mandate's shared responsibility penalty payment and the limitation that the older enrollees would pay only three times the premiums of younger enrollees represented a major wealth transfer from the young to the old. When the young did not cooperate as desired, the premiums for individual policies on the exchanges soared.

Professional Versus Institutional Responsibilities for Process Development and Improvement

In the past, when new findings were generated more slowly and there was little concern about cost, we relied on individual professionals to stay abreast of

the new developments and decide when and where to adopt them. We relied on continuing professional development to lead to improvements, but change often proved agonizingly slow. Many of the recommendations of management experts now call for reliance on improved learning by provider organizations on top of professional competency. Because the healthcare marketplace is highly fragmented, most provider organizations cannot undertake research and development unless it results in a product that can be patented, providing a legal monopoly. Local providers can only amortize research and development costs over their own client base, and it would take too long to recoup their investment. Alternatives are to turn to the federal government or to vendors that support multiple providers. However, vendors are not disinterested parties. Various provisions of the ACA attempt to deal with this by setting up new institutes and boards to stimulate innovation, but these provisions seemed to lack the support of a broad consensus and have proved difficult to maintain and fund in the face of determined lobbying efforts.

The following new agencies and boards were included under the ACA:

- Independent Payment Advisory Board
- Center for Medicare & Medicaid Innovation
- National Prevention, Health Promotion, and Public Health Council
- National Health Care Workforce Commission
- Interagency Working Group on Health Care Quality
- Patient-Centered Outcomes Research Institute

The role of the Medicaid and CHIP Payment and Access Commission (MACPAC) was also expanded to parallel the functions of the Independent Payment Advisory Board concerning Medicare.

Areas of research and development where government already plays some role in the United States include:

- Basic science
- Clinical applications
- Testing for efficacy and safety

► **Impact of Societal Values on Policy Decisions**

Healthcare policy does not develop in a vacuum. It is profoundly influenced by value-driven issues that cut across the entire U.S. policy landscape. These include debates over the role of free versus managed market mechanisms and pro-life and right-to-die issues. The battle over embryonic stem cell research is a case in point. The idea of using cells from fertilized eggs that were going to be thrown out anyway might not have attracted attention if it were not for the continuing debate about abortion, much of which turns on the definition of when life begins. If “life” begins at birth, then opposition to early abortion—and the objection to using embryonic stem cells—is greatly weakened. If “life” begins with the union of the egg and sperm, then there is a logic to protecting embryos. Strong clashes among value frameworks affect other healthcare issues such as physician-assisted suicide and criminal executions, contraception for minors, morning-after pills, concerns about informed consent, and direct-to-consumer pharmaceutical marketing.

BOX 1-8 Selected Objectives for *Healthy People 2020*

- Increase the proportion of population of all ages with a specific source of ongoing care from 86.4% in 2008 to 95.0%.
- Reduce U.S.-acquired measles cases from 115 in 2008 to 30.
- Reduce age-adjusted deaths from HIV infections in those over 13 from 3.7 per 100,000 population in 2007 to 3.3.
- Reduce age-adjusted smoking rates by persons over 18 from 20.6 in 2008 to 12.0.
- Reduce the rate of infant deaths in the first year of life from 6.7 per 1,000 live births in 2008 to 6.0.
- Increase the age-adjusted rate of adults 18 and older whose hypertension is under control from 43.7% in 2005–2008 to 61.2%.
- Reduce age-adjusted coronary artery disease deaths from 126 per 100,000 population in 2007 to 100.8.
- Increase the proportion of adults who engage in aerobic physical activity of at least moderate intensity at least 150 minutes/week or 75 minutes of vigorous intensity or an equivalent combination in 2008 from 43.5% to 47.9%.
- Decrease the proportion of adults 18 and over who experienced a major depressive episode in 2008 from 6.8% to 6.1%.
- Decrease age-adjusted death rate due to fatal injuries from 59.2 per 100,000 in 2008 to 53.3.
- Increase the age-adjusted rate of adults receiving colorectal cancer screening according to most recent guidelines from 52.1% in 2008 to 70.1%.
- Increase the proportion of cancer survivors living 5 years or more after diagnosis from 66.2% in 2007 to 72.8%.

Reproduced from 2020 Topics and objectives—objectives A–Z. Retrieved October 15, 2017, from www.healthypeople.gov/2020/topics-objectives2020/

► Politicization of Science and Limiting Role of Expertise

Values have always affected the choice of metrics to use in policy analysis. The U.S. Congress placed constraints on the use of quality-adjusted life years for key resource allocation decisions in government programs. Early in the Trump administration, Environmental Protection Agency staff were prevented from attending meetings on climate change. In December 2017, the *Washington Post* reported that the director of the CDC had ordered staff to restrain from using the words: evidence-based or science-based in budget documents. The following was reported to be a suggested substitute: “CDC bases its recommendations on science in consideration with community standards and wishes,” (Sun & Eilperin, 2017).

► Conclusion

Health policy analysis involves using the many disciplinary skills of team members to reach a recommendation and suggest how best to implement and evaluate it. The role of a policy analyst was discussed and an example given. Several examples were given of the impact of social issues on health policy analysis.



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CHAPTER 2

American Exceptionalism— Structural and Conceptual

Healthcare reform in the United States has been a roller coaster ride for 50 years or more. Presidents Truman and Nixon made proposals for reform that were not really acted on. Lyndon Johnson's 1965 Great Society legislation establishing Medicare and Medicaid set the stage for the current pulling and hauling. Since the Clinton "health security" proposal in 1993, we have seen many attempts by Congress and successive administrations to address the mounting problems of access, coverage, and cost.

Even the George W. Bush administration pushed a significant change in the delivery system based on subsidized interoperable digital health records. Then the Affordable Care Act (ACA) became the law of the land. After the 2016 election, Congress and the Trump administration were committed to unwinding it and moving back toward the prior system, but promising to hold down the proportion of the population without health insurance. Their efforts to repeal and replace it in 2017 failed because of splits within their party in the Senate.

The roots of the turmoil have been structural, conceptual, historical, and political.

The emergence of a generalized concern about the issues of healthcare policy has paralleled:

- The ability of the medical system to treat more and more diseases and syndromes
- Higher than average rates of inflation in professional service sectors, including higher education and health care
- Freeing up the health sector to marketing and advertising, triggering rising expectations and demand
- An aging population with fewer full-time workers to support Medicare and Medicaid
- The obvious successes of other national health systems in providing as much or more with less

This, coupled with a period of declining personal income growth, has put many families in a bind and contributed to a feeling of being left behind. Health

care with its growing share (18% in 2015) of our economy and employment cannot be ignored.

This chapter deals with a number of structural and conceptual issues. Subsequent chapters deal with historical and political ones.

► Key Structural Issues

There are a number of structural issues virtually unique to the United States' situation:

- Constitutional guarantees of states' rights
- The bureaucratic dispersal of healthcare programs
- The separation of healthcare demand and health professions supply
- Uneasy balance between public and private financing
- Conflicts between consolidation and market competition

Constitutional Guarantee of States' Rights

If everyone is in charge, then no one is in charge. Health policy is problematic throughout the world, but it is particularly challenging in the United States where there is no consensus about which government agency or social institution, if any, has an accepted, legitimate role of developing or implementing national health policy. The U.S. Constitution is silent on the subject of health and health care. Although its preamble promises "to promote the general Welfare," the Tenth Amendment states, "The powers not delegated to the United States by the Constitution, nor prohibited by it to the States, are reserved to the States respectively, or to the people." The omission of health, however, cannot be attributed solely to the framers' intent, despite the presence of three physicians at the Constitutional Convention. They lived in a world of "evil humours" where one visited "barbers and surgeons."

Constitutional issues almost derailed the ACA before the Supreme Court. In 2012, the Court, by a 5–4 vote, upheld the constitutionality of the "individual mandate" provisions that require most individuals to carry basic health insurance or pay a penalty on their income tax return. At the same time, it overturned a provision requiring states to expand Medicaid access, ruling that it was unconstitutional because it coerced the states to provide coverage.

The Trump administration and the Congress opposed the federal insurance exchanges under the ACA and sought to block-grant Medicaid to the states. The latter has always been a state program with close federal supervision and a lot of experimentation going on under Section 1115 and 1123 waivers. Repeal and replace legislation would have given even more flexibility to the states, including a possible work or service requirement for able-bodied individuals, but would have sharply reduced the federal portion of Medicaid funding over time.

Bureaucratic Dispersal of Healthcare Programs

Given the highly visible opposition of organized medicine to Lyndon Johnson's War on Poverty, the health components of these new programs were housed outside of the U.S. Public Health Service. For example, the Office of Economic Opportunity

started neighborhood health centers, and its Head Start program provided health assessment and healthcare components for children.

When the Johnson administration finally secured passage of the Social Security Amendments of 1965, it accommodated American Medical Association (AMA) concerns by offering three separate programs: (1) Medicare Part A, which provided hospital coverage for most older persons, mirroring the existing insurance relationships; (2) Medicare Part B, a voluntary supplementary medical insurance program that paid doctors directly on a fee-for-service basis; and (3) Medicaid, which expanded the Kerr-Mills welfare programs under the Welfare Administration with its traditional federal-state partnership. Meanwhile, the Department of Defense, the Veterans Administration and the Indian Health Service continued to operate separately. Furthermore, the Employee Retirement Income Security Act of 1974 included a provision called the deemer clause that exempted self-insured employer health plans from the usual state regulation of health plans.

Public health services are split between the federal, state, and local government levels. Research and oversight are at the federal level, including the Centers for Disease Control and Prevention (CDC), National Institutes of Health (NIH), and Food and Drug Administration (FDA), while state laws govern public health practice and professional licensure, and local health departments deliver public health services and sometime care.

Separation of Healthcare Demand and Health Professions Supply

In most countries, a Ministry of Education or the like determines how many doctors and other health professionals to train annually and allocates resources to educational institutions for that task. There is the opportunity to coordinate that process with the perceived demand for health services. Admittedly, the linkage is often not what it could be due to bureaucratic and interpersonal issues and budget constraints. U.S. educational institutions make their capacity decisions quite independently. Congress and the states step in occasionally to support expansion of programs to cover acute shortages or to promote new approaches, but the supply is left pretty much to the free market.

There is little or no impetus for reducing the output of specific types of providers, and this upward bias contributes to the sector's cost inflation. In fact, some efforts aimed at limiting numbers of specialists trained and credentialed have given rise to suits charging a conspiracy in restraint of trade.

Uneasy Balance Between Public and Private Financing

The U.S. system is usually described as a private healthcare system. That is true in the sense that most providers are employed in the private sector. But financing of health care is split pretty evenly between the governmental and private sectors. This has produced the unique situation that the U.S. "private" system spends more public money per capita on health care than most other developed countries spend in private and public funds combined (OECD, 2017).

Conflicts Between Consolidation and Market Competition

Consolidation of healthcare organizations continues at a rapid pace. This has been driven by numerous factors:

- Needs for better coordinated care
- Economies of scale
- Enhanced bargaining power
- Weak antitrust legislation and enforcement

Poor coordination of care is known to be a major factor driving up costs and reducing quality. It has motivated case management, medical homes, bundled payments, and investments in information technology. But one effective way to achieve coordination and align incentives is to have a single manager over the entire process. The decades of success of financially integrated systems, such as Kaiser Permanente and Intermountain Health, are a case in point.

Economies of scale are a frequent justification for consolidation—touted benefits include the elimination of duplicated managerial and staff positions, improved access to capital, and reductions in redundant services. So far, it is hard to notice the cost impact of these effects in the face of higher executive salaries and increased prices.

The most visible impact of consolidation is increased pricing power (Dafny & Lee, 2015; Scheffler & Arnold, 2017). Consolidation has left midsized cities and even some large ones with only two or three hospitals, which insurers need to make their networks competitive. This applies elsewhere in health care (Fowler, Grabowski, Gambrel, Huskamp, & Stevenson, 2017). The most egregious example of pricing power has been the large and rapid price increases for low-volume drugs when a single supplier is left in the marketplace. Patents and regulations can enable legal monopolies and oligopolies. Havighurst (2017) observes that monopolies and oligopolies are especially pernicious when they are paid through insurance. Gaynor and Town (2012) estimate price increases with hospital mergers in markets with few hospitals at up to 20% and found little evidence of quality improvement.

Weak healthcare antitrust laws make remedies hard to find. They are further weakened by state and local resistance to enforcement and by political support for local institutions. Underfunding for enforcement and the large number of potential cases is also a problem (Greaney, 2017). Effective judicial remedies are hard to develop given existing regional flows of people and information, as well as regulatory and capital investment barriers to entry.

► Key Conceptual Issues

There are many differences in how individuals and groups see the possibilities for improving the system. Among the key conceptual issues are:

- The extent to which health care is an entitlement
- Contending visions for controlling quality and cost
- Ownership and enhancement of intellectual capital

Extent to Which Health Care Is an Entitlement

The position that everyone should have coverage regardless of income has steadily gained traction in American society. Paradoxically, no assumption gets more attention than whether health care is a right. A yes or no answer gets us nowhere

(Gawande, 2017). Is one talking about antibiotics for a serious illness, cosmetic surgery, or *in vitro* fertilization? Again, there is a need to try to define what one is arguing about, rather than repeating assertions based on undefined assumptions.

But coverage to what extent? That debate has been evident in policy decisions about access and coverages before, during, and after the ACA. The ACA included a provision requiring plans subsidized on the exchanges to meet “minimum essential benefit” requirements. Qualifying plans under the ACA had to offer a basic package that included preventive services, maternity benefits, and behavioral health care. This concept of a “minimum essential benefit” was a lightning rod in the bill. It led to the cancellation of some less expensive existing policies and upset individuals who believed these cancellations broke election promises that they could keep their current insurance. Proponents of ACA repeal have talked in terms of assuring that all people can get the benefits they can afford. That has been linked to tax credits that seem quite small compared with the amount necessary to purchase a package like those meeting ACA requirements. Should we allow and subsidize plans that fail to provide extensive support in case of a major adverse event—the purpose of insurance in most other settings?

The 2017 Republican American Health Care Act submitted to the House and the subsequent Senate version included major changes to Medicaid, essentially limiting the federal government’s liability and increasing both the liability and flexibility of state programs. One change insisted on by the conservative caucus was to allow a work or service requirement for able-bodied recipients. This, however, lost votes from more centrist Republicans.

McClure, Einthoven, and McDonald (2017) argue that the right vs. left arguments over universal health coverage miss the point. They argue for universal coverage not as a right, but as an important public investment. “If you want to outcompete a billion Chinese, you better have a workforce not only better-educated, but healthier than anyone else.” They back up their arguments by making reference to the “general welfare” clause of the Constitution. They go on to suggest that if the system is treated as a public investment, incentives should be realigned to reduce costs by 15%–20%.

Gawande (2017) observes that compared with Medicaid, support for Medicare is not as conflicted. A long-term set of contractual rules have bound the public to Medicare. He cites the first attempt at universal coverage in the United States, the Vaccine Act of 1813, which provided cowpox-based smallpox vaccine from a private supplier to everyone free of charge. However, that law was repealed in 1822 after a batch infected with smallpox caused an outbreak in Tarboro, NC. But gradually we have achieved near-universal coverage through cooperative activities between the federal government and state and local governments.

What we agree on, broadly, is that the rules should apply to everyone. But we’ve yet to put this moral principle into practice. The challenge for any plan is to avoid the political perils of a big, overnight switch that could leave many people with higher costs and lower benefits.

Gawande (2017, p. 54)

Contending Visions for Controlling Quality and Cost

There are a number of ways of looking at our healthcare system and how it might be organized and controlled. **BOX 2-1** suggests how someone happening onto our health policy landscape might end up quite confused.

BOX 2-1 Arriving at Healthville for the First Time

In late 2017, a passenger airliner was on the landing approach to the new Healthville airport. This was the pilots' first visit here and they seemed confused.

Pilot to Healthville Control Tower: What approach are we on? We are getting three sets of signals at once!

Control Tower to Pilot: That is the way we see things around here. You will just have to get used to it.

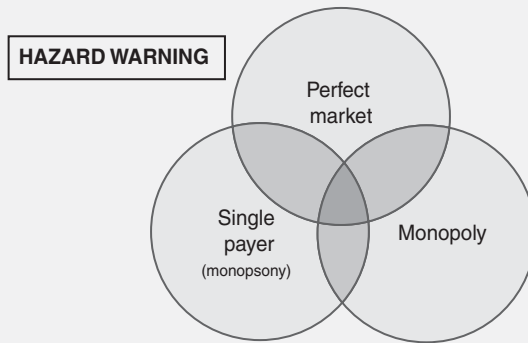
Pilot: What do I do next?

Tower: Circle and consult the Manual.

Pilot: Under which topic?

Tower: Lenses

They circle while the copilot frantically searches and finds following page under Lenses.



MANUAL SECTION 2.1.23.B.I

They look down and select Runway CL3 which looks most familiar. They land smoothly and begin to taxi.

Pilot to Ground Control: Taxiing on CL3, awaiting Ramp Instructions.
No response.

Pilot to Tower: I cannot rouse Ground Control!

Tower: There isn't any Ground Control here!

Pilot: Need ramp to gate instructions.

Tower: There are three clusters. Look around. They crisscross each other everywhere and most any route will get you there. Just feel your way in. Finally. At the gate.

Copilot to Pilot: That was a great job navigating through the short-term parking lot.

Pilot: Thanks. It was OK given that we're paid by time and mileage. But next time we need a roadmap to avoid the parking lots.

Conflicts between different visions of how the health system should operate have dominated U.S. healthcare policy making for almost a century.

The PPACA is in some respects a conglomeration of some of the pieces of past proposals for major healthcare reform. The provision of health insurance coverage through private insurers instead of directly through the government, employer mandated health insurance, the creation of state, regional or national clearinghouses for insurance, federal subsidies for low income individuals, and “guaranteed eligibility” have all been proposed in previous attempts at reform.

Taylor (2014)

The three distracting lenses represent ideological abstractions.

Three related but significantly different lenses have emerged and been more or less dominant at different times. Yet there has not been a single dominant viewpoint since the 1960s, and all three contending approaches have remained on the table. We have labelled them: Corporate Lite, Managed Competition, and Consumer-Driven Care.

Another possibility, a provider monopoly, has been ruled out by our legal system, even though there are current problems with antitrust enforcement in the healthcare sector. The extreme monopsony position, the *single payer*, can be represented by the original version of the United Kingdom’s National Health Service. This model is not currently a realistic contender for adoption in the United States.

Corporate Lite

Corporate Lite dominates the status quo as health care industrializes and consolidates. Many of the compromises in the ACA were made in deference to the lobbying power of established payers and providers to get them on board. The 2017 attempts to repeal and replace Obamacare failed to recruit industry backing and were heavily opposed.

This oligopolistic competition model involves a market dominated by a few large sellers and is a characteristic of many U.S. industrial sectors including health care. Usually, three or four major sources for goods or services exist, and those sources together control at least 40% of the market. In health care, two, three, or four providers often control state or local markets in the absence of a national market. In health insurance, there is a near monopoly in many markets. National oligopolies appear to exist in many other health-related markets, such as Medicare managed care, replacement joints, imaging equipment, and pharmaceuticals distribution. Concentration in hospital markets has been increasing sufficiently to become a concern of the Federal Trade Commission (FTC). Although available studies of hospital concentration can yield conflicting findings (Gaynor & Town, 2012), there can be little doubt that concentration increases pricing power. In many state markets, the same is true of health insurance providers. It is widely believed that market power has shifted in recent years from insurers to providers, especially larger hospitals and their associated group practices.

Examples of the approaches often associated with each lens are include in **TABLE 2-1**.

Many of the policy approaches associated with Corporate Lite maintain the status quo (employment-based insurance, current pricing methods, self-regulation).

TABLE 2-1 Approaches Associated with the Lenses

What Is Managed Competition ?	What Is Consumer-Driven Care ?	What Is Corporate Lite ?
■ “Sponsor” to manage system ■ Evidence-based medicine ■ Competitive bidding among health plans ■ Penalties for readmissions, HACs, etc. ■ Administered prices ■ Reference pricing ■ Value purchasing (bundling) ■ Certificate of Need	■ High deductible insurance with tax-sheltered gain-sharing ■ Consumer ratings ■ Quality transparency ■ Price transparency ■ Multistate insurance ■ Competition ■ Insurance exchanges ■ Premium tax credits replacing employer payments ■ Individual mandate	■ Employed physicians ■ Self-regulation or “get the incentives right” ■ Employer-based insurance ■ Organizational learning and peer enforced change ■ Competition on reputation and amenities ■ Reasonable & customary + Relative value scale (the physician’s Red Book) ■ Consolidation/market dominance ■ Lip-service about disruptive innovation

Consolidation has led to many more employed physicians and to attention to institutional learning and control of practice methods.

Managed Competition

Managed competition implies that there are multiple suppliers but that the market is strongly influenced by a powerful buyer or “sponsor,” usually a government creation (Einhoven, 1993). When the sponsor is a government, this is sometimes referred to as administered competition. It may involve universal coverage, a single disbursement agent, and/or a single underwriter. Somewhere in the mix is a “sponsor” with sufficient clout to keep the system in line. Currently, price competition is avoided, but increased share of market is prized. Access is gladly increased and competition is based on reputation, amenities, and availability, but seldom on price.

The size of governments’ stake in financing health care has been one driver of this approach. Health policy makers have also observed that many other developed countries have resolved their access issues by having tightly managed universal health coverage. Yet another factor is the belief that there are attributes of health care that make it an imperfect market and that a managed approach is necessary to overcome the results of market failure (Arrow, 1963).

Those in charge of payment want rational decision-making. So, measures are put in place to restrain the arms race among hospitals and to penalize poor care such as unnecessary readmissions and hospital-acquired conditions. Proponents of managed competition act to align incentives to encourage price competition among all

providers and insurers. Given sufficient market power, these managers might want to have clinical decisions made on the basis of evidence, support value purchasing, promote reference pricing, and ultimately set prices.

Consumer-Driven Care

Consumer-driven health care is more of a free-market approach that assumes consumers' choices will help shape the market if consumers have accurate and adequate information and are not subject to perverse incentives. This approach is driven by a free-market ideology reinforced by American individualism. We want our choices among doctors, hospitals and insurers.

Perfect (free-market) competition assumes the following conditions:

- There are large numbers of buyers and sellers so that no one controls prices.
- All buyers and sellers have complete and accurate information about the quality, availability, and prices of goods.
- All products have available perfect substitutes.
- All buyers and sellers are free to enter or leave the market at will.

Free-market ideology has been playing out in health care even in the absence of a true free market. It goes by several names—*consumer-driven health care* is one example, as is *market-driven health care*. Supporters of this approach call for much greater transparency and more consumer choice and responsibility. It has been implemented, in part, through innovations such as health savings accounts (HSAs) and private options for Medicare. Insurance exchanges are another manifestation of this approach and were initially suggested by conservative think tanks that support a free-market approach.

Early attempts at managed care ran into strong resistance. A more recent approach has been the movement to empower consumers to be much more involved in making choices about their care. Proponents of consumer-driven care typically support high-deductible health insurance with HSAs. They were also early supporters of the individual mandate and insurance exchanges, especially ones that allowed multistate insurance competition. Consumer choice is supported by efforts to overcome information asymmetry and make available data for comparisons of consumer experience, outcomes, and cost.

More recently, strong proponents of market solutions have revived the concept of nonrefundable premium tax credits. For the poor, they would allow individualized purchase decisions. They would require high-risk pools behind them to assure affordability. As a strategy for universal coverage, the credits would replace the employer contribution, partly financing the credit by eliminating the tax deductibility of employer contributions. This strategy would do away with a standardized coverage and allow the individual to pick and choose from a wide variety of plans. The primary objective would be to unleash competition to reduce costs. This, however, would require enhanced anti-trust enforcement and having the credit rising at a slower pace than healthcare costs.

Living with All Three at Once

So how do these three lenses relate to the confusing picture of healthcare reform?

FIGURE 2-1 provides a very rudimentary roadmap. In it, all three relevant visions have

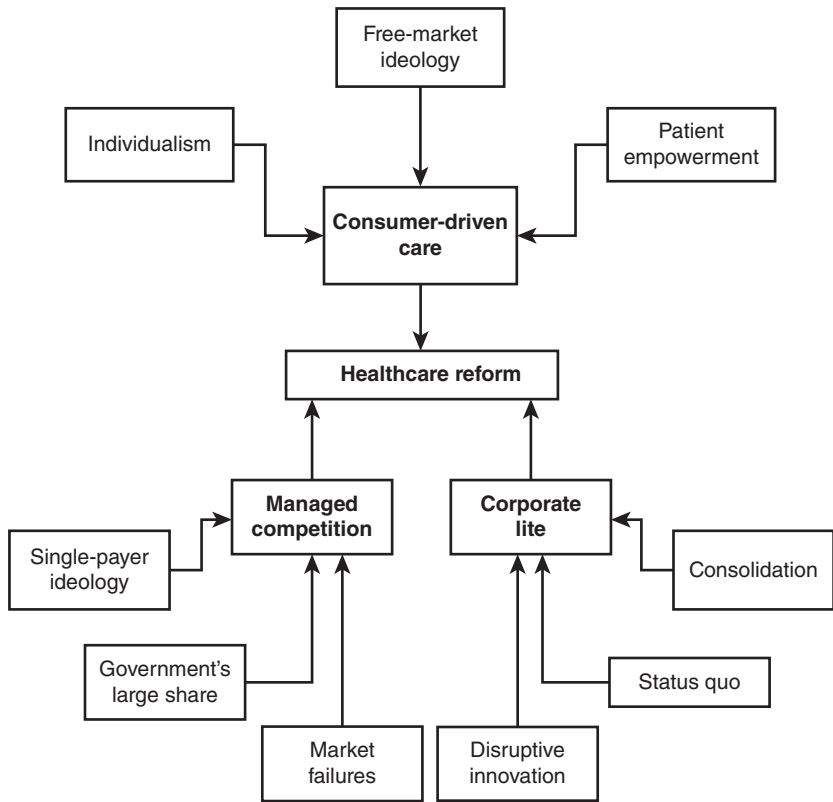


FIGURE 2-1 A roadmap of major influences on healthcare reform.

had a place in recent healthcare reform attempts. They are influenced by many factors, a few of which appear in the drawing in Figure 2-1.

FIGURE 2-2 represents our current view of the influence distribution between the three relevant lenses.

► Industrialization and Corporate Lite

The current status quo, dominated by Corporate Lite, cannot be traced to a specific policy or period. Medicine before World War II has been described as a craft/guild system, which implies that medicine was primarily an art lacking decision rules that could be communicated effectively (tacit knowledge) (Ferdows, 2006). With more and more scientific and/or codified knowledge, it was possible to differentiate between cases and processes. Simple industrial activities can be turned into mass production systems that repeat the same process over and over. If the knowledge is still pretty much an art but the task simple, the work can be delegated to less experienced or less trained personnel as in the *apprentice system*, in which much of the simpler work was delegated to others but the master craftsman maintained control and handled the trickiest parts (or conducted rounds in the teaching hospital).

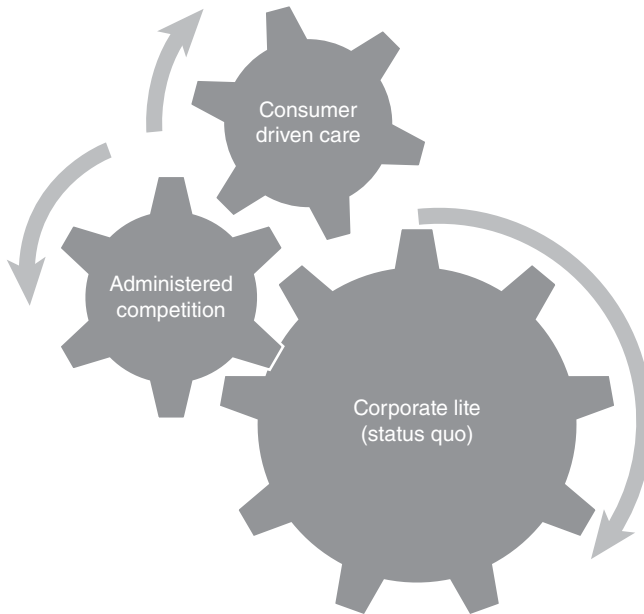


FIGURE 2-2 Current distribution of influence.

Many healthcare tasks can be done by more than one level of healthcare worker. For example, midwives, family physicians, and obstetricians can deliver babies. The practice of midwifery nearly disappeared in the United States but is now undergoing a resurgence. Nurse practitioners and physician assistants now are the first level of care for many patient encounters. In many psychiatric practices, the psychiatrist handles the patient's medications but delegates most other care activities to psychologists, social workers, or other counselors. Pharmacies now use pharmacy technicians as well as pharmacists. Dental practices have their own dental hygienists and technicians working in parallel with dentists. Primary care physicians perform procedures once limited to specialists. The key to further substitution is whether the alternative workers are qualified for the problem at hand and whether their unit cost is less. Most substitutions were initially proposed to overcome a shortage of personnel in one area, but after the experiment worked, more and more organizations have implemented it to increase access and reduce cost.

Mass production exists in cataract surgery centers and other “centers of excellence,” but in general there is widespread desire to avoid mass production of medical services. That desire is legitimate given the inherent variability in patient anatomy, physiology, and psychological needs and preferences. Health care differs from industrial production in the sense that patients present with both simple and complex problems (multisystem problems or comorbidities). Problems that have a clearly optimal treatment regimen and those for which medical knowledge is limited can appear simultaneously in the same individual.

Ownership and Enhancement of Intellectual Capital

As work is industrialized, work methods are specified by the organization rather than the individual artisan. In health care, we have historically assumed that intellectual capital resides with the professional. But as competition increasingly depends on the implementation of evidence-based practices by an institution and on rapid dissemination and adoption by practitioners, organizational rather than professional learning comes into focus. That raises new questions about management–provider conflicts (often called *suits versus coats*), the nature of continuing graduate medical education, and access to clinical records and research outputs.

The drivers of industrialization in health care have been the expansion of the science base of medicine and the codification of product definitions and process specifications. An example of the trend toward codification by medical institutions and professions is the effort by the Institute of Medicine to support the “learning health care system.” Professionals must be prepared to take leadership in issues around developing, disseminating, and compensating for intellectual capital or they will lose even more autonomy.

► Conclusion

Health policy analysts must be aware of the structural history of health care in the United States and the conflicting visions of how the system should work. The analyst cannot expect consistency from individuals or groups, but he or she can benefit from recognizing where people are coming from in order to deal with both facts and fantasies.

Three visions played out in portions of the ACA, producing Taylor’s “conglomeration.” Starr (2011) describes the process leading up to the passage of the ACA as one of reaching a compromise between managed competition and consumer-driven health care, but the legislation was crafted to be “minimally invasive” to avoid public counterattacks from established corporate interests such as hospitals, pharmaceutical companies, and the insurance industry. It is likely that they will ultimately worm their way into any Republican replacement as well.



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CHAPTER 3

American Exceptionalism— Historical and Political

This chapter focuses on the parallel evolution of two separate, but linked, U.S. health systems—one for delivering medical care and one for financing it. The health policy analyst needs to be able to answer these questions about these systems: Where are we and how did we get there? and What are the societal issues we are wrestling with?

► A Chronology

Centuries ago, medical care was a religious calling, not a scientific field. The term *hospice* was more representative of healthcare institutions than *hospital*. Gradually, health care has become a calling *and* an industry. Well into the 20th century, U.S. physicians often took whatever people could pay. Teaching institutions provided free care in return for allowing learners to work on those who could not pay.

A history of healthcare policy-making in the United States could well start in 1791 with the passage of the Bill of Rights. The Tenth Amendment to the U.S. Constitution declares that those powers not expressly given to the federal government belong to state and local governments. Health and education were not expressly given to the federal government. In 1910, the Supreme Court ruled that a federal workers' compensation system was unconstitutional. Each state then established its own system. Hadler (2013) cites this as the regulatory template for U.S. health insurance programs.

The Health “Insurance” Approach: Moving from Provider Monopoly Toward Provider-Insurer-Oligopolies

Health insurance systems in the United States were implemented during the Great Depression to stabilize cash flows of providers. The concept existed in Europe much

earlier (Starr, 1982). Many of the early insurance efforts in the United States evolved into the nonprofit Blue Cross/Blue Shield organizations.

Dr. Justin Ford Kimball, the administrator of Baylor Hospital in Dallas, is often credited with starting the U.S. medical insurance movement in 1929. He conceived of the idea of collecting “insurance premiums” in advance and guaranteeing the hospital’s service to members’ subscribing groups. Furthermore, he found a way to involve employers in the administration of the plan, thereby reducing expenses associated with marketing and enrollment. The first employer to work with Baylor Hospital was the Dallas school district, which enrolled schoolteachers and collected the biweekly premium of 50 cents (Richmond & Fein, 2005).

About the same time, prepaid group practices began in Oklahoma. They too were established to provide stable cash flows, but remained a relatively minor factor for decades because of medical society opposition. For example, Group Health Association of Washington, DC, a prepaid group practice, was established in 1937, but it faced strong opposition. In 1943, the Supreme Court (*AMA v. U.S.*, 1943), hearing a case brought by the Federal Trade Commission, upheld a lower court finding that the American Medical Association (AMA) and the DC Medical Society were guilty of “a conspiracy in restraint of trade under the Sherman Anti-Trust Act” and had hindered and obstructed Group Health “in procuring and retaining on its staff qualified doctors” and “from privilege of consulting with others and using the facilities of hospitals” (Richmond & Fein, 2005, p. 34).

Expanding Participation

World War II led to the industrialization of all available nonmilitary hands, breaking the Great Depression, inducing migration from rural areas to industrial cities, increasing the power of industrial unions, and inaugurating an era of big science. It also led to an era of optimism that Americans could accomplish anything they wanted, if they worked together collectively (Strauss & Howe, 1991).

Many employers established health services to support their employees and the war effort. Some of these services evolved into prepaid group practices. Most notably, Kaiser Industries’ medical department became the Kaiser Permanente system, which opened up to outside enrollees after the war.

The federal government imposed wage and price controls during World War II. As labor became scarce and the war turned in the Allies’ favor, workers pressed for better compensation. The Office of Price Administration held the line on wage increases, but allowed improved benefits through collective bargaining. This led to a rapid expansion of health insurance among unionized industrial and government workers. This trend was also consistent with the provision of medical benefits to the vast military establishment. Health insurance costs were not yet a serious concern for corporate managers or the government.

In 1948, the National Labor Relations Board ruled that refusal to bargain over health benefits was an unfair labor practice. Collective bargaining became the basic vehicle for determining health benefits. Because union officers were elected by their membership, they sought to maximize the visibility of benefits to their rank-and-file (voting) members. They did not choose catastrophic coverage. Rather, they bargained for first-dollar coverage for their members, but allowed lifetime limitations

on benefits for those who were born with or developed catastrophic or high-cost chronic conditions. They also emphasized employment-related coverage for dependents, so young and healthy union members would see payments for services such as obstetric and pediatric care. Employers did not much care how their workers divided the contract settlements among wages, health benefits, and other offerings. Employers saw health insurance as an inconsequential component of the overall labor costs established through collective bargaining. If workers and their families already had individual health coverage, they still gained a tax advantage if the employer paid the premium directly. Blue Cross enrollments tripled between 1942 and 1946, while enrollment in commercial health insurance plans more than doubled (Becker, 1955).

Postwar Responses

Following World War II, most presidential administrations sought healthcare reforms of some sort. The Hill-Burton Act of 1946 expanded hospital facilities. President Truman suggested developing a system of universal health insurance based on the report of the President's Commission on Health Needs of the Nation; however, his proposal was opposed by entrenched interests and ignored when President Eisenhower was elected. In 1950, Congress approved a grant program to the states to pay providers for medical care for people receiving public assistance.

The 1960 Kerr-Mills Act created a program administered by the Welfare Administration and the states for "Medical Assistance to the Aged," which also covered "medically needy" older persons who did not necessarily need to qualify for public assistance. Richmond and Fein (2005) described Kerr-Mills as an attempt to stave off Medicare-type programs.

The Joint Commission on Mental Illness and Health, formed under Eisenhower, did not issue its final report until 1961 under the Kennedy administration. It led to the passage of the Mental Retardation Facilities Construction Act of 1963 and the Community Mental Health Centers Act of 1963.

Early in his term, President Johnson announced the formation of a Commission on Heart Disease, Cancer, and Stroke. Its recommendations led to the Regional Medical Programs legislation to advance training and research. Congress, however, added a provision that this work was not to interfere in any way with "patterns and methods of financing medical care, professional practice, or the administration of any existing institutions" (Richmond & Fein, 2005, p. 44).

While the Medicare debate continued, Congress passed many health measures as part of Johnson's War on Poverty. Given the highly visible opposition of organized medicine, the health components of these new programs were housed outside of the U.S. Public Health Service. For example, the Office of Economic Opportunity started neighborhood health centers, and its Head Start program provided health assessment and healthcare components for children.

When the Johnson administration finally secured passage of the Social Security Amendments of 1965, it accommodated AMA concerns by offering three separate programs: (1) Medicare Part A, which provided hospital coverage for most older persons; (2) Medicare Part B, a voluntary supplementary medical insurance program; and (3) Medicaid, which expanded the Kerr-Mills program to help with out-of-pocket expenses such as nursing home care and drugs and extended potential eligibility to families with children, the blind, and the disabled under the Welfare

Administration. Starr (2011) cites this set of programs as the beginning of the “policy trap” that haunts us today:

The key elements of the trap are a system of employer-provided insurance that conceals true costs from those who benefit from it; targeted government programs that protect groups such as the elderly and veterans, who are well organized and enjoy wide public sympathy and believe, unlike other claimants, that they have earned their benefits; and a financing system that has expanded and enriched the healthcare industry, creating powerful interests averse to change.

Starr (2011, p. 123)

There were other compromises in the Social Security Amendments. For example, at the time, hospital-based physicians were being placed on salary so that hospitals could use some of their fee revenue to cover the capital costs of their practices. The 1965 Medicare bill specifically required that anesthesiologists, radiologists, and pathologists be paid directly, not through the hospital. That law also stated, “Nothing in this title shall be construed to authorize any federal officer or employee to exercise any supervision or control over the practice of medicine.” Bodenheimer and Grumbach (2005) labeled the years 1945–1970 as those of the “provider-insurer pact” (p. 167). Starr (1982) argued that the period before 1970 was characterized by an accommodation between the insurance industry and the medical profession. He noted that it was a period in which most employed Americans were covered because union shops were dominant. “The government supported this private tax system by making employers’ contributions into an exemption from the government’s own taxes. Private voluntary insurance was neither strictly voluntary, nor strictly private, but its compulsory and public features were hardly noticeable” (p. 334). That system, however, left out the poor, the unemployed, agricultural and domestic workers, most farmers, the disabled, and older persons. The 1965 Great Society legislation addressed the needs of some of these uninsured populations.

The Great Society

When implemented in 1965, Medicare mirrored the structure of health insurance in the industrial sector, but without lifetime limitations. It did not provide adequate coverage for drugs or for long-term care (i.e., nursing homes and home health), nor did it allocate much for prevention. Many employment-based health plans paid for prescription drugs, but not for long-term care. Medicare did not cover prescription drugs until 2006.

Before 1965 academic medical centers delivered large amounts of charity care. Local volunteer physicians supervised the clinics, and patients received care at no charge or for a nominal fee in return for letting learners practice on them. Many people covered by Medicare and Medicaid had been receiving charity care, but Medicaid and Medicare paid in full for services once provided free or with income-based discounts. The new coverage also gave the urban poor a choice of institutions, which they quickly exercised. Through its association with Social Security, Medicare became viewed as an entitlement earned by years of paying into a system, and something seen as sacred as Social Security. However, all enrollees received the same

benefit regardless of their payment history. More recently Medicare Part B has begun to require tiered premium payments tied to one's prior year's income. Medicaid, although intended to be a comprehensive care plan for low-income families, ended up for the time being primarily a long-term care program, and did not cover low-income families without children. The expansion of Medicaid under the Affordable Care Act (ACA) remedied part of that. The fact that Medicaid expansion did not go forward in many states indicates the weak political clout of that population.

Rapid Expansion of Capacity

The fee-for-service payments for visits once provided for free, or nearly so, increased physicians' incomes without increasing their numbers. At the same time, availability of insurance coverage for underserved populations increased the demand for services. Academic medical centers added new, full-time staff and billed all insurers for their services to subsidize education and research. Heavy investments in medical research increased the variety, cost, and effectiveness of what providers could offer. A limited supply of resources, increased demand, and rapid technological advances led to rapid price inflation.

One policy response in the 1970s and 1980s to this increased demand was to increase the supply of resources. For example, Congress launched the Community Health Center and Migrant Health Center programs, which offered subsidized services in underserved communities. In 1970, it established the National Health Service Corps to increase provider supply in underserved areas via scholarships and loan forgiveness. Many new programs provided training for health professionals, and existing ones were expanded with financial assistance from state and federal governments.

The Private Sector Responds

At the end of World War II, the healthcare sector accounted for 4.5% of GDP. By the mid-1980s, its share was up to 11%. With the cash flows from private insurance and Medicare and Medicaid, community hospitals expanded rapidly but no longer relied on philanthropy for capital. Wall Street was happy to finance their expansion by selling bonds. Interest was considered a reimbursable cost by rate setters. Health care attracted entrepreneurs, and for-profit hospital chains, home health, and kidney dialysis grew rapidly. The medical establishment was relatively helpless. The AMA's stance against Medicare and Medicaid had cost it credibility, and its constituency was now spread out between the AMA, specialty and subspecialty societies, and the academic medical centers, each of which had its own interests.

Costs and Concerns Mount

As healthcare costs mounted and the healthcare sector accounted for a much more significant share of the economy, more and more observers began suggesting ways to control costs. One was prepaid group practices, or health maintenance organizations (HMOs), which share some of the cost risk with the employer, thereby inducing reduced costs. The successes Kaiser Permanente and similar organizations had in delivering care at a lower premium cost without evident diminution of quality drew much attention. This led the Nixon Administration to support the Health

Maintenance Organization and Resources Development Act of 1973. Although that legislation had little immediate impact, later amendments opened the way for an explosion of HMOs and other vertically integrated healthcare systems in the 1980s. In 1974, the Nixon administration proposed the Comprehensive Health Insurance Program, which sought to provide health insurance to all employees. Congress debated this and a similar measure, the Kennedy-Mills bill, but did not enact either. Richmond and Fein (2005), writing before enactment of the ACA, argued that 1974 was the closest the nation ever came to universal health insurance, and that Nixon's proposals, although eclipsed by the Watergate cover-up and his resignation, were the basis for successive calls for congressional action by presidents Carter, Clinton, and Obama.

Cost Shifting and Bloated Charges

Over time a provider institution must cover its operating and capital costs. Prices are set by marking up estimated costs or by observing what is charged by others in the same market area. When individuals or insurers pay less than the breakeven price, the institution marks up the prices charged to those it figures can pay more or that lack sufficient bargaining power.

Originally, Blue Cross organizations, which were owned by state hospital associations, were interested in a management cost-finding system that fairly allocated the full costs of services among the users of those services. Because they understood that most costs in a hospital system are (1) fixed and (2) joint, they did not attempt to find out the marginal cost of a service (*marginal cost* is the additional cost of producing one additional unit of a product). They established an estimated average direct cost for each unit of service (e.g., bed day, laboratory test, operating room hour, X-ray) and then allocated the overhead costs based on the number of units consumed by the payer's enrollees. The largest expense in the institution, nursing time, was treated as an overhead and not allocated to the individual patient or diagnosis. If the patients in a health plan used a quarter of the X-rays produced by the radiology department, the plan paid a quarter of the full costs of that department (including allocated overheads). If hospitals and other healthcare institutions offered discounts, they tended to favor the Blues, not the other insurers, and certainly not direct-paying patients. This resulted in what is called *cost shifting*.

As more patient care costs went uncovered under insurance contracts, hospitals added the cost of this uncompensated care to the overhead rate and increased charges accordingly. It was easy to manipulate charges to mark up costs and either make a profit or provide deeper discounts to preferred customers. First the Blues and then the federal government exerted pressure on providers, obtaining substantial discounts in return for their business. This shifted the costs of uncompensated care to private insurers and the uninsured. Reinhardt (2006, p. 64) observed, "What prevailing distributive ethic in U.S. society, for example, would dictate that uninsured patients be billed the highest prices for hospital care and then be hounded, often mercilessly, by bill collectors?"

Cost Shifting Hits Private Plans

Commercial insurance companies had contracts that discounted the charges. The Blues and large HMOs enjoyed even bigger discounts, and the federal government

got the biggest discount because it demanded the lowest rate allowed to any customer. Because of the inflated charge figures posted to most bills, the public assumed that the actual costs of care were a great deal higher than they really were, and that insurers were picking up a higher proportion of the cost. Real transfer prices for medical services were kept under wraps. This made deductibles and co-payments appear to be a much smaller proportion of actual costs than they really were. Under pressure from the public for greater transparency, this has gradually changed. The public now sees more of what is actually paid and by whom, but real transparency is still lacking. **TABLE 3-1** shows Medicare charges billed by hospitals and payments made by Medicare and the patients to Miami-area hospitals for a specific procedure type. Making such data available is part of the federal government's efforts to promote price transparency. The payments are the full payments and include added payments such as the disproportionate share subsidy for safety net hospitals.

The financial reports of healthcare institutions offer a picture of the size of these discounts. Many institutions book their full charges as revenue and then deduct trade discounts under discounts and "allowances" and reflect charity care costs under bad debt written off and under "uncompensated care." Tomkins, Altman, and Eilat (2006) reported that the ratio of gross revenue (charges) to net revenue (payments received) had grown from 1.1 to 2.6 over a 25-year period. In 2009 markups in Atlanta hospitals ranged from 157% to 702%, and the average had moved from 124% to 319% over the preceding 10 years (Pell, 2011). Even then, according to the same article, an Atlanta hospital offered an uninsured individual a 40% cash discount. In fact, one for-profit hospital with a markup of 702% reported that no patients paid the full markup.

Currently, every hospital has an official price list called a *chargemaster*, which may have as many as 20,000 items. These are the charges that the patient usually sees. Terms are not standardized, and some items are really bundles of services, so patients still have trouble comparing prices between institutions (Brill, 2013). In California hospitals, reported charges for the same procedure at one hospital might be four times that of another. Reinhardt (2006) argued that pricing practices would have to change radically if patients were to be able to make rational buying decisions. He seemed to support the recommendation of Porter and Teisberg (2006) that hospitals post one set of bundled prices per disease entity and charge the same to everyone. However, Altman, Shactman, and Eilat (2006) wondered whether transparent pricing and customer sensitivity to pricing might destabilize the hospital sector, bringing average prices down and forcing some into bankruptcy. The ACA provides for a number of demonstrations of bundled pricing.

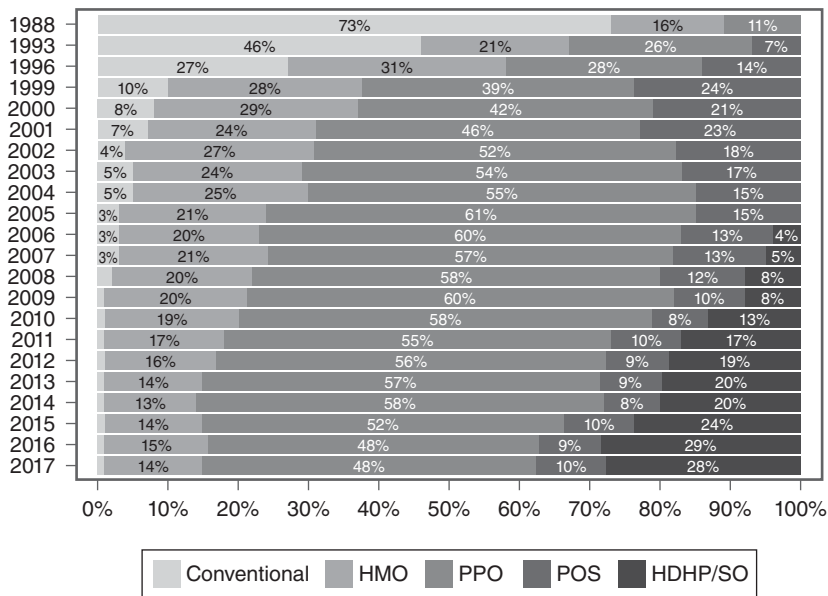
Responding to Cost Shifting

As employers and private insurers became increasingly aware of the effects of cost shifting, they adopted a number of measures to counter it and combat the overall inflation in the costs of care. As early as the 1970s, employers demanded that insurance companies begin to control premiums (Mayer & Mayer, 1985; Starr, 1982). These measures fit under the general heading of *managed care*. Most of these measures spread slowly until the 1980s. Employers moved away from contracts that accepted provider-established fees from any provider, and instead

TABLE 3-1 FY 2015 Medicare Charges and Total Payments at Selected Miami-Area Hospitals for DRG 470 (Major Joint Replacement or Reattachment of Lower Extremity w/o MCC)

Hospital	City/Suburb	Number Performed	Average Charge	Average Total Payment
Kendall Regional Hospital	Miami	62	\$102,206	\$16,444
Aventura Hospital & MC	Aventura	68	\$97,128	\$13,067
Homestead Hospital	Homestead	29	\$91,749	\$17,777
Palmetto General Hospital	Hialeah	53	\$90,937	\$16,362
South Miami Hospital	South Miami	67	\$87,806	\$16,656
Coral Gables Hospital	Coral Gables	29	\$85,150	\$16,341
Baptist Hospital of Miami	Miami	267	\$79,573	\$16,075
Doctors Hospital	Coral Gables	163	\$79,097	\$15,367
University of Miami Hospital	Miami	90	\$77,834	\$16,001
Memorial Regional Hospital	Hollywood	54	\$67,091	\$15,157
Memorial Regional West	Pembroke Pines	73	\$65,809	\$14,551
Memorial Hospital Miramar	Miramar	30	\$64,970	\$15,689
Memorial Hospital Pembroke	Pembroke Pines	23	\$61,594	\$13,587
Jackson Memorial Hospital	Miami	44	\$56,908	\$25,361
Mt. Sinai Medical Center	Miami Beach	97	\$59,176	\$16,871
Larkin Community Hospital	South Miami	172	\$52,984	\$21,657
Cleveland Clinic Hospital	Weston	291	\$45,813	\$11,969

Reproduced from Inpatient Prospective Payment System (IPPS) Provider Summary for the Top 100 Diagnosis-Related Groups (DRG). Inpatient Charge Data. Retrieved March 7, 2014–October 23, 2017, from www.cms.gov/Research-Statistics-Data-and-Systems/Statistics-Trends-and-Reports/Medicare-Provider-Charge-Data/Inpatient2015.html; <https://data.cms.gov/Medicare/Inpatient-Prospective-Payment-System-IPPS-Provider/97k6-zzx3>



NOTE: Information was not obtained for POS plans in 1988 or for HDHP/SO plans until 2006. A portion of the change in plan type enrollment for 2005 is likely attributable to incorporating more recent Census Bureau estimates of the number of state and local government workers and removing federal workers from the weights. See the survey design and methods section from the 2005 Kaiser/HRET survey of employer-sponsored health benefits for additional information.

FIGURE 3-1 Distribution of health plan enrollment for covered workers by plan type, 1988–2017.

Reproduced from Employer Health Benefits 2013–2017 Annual Survey—Chartpack. The Henry J. Kaiser Family Foundation and Health Research & Educational Trust.

signed up with HMOs. **FIGURE 3-1** illustrates the roughly 60% decline in market share for traditional indemnity.

The HMO/POS (health maintenance organization plus point-of-service) and PPO (preferred provider organization) plans appeared much better able to control healthcare costs by exacting their own discounts and by constraining what patients and providers would be able to do. The big shift, however, was to high deductible health plans and health savings accounts. The Blues began to shed their nonprofit identities and their focus on being community-based cooperative organizations as they competed with the newer for-profit insurers. The concept of the HMO was no longer a prepaid group practice. It had become an organization that managed the insurance risk and the delivery of care either directly or through a designated provider network. HMOs (for-profit and nonprofit) continued to negotiate with all types of providers for deeper and deeper discounts.

► The Current “Era” Emerges

Fox (2001) described three eras of managed care:

- Pre-1970, the early years
- 1970–1985, the adolescent years
- 1985–2001, the years when managed care came of age

Richmond and Fein (2005) described the period 1965–1985 as a time of emerging tensions between regulation and market forces and called the period after 1985 the “Entrepreneurial Revolution.” Bodenheimer and Grumbach described the 1970s as a period of developing tension, the 1980s as the “Revolt of the Purchasers,” and the 1990s as the breakup of the provider–insurer pact. These changes did not occur in an intellectual vacuum. The lens of administered competition gained powerful adherents in the 1960s especially after it was articulated by Ellwood (Ellwood et al., 1971) and Einthoven (1993).

Then in the 1980s, policy analysts began to crack the biggest barrier to managed care, the lack of common product definitions and labels that had made it nearly impossible to compare either quality or cost among providers. The introduction of diagnostic-related groups (DRGs) led to *prospective payment* (payment per admission by diagnosis) systems that eliminated some cost *outliers*, first for Medicaid and Medicare and then for the HMOs. Having a uniformly defined cluster of cases to follow allowed for the development of classification and information systems and for internal and external oversight of care. Utilization review became a major activity of insurers, and decisions about whom to retain in the service network could be based on *profiles* of the cases treated by providers and institutions. The intent of the developers of DRGs was to enable quality comparisons, but the system was soon adopted for pricing as well.

Some Movement

The changeover to managed care slowed the growth of premiums from the mid-1990s into the first years of the new century, but then premiums took off again. In the meantime, both providers and patients expressed displeasure with HMO constraints on treatment choice and provider choice. New state laws sprang up limiting the ability of insurers and HMOs to control professionals and patients. Some thought managed-care control mechanisms had already picked the low-hanging fruit and had stopped the most egregious cases of inappropriate utilization. One control mechanism, *capitation* (a fixed payment per enrollee per time period), had been promoted because it shifted some of the cost risk from employers and insurers to providers, but it became less fashionable when providers were unable to manage it or lacked sufficiently large risk pools and capital reserves to handle it. Insurers moved toward preferred provider plans that allowed patients and providers more freedom of choice; however, providers gave deeper discounts, and enrollees were subject to higher premiums and greater deductibles and co-payments.

Incremental changes in health insurance regulation occurred during the Clinton administration. The Health Insurance Portability and Privacy Act (HIPPA) sought to relieve job lock. More children were covered by Medicaid and the State Children’s Health Insurance Program (SCHIP), and more reimbursement became prospective.

A new Republican majority under President George W. Bush passed the Medicare Prescription Drug, Improvement, and Modernization Act (MMA), which had three major components:

- Prescription drug coverage under Medicare Part D, which created major additions to the federal deficit
- Health savings accounts (HSAs) linked to high-deductible insurance
- Privatized and subsidized Medicare plans (known as Medicare Advantage)

Although the latter two programs were slow to take off, they have become increasingly significant over time. They have allowed the consumer to become more of a decision-maker about health expenditures.

The Massachusetts Model

As stalemates in Washington became more intractable, costs mounted and the problems of the working poor got worse. Many people were being forced to forego necessary care. States were faced with increased welfare burdens, including Medicaid, and providers were supplying more uncompensated care.

Massachusetts had been trying to set up its own universal coverage system since 1988. Finally, in 2006, Massachusetts enacted near-universal health insurance. This was based on a proposal that had come in large part from the Heritage Foundation via Governor Romney's office. The final bill reflected the concerns and proposals of both the Republican governor and the Democratic legislature. It included an insurance exchange, individual mandates, premium subsidies for low-income individuals, and an employer mandate. It did little to try to control costs. Almost half of the uninsured in the state were soon covered, but costs rose and the legislature had to pass cost-control legislation in subsequent years. Public reaction to the program in the state has been generally favorable, but in the 2012 presidential election, Governor Romney opposed using it as a national template.

The Obama Administration Makes Reform a High Priority

Once President Obama was inaugurated, he made energy reform and health-care reform his highest priorities. As it became clear that energy reform would split the Democrats' narrow congressional majorities, he focused on healthcare reform. He encouraged Congress to take a much stronger role in formulating the new law than President Clinton had when he developed his Health Security legislation. Many compromises were made on cost control to accommodate interest groups, especially hospitals, insurers, and drug manufacturers. Congress scrapped the "public option," a government-sponsored insurance program that would have been offered through the exchanges. It included mandatory increases in Medicaid coverage and voluntary long-term care insurance. The Supreme Court weakened the former, and the administration put off the latter as unworkable.

Although much of the bipartisan coalition congressional leaders had built to support the measure ultimately fell apart, differing versions of the ACA narrowly passed both houses of Congress in March 2010. With the death of Senator Ted Kennedy and the election of a Republican to replace him, the administration no longer had 60 senate votes. The legislation became law after the two houses resolved the differences through the budget reconciliation process. In the wake of the 2012 Supreme Court ruling, many states refused to expand Medicaid with federal money, which meant many households between 100% and 138% of the federal poverty level were not covered.

The ACA did not establish the right to health care. It even allowed discrimination in premiums based on age; the oldest workers could have premiums up to three times higher than those paid by the youngest ones. But the effect of this provision

was blunted, in part, by income-based subsidies. The law was voluminous and complex, and nearly half had little to do with financing medical care. It would take much of this book just to enumerate the Act.

The ACA encouraged experimentation by the states. Oregon and Vermont sought waivers to reallocate state and federal funds to provide a comprehensive coverage system for their residents. Other states, including New York, worked to provide medical homes for chronically ill individuals who are disabled and qualify for both Medicare and Medicaid. These dual eligibles account for a surprisingly large proportion of those programs' costs.

► Employers Want Out: Backing Consumer-Driven Health Care

Throughout the 1990s, observers argued that the United States should move rapidly in the direction of a less regulated national market in health care, as the Reagan Revolution led economists and politicians to seek deregulation and consumer sovereignty in all areas. Commentators, such as Herzlinger (1997), pointed to the disappointing results coming out of managed care contracts and argued that the only way to control healthcare costs would be to motivate consumers to take more responsibility for their own buying decisions. Management experts noted that neither patients nor providers were fully aware of what things cost, and patient pocket-books were not affected significantly by the choices made. Providers were likely to benefit from waste and overutilization, which were of no concern to fully insured consumers. The only way to get costs under control, they argued, was to have as much of a market system in health insurance and health care as for other professional services. At the same time, the Internet was opening relatively painless access to medical information for consumers.

Payers and insurers established standard sets of provider report cards that purported to rank local providers in terms of their quality of care and costs. They increased deductibles and co-payments and launched experiments to test various pay-for-performance schemes that rewarded desired quality and cost-related behaviors. The federal government and some states set up websites to display comparative quality data on institutions.

Corporations began to assess the impact that high employee and retiree health-care costs had on their ability to price goods competitively. Increasingly, their competition came from countries where the overall tax system supported much of the costs of health care. They began to support strongly the notion of a defined benefit package (a constrained dollar amount) and moved toward a healthcare market in which employees would take more responsibility for expenditures and for selecting effective care. Labor unions often opposed this, since the benefits had been won through collective bargaining, but by then their power had substantially weakened.

Consumer-driven health insurance contracts fall into two groups: tiered programs and *health reimbursement accounts*. Tiered programs are of two types: (1) tiered premiums and (2) tiered point-of-care cost sharing. The first type gives employees premium benefits in return for accepting higher co-payments and deductibles, a more restrictive network, or less freedom from utilization review. The second type allows cost sharing for those who choose providers deemed to be

preferred providers based on cost and/or quality measures. Media coverage and marketing efforts have focused on the health reimbursement account arrangements, especially tax-sheltered HSAs. Typically, the employer establishes an account for the employee to spend on health care, and then a large deductible comes into play. Insurance kicks in when the total of these two is exceeded. Unexpended money in the initial account often can roll over from year to year. So far, employer payments under these programs seem to be considerably less than under traditional health plans. It is unclear how much of the difference comes from reduced utilization, from higher out-of-pocket payments, or from more knowledgeable purchasing decisions (Rosenthal & Milstein, 2004). Employers and employees were slow to adopt such policies. However, the enrollment of covered workers in such plans rose from 8% in 2009 to 28% in 2017, and the majority of employers are offering such plans to their employees. Recent Republican proposals have even included expanding HSAs to Medicare and Medicaid because they believe that much of the cost problem is a moral hazard problem.

► The Law of the Land: The ACA (Temporarily?)

The individual mandate of the ACA was not popular with the public. Public confidence in the ACA was not helped by the flawed implementation of the Healthcare.gov exchange. That probably contributed to the lower than expected enrollments of young, healthy “invincibles.” Their absence led to a high average cost per enrollee, rapidly increasing premiums, and withdrawal of insurers in some areas.

Many states refused to establish their own exchanges and relied on the federal exchange. But policies offered on the exchanges remained unaffordable to many. Employers moved toward self-insured plans under the Employment Retirement and Income Security Act (ERISA) to avoid the generous coverages required for exchange offerings.

Repeal and Replace: Health Policy in a Climate of Chaos

Republicans in the House of Representatives had proposed and passed more than 60 bills over the years to repeal the ACA. During the 2016 presidential election, candidate Trump promised that he would act to repeal the ACA on “Day One.” As President, he acted on January 21, 2017, the day of his inauguration, with an Executive Order that stated:

To the maximum extent permitted by law, the Secretary of Health and Human Services [HHS] and the heads of all other executive departments and agencies with authorities and responsibilities under the Act shall exercise all authority and discretion available to them to waive, defer, grant exemptions from, or delay the implementation of any provision or requirement of the Act that would impose a fiscal burden on any State or a cost, fee, tax, penalty or regulatory burden on individuals, families, health care providers, health insurers, patients, recipients of health care services, purchasers of health insurance, or makers of medical devices, products or medications.

His new Secretary of HHS was Rep. Tom Price (R-GA), a former orthopedic surgeon, who had consistently submitted bills in the House offering alternatives to the ACA. These focused on portability of health insurance and tax subsidies for the purchase of individual insurance, and malpractice reform. He was confirmed by the Senate in early February after facing a number of ethics questions pertaining to investments in pharmaceutical companies.

Yet with all of these pronouncements, there was no proposal from the White House for the repeal and replacement of the ACA. This policy free-for-all was also evident in the Congress during the initial 2 weeks of the administration. As jubilation over the election results subsided, Congress appeared divided over what to do next. One group of about 40 conservative House Republicans wanted to pass repeal immediately. Others felt that the result of rapid repeal would lead to a political disaster. Senate members wanted to go slower. Their seats were not gerrymandered, and they were more concerned about the reelection consequences.

When President Trump addressed a joint session of Congress on February 28, he stated:

So I am calling on all Democrats and Republicans in the Congress to work with us to save Americans from this imploding Obamacare disaster.

Here are the principles that should guide the Congress as we move to create a better healthcare system for all Americans:

First, we should ensure that Americans with pre-existing conditions have access to coverage, and that we have a stable transition for Americans currently enrolled in the healthcare exchanges.

Secondly, we should help Americans purchase their own coverage, through the use of tax credits and expanded Health Savings Accounts—but it must be the plan they want, not the plan forced on them by the Government.

Thirdly, we should give our great State Governors the resources and flexibility they need with Medicaid to make sure no one is left out.

Fourthly, we should implement legal reforms that protect patients and doctors from unnecessary costs that drive up the price of insurance—and work to bring down the artificially high price of drugs and bring them down immediately.

Finally, the time has come to give Americans the freedom to purchase health insurance across State lines—creating a truly competitive national marketplace that will bring cost way down and provide far better care.

Everything that is broken in our country can be fixed. Every problem can be solved. And every hurting family can find healing, and hope.

Communications took place between White House staff and congressional staff, but President Trump seemed to leave it up to the House under Speaker Ryan to draft and pass legislation.

The House product was the American Healthcare Act, which included:

- Removal of the individual mandate and penalty. The employer mandate was pushed even further into the future.
- Age-based refundable tax credits ranging from \$2000 to \$4000 (beginning in 2020) replaced the ACA's income-based subsidies. Each household would be

limited to \$14,000. Credits would be phased out for individuals earning at least \$75,000 and families earning at least \$150,000.

- Elimination of ACA taxes, including the medical device tax, the drug industry excise tax, and the tanning salon tax, beginning in 2018; delayed implementation of the “Cadillac” tax (on employer sponsored insurance health benefits) until 2025, and pulling the medical and dental tax deduction floor back to 7.5%.
- Repeal of essential benefit plan requirements for the Medicaid expansion population beginning December 31, 2019.
- Limits on Medicaid payments to states based on the prior year’s number of Medicaid enrollees (a per capita cap) beginning with fiscal year 2019. The base year would be 2016, and the growth rate will be based on the medical care component of the consumer price index.
- Elimination of the enhanced federal portion of Medicaid for the expansion population who enroll after 2019. “Grandfathered” expansion population enrollees (prior to 2020) as long as they remain continuously enrolled. They would have to maintain coverage or be subject to a substantial penalty on resuming coverage.
- Funding for states to stabilize the individual market. These funds could be used for reinsurance programs or for subsidizing high-risk health insurance pools.
- Repeal of the requirement that insurers charge their oldest customers no more than three times as much as younger enrollees and substitute a five-to-one ratio.
- Reinstatement of Medicaid Disproportionate Share Hospital (DSH) payments.
- General tightening of Medicaid eligibility with more frequent eligibility determinations, more citizenship documentation, and reduced hospital presumptive eligibility determinations.

The Congressional Budget Office “scoring” of that bill reported on March 13 was summarized as:

CBO and JCT (Joint Commission on Taxation) estimate that enacting the American Health Care Act would reduce federal deficits by \$337 billion over the coming decade and increase the number of people who are uninsured by 15 million in 2019 and 24 million in 2026 relative to current law.

Congressional Budget Office

Attempts to meet the demands of the conservative Freedom Caucus only served to generate opposition from moderate House members who wanted to preserve their constituents’ insurance. As the ACA was threatened with repeal, it seemed to gain in popularity. Both the White House and Speaker Ryan focused on a vote on March 24th, but at the last minute, Speaker Ryan did not call for a vote and President Trump said that they would move on to other campaign issues. However, House members continued to offer potential compromises around allowing states to modify the minimum essential benefit and allow higher charges for preexisting conditions, but with subsidies from high-risk pools which would be either visible or invisible to the enrollees. Invisible risk pools were already in use in Maine and Alaska.

The Senate made a fresh start with a bill proposing a per capita cap on Medicaid spending with more money in the short run, but less in the long run by using the overall inflation rate rather than healthcare inflation and cutting off funding for the ACA Medicaid expansion. While the Senate bill had more generous subsidies,

it also eliminated subsidies on incomes over 350% of the poverty level and allowed states to install a work requirement for Medicaid. That did not pass. After the Senate deadline passed, the White House tried to keep the process open using the leverage of cutting off the ACA's subsidies to insurers.

The Trump Administration also issued a number of executive orders and Medicaid waivers favoring their preferences. Then in late December, the major tax overhaul passed and became law. It included the removal in 2019 of the ACA tax penalty for not purchasing health insurance. This led to a Kafkaesque situation: Americans would be required by law to purchase health insurance meeting specific requirements, but there would be no consequences when they didn't. This came about because the reconciliation process, which was used to avoid the Senate rule that required a 60-vote majority to overcome a filibuster, could only apply to budget and tax matters. This led to predictions that the individual health insurance market would go into a death spiral with fewer younger and healthier enrollees.

The Resulting Picture

Currently, the U.S. government and the private sector are operating with a hodge-podge of approaches. Medicare and Medicaid are monopsonistic, government-administered systems. President George W. Bush left a legacy in the form of policies that emphasize consumer-driven health care, particularly a focus on HSAs. Federal healthcare policy since the Nixon administration has tended to support development of large HMOs, which started out as administered competition but often became examples of oligopolistic competition, and the ACOs envisioned in the ACA could easily go the same way. Consolidation into larger multisite firms continues to take place both locally in hospital markets and nationally in subsectors such as kidney dialysis centers, nursing homes, pharmaceutical distribution, medical oxygen distribution, and rehabilitation centers. Two firms had 39% of Medicare Advantage enrollment in 2015, with even greater concentration in many local markets. At the same time, successive congressional budgets have reduced funding for Medicare and Medicaid, creating new concerns about cost shifting to private insurance programs and institutions already burdened with the costs of the uninsured and underinsured. The ACA tried to partially reverse that trend, but future budget battles involving entitlements are likely to reestablish that downward trend in provider payments.

► Conclusion

Each of three contending lenses—managed competition, consumer-driven care, and corporate lite—is rooted in our health system's past and present. Each has helped guide our attempts to deliver quality care and constrain costs. At present, the latter, corporate lite (provider oligopoly), is in the driver's seat. In the future we will continue to wrestle with the question, "Who is in charge?"

Like any other democracy, the United States has a system of health care that evolved through a political process influenced by trends in culture, technology, demographics, political ideology, and economic development. It also evolved through experimentation. When something did not work or stopped working,

the country tried other things. We might take solace in a statement attributed to Winston Churchill, “You can always count on the Americans to do the right thing—after they’ve tried everything else.”

Countries that have tried government-administered medicine have been moving toward decentralization and leaving more to the private sector. Countries that started out with a private insurance system have had to add more and more government funding to deal with aging populations and burgeoning technology. What has worked in one time and place will not necessarily work in another, or that what has not worked in one time and place could be made to work in another. There is no magic bullet. Many measures and many accommodations are needed to achieve good care at reasonable cost. The United States has paid a high price for its ongoing cycles of health policy development, all without reaching agreement on the objectives of its health system and without adequately studying the effectiveness of new measures before the political system becomes disappointed and acts, as it often does, without sound outcome assessments.

CASE 3: International Comparisons: Where Else Might We Go?

It can be useful to consider the roads not taken—the measures that have evolved in other, similarly developed countries. This case study briefly reviews the health systems of six roughly comparable countries: Canada, the United Kingdom, Australia, Germany, the Netherlands, and Japan. The approaches are varied, yet all seem to be producing similar results (except costs). Satisfaction surveys for the four English-speaking countries show similar ratings of consumer satisfaction and quality of medical and hospital care; however, self-reported access and expenditures differ widely.

Different Cultures, Different Systems

Canada

Canada’s health system initially paralleled the U.S. system, but the country switched to a single-payer system incrementally, beginning with initiatives in two provinces. Federal legislation passed in 1957 offered to pay 50% of costs if provinces provided universal hospital coverage; by 1961, all ten provinces were participating. The 1966 Medical Care Act extended this cost-share inducement to universal coverage programs, and universal coverage was fully implemented nationwide in 1971. The federal government’s share exceeded 76%, before being lowered.

Key aspects of the Canadian system are:

- Universal coverage under provincial health plans is financed through payroll and income taxes. Many Canadians have private insurance to cover costs the government does not pay and to provide rapid access to scarce services.
- Physicians are typically in private practices, and are paid per visit according to a government fee schedule. Most hospitals are independent public entities that operate within budgets established by the provincial government. Government regulations also affect the prices of prescription drugs.

- Hospitals finance new technology or facilities through the provincial budgeting system, not capital markets. Adoption of new technology such as imaging equipment and surgical capacity is slower than in the United States.
- Rationing occurs through delays in elective services, rather than ability to pay. Lengths of hospital stays have not dropped as rapidly as in U.S. hospitals. Physician visits per person are similar to U.S. rates, but percentage of GDP devoted to health care has grown much more slowly. Canada has a similar number of physicians per 1000 population as the United States (2.7 vs. 2.6).
- Concerns exist about access to specialists and primary care after hours. Canadians' levels of satisfaction with their health care are similar to those of U.S. respondents, but they complain a little more about the shortness of physician visits.
- Although it slowed sharply after 1971, growth in per capita spending has picked up since, despite long waits for scanning procedures and "elective" surgery such as hip replacements, cataract removal, and cardiovascular surgery.
- The system has become increasingly fragmented as the provinces have modified their individual systems over the years. Lewis (2015) argued that it is "a system in name only."
- Per capita healthcare spending is somewhat lower than in the United States and health outcomes slightly better. No one is sure how much leakage of services and expenditures takes place across the border between the two countries, with U.S. citizens purchasing pharmaceuticals in Canada and Canadians purchasing scarce physician and hospital services in the United States.

England

The British National Health Service (NHS) became a socialized system in 1948 after a gradual movement through voluntary and then mandatory health insurance. Until quite recently, NHS was a government program housed in the Department of Health. Ten strategic health authorities (SHAs) implemented national policies at the regional level. Each general practitioner (GP) operated through a local primary care trust (PCT). PCTs served about 100,000 people each and were responsible for disbursing tax revenues dedicated to health within their service areas. In addition to paying GPs through a system of capitation, allowances, and incentives, PCTs contracted with local consultants (specialists) and the government-owned hospitals. A very small private insurance market was allowed. It has grown in recent years and some physicians practice outside of the NHS. In London, so-called Harley Street physicians cater to the wealthy.

The Health and Social Care Act of 2012 significantly reorganized the NHS. Ostensibly, the reforms were designed to make the system more patient-centric, empower medical providers, increase the focus on clinical outcomes, and provide more local autonomy. There are really four different health systems: England, Scotland, Wales, and Northern Ireland, with the latter three functioning through block grants from the national government. So, the systems have gradually diverged.

- At the center of the reformed NHS is NHS England with 80% of the population, which was established as the Commissioning Board in October 2012. This independent quasi-governmental agency pushes public funds out to local Clinical Commissioning Groups (CCGs), which replaced PCTs on April 1, 2013. It also promotes quality of care and improvements in health outcomes. The role of the national government has been limited to general oversight of the system and

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combined strategic leadership for the health and social services systems. SHAs were abolished along with the PCTs.

- CCGs are primarily composed of GPs, but nurses and other providers are represented. The CCGs can commission any health services that meet government standards, including NHS hospitals, consultants (who are typically hospital-based), mental health services, urgent and emergency care, rehabilitation, and community health services.
- The Health and Social Care bill made several other changes to improve coordination, increase democratic input, improve quality and address community public health. For example, leaders of public health, adult social services, children's social services, a consumer representative, and an elected community representative comprise health and wellbeing boards, which promote coordination across sectors and advise CCGs. An organization called Monitor has authority to license providers beginning April 2014 and is responsible for overseeing the transition of NHS hospitals from government entities to foundations. Public health has become a local responsibility, but a new government agency, Public Health England, performs a national role similar in ways to the role the Centers for Disease Control and Prevention plays in U.S. public health.
- The number of physicians per thousand patients is slightly higher in the United Kingdom than in the United States (2.8 vs. 2.6). British nurses do many things physicians would handle in the United States, including delivering babies.
- Rationing has been based in part on waiting times for treatments for nonacute conditions. These have included cataract removal, hip replacement, and coronary artery bypass surgery, for which patients may wait as much as a year.
- In 2004, the NHS adopted a pay-for-performance system for family physicians that involved 146 quality performance measures. According to Doran et al. (2006), primary care practices met targets for 83% of patients and achieved 97% of the possible points, much more than the 75% anticipated in the budget, resulting in an average of more than \$40,000 in additional payments per physician. The result was a substantial budget overrun. Because a major baseline study was not performed, how much of the improvement was due to changed medical care and how much was due to improved documentation is not knowable.
- Long queues were a major political issue in the 1997 elections that brought back the Labour government. That government increased NHS funding, and waiting times dropped. Some management decision-making was also decentralized from the regional SHAs to the local hospitals, whose accountability for quality and cost was increased. At the same time, the government established the National Institute for Health and Care Excellence (NICE) to evaluate procedures, treatments, and technologies and to speed their adoption if the evidence is adequate and favorable. This was in response to reliable evidence of differences in treatments and outcome differences among various geographic areas, regional health authorities, and fund-holder groups.

Australia

Australia has a hybrid public–private healthcare system. A national healthcare system called “Medicare” is financed out of taxation. When established in 1984, Medicare supported government hospitals, medical care, and prescription drugs for the indigent. It also provided grants to state and territorial governments to operate hospitals. The 1999 addition of a Medicare levy—1.5%–2.5% depending on income level—extended

these benefits to the general population. Government incentives encourage private insurance, which pays “cost sharing” fees and provides access to private hospitals, specialists, and physicians. About 50% of Australians have private insurance, which pays 11% of healthcare costs.

- Australia has a federal system with the states and territories actually managing the public hospitals and a number of other services, but most primary care and pharmaceutical coverage is paid by the central government.
- Australians seem to have less access problems overall than Canadian and U.S. patients, but they report problems accessing care on nights and weekends and difficulties paying for prescription drugs.
- Australians with insurance entering local public hospitals decide whether to do so as public or private patients. Public patients receive free hospital and physician care. Private patients can choose their doctors. They pay minor charges, but most charges are covered by a combination of Medicare and private insurance.
- Subsidies are limited to pharmaceuticals approved for cost-effectiveness by the independent Pharmaceutical Benefits Advisory Committee.
- Under a program called Lifetime Health Cover, those who join a private health plan before the age of 31 pay a lower premium over their lifetime. Two percent is added to the premium for each year of delay. This is to prevent “hit-and-run” enrollment when people anticipate major expenses and to maintain a larger, healthier risk pool.
- Community rating is mandatory for private health insurance. A “reinsurance” system redistributes the costs of claims among insurers to avoid winners and losers.
- To reduce reliance on public funding, the government provides a 30% rebate on private health insurance costs.
- A government subsidy for the long-term care of older persons includes institutional, community-based, and in-home support. In return, the government controls the supply of long-term beds.
- Australia has 3.5 physicians per 1000 population compared with 2.6 in the United States.

Germany

Chancellor Otto von Bismarck is credited with starting the first national health insurance program in the 1880s. Today, it is built around hundreds of not-for-profit sickness funds that negotiate with labor unions, employers, and providers. The various parties interact quite formally.

- All individuals must have health insurance. A federal unemployment insurance fund pays premiums for the unemployed. A worker’s pension fund pays premiums for retired workers. Workers have choices among funds, but funds tend to be linked to an industry or locale.
- Management of the system is split between the states, the Federal government and a number of independent organizations representing employers, providers and insurers. Yet the system is considered very efficient. However, consumer satisfaction tends to be low as well.
- Funds assess premiums on a graduated scale based on income. Co-payments have increased in recent years to cover revenue shortfalls.
- Physician associations receive a fixed amount per person per year, as do hospitals. Hospitals pay hospital-based physicians salaries from their capitation

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income. Hospitals are reimbursed under a DRG system similar to the U.S. system except that it includes physician services and aftercare for 30 days post discharge. Ambulatory-care physicians are paid either a fee for service or the physician associations pay them a salary from capitated revenues. They generally cannot follow patients into the hospital. Germany has recently started to develop specialized ambulatory care centers for specific diseases with highly integrated services.

- Doctor visits are shorter and more frequent than in the United States, and hospital stays are longer; however, the hospital staffing ratios are much lower. The average cost of a hospital stay in the United States is much greater than that of Germany despite a much longer average length of stay.
- Germany has 4.1 physicians per 1000 population compared with 2.6 in the United States.
- Among developed countries, Germany has the fourth-highest percentage of GDP devoted to health care after the United States, France, and Denmark. Because of cost increases, high unemployment, and an aging population, a 2006 political compromise increased premiums to an average of 15.5% of salaries beginning in 2009. Premiums are pooled, and each insurer receives the same premium per enrollee in an attempt to motivate efficiency improvements.

Netherlands

The Netherlands is considered one of the best developed-country health systems. It has universal coverage, but its delivery system is private with many not-for-profit insurers and independent primary care providers. On top of that there is a mandatory government insurance system for long-term care. Despite premium and co-pay requirements, out-of-pocket costs are low.

- Primary care providers are the gatekeepers in this system. Access to specialists is virtually available only through referral from a GP, including mental health issues. A special program brings mental health services to youth through their schools.
- The Dutch health system is not cheap despite low drug prices and low hospitalization rates.
- This system has focused a great deal on integrated (health and mental) care and coordination of services. Recent legislation (2014 & 2015) has focused on long-term care in the community, on a special program for integrated care for the frail elderly, and psychosocial youth care. These reforms required more involvement by municipalities which had to scramble to build capacity.
- Attempts to allow insurers to limit their networks failed in Parliament in 2014.
- Netherlands institutions while independent are under pressure from the government to follow evidence-based practices.
- The Netherlands has an unusually small number of pharmacists. In 2014 it had 4.27 physicians per 1000 compared to 3.35 for the U.S. International comparisons for nurses are difficult because their roles vary so much. In the United States many nurses have roles outside direct delivery due to the complexity of our insurance systems.
- The Dutch public reports more satisfaction with their health system than do their neighbors.

Japan

Employment-based health insurance is the core of Japan's health system, and it continues to produce the best health outcomes of any of the systems mentioned here;

however, some ascribe much of the differential outcomes to demographic and lifestyle issues, especially diet.¹ There is also a national health insurance program financed with national and local taxes. Premiums are scaled to family income. Households not covered by employment-based insurance must belong to community insurance programs under the national plan. Retirees are covered by their former employers or their community plans. There is a high-risk pool available to literally thousands of insurance schemes. The co-pay for most services is 30%.

- The government sets fee schedules for physicians at a level much below U.S. rates. Fees are identical for all plans; however, patients often add 3% or 4% “gifts” to their payments. Fee levels are modified based on utilization. If too many of a procedure are done, the fee is lowered.
- There are both nonprofit and for-profit hospitals, and hospitals may be owned by doctors.
- Most physicians work out of large clinics, some of which are associated with hospitals, and are reluctant to send patients into the hospital because they cannot follow them once they are admitted. Japan has 2.4 physicians per 1000 population compared to the United States with 2.6.
- Specialists are hospital employees and earn less than primary care physicians.
- Clinics usually dispense their own drugs.
- Japanese patients have many more, briefer visits and many more prescriptions than their U.S. counterparts. They also have many fewer admissions, although lengths of stay tend to be much longer.
- There are fewer doctors per capita in Japan than in the United States, and waiting lines tend to be managed on a first-come, first-served basis.
- Japanese hospitals are considered by many to have somewhat outdated equipment and shabby facilities. Physicians do not seem to be customer oriented or highly motivated to meet patients’ affective needs.

The government is aware of the collision course the health system is on with a decreasing and aging population.

Other OECD Countries

Most other Organization for Economic Cooperation and Development (OECD) countries have more physicians and nurses per thousand population than the United States. The Netherlands has one of the highest ratios of nurses and recently increased the roles of nurses in primary care. France, Sweden, and Spain had high ratios of physicians per 1000 population (3.3, 4.2, and 3.92, respectively). All physician per 1000 population statistics cited herein come from the OECD data base.

Some Repeating Themes

A number of themes recur in the systems of these various countries. Some represent ideas that have been tried in the United States (under the ACA, in some instances), but all might warrant further consideration as the U.S. system changes over the next 10–20 years.

Universal Coverage

Health care is provided to all. Often it is through a patchwork of public and private funds, but every effort is made to have everyone in the system. General tax revenues (income, payroll, and value added taxes) are used extensively to fund health care, but in most cases, there is a mixture of additional revenue sources, including patient

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co-payment, employment-based insurance, retirement funds, local government revenues, and private insurance. This patchwork of payment mechanisms does not leave large gaps of uninsured or underinsured citizens. Private insurance and private care are available to those who choose to pay more. Where co-pays are required, a careful effort is made to make sure that ability to pay does not control access to basic care. In the United States, there have been numerous attempts to expand coverage over the last several decades, with ACA being the most ambitious attempt to approach universal coverage.

The Public-Private Pendulum

Anell (2015) used the “public-private pendulum” analogy with respect to Sweden. But similar issues have been playing out in England, France, The Netherlands and especially in Australia which has seen at least five changes in the way health care is financed since 1976 (Hall, 2015). For example, while France has universal coverage, 96% of its population also carries complementary health insurance. Vouchers to purchase the private complementary insurance are available to the poor (Steffen, 2016).

Centralized Versus Decentralized Delivery System Controls

Country after country that started out with a centralized healthcare system has decentralized operational control to district or regional authorities. Sometimes their boards are elected separately from the local administrative body. The objectives here are multiple. Sometimes it is privatization, sometimes it is seeking more patient-centered care, and often it is aimed at adding regional and local revenue sources to the mix.

Hospitals Are Budget Constrained

Since the introduction of prospective payment based on diagnostic-related group classifications, hospitals in the United States have operated more as cost centers than revenue centers. A number of countries have established global hospital budgets or capitation budgets for hospitals, often administered through local authorities or trusts. Capital investment is constrained to avoid a hospital arms race.

Specialists Are Salaried and PCPs Incentivized

Income of the universal coverage system is used to pay the salaries of specialists, whereas fee-for-service payments reimburse the primary care providers. They serve as gatekeepers for referrals to specialists and hospitals and do not follow patients into the hospital. They are motivated, therefore, to avoid unnecessary hospitalizations. The British experiment with pay-for-performance was sufficiently successful that Epstein (2006) argued that its time has come for the United States. One might also see it as a way to boost the incomes of primary care physicians in the United States sufficiently to attract new practitioners and bolster the currently dwindling supply (Basch, 2006).

Large Premium and Risk Pools Are Maintained

Individuals are compelled to belong to one health plan or another. Young and healthy individuals cannot opt out, or where they can, incentives are provided to try to keep them in. Trusts serve very large employers, but the needs of small

businesses and individuals are met through required community rating, local community health plans, and tax subsidies. Large premium and risk pools are built in to level the playing field and hold down administrative and marketing costs.

Systems Integration

The integration of the system is provided at the governmental rather than the institutional level. Circuit breakers in the system, especially between hospitals, specialists, and primary care practices, keep individuals and institutions from maximizing utilization. Incentives focus on motivating primary care physicians to control costs and improve quality.

Rationalization and Standardization

A trend toward decentralization of healthcare services is offset in part by setting up staff units that analyze and report on current medical technology, evidence about best practices, and evaluation of the cost-effectiveness of common interventions. These recommendations will probably be worked increasingly into pay-for-performance systems. For example, the Netherlands started with a personal healthcare budget for those in long-term care, but has had to rein in expenses after an expenditure growth rate of 23% (van Ginneken, 2015).

Labor Substitution

Many countries with lower costs seem to have not only lower professional incomes, but also substitute nurses and pharmacists for physicians, and physician generalists for specialists in their delivery systems.

Pharmaceutical Costs

Most developed countries except New Zealand and the United States constrain or ban direct-to-consumer advertising for prescription drugs (a cost that reached \$4.5 billion annually in the United States in 2009), and rely on recommendations by physicians for decision-making. The profit margins of pharmaceutical companies are constrained through a number of mechanisms, depending on what alternatives exist for payment in the national system. In a few cases, physicians may supplement their revenue by dispensing in their practices.

► Discussion Questions

We invite you to discuss themes in the case that relate to the following questions:

1. In most of these countries, does universal coverage provide the gold standard of care?
2. Does rationing occur in these countries? How is it different from rationing in the United States?
3. Do revenues used to pay for health care tend to come from a single source or many?

4. What steps do these countries take to assure that payments required of individuals do not become a barrier to access?
5. To what extent do you think the crafting of the ACA relied on experiences in other countries? Can you think of examples of policies or countries that may have been influential?
6. Which of these policies would you like to see the United States adopt next? Why?



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CHAPTER 4

Where Do We Want to Be?

Even in a country that lacks a cohesive health policy, it is useful to ask: How unhappy are we with our health care? What do we want to change? Do not expect consistent responses from the American public. When the nation was debating the Clinton health plan, numerous organizations surveyed the public. Respondents reported they believed that the healthcare system was in trouble. At the same time, they expressed satisfaction with their own largely employer-financed healthcare programs. Public support for universal coverage was strong, but individuals did not want to pay higher taxes to support it (Peterson, 1995). An *ABC News/Washington Post* poll in October 1993 showed the following (Schick, 1995):

- 51% of the public favored the Clinton health plan.
- 59% thought that it was better than the existing system.
- Only 19% thought that their care would get better under it, and 34% thought worse care would result.
- However, 57% were against tax increases to pay for it, whereas 40% would be willing to pay.

It did not survive intact.

The American public was split over the Patient Protection and Affordable Care Act (ACA) as a whole. Some opposition came from people who wanted a public option, for example, or a single-payer system. Overall, the public was negative about the individual mandate and the employer mandate, but was much in favor of the insurance changes, especially the ban on excluding preexisting conditions and allowing children to stay on their parents' insurance until they turn 26. One poll reported that 42% of respondents did not know that the ACA was still the law of the land. In the same poll, 12% believed it had been repealed by Congress, 7% believed it had been overturned by the Supreme Court, and 23% didn't know whether it was still in effect or not (Kaiser Family Foundation, 2013). By the time the House had passed its repeal version and the Senate had made multiple failed attempts, the majority of the public favored the ACA with an overwhelming majority favoring coverage of preexisting conditions.

► Where Are We?

Americans report being in good health more than any other Organization for Economic Cooperation and Development (OECD) country. A 2010 study of six

developed countries showed that Americans were satisfied with their doctors and the availability of effective care, but were also more likely to report that the system needed to be completely rebuilt (Papanicolas, Cylus, & Smith, 2013). Their complaints are mostly about financial risks and to some extent access and waiting. The same concerns continued to dominate after the ACA went into effect (Osborn, Squires, Doty, Sarnak, & Schneider, 2016).

Despite the positive self-reporting, America's health system is not delivering better results overall. A 2017 Commonwealth Foundation study ranked the U.S. health system with those of 11 high-income countries. It concluded that:

The U.S. ranked last on performance overall, and ranked last or next to last or near last on Access, Administrative Efficiency, Equity and Health Care.... Based on a broad range of indicators, the U.S. health system is an outlier, spending far more but falling short ... The results suggest U.S. health system should look at other countries' approaches if it wants to achieve an affordable high-performing health care system that serves all Americans.

Schneider, Sarnak, Squires, Shah, & Doty (2017, Abstract)

Top overall rankings went to the United Kingdom, Australia, and the Netherlands. The most important measure, outcomes, shows mixed results with high performance in 30-day in-hospital mortality after a heart attack and stroke, and in selected cancer survival rates.

TABLE 4-1 illustrates this by comparing 11 countries on two resource-input dimensions and on two outcome dimensions: overall life expectation and infant mortality rates. This incongruous combination of high U.S. costs and low population health outcomes does not seem to be associated with any one specific organizational or financial approach. Some high performers have a single-payer system, some don't. The most important variable for population health outcomes is universal access, but that can be achieved through either public or private health insurance and by varied methods of payment.

The U.S. cost problem has been building for a long period with the inflation rate for health expenditures roughly the overall inflation rate plus two percent. On top of that, these costs have impinged even more heavily on individuals and families as employers have negotiated insurance contracts with larger and larger deductibles and co-pays. **FIGURE 4-1** shows these trends. Between 1999 and 2017 overall inflation had risen 47% and wages 64%, while family health insurance premiums rose 224% and worker contributions rose 270%.

It will come as a surprise to most Americans that their publicly financed health-care expenditures are about equal to the private sector contributions. Is it a private or a public system and does it matter? This is apparent in **TABLE 4-2** showing that the United States spends about the same percentage of Gross Domestic Product (GDP) on government support of health care as the other high-income countries, but they manage to provide universal coverage with a similar government investment and a lot less total investment. Those other countries do have private insurance sector financing but it is very much smaller.

TABLE 4-1 Selected International Comparison of Health Inputs and Resources, 2015

Developed Country	Per Capita US \$	% GDP	Life Expectancy at Birth	Infant Deaths per 1000
	2015	2015	2015	2014
United States	9451	16.9	78.8	6
Switzerland	6935	11.5	81.3	2.2
Australia	4420	9.3	82.4	3.4
Netherlands	5343	10.8	81.8	2.2
Germany	5267	11.1	81.2	3.2
Sweden	5228	11.1	82.3	3.9
Belgium	4611	10.4	81.4	3.4
Canada	4608	10.1	81.5	4.8
France	4407	11	82.8	3.5
Japan	4150	11.2	83.7	2.1
United Kingdom	4003	9.8	81.4	3.9

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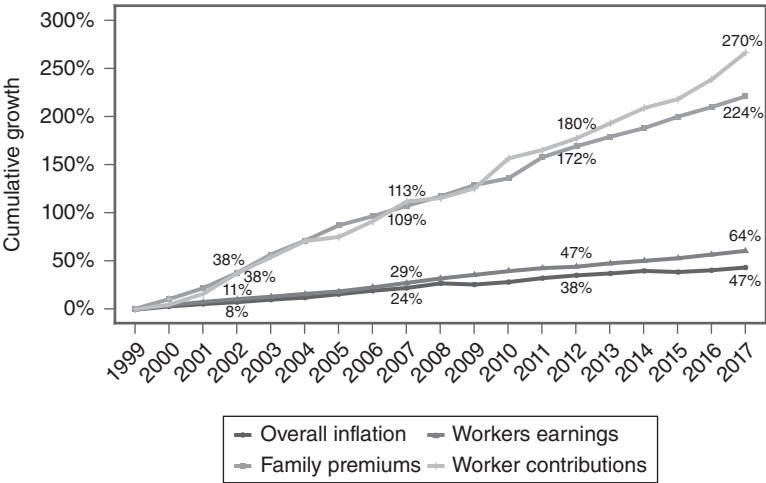


FIGURE 4-1 Cumulative family health insurance premiums, worker contributions to premiums, inflation, and workers' earnings, 1999–2017.

Reproduced from Kaiser/HRET Survey of Employer-Sponsored Health Benefits, 1999–2017. Bureau of Labor Statistics, Consumer Price Index, U.S. City Average of Annual Inflation (April to April), 1999–2017; Bureau of Labor Statistics, Seasonally Adjusted Data from the Current Employment Statistics Survey 1999–2017 (April to April). Accessed October 20, 2017.

TABLE 4-2 Comparison of Public and Total Expenditures in Selected Countries, 2015

Developed Country	Per Capita US\$ 2015*	Gov't + Compulsory Insurance	% Gov't + Compulsory Insurance
Australia	4420	2950	66.74
Belgium	4611	3576	77.55
Canada	4608	3262	70.79
France	4407	3474	78.83
Germany	5267	4477	85.00
Japan	4150	3523	84.89
Netherlands	5343	4312	80.70
Sweden	5228	4375	83.68
Switzerland	6935	4711	67.93
United Kingdom	4003	3163	79.02
United States	9451	4672	49.43

* Using Purchasing Power Parity adjusted international dollar rate.

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► Alignment with the Rest of Society

The democratic process is likely to generate many more policy experiments as we cope with advancing technology, political events, changing demographics, and economic fluctuations. These experiments will continue to stir debate about the merits of the many delivery and payment alternatives available in the United States and elsewhere. Unfortunately, the observed effects are usually mixed and the distributional effects produce winners and losers.

Even the most prestigious institutions are affected by these external drivers. For example, the Finnish national orthopedic hospital, the Orton Hospital in Helsinki, had to downsize and reach out to private-pay individuals when the Finnish federal government chose to decentralize its jointly financed government healthcare program and pass administration on to local governments (Masalin, 1994). These local governments then attempted to control the rising cost of health care by reducing

referrals to central specialized hospitals. Orton Hospital was a national resource of high-quality care, but as local politics changed the referral patterns, it too had to change the way it functioned to survive.

There is no universal, monolithic “we” in health policy. There are interest groups, each of which has its own point of view. Within each group are many individuals with some diversity of views. They may be willing to compromise on some issues, but not on others. In the United States, much progressive legislation has been built by reaching agreement on means rather than battling over ends.

What Do Providers Want?

Providers are aware of their responsibility to act in the best interests of their patients. They are also inculcated with the “first do no harm” dictum. Even among the “disinterested” parties, some care most about individuals, whereas others focus on populations. This can be a vexing problem for those clinicians who, although committed to the needs of individual patients, are also trained in statistical thinking and population-based approaches.

Providers want professional autonomy, income stability and income growth comparable to their peers, successful outcomes for their patients, a sense of mastery of their field, and the respect of the public. They know that they will make some mistakes, but they will work very hard to avoid them. They do not want to put their careers on the line with every decision. They do not want to waste energy on bureaucratic exercises that consume resources and distract them from effective care. They also would like to see provisions to pay for care for the uninsured. They are aware that these individuals often forego preventive care and may end up later with more serious and costly problems. Because of the low reimbursement rates, however, some providers refuse to participate in federal and state programs, further reducing access and availability. That is why some hospitals and health maintenance organizations have strongly endorsed state plans to cover the uninsured, even when they involve adding a tax on their charges to paying patients.

The American Nurses Association (ANA) has expressed some of these desires in its Bill of Rights for Registered Nurses (ANA, 2005). It addresses not only working conditions for nurses but also the right to provide “services that maintain respect for human dignity and embrace the uniqueness of each patient and the nature of his or her health problems, without restriction with regard to social or economic status.” The document says nurses must have the ability to meet their obligations to society and their patients, as well as meet their own needs. In addition, they have a right to workplaces that allow them to meet professional standards, act within their scope of practice, perform their duties in accordance with the ANA code of ethics, advocate for themselves as well as their patients, be safe, keep their patients safe, and negotiate their conditions of employment.

Professional Autonomy

The professional mystique of physicians has rested on their unique expertise and control of information. Those who favor a consumer-centric, free-market approach to healthcare decision-making want to maximize the amount of information available to consumers. This has led many physicians to argue for privacy in the conduct of their practices, often in the name of protection of business

secrets and personal privacy. Many physicians object, for example, to the fact that a drug company's local sales representative has data on their prescribing behaviors (Saul, 2006). Insurers certainly profile physicians and institutions for costs and outcomes regularly, and aggregated data are increasingly available to employers, the National Committee for Quality Assurance, the Joint Commission, the federal government, and the general public. Employer representatives want more information to be available to consumers. One thorny issue is information on individual physicians. President George W. Bush called for "transparency in the marketplace" and urged private insurers to disclose data on physician costs and outcomes; however, when the Business Roundtable called on the federal government to make its Medicare databases available, his administration cited a 1979 court ruling protecting the privacy of physicians and prohibiting disclosure of Medicare payments to individual physicians (Pear, 2006). The Centers for Medicare & Medicaid Services (CMS) only started releasing specific physician income data in 2014.

This is a difficult area. Professionals, like other businesspeople, have some rights to privacy and protections from the prying eyes of competitors, but some observers see the current tensions as the last gasp of a professional monopoly and an attempt to withhold information that bolsters purchaser sovereignty at all levels. The public does have difficulty interpreting this information effectively. CMS studies show that patients have a low degree of trust in government-sponsored sites and often do not find the specific quality information they are seeking (Williams, 2016). Current techniques for evaluating case mix and adjusting for risk are crude at best. Measuring the outputs of medical interventions is difficult unless one knows that the inputs are comparable and unless there is a way to adjust the data to reflect differences in inputs, especially the condition of the patient going in. Yet despite all the questions about the accuracy of these measures, the majority of primary care physicians participate in one or more pay-for-performance schemes.

Other professions fight to overcome the dominance of physicians. In many countries, pharmacists are freer to dispense independently. Nurse practitioners and midwives have fought state by state for the right to practice independently. Psychologists have been fighting similar battles with respect to prescribing for the mentally ill, and more and more types of counselors want to be able to bill Medicare, Medicaid, and private insurance.

What Do Patients and Their Families Want?

Patients want to beat the odds. They and their families want the best possible outcomes, and they want to know that everything possible was done to ensure recovery (or a comfortable and dignified death) for their loved ones. Some want miracles. Most know that they need experts to look after their interests, but still want to be kept informed of what is going on so they can make sense of what is happening and avoid serious medical errors. Again, the issues are complex. Patients and families want to have access to quality information if they have the time and energy to make their own decisions. At the same time, they employ the provider as their agent, and the sicker they are, the more they rely on the clinician's judgment. They want choices, but often don't feel empowered to make them.

When they are not terribly sick, they also worry about the cost of their care. They do not want to spend a lot of time in the waiting room or figuring out how to

fill out paperwork. That is a nonmonetary cost, but a cost to them, nevertheless. It can also be a monetary cost if they lose work hours or reimbursement opportunities because of it.

They want to know that they were not treated unfairly by any part of the health-care system and that their treatment was not affected by their gender, their ethnicity, or the color of their skin.

They also become apprehensive when they believe that profitability concerns or payment mechanisms are influencing which treatments they receive. An example has been the debate over whether the drugs chosen by oncologists for outpatient treatment have been chosen for their effectiveness or their profitability (Abelson, 2006a; Jacobson, O'Malley, Earle, Gaccione, & Newhouse, 2006). Increasingly, patients are aware of the financial incentives affecting providers that in the long run can undermine provider legitimacy (Schlesinger, 2002).

Patients want affordable health insurance. They do not want to be denied insurance on the basis of prior medical conditions over which they had little or no control. They do not want to be harassed by debt collectors, have their wages garnished, or be bankrupted by medical bills.

The ACA dealt with this in several ways, including barring lifetime caps on claims. Yet consumers also worry about what the premium costs will do to their disposable income. Although the ACA tried to deal with that by subsidizing the premiums of low-income workers, many saw their premiums become less affordable due to adverse selection.

What Do Insurers Want?

Insurers want to stay in business. That is why they fought so hard against any hint of a single-payer system and against Obama administration proposals for a government health plan that would appear in the exchanges alongside the existing private products (the public option).

They want to be able to make an acceptable level of net revenue whether they are a for-profit or a nonprofit organization. They want to be able to compete in the marketplace on a "level playing field." Their customers are the payers—the employers and the group and individual enrollees—and they want to maintain a good reputation with them. In the world of HMOs and preferred provider organizations, insurers want the biggest possible discounts from providers and to keep their medical loss ratios competitive.

Insurers also want to avoid adverse selection. They want protection against having those who know they have a higher-than-average probability of a claim from joining their system in disproportionate numbers. They also want to be able to attract those with a below-average probability of a claim. They would like to continue to compete on marketing skills, on underwriting ability, on investment returns on their reserves, and on operating efficiencies.

Insurers, however, are very sensitive to market shifts. For example, many are currently developing new insurance products for individuals and small groups as the notion of consumer-oriented care and insurance exchanges increase consumer involvement in choosing products (and as employers reduce their contributions and coverage). They suddenly seem interested in individual subscribers that they ignored a few years ago. They are also interested in offering self-insurance plans to the small employers that they ignored previously.

What Do Employers Want?

Employers want to be able to recruit and retain competent, productive employees, and they want competitive cost structures. They are not in the healthcare purchasing business for any other reason. They are generally supportive of consumer-driven health care that allows individual consumers, rather than employers, insurers, or provider organizations, to make more decisions than in the past. This effectively shifts more of the costs onto the employees. With the exchanges and the play-or-pay provisions of the ACA, they had many more options to consider.

The employer mandate in the ACA was delayed and delayed and does not apply to small employers. Frankly, it is not as critical to the effectiveness of the bill as the triumvirate of (1) subsidized premiums, (2) community rating (no avoiding preexisting conditions), and (3) minimum essential benefits.

Large Employers and Unions

In unionized firms, premium payments are set through collective bargaining between the company and the union. This bargaining can expand or contract the healthcare benefits, depending on the wants and needs of the employer and union leadership. After a period during which many of our bitterest strikes were waged over health benefit issues, both sides are now recognizing that employment-related healthcare costs can reduce domestic employment by encouraging companies to shift production to other countries. Large firms and their unions are increasingly aligned in their desire to hold down costs and maintain coverage. Employers are rapidly limiting their risk by shifting to defined dollar contributions toward health care for employees and retirees, rather than defined benefits.

Small Employers

Because healthcare insurance bargaining power can be increased by pooling large numbers of beneficiaries, and because administrative costs and insurance prices are sensitive to the number of individuals covered, small businesses find it hard to provide competitive health benefits to their workers. They need either subsidies or effective ways of pooling their people with others to make a viable enrollee population.

► What Do Governments Want?

They want a satisfied public. They want to look like winners. They want healthcare expenditures to be predictable and at a level that does not disadvantage economic growth in both domestic and international competition. They believe the system should work within the parameters of accepted cultural norms of equity and fairness so it does not foment voter dissatisfaction.

All levels of government want to keep costs down to avoid crowding out other programs or increasing taxpayer discontent. Federal, state, and local governments are also concerned about their longer-term liabilities for the viability of Medicare trust funds, Medicaid costs, and their accrued liabilities for government retiree health benefits.

Unfortunately, the forecast of growth in overall U.S. health expenditures by the Office of the Actuary of CMS, shown in **TABLE 4-3**, does not bode well for this objective.

TABLE 4-3 U.S. National Health Expenditures (NHE) and Percentage of GDP
Selected Years 2007–2024

Selected Years	2007	2014	2017*	2019*	2024*
NHE (\$ billions)	2,495	3,029	3,539	3,966	5,255
NHE per capita	8,141	9,515	10,833	11,912	15,800
NHE as % GDP	15.9	17.4	18.3	18.6	19.7

* Estimated

CMS, Office of the Actuary, *National Health Expenditure Data Projected*. Retrieved October 23, 2017, from www.cms.gov/Research-Statistics-Data-and-Systems/Statistics-Trends-and-Reports/NationalHealthExpendData/NationalHealthAccountsProjected.html

Federal Government

The federal government purchases health care on behalf of special populations: the poor, older adults, veterans, active duty military, Native Americans, members of Congress, and so forth. It pays for more than 40% of healthcare purchases (and covers more than 60% of patients in some markets). Children have actually fared better in federally purchased health care since 2000 because of the State Children's Health Insurance Program (SCHIP). Federal programs, which are highly sensitive to political pressures, have tended to rely on their purchasing power to garner deep discounts. Only recently have governments begun to encourage evidence-based medicine and to refuse to pay for avoidable costs such as hospital-acquired infections.

State and Local Governments

The interests of state and local governments are much more limited, involving Medicaid and local programs for the uninsured poor. Local revenue streams are limited and inflexible, often being tied to real property taxes that cannot adapt rapidly to changing economic conditions. However, revenue restraints have encouraged innovation. State and local governments cannot print money to cover deficit spending. Legislators get upset at the inability of staff to accurately forecast expenditures a year or two out. Despite all of the rhetoric at the national level, most experimental attempts to control health costs have occurred at the state level. Maine, Vermont, and Massachusetts have all experimented with systems to provide universal coverage using a variety of funding sources.

Prior to passage of the ACA, the governors of more than half the states had proposed measures that included one or more of the following approaches:

- Mandated insurance coverage for all citizens
- Mandated employer coverage or payments into insurance pools in lieu of coverage
- Mandated lower premium individual policies from insurers

- State coverage for all uninsured children
- Tax credits and deductions for individual insurance premiums
- Special discounts for prevention and health lifestyles
- Expansion of Medicare to cover the uninsured (Barry & Basher, 2007; Solomon & Wessel, 2007)

State-by-state experimentation is expected to continue and expand under the Republican administration. The Section 1115 and 1332 Waiver programs allow considerable flexibility for the states to experiment with demonstration programs. Examples are waivers approved in 2017 allowing Alaska and other states to stabilize their insurance markets.

Politicians

The fact that we have to ask that question is emblematic of the politicization of health policy. Senator Chuck Grassley (R-Iowa), one of the Senate's most prominent health-care experts, said of the Graham-Cassidy bill, according to multiple news reports, "You know, I could maybe give you 10 reasons why this bill shouldn't be considered. But Republicans campaigned on this so often that you have a responsibility to carry out what you said in the campaign. That's pretty much as much of a reason as the substance of the bill."

The politicians also have their three frames for looking at healthcare legislation.

Conservatives: Believe that the federal government has become too big and too intrusive. They want to return control of health insurance back to the state. They also want to cap the federal liability for Medicaid, freeing up funds for future tax cuts (Oberlander, 2017).

Progressives: Believe that health care is a right and are willing to raise taxes to achieve universal coverage. They believe in further regulation of healthcare delivery controls on utilization and prices.

Moderates: Believe that the ACA is deeply flawed, but salvageable through cost cutting and informed consumer decision-making. They would expand subsidized premiums to cover the remaining uninsured and support a number of interventions to cut costs. Their approach would be incremental and measured over time.

What Does the General Public Want?

Members of the public want to feel that they and their families are safe and that the system will treat them fairly and effectively, if and when needed. This requires access to health care, which means health insurance coverage. They also do not want to feel guilty about the plight of their uninsured fellow citizens. At the same time, they are not enthusiastic about using the tax system to cover the needs of others. They do not want to be treated in a way that marks them as a member of any underclass, but want to be treated as middle class or above.

Rosenthal (2017) and Havighurst (2017) both express a populist view that the current health system with a requirement for purchasing insurance and the near monopolies of providers represents a totally unfair extraction of wealth from middle- and low-income households. They observe that the employment benefit exists at the expense

of wage earners and that the lack of user involvement in purchasing decisions supports the upward bias in the use of expensive technology and rising prices and profits.

Americans seem unwilling to pay more than the current proportion of national income (about 18%) for health care, yet the inflation rate in health care continues to exceed the rate of growth in our overall economy (growth in GDP). International economic comparisons have shown a close relationship between national income and per capita health expenditures, but with the United States as an outlier. Wealthy nations do spend more per person. This would seem to indicate that part of the cost of health care is related to need, part is related to availability, and part is related to decisions people make as consumers.

Individual Insurance Purchasers

The problems affecting small businesses are even worse for the self-employed seeking coverage. Premiums are highest and rejections frequent. Many end up going without coverage. Yet the self-employed often are in the same boat as the working poor who are unable to obtain insurance through their employer—because it is not offered, because they only work part time and are not eligible, or because they cannot afford the added expense. The ACA helped these two populations a lot and it remains to be seen how any “replacement” of its subsidies works for them.

What Do Policy Wonks Want?

Policy wonks want a system that is efficient, coherent, and rational and that provides effective care. More than most actors, they tend to compare the United States against the rest of the developed world. Health care competes with other services for scarce resources. Money spent on health care cannot be spent on transportation or public education; therefore, there must be some calculus for allocating scarce resources to health care and other meritorious causes on a consistent basis. They want the public to get the best care possible with the available resources.

Analysts are sharply divided on many other issues. For example, although many argue for measures to forestall continued growth in healthcare expenditures, others say, “Don’t worry—the U.S. citizenry can afford it and it constitutes economic growth and increasing employment even if it were to approach 25% of our economy.” Others say, “Yes, that is growth, but it represents a transfer of assets from the young to the elderly that is not sustainable.”

Much care is effective and worth doing, but there is agreement that there is considerable waste and inefficiency, even though waste and inefficiency are income-enhancing for someone. Examples of other areas of concern to policy analysts include:

Free riders. These are individuals who could otherwise pay, but avoid doing so while still relying on the system for help in case of a catastrophic event. An even thornier problem has been the *moral hazard* of those who knowingly indulge in high-risk behaviors for which the general public has had to pick up much of the costs. The primary rationale for mandatory helmet laws, for example, was not just to protect motorcyclists; it was to shelter the public from incurring the tremendous ongoing costs of caring for people with serious brain damage from motorcycle accidents.

Overserved and underserved areas. Not only are healthcare resources limited by the willingness and ability of governments, firms, and individuals to pay, but available resources can be poorly distributed, resulting in surpluses in some areas and shortages in others. Governments and payers have attempted to regulate the supply of health services as well as the demand. One example of the regulation of supply is *certificate of need* (CON) legislation. In an attempt to regulate against an oversupply of healthcare capital investments, many states have kept CON legislation that requires independent review to determine whether additional investments are warranted. If not, the services they provide cannot be compensated for from state and federal funds. While CON helps avoid the arms race among providers, opponents see it as restricting freedom of entry by potential competitors, raising prices.

Withdrawal of services. Demand is constrained when payers restrict what they will pay for and how much they will pay. The absence of sufficient demand or reimbursement for services can lead professionals and provider institutions to withdraw from the market, downsize, or file for bankruptcy. Because of high malpractice insurance costs for obstetric procedures, for instance, many obstetricians have stopped doing deliveries and only provide gynecologic services. Many hospitals have reduced their psychiatric beds, shifting the burden to the state institutions. Some withdrawals are responses to overcapacity in an area, but others are the result of decisions that a particular line of services is losing money.

► Where in the World?

There are many sources of international comparative information. For domestic assessments of how well the U.S. medical care system is working, you can follow the reports of the Kaiser Family Foundation and the Commonwealth Fund. International comparisons are available from the OECD, the World Health Organization, and the European Observatory on Health Systems and Policies.

► Conclusion

Each group of actors wants specific things from the healthcare system. For the most part, there are clear majorities favoring insurance for all, controlling the rate of inflation, eliminating waste, improving quality care, investing in the most beneficial programs, taking care of children and the older population, and pushing ahead with research to find cures for diseases. Yet there will be a vocal minority on just about every issue—largely people representing special interests and those with strong conservative economic and social ideologies. All of the areas of general agreement were reflected to one degree or another in provisions of the ACA, but these provisions represented compromise, incrementalism, and extreme political pragmatism.

We still have much to learn about how theory translates into practice. That is why we believe in the continuing importance of policy analysis promoting rationality in decision-making, making fewer ungrounded assertions, and increasing ability to deal with new evidence and new opportunities.

CASE 4: National Standards on Culturally and Linguistically Appropriate Services in Health and Health Care (CLAS)

In 1997, the Office of Minority Health (OMH) in the U.S. Department of Health and Human Services began work on national standards for culturally and linguistically competent health care. The stated goal was to reduce health disparities. In December 2000, OMH released the standards. Although the standards were primarily directed at healthcare organizations, OMH encouraged their use by individual providers as well as by policy makers, accreditation and credentialing agencies, purchasers, patients, advocates, educators, and the healthcare community in general (OMH, 2001).

OMH began reviewing and revising the standards in fall 2010. The National CLAS Standards Enhancement Initiative included three lines of inquiry: a literature review, a public comment period, and consultation with an advisory committee. The purpose was to revise the standards “to reflect the past decade’s advancements, expand their scope, and improve their clarity to ensure understanding and implementation.” Precipitating factors, according to OMH, were increasing diversity in the U.S. population and rapid growth in the fields of cultural and linguistic competency.

This process culminated in the April 2013 release of a set of enhanced standards (OMH, 2013a). In the supporting documents, OMH explicitly defines health to include physical, mental, and social well-being. Cultures deserving of appropriate services include those identified by geographic, religious and spiritual, biological, and sociological characteristics, in addition to racial, ethnic, and linguistic groups addressed by the original standards. OMH also expands the target audiences and the target recipients—from “health care organizations” to “health and health care organizations” and from “patients and consumers” to “individuals and groups,” respectively (OMH, 2013b).

The original 14 standards included a mix of recommendations, mandates, and guidelines. The 15 enhanced standards are all guidelines of equal weight.

To encourage successful implementation of the enhanced standards, OMH published *The Blueprint*, an extensive implementation guide with explanations of each standard and listings of resources (OMH, 2013b). OMH declared that completion of the enhancement initiative means that for the next decade, the 2013 standard will “serve as the cornerstone for advancing health equity through culturally and linguistically appropriate services” (p. 155)

Principal Standard

1. Provide effective, equitable, understandable, and respectful quality care and services that are responsive to diverse cultural health beliefs and practices, preferred languages, health literacy, and other communication needs.

Governance, Leadership, and Workforce

1. Advance and sustain organizational governance and leadership that promotes CLAS and health equity through policy, practices, and allocated resources.
2. Recruit, promote, and support a culturally and linguistically diverse governance, leadership, and workforce that are responsive to the population in the service area.

3. Educate and train governance, leadership, and workforce in culturally and linguistically appropriate policies and practices on an ongoing basis.

Communication and Language Assistance

1. Offer language assistance to individuals who have limited English proficiency and/or other communication needs, at no cost to them, to facilitate timely access to all health care and services.
2. Inform all individuals of the availability of language assistance services clearly and in their preferred language, verbally and in writing.
3. Ensure the competence of individuals providing language assistance, recognizing that the use of untrained individuals and/or minors as interpreters should be avoided.
4. Provide easy-to-understand print and multimedia materials and signage in the languages commonly used by the populations in the service area.

Engagement, Continuous Improvement, and Accountability

1. Establish culturally and linguistically appropriate goals, policies, and management accountability, and infuse them throughout the organization's planning and operations.
2. Conduct ongoing assessments of the organization's CLAS-related activities and integrate CLAS-related measures into measurement and continuous quality improvement activities.
3. Collect and maintain accurate and reliable demographic data to monitor and evaluate the impact of CLAS on health equity and outcomes and to inform service delivery.
4. Conduct regular assessments of community health assets and needs and use the results to plan and implement services that respond to the cultural and linguistic diversity of populations in the service area.
5. Partner with the community to design, implement, and evaluate policies, practices, and services to ensure cultural and linguistic appropriateness.
6. Create conflict and grievance resolution processes that are culturally and linguistically appropriate to identify, prevent, and resolve conflicts or complaints.
7. Communicate the organization's progress in implementing and sustaining CLAS to all stakeholders, constituents, and the general public (OMH, 2013a).

Reproduced from Office of Minority Health, Department of Health and Human Services. (2013). *National standards for culturally and linguistically appropriate services (CLAS) in health and health care*. Washington, DC: U.S. Department of Health and Human Services. Retrieved September 4, 2017, from www.thinkculturalhealth.hhs.gov/clas/standards.

► Discussion Questions

1. OMH has said the enhancement initiative followed a decade of successful implementation of the 2000 standards, but the news media have reported on implementation problems, and the 2013 standards were released with an implementation manual almost 200 pages long. What do you think might be some barriers to implementation, and what would you suggest to encourage implementation?

2. What do the expanded definitions of health and culture say in general about the way government framed health in 2013 as compared to 2010? What might have contributed to the change?
3. How do you explain the differences between recommendations, guidelines, and mandates? What are the possible explanations for making all enhanced standards guidelines?
4. Which are the most important standards in your opinion? Which are the least important? Explain your reasoning.
5. What would you change about these standards if you were in charge of the U.S. Department of Health and Human Services?
6. Does it seem unusual to you for a government agency to wait more than 10 years to review and revise standards like these? What might be the reasons for waiting? What arguments could be made for more frequent review?



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CHAPTER 5

Representative Policy Options

This chapter presents a number of policy choices governments, institutions, organizations and individuals might support or oppose. These are not mutually exclusive nor collectively exhaustive; rather, they illustrate the breadth of health policy choices.

BOX 5-1 presents an array of federal-level alternatives organized by their primary objectives—access, quality, or cost. Many of these alternatives were in the Affordable Care Act (ACA) or in subsequent proposals from the Republican Congress and the Trump administration. **BOX 5-2** displays additional policy alternatives for state and local government levels, and **BOX 5-3** addresses policy options for institutions, organizations, and individuals.

The discussion below of many of these policy alternatives has been grouped under four headings:

- Access to Care
- Quality of Care
- Costs of Health
- Relationships with the External Environment

There is, of course, much overlap between these options. Many policy options assigned to one category could easily appear in another. Health care is a complex activity.

► Access to Care

Governments respond to political pressures, so government-administered systems focus first on access needs and then on costs, and these two foci tend to take priority over other quality criteria such as continuity of care (McLaughlin, 1998). While the United States is nominally a private system, its public sector makes very heavy expenditures in health care. In fact, the Organization for Economic Cooperation and Development (OECD) data indicate that the United States not only spends more per capita on health care than all other countries, it spends more public money per capita than most other countries that have universal public coverage, including Canada and the United Kingdom.

BOX 5-1 Illustrations of Federal Government Health Policy Options**Access to Care**

- Removing financial barriers
 - Universal coverage
 - Subsidize payment of bills
 - Modify insurance market
 - Specify basic coverage
 - Subsidize payment of premiums
 - Mandate enrollment
 - Mandate employer payments
 - Issue vouchers for premiums
 - Provide tax incentives for premiums
 - Provider high-risk reinsurance pools
- Modify supply to assure access
 - Increase number of professionals trained
 - Provide services directly
 - Support services for underserved populations

Quality of Care

- Incentivize quality improvement
- Publicize quality performance
- Remove regulatory barriers to innovation
- Conduct research on impact of information technology on satisfaction/outcomes
- Train professionals how to involve patients more in decision-making
- Support continuity of care
 - Provide medical homes for high needs patients
 - Support interoperability of information systems
 - Require culturally appropriate services

Costs of Health

- Modify demand
 - Reduce excess insurance coverage
 - Increase uptake of preventive services
- Modify supply
 - Reduce excess provider capacity
 - Limit funding for specific services or populations
 - Encourage disruptive innovation
- Constrain prices
 - Use reference pricing or limit network
 - Encourage competitive bidding
 - Use price controls
 - Constrain medical loss ratios
- Align incentives
 - Put someone in charge of all pricing and payments
 - Have just one price for all patients with same diagnosis
 - Use bundled pricing
- Change processes
 - Assign work to higher value organizations
 - Adopt nonmedical approaches to specific health problems

(continues)

BOX 5-1 Illustrations of Federal Government Health Policy Options (continued)

- Undertake regulatory reform
 - Allow more flexibility in labor substitution
 - Repeal ERISA exemption
- Enhance competition
 - Enhance antitrust law enforcement
 - Block hospital and insurer consolidation
 - Modify patent law to limit period of monopoly
 - Buy out patents to encourage generics sooner
- Reform tort law
 - Cap awards for pain and suffering
 - Use administrative courts to settle claims

Relationships with the External Environment

- Scan the environment
- Lobby
- Maintain public relations campaigns

BOX 5-2 Illustrations of State and Local Government Health Policy Options**Access to Care**

- Administered system
 - Universal coverage using general revenues
 - Expanded/reduced eligibility and benefits
 - Mandated coverages and services
 - Captive providers (e.g., health department clinics)
 - Increase funding to enable full enrollment of eligible populations
- Consumer-driven competition
 - Individual mandate for health insurance
 - Modify medical practice constraints
 - Support cooperative buying arrangements for smaller businesses
- Oligopolistic competition
 - Mandate employer participation/play-or-pay
 - Impose special taxes on providers and insurers to subsidize low-income uninsured
 - Increase primary care provider supply
 - Support pooled insurance risks
 - Make reinsurance more widely available

Quality of Care

- Administered system
 - Encourage “medical home,” especially for special needs enrollees
 - Use pay-for-performance approach in state purchased plans
 - Mandate installation and use of health information technology
- Consumer-driven competition
 - Support reporting of quality outcomes and quality survey data
 - Support interoperability and transferability of personal health records

- Oligopolistic competition
 - Support regional consumer information reporting and databases
 - Support pay-for-performance in private as well as public sector
 - Support training of providers in evidence-based practices

Cost of Care

- Administered system
 - Integrate Medicare and Medicaid services for dual eligibles
 - Negotiate program fees and discounts
 - Require disease management for special populations
 - Reduce/enhance primary care provider payments
- Consumer-driven competition
 - Strengthen antitrust laws and regulations
 - Remove insurance barriers to medical tourism
- Oligopolistic competition
 - Facilitate exchanges to enable access to more than one insurer's plans
 - Modify medical practice laws and constraints, as necessary, to encourage licensing and credentialing of new providers
 - Use certificate-of-need procedures
 - Encourage managed care and disease management
 - Enforce antitrust laws and regulations

Other Interventions

- Research and development (e.g., embryonic stem cell research)
- Capital investment
- Education of professionals
- Increase primary care provider supply
- Distribution of professionals and services
- Public health functions and departments
- Malpractice (tort law) reform

Broad approaches to improving access to care include:

- Removing financial barriers
- Removing insurance barriers
- Assuring availability of appropriate services

Americans' greatest concerns about health care consistently are its cost and how to pay for it. Those concerns did not go away with the ACA. The greatest financial risks have fallen on the working poor. Patients without insurance faced heavy debts and the risk of bankruptcy. Millions have gone without care or prescriptions or faced unnecessary adverse events. Many of the most important and most controversial reform proposals have sought to overcome those barriers. Some of the numerous possibilities are discussed in the following sections.

Providing Universal Coverage

Most developed countries have achieved this objective. It can be accomplished in different ways. It can be provided by the public sector as in the United Kingdom or through mandated private insurance as in Switzerland. It can be financed through general or special taxation or through employer contributions or through individual payments. Most countries use a combination of sources. The United States uses a wide variety of

BOX 5-3 Illustrative Responses and Initiatives of Organizations and Individuals**Common Approaches**

- Public relations
- Marketing and education
- Lobbying

Payers

- Employers
 - Eligibility
 - Subsidy offered
 - Plans offered
 - Relationship with insurers/self-insurance
 - Worker education and training
- Insurers
 - Method of organization
 - Method of payment
 - Plans offered
 - Case management/carve-outs
 - Utilization constraints
 - Consumer education

Providers

- Professionals
 - Organization of practice
 - Services offered
 - Incentives
 - Pricing
 - Patient relationships
 - Primary versus specialty care
 - Efficiency
- Institutions
 - Organizational structure
 - Scope and scale of services
 - Pricing/discounts
 - Efficiency
 - Quality improvement
 - Consumer information
 - Credentialing decisions
 - Involving payers in change processes
- Professions
 - Quality improvement
 - Provider education
 - Consumer education

Consumers

- Plan selection
- Provider selection
- Self-help

financing arrangements, funding Medicare with federal employment taxes, Medicaid with general state and federal revenues, and private insurance for most of the workforce through a mixture of employer and employee contributions and government premium subsidies. One potential option for universal coverage would be Medicare-for-All.

Employer contributions and individual insurance payments are often mandated. A feature of the ACA was the requirement that individuals purchase health insurance meeting certain coverage standards or pay a “shared responsibility payment” to the IRS for each uninsured member of the household. Families received premium support based on their income. Australia offers a two-tier system in which basic care is provided in government hospitals, but most Australians buy private insurance that allows them to use private hospitals with private rooms and other amenities.

Policy decisions associated with universal coverage include:

- How do we pay for it?
- What services should be covered?
- Should some populations like resident aliens be in or out?

Medicaid expansion, especially, has become a place where values clash. The ACA attempted to expand coverage to include those who are single and without children. The Trump administration has been committed to pulling that back with or without congressional involvement. HHS Secretary Tom Price and CMS Administrator Seema Verma (2017) expressed this point of view in a letter to all state governors which included the following: “The expansion of Medicaid through the Affordable Care Act (ACA) to non-disabled, working-age adults without dependent children was a clear departure from the core, historical mission of the program. Moreover, by providing a much higher federal reimbursement rate for the expansion population, the ACA provided states with an incentive to deprioritize the most vulnerable populations.”

They added, “We are going to work with both expansion and non-expansion states on a solution that best uses taxpayer dollars to serve the truly vulnerable” (p. 1).

Meanwhile individual states have been making stabs at universal coverage. For example, Maine and Vermont had legislation, but could not fund such a program. Washington State and Colorado considered proposals, Massachusetts has had a program in place that strained its finances. More recent proposals have gained traction in New York, Nevada, and California but have been placed on hold. The desire is there, but the resources are not readily available at the state level. Presumably, states will continue to experiment with the idea (Hervey, Mullin, & Bordelon, 2017). If Medicaid becomes a block grant, more states might consider that option more seriously.

Removing Financial Barriers

Partially Subsidizing Purchase of Insurance or Payment of Bills

Societies may also decide to cover in part:

- Specific healthcare activities
- Selected populations
- Specific health events

For example, Medicaid requires dental coverage for children, but leaves adult dental coverage as an option that most states do not implement. The federal government pays for care for specific populations through the Ryan White HIV/AIDS and End Stage Renal Disease programs. Some employers make smoking and weight reduction programs available to at-risk employees. Some communities modify their environment to promote walking and other exercise opportunities.

Removing Insurance Barriers

Insurance: The Usual Starting Point

Most health reform in the United States has been health insurance reform. Insurance is customarily a means of protecting against events that have a low probability of occurrence but a major financial impact. But health insurance, resting on a history as an employer-managed, tax-exempt component of employee compensation, includes a wide spectrum of care: emergent, chronic, and preventive. The consequences of decision-making are screened from the patient and many providers. Making rational choices in an environment of high uncertainty hard to achieve (Arrow, 1963). As **FIGURE 5-1** indicates, the insurance cycle involves many actors and many inputs and responses. There are no one-shot fixes to its problems, but there are many interacting linkages for health policy analysts to consider and for policy makers to manipulate.

Basic Coverage. The contract between the insurer and the insured specifies what is or is not covered and what the premium will be. However, these contract terms are covered by both state and federal laws and regulations. Employers present a limited set of plans, as do the insurance exchanges. Then the employee chooses one for which he or she would be willing to pay a share.

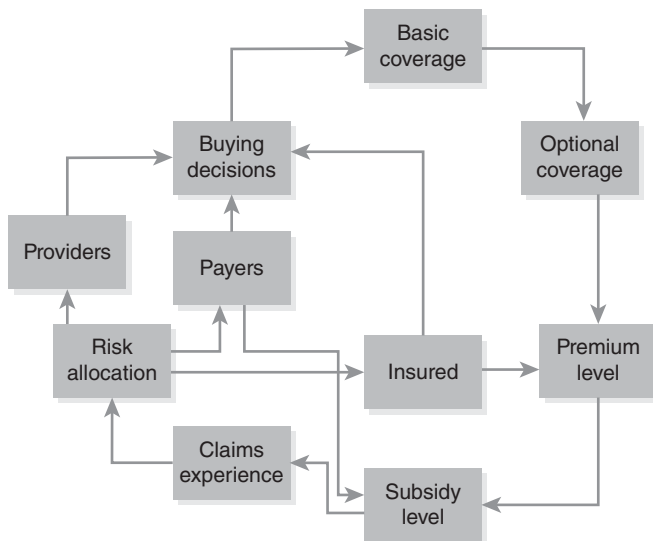


FIGURE 5-1 The health insurance purchasing cycle.

One of the most contentious issues associated with the ACA was the requirement to have health insurance that provided “minimum essential coverage.” It had to include ten specific services:

1. Outpatient care
2. Emergency room care
3. Inpatient care
4. Maternity care, pre- and post-
5. Mental health and substance use disorder services: This includes behavioral health treatment, counseling, and psychotherapy
6. Prescription drugs
7. Rehabilitation therapies
8. Laboratory tests
9. Preventive services—acute and chronic
10. Pediatric services including dental and vision care.

Other examples of covering specified services are “the basket” in Israel’s system or “medically necessary” under Medicare.

President Obama said people could keep their existing policies, but many received cancellation notices when their plans failed this test. Replacement plans with these coverages were often more expensive. When individuals with preexisting conditions joined the pools, the premiums went up further and real questions about “affordability” emerged.

Republican efforts to repeal the ACA included allowing plan options that excluded services such as preventive, mental health, contraception, and maternity care. During debates over these provisions, supporters used the term “affordable access,” which seemed to mean bringing back a much wider choice of plans in terms of what people can choose to insure for or not. These less expensive plans do not fully resolve the issue of affordability, because these plans have limited coverages, high deductibles and small payouts. This debate has exemplified the ideological tug of war between “shared responsibility” and free-market individual choice.

Optional Coverage. Governments, employers and individuals may add covered services, provide amenities, or insure additional types of care, including vision and dental. Mental health and maternity coverages are not considered as essential by some.

Premium Level. Once the coverage is defined, the insurer puts its underwriters and actuaries to work assessing the risks and determining the premium on the basis of estimated claims for that individual or pool of individuals plus operating costs and a profit margin (even if the insurer is a not-for-profit). Premium levels also reflect the bargaining power of the payer in the local market. Medicaid payments by the states tend to be the most limited, motivating many providers not to service that population.

Many provisions of the ACA curtailed the flexibility of health insurers to underwrite and otherwise limit their risks. Men and women were treated equally and the options for age-adjusting premiums were limited.

Other ACA provisions required insurers to offer coverage for children up to the age of 26 and prohibited exclusion of preexisting conditions, rescinding policies for people with high-cost illnesses, putting annual or lifetime limits on payments, and establishing long waiting periods before new enrollees received coverage. The ACA’s

underwriting approach is close to what is called community rating. All members of the community pay the same premium. The ACA did allow high premiums for older enrollees.

These constraints were very popular with the public. Their disadvantage was that insurance companies withdrew from areas where costs were high due to an older, sicker population. One way around that would have been to make participation in profitable markets conditional on taking on a pool of less profitable patients. The original plan was to offset the costs of these patients by having the individual mandate bring in premium revenue from younger, healthier individuals, but their response was underwhelming.

Several proposals have been put forward to segregate risk and offer lower-cost plan choices. These involve special subsidized pools for individuals with high risk factors and conditions. Risk pools have been used with some success for auto insurance, but health insurance costs are of a different order of magnitude and frequency. The failed 2017 Cassidy-Graham Senate bill to repeal the ACA did not allow insurers to reject enrollees with preexisting conditions, but it did allow insurers to charge them more.

Subsidy Level. The premium paid out of pocket by a household is often less than the premium cost. Partial subsidies come in many forms including:

- Employer contributions
- Premium payments direct to insurers or offsetting payments made to insureds
- Vouchers or other defined payments
- Income tax credits/healthcare and insurance tax deductions/tax deductible accounts
- Offsetting payments to safety-net providers
- Reinsurance, either prospective or retrospective
- Providing services below cost

Which methods to choose is a frequently debated topic. Two underlying concerns are impact on purchasing behavior and degree of wealth transfer. Tax credits and tax deductions favor the wealthy unless they are income-adjusted. For example, Medicare Part B premium deductions from Social Security payments are income-adjusted. Proponents of consumer-driven care want subsidy payments to go through the insured so that they make better-informed purchasing decisions and avoid unnecessary coverages.

The ACA had two types of subsidies: (1) one went to help families with 400% of poverty level or less for their premium payments, and (2) the second went to insurers to make premiums on the exchanges more affordable (referred to as cost-sharing reduction payments in the ACA). The first set of subsidies was funded in the bill, but the second was not. Lawsuits have been filed over whether President Obama exceeded his executive authority when instituting the second set of payments, and President Trump refused to pay them unless Congress authorized them.

Some conservative proposals for the repeal and replacement of the ACA included a subsidy in the form of a tax credit to try to make the plans “affordable.” Employers offer a health “benefit” that covers much of employee premium cost. While the employee may see it as a benefit, the employer sees it as part of the wage cost associated with the employee. The ACA premium subsidies were substantial for low-income families, but subsequent proposed tax credits, while related to incomes, were far less generous.

Conservative legislators, seeking to reduce the premium cost and encourage more “affordable” health care, have suggested a number of reinsurance programs.

Retrospective reinsurance would pick up the tab for extremely high-cost cases after the fact. Prospective reinsurance, sometimes called “invisible risk sharing,” would seek to identify patients expected to be costly and cover part of their premiums from a risk pool. The hope is that with more affordable premiums, more individuals would sign up and that would lead to a net increase in revenue.

CMS and the U.S. Treasury’s checklist for a 1332 waiver emphasized the desirability of reinsurance pools. “State-operated reinsurance programs have a demonstrated ability to help lower premiums, and if the state shows a reduction in federal spending on premium tax credits a state could receive Federal pass-through funding to help fund the state’s reinsurance program” (CMS, 2017, Introduction).

In 2017, for example, Iowa’s 1332 waiver application proposed two innovations. They would take Medicaid money intended for ACA premium subsidies and reinsurance claims between \$100,000 and \$3 million. They would also use those funds to subsidize the individual insurance market with a premium credit toward all individual policies under a single silver plan with high deductibles and coinsurance (Jost, 2017). This waiver request was rejected by CMS.

The withdrawal of the individual mandate as part of the tax reform legislation enacted at the end of 2017 is very likely to send the individual health insurance market under the ACA into a death spiral. Near universal coverage under the ACA rested on three pillars:

- Individual mandate
- Premium subsidies (direct and indirect) to assure affordability
- Standardized benefit package (minimum essential benefit)

Individual policy premiums were already rising rapidly in 2017–2018 because (1) fewer younger and healthier people signed up (choosing to pay the penalty instead) and (2) insurers anticipated partial or total ACA repeal.

Claims Experience and Allocation of Risk. How are the costs of care distributed under the insurance contract? This may be retrospective based on claims experience, or prospective through mechanisms such as capitation, or mixed such as through reference pricing, bundling, and diagnosis-related groups.

This is a tug of war out of sight of most insureds. When a health event occurs, someone must ante up the negotiated payment for the service. The patient usually has deductibles and co-payments. Providers’ billings are reduced by discounts, uncompensated care, and occasionally by penalties. They are often affected by risk-sharing agreements for actions that avoid costs and by quality and consumer satisfaction metrics.

Payers take many steps in addition to those previously mentioned to reduce their share of risks. They choose the providers in their networks based on quality, cost, and geographic coverage. They work to align the incentives in their contracts with their objectives.

Health policy debates have long involved the issue of moral hazard induced by health insurance. Conservatives often blame the high cost of care on the fact that employers pay the premium and insurers pay the bills and patients may not have much skin in the game. Providers often say that they don’t want to know the costs and they seldom introduce the subject of costs when patients are making decisions. There are many alternatives that try to get patient and provider to pay attention to cost. Most drug plans have tiered co-pays that encourage use of generics and other

lower-cost alternatives. A number of tax-sheltered alternatives are available that directly and indirectly reward conscious purchase behavior, including Health Savings Accounts and Cafeteria plans. Employers have increasingly switched employees to defined contribution plans and encouraged use of high deductible plans. A key question is how much these tax incentives discourage patient investment in postponable care, especially preventive care. A number of state Medicaid plans, including Indiana under then-Governor Pence's plan, require a monthly, nominal premium payment in hopes of focusing patient attention on costs. There are political constraints on the federal government bargaining directly on price, but other payers, including Medicare Advantage insurers, have long done so.

The public is suspicious of measures that restrict choice. That suspicion undermined the Health Maintenance Organization (HMO) efforts during the Nixon administration and weakened the role of the gatekeeper. However, these managed care concepts are gaining favor again, especially with states that are privatizing their Medicaid programs.

Outright rationing is rare, but barriers to access are not. The UK National Health Service tried to limit the use of new, high-cost, but often curative Hepatitis B drugs to only the most urgent cases. It ran into strong resistance. More often, hurdles are used. Long waits for optional treatments are evident in many national systems. Doctors' offices complain about the time spent on the phone with insurance companies to get authorizations. Long waits in offices can discourage patients from seeking emergent and even acute care. Some U.S. states limit the number of nursing home beds they license to match their acceptable growth rate for Medicaid expenditures. Insurers can also restrict their provider networks to remove cost outliers and direct enrollees to "centers of excellence."

Buying Decisions. The cycle is complete with the next choice of a plan. This can be influenced by all the payers as insurers design the plans to be offered, insurers negotiate with providers over their fees and roles in the network, employers and insurers offer the households a menu of plans, and the households choose plans. They make their decisions based on recent experience.

One decision is whether to expand or contract coverage. Insurers enable this all the time through their multiple offerings to satisfy employers and individual purchasers. Medicaid coverage for children was increased incrementally, 1 year of age at a time, in the 1970s and 1980s without much fanfare. Many states obtain Medicaid waivers to add or modify services. The Trump administration has increased considerably its flexibility in granting Medicaid waivers, especially those that one way or another reduce coverage and participation under that program.

Modifying Supply to Improve Access

Access issues are not limited to insurance coverages and payment mechanisms. They also relate to the supply of services available.

Increase Overall Supply of Services

The number of health professionals generated by individual educational institutions is subject only to student demand and funding. This is a pretty haphazard system for

matching short-run supply and demand. The supply of health professionals can also be manipulated by loosening or tightening state licensure requirements and federal immigration rules. States have set up bonding authorities making tax-exempt bond revenue available at lower rates to nonprofit health institutions.

Provide Services Directly

The Veterans Health Administration, rural and urban health centers, many public health departments, and the City of San Francisco have all provided examples of direct delivery of care subsidized by various levels of government. Since most care of at-risk populations takes place in private sector institutions, there are special government payments intended to subsidize the delivery of services below cost. Nonprofit providers also use grants and donations to support their free or underpriced services.

Subsidize Facilities and Services for Underserved Populations

The United States has long provided loan forgiveness for medical graduates who go to underserved areas through programs such as Public Service Loan Forgiveness (PSLF). There has been grant funding for community health centers in high-needs areas. The U.S. Indian Health Service provides care on Indian reservations and the military cares for personnel and their dependents at home and abroad.

► Quality of Care

Poor quality care is dangerous and wasteful, as Institute of Medicine studies have shown (IOM, 2000, 2001). Much has been done recently to try to remedy systemic quality problems. This effort has affected all levels of the healthcare system—governmental, institutional, professional, provider, and patient. Donabedian (1980) suggested the four categories for assessing quality of care used here: access (previously addressed), technical quality, patient experience, and continuity of care.

Improve Technical Quality

Many efforts are under way to improve and assure the technical quality of care delivered. They include generating new knowledge, better implementing existing and new knowledge, and motivating quality-conscious behaviors.

Generate, Disseminate, and Encourage Use of Evidence-Based Medicine

Policy analysts and managers throughout health care have shown a strong interest in finding and adopting evidence-based procedures and treatments. Failing to use them usually means that care will be less effective and more wasteful. But this movement still has a long way to go. One hopeful sign is the increased willingness of payers and institutions to recognize that developing this intellectual capital is their responsibility as much as it is the individual professional's.

Incentivize Quality Improvement

Payers and regulators are increasingly willing to use both carrots and sticks to improve quality of care. Private insurers have been out in front of governments, institutions, and professions in this area. More than half of the contracts between private insurers and physicians now have incentive components. Since it is very hard to measure outcomes quickly, most of the incentives relate to process variables. However, some adverse events like hospital-acquired infection now carry penalties in government programs. In fact, the implementation of so many different sets of measures by different payers now represents a challenge to providers.

Publicize Quality Performance

Increasingly patients and their families as well as insurers have access to quality data. Much of it is not very fine-grained nor specific to individual providers, but the trend is toward more and more disclosure. There are still problems in determining who is responsible for quality in specific situations and in standardizing results for patient differences. This movement took off in 2006 when President George W. Bush issued Executive Order 13410 entitled “Promoting Quality and Efficient Health Care in Federal Government Administered or Sponsored Health Care Programs.” HHS Secretary Michael Leavitt noted that there were “Four cornerstones” to the Executive Order:

- Interoperable health information technology
- Transparency of quality
- Transparency of price
- Incentives for high value care. (Leavitt, 2006)

The American Reinvestment and Recovery Act of 2009 and the ACA continued this initiative. Although the large regional data bases envisioned at the time failed to materialize, the financial and regulatory support for interoperability and transparency enabled collection of data direct from clinical records and reporting of provider performance similar to existing hospital performance ratings.

Remove Regulatory Barriers to Innovation

Regulatory and standards-setting bodies do not deal very well with moving targets. The development of new classes of caregivers has usually been opposed by existing licensing and credentialing organizations. Many states have laws restricting the practice of medicine. This often constrains new ways of delivering care. The 21st Century Cures Act of 2016 substantially expanded the types of information acceptable in the approval of new applications of existing drugs. As genomic information makes the defined population with a diagnosis narrower, the approval of new drugs will have to be made faster and less costly.

► Enhance Patient Experience

Television has given us the extremes of Marcus Welby and Doc Martin. Both held the public’s attention for a long time. Neither represents a realist expectation. Face time

is limited and costly, but trust and care is important. After financial risks, lack of face time is one of patients' principle concerns. All organizations in competitive markets have to pay attention to it, because patients have many choices of providers and feel increasingly free to pick and choose.

Collect and Share Information About Patient Satisfaction

Patient satisfaction information is now widely collected and shared due to government and insurer efforts. It is often not very specific, however. People want to know more at the individual level as well as the levels of institutions and practices. Buyers will act on these findings more in network inclusion decisions should the information prove reliable. Research linking patient satisfaction to outcomes could spur further quality interest at all levels.

Conduct Research on the Effect of Information Technology on Patient Experience

Patients do not always report that technology improves their experience. While they may appreciate being able to look up their lab results prior to their visit, they may find that clinicians do not respond to their concerns. Many patients report that their provider spends more time looking at the screen than at them and that makes them feel uncomfortable. We need good research on how to improve patient experience in the face of new technologies.

Train Health Professionals on How Best to Involve Patients and Families in Decision-Making

The trend toward making the treatment selection a cooperative venture between patient and provider is still young. It involves the melding of statistical findings and personal preferences, which is sometimes difficult to combine and interpret. It may produce decisions that run counter to existing procedures and protocols developed based on populations rather than the current patient. We will need to continue to make substantial changes in provider training programs to make that cooperation more effective.

Support Continuity of Care

The concept of continuity of care implies that there is a provider who knows the patient well enough to engage over time in meaningful cooperation with the patient and coordinate the patient's care from various services. Similar concerns are behind the current emphasis on making sure that individuals with chronic illness have a medical home.

Provide Medical Homes for High-Needs Individuals

The fragmented nature of health care with various specialists and institutions serving the high-needs individuals often leads to wasteful and ineffective care. This has been especially true of individuals who are dually eligible for Medicare and

Medicaid. State after state has been taking steps to somehow integrate their support and care through designation of a medical home and case management.

Support Information System Interoperability

While information systems are not really the equivalent of a human relationship, they can support that process. They can also help enhance the memory of the provider or provide better support when the primary caregiver is not available. To do so, the provider must be able to see records from all relevant providers.

Provide Culturally Appropriate Services

Individuals in need will not seek services where they do not feel welcome, except during their most acute episodes. Federal agencies have had offices devoted to minority affairs that set standards for translation and cultural sensitivity. The degree of support for this effort waxes and wanes with support for multiculturalism. Changing demographics also pique the interest of provider groups and institutions as they are always on the lookout for their expanding share of market.

► Costs of Health

Of all the issues surrounding health care, cost seems to be the hardest nut to crack. The rate of growth ebbs and wanes, but it averages 2% greater than that of the economy as a whole. Some of this is the aging of the population, but much of it is the addition of new technology and the misaligned incentives that motivate doing more and avoiding price reductions. One contributing factor has been health professionals' resistance to economic interpretations of their sphere of influence. But it is still a powerful way to view cost-related policy issues.

The broad categories used here are:

- Modify demand
- Modify supply
- Constrain prices
- Align incentives
- Change processes
- Undertake regulatory reform
- Enhance competition
- Reform tort law

A useful class exercise would be to define additional or different categories. Maybe you and your instructor can come up with a categorization that bypasses classical economic models.

Modify Demand

Where we used to have cigarette ads we now have healthcare ads. There is continuous pressure on Americans to increase their consumption of health care. Efforts to increase access also increase demand. There was major cost inflation after 1965

when Medicaid and Medicare were introduced. Some of the many possible ways to modify demand are discussed in the following sections.

Limit Advertising of Products of Limited Health Impact

Most countries constrain direct-to-consumer advertising. The ability of the United States to follow is severely limited by Supreme Court rulings on “commercial speech.” The FDA focuses on safety and efficacy, but does not require material relative improvement to warrant approval. One might go even further and question why we treat so much that is part of normal living and aging. Many of the products most highly promoted on television are not aimed at physically threatening conditions. Erectile dysfunction and mild heartburn due to indulgence are examples.

Reduce Excess Insurance Coverage

Our tax laws make employer health insurance payments a tax-sheltered source of income. The ACA included a “Cadillac” tax on expensive insurance plans. This was opposed by labor unions who had taken this benefit in collective bargaining in lieu of taxable income. Critics argue that the “minimum essential coverage” requirements of the ACA raised the cost of care because they included services that individuals would not ordinarily choose to cover. One might debate whether paying for a plan with maternity benefits by a 27-year-old single male was inappropriate or was a legitimate sharing of responsibility. The Republican proposed replacements for Obamacare referred as “universal access,” allowing individuals to avoid paying for coverages they did not want, but also allowing them to buy policies with little catastrophic coverage. So has President Trump’s Executive Order expanding availability of association health plans and plans without contraceptive coverage. Some Republican proposals would make preventive services optional.

Further Involve Patients and Families in Decision-Making

There is ample evidence from studies of end-of-life care that involvement of patients and their families leads to less heroic interventions and reduced cost. Patients who ask if the next procedure is really worthwhile may decline the honor. They are not concerned about malpractice suits, only about the effect on their future safety and comfort. Evidence is increasing that informed, involved patients achieve better outcomes. Provider attitudes are slowly changing and we will see more patient involvement, but there is still much to learn about the best ways to do it.

Increase Uptake of Preventive Services

Providing preventive services is another way of reducing demand. The ACA insisted that these be provided without co-pays on grounds that cost would be averted later. Private insurers have been of two minds. The insurer who bears the cost may not reap the benefit down the road. But insurers want to be responsive to employers who take a somewhat longer view and support preventive activities.

Modify Supply

Increases in the supply of services are usually celebrated by communities. Health care provides employment and is the only route to upward mobility open to many. Reducing the available supply is another story.

Avoid or Reduce Excess Provider Capacity

Institutions will often close down service centers that fail to cover their expenses despite the pressure to offer a full range of services and keep people from going elsewhere. The New York State legislature reduced excess hospital beds by a process modeled after the congressional Base Realignment and Closure process. A bipartisan commission reviewed the needs of the state and recommended reductions. Its recommendations became law when the Legislature took no further action (Cooper & Chan, 2006). That kept legislators from having to go on record about specific closings.

In the 1960s, certificate-of-need legislation appeared requiring providers to obtain state approval for additional major capital investments, such as imaging equipment and additional beds, if they were to receive state reimbursement. It was an attempt to mediate arms races among provider institutions; however, it is typically a highly political process that has often lacked effectiveness. Opponents argue that it stifles competition by limiting entry into the field and by stopping efficient competitors from expanding to meet demand.

Limit Funding for Specific Services or Populations

This has happened at both the state and federal levels at different times with Medicaid. Republican administrations have capped federal payments to the states and Democratic administrations have removed them. States have closed enrollment when funds ran out. Sometimes enrollment was reopened despite limited funds and a lottery was used.

Insurers may also reduce the size of their network, eliminating high cost providers. An alternative is to establish “centers of excellence,” which are known to avoid overutilization and to provide high quality of care. Use of these centers is usually optional, but the insurer may also demand that physicians in their network refer specific cases to those centers.

Rationalize Education and Training of Professionals

Most professionals are licensed by state boards. Most specialists are certified by national professional boards. These processes represent an opportunity for restricting entry. The Federal Trade Commission (FTC) has been aware of this issue and has moved decisively against the professions’ restrictive measures such as rules against advertising. While antitrust considerations probably limit the ability of boards and institutions to restrict enrollment to match demand, analysts must be aware that overexpansion is a risk, especially in as much as the financial dynamics of institutions that train providers and produce services usually favor expansion over contraction.

Encourage Disruptive Innovation

Disruptive innovation usually involves simplified delivery of a less complex, less expensive product (Christensen, Bohmer, & Kenagy, 2000). The introduction of a highly dilute form of a cancer drug to fight wet age-related macular degeneration is one example.

Since much of the differential between the cost of care in the United States and that in other countries is in our high wages, there have been many efforts to encourage labor substitution at the governmental and institutional levels. Any attempt to introduce a new type of provider that performs a more limited range of services at lower cost has usually been resisted by entrenched professions. In general, legislatures have had to intervene, citing the needs of underserved areas or populations. Examples include nurse practitioners, physician assistants, and surgicenters.

Constrain Prices

Health care is not as price sensitive as many other sectors. It is a service for which value is difficult to judge, so price level and provider reputation are often used as value proxies. Furthermore, the lack of geographic competition and consolidation coupled with the deep pockets of insurers have given prices a strong upward bias. For example, a 2015 Government Accounting Office (GAO) report recommended adjustments to Medicare fee schedules that were encouraging consolidation of physician practices, in part to get outpatient department fees rather than normal office visit fees (GAO, 2015).

The policy alternatives that are used or proposed to constrain prices of goods and services in health care are illustrated in **FIGURE 5-2**. A few of the more important ones are included in the discussion in the next sections.

Other policy alternatives affecting price are considered under “Enhance Competition.”

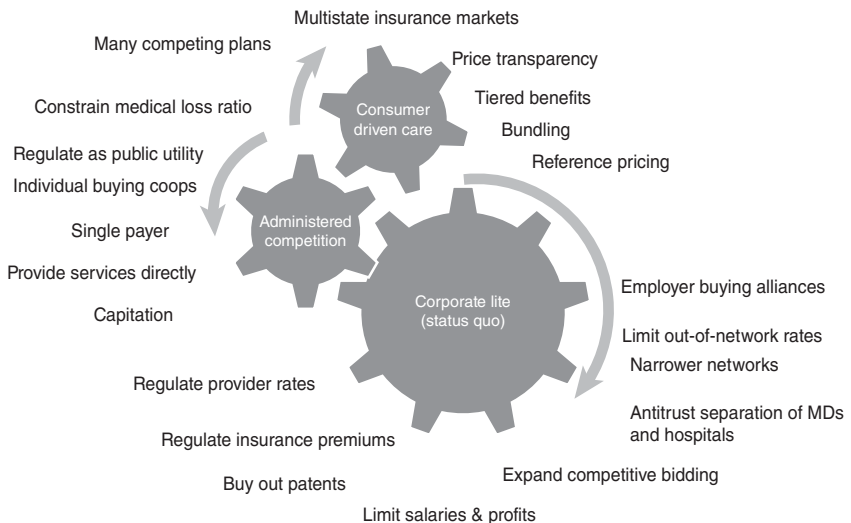


FIGURE 5-2 Pricing countermeasures under current influence distribution.

Data from Brill (2015), Ginsberg & Powell (2014), Havinghurst (2017)

Expand Competitive Bidding for Goods and Services

Congress and the major U.S. health programs have been very reluctant to embrace competitive bidding. One reason is that competitive bidding often restricts choice. If the buyer is using a service provider that loses out, he or she must then switch to a winner. The equivalency of service quality is hard to prove in many cases. Yet competitive bidding in the absence of collusion is very effective at bringing costs down. So cost-conscious organizations use it in those areas where it is not very risky. A demonstration program in which Medicare introduced competitive bidding for durable medical equipment appeared to produce a 35% saving (Newman, Barrette, & McGraves-Lloyd, 2017).

Set Limits on Prices Including Reference Pricing

Most countries that have large numbers of private providers in their national plan set prices. The United States does so for its largest programs. The federal government sets Medicare payments and states set Medicaid payments. Private insurers negotiate set fees with provider groups and institutions. This has led to the phenomenon of “cost shifting” in which list prices are marked up to cover lower prices charged to powerful purchasers.

An intermediate way of limiting payment is reference pricing in which payment is limited to some target level within the distribution of prices charged by a reference group. The patient who chooses a provider who charges more than the reference price is liable for the difference. Germany sets its payments for drugs with reference to the prices paid by nearby countries (Lauterbach, McDonough, & Seeley, 2016). For example, grocer Safeway found that the prices for colonoscopies in the San Francisco area ranged from \$898 to \$5984. The company determined that in 2009 it could achieve coverage with practices charging \$1500 or less and limited its payment to that. It reduced its payment to \$1250 in 2010 (Robinson & MacPherson, 2012). California Public Employee Retirement System limited payment for knee replacements with reference to other hospitals in the same region. It, in effect, allows high cost providers to stay in the network, but limits their payment. Boynton and Robinson (2015) report that the high-priced providers tended to reduce their charges to be in line with the competition.

Encourage New Market Entrants

There has been much public outrage over companies who rapidly increase the price of existing drugs when competitors leave the market. Their monopoly is not illegal. But one policy option would be to tax away price increases that exceed a specified rate of growth (Hwang & Kesselheim, 2017). Governments could subsidize and offer regulatory relief to new entrants who agree to provide the product at lower prices. Governments could also subsidize the development and approval process for products and services with significant potential impact on limited populations. Orphan drug laws in the United States and the European Union are a model that could be extended to those willing to reduce prices. Governments of large but poor countries have successfully driven key drug prices down by threatening to withdraw patent protection and develop local suppliers.

Remove Insurance Barriers to Cost-Effective Purchasing

Interstate sale of health plans is a reform intended to bring about competition among health insurers. It has been opposed by state insurance commissioners who would lose power and are legitimately concerned about a race to the bottom among the states. One could also take an international approach and support coverage of specific procedures under medical tourism.

Reduce Impact of Patent Monopolies

Patent rights are written into the U.S. Constitution. However, there are also laws concerning patent limitations for public use, such as those governing generic drugs and biosimilars. Congress can change the length of the monopoly, insist on the formation of patent pools, and bar agreements to pay generics manufacturers to withhold generics for a specified period. Silver (2017) suggests a variant of patent pools in which a nonprofit organization buys the intellectual property rights to one of several competing patented drugs and relicenses it to generic manufacturers.

Constrain Medical Loss Ratios

Medical loss ratio is a term that refers to the proportion of insurance premiums spent on provision of care rather than administrative and marketing costs and profit margins. Many states have set minimum medical ratios. So did the ACA.

Align Incentives

The fee-for-service system is generally accused of being a volume-maximizing system. Furthermore, the fee schedules are skewed toward procedures and away from cognitive activities. So almost every reform proposal attempts to align incentives more effectively, often by putting one decision-maker in charge or putting every involved provider in the same boat.

Put One Decision-Maker in Charge

We have seen many attempts to put someone in charge including gatekeepers in the British Health Service, HMOs managed by insurers, ACOs managed by providers, and large, integrated group practices with an insurance arm. Where the organization lacks expertise, carve-outs are assigned management and for complex and high-cost diagnoses case managers are used.

The Independent Payment Advisory Board established by the ACA was empowered to recommend reduced prices (but not services) for Medicare services if certain components of healthcare inflation failed to achieve certain targets. If Congress did not act promptly against the changes, they would go into effect. There has been continuous provider lobbying against this provision, although some primary care providers have supported it as a way of overcoming the low prices for cognitive work when compared to doing procedures.

Put Everyone in the Same Boat

Two popular ways of doing so are capitation and bundling. Capitation involves a single payment for specified care over a stipulated time period. Bundling involves a single payment for a single episode of care. Capitation is widespread in the private sector. Bundling is still somewhat experimental and has been slow to take off given lack of an accepted rule for dividing the pie among practices and institutions. A frequent target for bundling is knee replacement including rehabilitation costs. In this area, problems of allocating revenue and handling atypical patients remain (Hoff, 2017). The Trump administration appears to be slowing the implementation of bundled payments in Medicare.

Align Compensation and Organization of Providers with Overall Objectives

The payer/employer wants to motivate providers to make decisions that enhance its organizational objectives. In economics this is called an agency issue. Healthcare providers face a second agency issue because the physician is already an agent for the patient, setting up a conflict of interest. A primary method of motivation is through practice organization and compensation arrangements. **TABLE 5-1** outlines various contract options used by insurers and also identifies the dominant organizational form associated with each payment method and the degree of control that each method exerts on providers. The organization tries to achieve its objectives through payment mechanisms. One way or another most physicians are involved in incentivized compensation schemes.

Change Processes

Some approaches to process change were presented during our discussion of technical quality. But many more are possible beyond evidence-based medicine and quality improvement efforts. Berwick and Hackbarth (2012) characterized many of these as involving eliminating waste in one of six categories that at minimum accounted for 20%, and more likely 34%, of the cost of health care.

1. Failures of care delivery—medical error, not following best practices in care and prevention.
2. Failures of care coordination
3. Overtreatment—overuse of antibiotics, excessive end-of-life measures, etc.
4. Administrative complexity—inefficient and misguided government, institutional, and professional rules and procedures
5. Market (pricing) failure, including lack of price transparency and competition
6. Fraud and abuse

ACOs and HMOs that have been successful in controlling costs have done so with:

- Focus on maximizing the population served rather than maximizing revenue
- A culture of operating comfortably with constrained resources, due to capitation, inability to cover the target population fully, or inadequate funding.
- Data driven analysis to identify cost outliers and opportunities for better coordination
- Top management support for steadily improving resource use.

TABLE 5-1 Compensation Arrangements for Physician Healthcare Services

Table	Associated MD Organization	Cost Risks Borne By	Payer Control	Comments
FFS	Independent partnership	Insurer	Weak	Rapidly disappearing
Discounted FFS	PPO	Insurer	Relatively little, some through review process	Includes Medicaid and Medicare under assignment
Discounted FFS with withhold	PPO or IPA-type network	Mostly insurer	Relatively weak	Often associated with gatekeeper roles
Discounted FFS with performance incentives	PPO or IPA network, heavy concentration in one payer ACO partnership	Mixed	Some through incentive structure	Frequently related to quality and not quantity
Mixed capitation FFS	HMO network ACO partnership	More to provider	Some through structure of payments	May include hospitalists and other specialists
Capitation	HMO contract ACO partnership	Provider	Must review quality	Few providers can tolerate risks for long
Salary and bonus	Staff model HMO or ACO	Mixed	Potentially high	Medical practice acts often limit control

FFS = fee for service. PPO = preferred provider organization. ACO = affordable care organization

- Highly visible incentives for the overall organization for producing value.
- Clarity that these rewards are being fed back into the organization to enhance its objectives.
- Encouragement for local operating units to undertake their own problem-solving efforts.

You can learn more about this by studying iconic HMOs like Kaiser Permanente or Intermountain Health and by following contemporary experience like that of NYC Health + Hospitals Medicare ACO (Stine et al., 2017).

Support Process Innovation

The private sector and the government conduct research on products and treatments, but relatively little work is done concerning process efficiency. For example,

we know that the level of waste is high in health care. One of the biggest sources of waste is low personnel utilization. Some of this is due to the uncertainties of demand levels. Waste reduction efforts have tended to be left on the back burner. For example, more could be done to apply job shop scheduling methods and process flow analysis to healthcare operations. Berry and Saltman (2007) provide an example of how this has worked at the Mayo Clinic.

Encourage Less Expensive Substitutes

The iconic version of this is adoption of generic drugs. As mentioned elsewhere, labor substitution is another example. Many hospitals unable to compete for U.S.-trained residents rely heavily on foreign-trained individuals. The rise of free-standing surgery centers is another example.

Assign Work to Higher Value Organizations

Sometimes the term “centers of excellence” is used, so is “focused factories.” Outcomes tend to improve with volume, all other things being equal. The time taken to undertake a procedure also goes down with volume due to individual and organizational learning. Proper selection of provider organizations can improve both quality and cost, but it takes considerable management initiative and control to achieve it. Presently, that task seems to rest with the insurers and a limited number of locally dominant employers.

Adopt Non-Medical Approaches to Some Health Issues

The provision of housing as part of addiction recovery by homeless individuals is a case in point. Modifying the built environment to encourage a healthier life style is another. Removing lead from interior paints is third. Neither healthcare causation nor treatment occur in a bubble and once the low-hanging fruit is picked, we may have to take a much greater interest in social causes of morbidity and mortality.

Undertake Regulatory Reform

Practice of medicine laws and the accompanying system of licensure and accrediting boards are a constraint on efforts to reduce cost through labor substitution. The rules and rulings in each state can be arbitrary and in conflict, limiting labor mobility and process improvement.

One issue that has confounded insurance reform efforts is the “deemer clause” of the Employee Retirement Income Security Act (ERISA) that essentially exempts self-insured plans from state regulation. Self-insured plans cover a substantial portion of employed workers, meaning that there are two highly fragmented systems of employer health insurance: one is highly regulated, the other is not.

Enhance Competition

Competition should lower prices and lead to cost-reduction efforts. There is an alternative: treat health care as a regulated public utility. That would require a total overhaul of our health system. Most regulated utilities provide simple services such as telephone, electricity and water supply. Even these simple systems have been difficult to regulate effectively. It is hard to see how that approach alone could work for our decentralized health system with literally hundreds, if not thousands, of products and mostly local markets.

Strengthen and Enhance Antitrust Enforcement

Earlier chapters have discussed the cost effects of consolidation and the role of federal antitrust laws in constraining combinations “in restraint of trade” and anticompetitive practices by professional groups. Fulton (2017) reports on the high level of concentration in healthcare markets. Using the FTC’s measure of concentration, he finds that from 2010 to 2016, concentration among primary care physicians increased the most and that in 2016 90% of Metropolitan Statistical Areas had high concentration in hospitals, 65% in specialists, 39% in primary care physicians, and 57% in insurers. Capps, Dranove, and Ody (2017) report that this consolidation of physician practices has taken place through acquisitions that are too small to trigger the oversight and legal authority of federal antitrust guardians, so action probably has to come from the states. These authors and others (Glied & Altman, 2017; Scheffler & Arnold, 2017) suggest that antitrust measures can be effective only when coupled with other regulatory measures such as price controls. Antitrust enforcement has managed to stop further consolidation of large insurers (Dafny, 2017).

Restrict Anticompetitive Practices

States can legislate against specific anticompetitive practices. For example, Pennsylvania law duplicates and supplements a number of the federal anti-kickback and Stark amendment provisions. One section prohibits hospitals from renting clinic or office space to physicians below market. Others require that patients be informed if their doctor has an economic interest in the facility to which they are being referred, and that their rights to choose an alternative facility are disclosed.

Reduce Barriers to Entry

Many HMOs require their providers to be board-certified. This is a marketing decision that adds another constraint to the local supply of providers. Professional organizations must also approve residency programs in their specialties, thereby exerting some control nationally over the quantity and quality of services available. To offset some of these possible anticompetitive effects, states have countered with antitrust actions and with “any willing provider” legislation. Such legislation often addresses two issues, namely (1) restrictions on the panel of providers that a patient can access within a profession and (2) restrictions across professions as to who can be compensated for a service. Under such a law, for example, if a person has acute low-back pain, the insurer cannot limit the patient to seeing a small number of pain experts and clinics, and it cannot limit payment to primary care physicians. It may also require allowing payments to other providers—chiropractors and acupuncturists, for example—if such treatment falls within their scope of practice. A counterargument is that such laws can hinder institutional cost-control efforts. State certificate of need processes are also considered barriers to entry.

As noted, state licensure and certification of specialists by national professional boards represent opportunity for restricting entry. The FTC has been aware of this issue and has moved decisively against the professions limiting training positions. The Interstate Commerce Commission has jurisdiction of hospital mergers and has acted in some cases. There is much debate over these mergers, which are supposed to improve efficiency and bargaining power but also appear to lead mostly to higher net prices in the affected markets.

Build Buying Combines to Offset Provider Concentration

Action generates reaction. In 2016, 20 major employers, including Coca Cola, American Express, Verizon, and Hospital Corporation of America, formed the Health Transformation Alliance to increase bargaining power. They started to pool claims and clinical information to support buying decisions. In 2017, the alliance announced contracts with CVS Health Corp., and UnitedHealth Group, Inc. to purchase pharmaceuticals through them, and with IBM to use its Watson system to analyze healthcare data. This is to be followed by contracts with specialized networks of local providers in selected cities for diabetes care, joint replacements, and low back pain.

Individuals and small businesses can also form purchasing groups that structure an array of contracts from which members can select. This would probably require the support of state insurance commissioners. The Trump administration, however, has opened up use of association health plans that operate under ERISA.

Conduct Tort-Law Reform

The ACA included a Sense of the Senate paragraph supporting experimentation by the states with various methods of resolving malpractice claims, but the legislation did not endorse a specific approach.

We hear a lot about malpractice reform, especially about frivolous lawsuits and unreasonable awards for pain and suffering. We see local television advertisements by law firms seeking to represent clients “wronged” by providers and insurers. State and federal legislatures debate whether to cap the size of awards. When Vermont tried to fund a single-payer system, it included malpractice reform when calculating systemwide savings to offset the cost of expanded coverage.

Funding flows freely into campaign coffers from organizations on both sides of the debate. Trial lawyers favor the Democratic party, whereas insurers and organizations representing health providers and facilities favor the Republican party. Putting politics aside, however, several values-related policy issues recur in these debates, including the following:

- The value of a human life lost and of other negative consequences suffered unnecessarily
- The allocation of the responsibility for error between the individual provider and the overall care system
- The amount of variability in outcomes and events that is unavoidable and how much is unacceptable
- The appropriate way to overcome inappropriate provider behavior
- How much to let the provider community police itself and how much and when to intervene in the public interest

Overall, massive awards seem to be going down, but the effect of these measures on the costs of defensive medicine is open to question.

► Relationships with the External Environment

All players manage their relationships with the external environment. They monitor demographic, social, and political trends. Then they attempt to influence that

environment by addressing the media and politicians and regulators directly and indirectly through lobbyists, trade associations, and professional groups.

Scanning the Environment

Organizations also monitor technological trends, search for disruptive innovations, and identify forces that ultimately might affect them. Some do this systematically as part of their policy analysis processes. Some do this internally, while others rely on the observations of government agencies, trade associations, and think tanks.

Lobbying

Lobbying is particularly intense around administered programs such as Medicare and Medicaid, especially when new legislation is under consideration. Lobbying also goes on continuously with relevant executive branch agencies as standards and rules are developed. The number of healthcare lobbyists assigned to Congress has exceeded the number of members.

Public Relations and Advertising

Public relations and advertising are used more assertively when regulators are considering changes, and marketing, especially advertising, is most intense where the market is less stable. Many nonprofit organizations such as hospitals seek community support in terms of financial contributions and donations of blood, organs, and volunteer time through public appeals.

Education

The term education gets applied to the many efforts to influence behavior. Public service announcements encourage worthy causes such as stopping teenage drinking. But “education” can also be used as a euphemism for commercial activities, such as direct-to-consumer advertising of prescription drugs or assertions that gynecological care and sports medicine are forms of primary care.

► Conclusion

Health care is very much influenced by governments in their roles as payers, insurers, regulators, employers, and providers of last resort. In the United States the private sector provides only about 51% of the payments for care (OECD, 2017), far less than many would believe.

Many types of organized groups try to influence health policy, putting forth a wide array of policy alternatives. So, we in the United States have ended up adopting a confusing and inconsistent array of measures designed to deal with the concerns of access, quality, and cost. For example, resources go out to enhance access to services and to expand the supply of providers and technology. At the same time other efforts seek to constrain consumption. No wonder professionals caught up in this maelstrom sometimes feel discouraged. Still most professionals and policy analysts continue to balance the interests of patients and their organizations conscientiously and successfully.

CASE 5: Global Medical Coverage

Background

Blue Ridge Paper Products, Inc. (BRPP) in Canton, North Carolina, was a paper company whose predominant product was food and beverage packaging (it continues to operate under a different name with new ownership). It was the largest employer in Western North Carolina in 2006, with 1300 covered employees in the state and 800 elsewhere. Started as the Champion Paper plant in 1908, it was purchased by the employees and their union (a United Steelworkers local) in May 1999 with the assistance of a venture capital firm. It operated under an employee stock ownership plan (ESOP). To purchase the plant, the employees agreed to a 15% wage cut and frozen wages and benefits for 7 years. From the buyout through the end of 2005, the company lost \$92 million and paid out \$107 million in healthcare claims. It became profitable in 2006. Maintaining health benefits for members and retirees was a very high priority with the employee-owners and the union, although retiree medical benefits were eliminated for salaried employees hired after March 1, 2005. The venture capital firm that financed the ESOP retained 55% ownership with 40% going to the employees and 5% to senior management. Profitability varied from year to year as the company expanded capacity and improved productivity of its single-serving drink carton lines and was caught up in a number of suits over water pollution problems at its Canton, North Carolina plant.

The majority of BRPP employees were male, older than age 48, and had several health risk factors. Most employees worked 12-hour, rotating shifts, making it extremely difficult to manage health conditions or improve lifestyle (Blackley, 2006). The ESOP worked hard to reduce its self-insured healthcare costs. Health insurance claims for 2006 had been estimated at \$36 million, but were closer to \$24 million, which was still 75% above the 2000 numbers. A volunteer benefits task force composed of union and nonunion employees worked to redesign a complex benefit system. After 2 years of 18% healthcare cost increases, the rate of growth dropped to 2% in 2003. It was 5% in 2004 and -3% in 2005.

Programs initiated in 2001 included a plan offering free diabetic medications and supplies in return for compliance, and a tobacco cessation plan with cash rewards. In 2004, the company opened a full-service pharmacy and medical center with a pharmacist, internist, and nurses. In 2005, it began a population health management program. Covered employees and spouses who completed a health risk assessment were rewarded with \$100 and assigned a "personal nurse coach." The nurse coach assisted those who were ready to change to set individual health goals and to choose from among one or more of 14 available health programs, which included reduced co-pays on medications, free self-help medical aids/equipment, and educational materials.

Where BRPP could not seem to make headway was with the prices paid to local providers. Community physicians refused deeper discounts. Even banding together in a buying cooperative with other companies could not move the local tertiary hospital to match discounts offered to regionally dominant insurers. This hospital was not distressed and had above-average operating margins.

Articles on "medical tourism" in the press and on television attracted the attention of benefits management. Reports were of high-quality care at 80% or less of U.S. prices with good outcomes. BRPP contacted a company offering services at hospitals in India, IndUShealth in Raleigh, North Carolina, and began working on a plan to make its services available to BRPP employees.

IndUSHealth

IndUSHealth provides a complete package to its U.S. and Canadian clients, including access to Indian superspecialty hospitals that are Joint Commission International accredited and to specialists and supporting physicians with U.S. or U.K. board certification. It arranges for postoperative care in India and for travel, lodging, and meals for the patient and an accompanying family member—all for a single package price. For example, it represents the Wockhardt hospitals in India, which are Joint Commission International accredited and affiliated with Harvard Medical International. Other Indian hospitals boast affiliations with the Johns Hopkins Medical Center and the Cleveland Clinic.

Mitral Valve Replacement

One of the first cases considered was a mitral valve replacement. IndUSHealth and BRPP sought package quotes from a number of domestic medical centers and could get only one estimate. That quote, from the University of Iowa academic medical center, was in the \$68,000 to \$98,000 range. The quote from India was for \$18,000 and included travel, food, and lodging for the patient and one companion. Testifying before the U.S. Senate Special Committee on Aging, Mr. Rajesh Rao, IndUSHealth's CEO (2006), cited the following costs:

Procedure	Typical U.S. Cost	India Cost
Heart bypass surgery	\$55,000–\$86,000	\$6,000
Angioplasty	\$33,000–\$49,000	\$6,000
Hip replacement	\$31,000–\$44,000	\$5,000
Spinal fusion	\$42,000–\$76,000	\$8,000

Employee Participation

To encourage employee participation, BRPP prepared a DVD on its medical tourism initiative, which it called Global Health Coverage. It outlined the opportunities and described the Indian facilities and credentials. The next step was to be a trip by an employee "due diligence" committee to India to inspect facilities and talk with doctors. Then they would discuss how to handle the option in the next set of union negotiations.

Senate Hearings

On June 27, 2006, the U.S. Senate Special Committee on Aging held hearings titled "The Globalization of Health Care: Can Medical Tourism Reduce Health Care Costs?" Both BRPP and IndUSHealth testified for the committee. When testifying to the Senate subcommittee, Bonnie Grissom Blackley, benefits director for BRPP, concluded:

Should I need a surgical procedure, provide me and my spouse with an all expense-paid trip to a Joint Commission International-approved hospital, that compares to a 5-star hotel, a surgeon educated and credentialed in the U.S., no hospital staph infections, a registered nurse around the clock, no one pushing me out of the hospital after 2 or 3 days, a several-day recovery period

(continues)

at a beach resort, email access, cell phone, great food, touring, etc., etc. for 25% of the savings up to \$10,000 and I won't be able to get out my passport fast enough.

Blue Ridge Paper Product's Test Case

The test case under the new arrangement was a volunteer, Carl Garrett, a 60-year-old BRPP paper-making technician who needed a gallbladder removal and a shoulder repair. He reportedly was looking forward to the trip in September 2006, accompanied by his fiancée. A 40-year employee approaching retirement, he would be the first company-sponsored U.S. worker to receive health care in India. The two operations would have cost \$100,000 in the United States, but would cost only \$20,000 in India. The arrangement was that the company would pay for the entire thing, waive the 20% co-payment, give Garrett about a \$10,000 incentive, and still save \$50,000.

However, the United Steel Workers Union (USW) national office objected strongly to the whole idea and threatened to file for an injunction. The local district representative commented, "We made it clear that if healthcare was going to be resolved, it would be resolved by modifying the system in the U.S., not by offshoring or exporting our own people." USW President Leo Gerard said, "No U.S. citizen should be exposed to the risk involved in travel internationally for health care services." The USW sent a letter to members of Congress that included the following (Parks, 2006):

Our members, along with thousands of unrepresented workers, are now being confronted with proposals to literally export themselves to have certain "expensive" medical procedures provided in India.

With companies now proposing to send their own American employees abroad for less expensive health care services, there can be no doubt that the U.S. health care system is in immediate need of massive reform.

The right to safe, secure, and dependable health care in one's own country should not be surrendered for any reason, certainly not to fatten the profit margins of corporate investors.

Parks (2006)

The union also cited the lack of comparable malpractice coverage in other countries. The company agreed to find a domestic source of care for Mr. Garrett, but considered continuing the experiment with its salaried, non-union employees. Carl Garrett responded unhappily. "The company dropped the ball ... people have given me so much encouragement," he said, "so much positive response, and they're devastated. A lot of people were waiting for me to report back on how it went and perhaps go themselves. This leaves them in limbo too" (Jonsson, 2006, p. 2).

In 2007, Rank Group, a New Zealand-based company, purchased the company by buying out the ESOP, paying the former employee-owners, on average, approximately \$20,000 each. Rank renamed the company Evergreen Packaging Group.

► Discussion Questions

1. What difference did it probably make that BRPP is an ESOP owned by the union members or that the national union is busy recruiting healthcare workers as members?

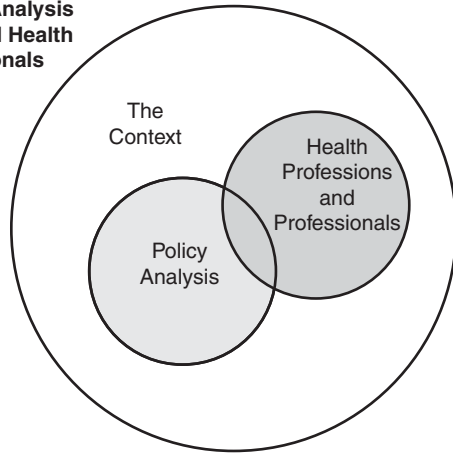
2. What are the ethical implications of a reward of up to \$10,000 for the employee to go to India for a major procedure?
3. If you were a hospital administrator, how would you react when a number of patients and companies began to ask to bargain about prices, including presenting price quotes from companies like IndUShealth?
4. What would be the difference in the bargaining position of an academic medical center and a large tertiary community hospital system?
5. How might state and national governments respond to this increasingly popular phenomenon?



PART II

The Policy Analysis Process

The Policy Analysis Process and Health Professionals



This section describes policy analysis processes applicable to health care. It begins in Chapter 6 with the need to develop an appropriate definition of the issues to be analyzed and then presents three major areas of analysis: technology (Chapters 7 and 8), political feasibility (Chapter 9), and economic viability (Chapter 10). This is followed in Chapter 11 by a review of values issues that impact many analyses and the presentation process and in Chapter 12 by consideration of implementation issues. Although each chapter covers a discrete topic, in the real world these are usually intertwined and are often addressed iteratively or in parallel.



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CHAPTER 6

The Policy Analysis Process: Identification and Definition

The right problem definition is critical when analyzing any healthcare policy issue. Imagination is required to create a definition that goes beyond what is already known and has been experienced. Learning from experience is a good thing, but only when it is relevant experience. There are a number of anecdotes about how well defeated generals were prepared to fight the previous war.

The stimulus for a policy analysis may come from external events and opportunities, political pressures, or scheduled periodic environmental scans. Some organizations use the term *horizon scanning*, while others use *environmental scanning*. The Canadian Agency for Drugs and Technology in Health (CADTH), an independent, nonprofit organization that advises Canada's federal, provincial, and territorial governments, provides an example of an organization's activities that are likely to trigger policy analyses.

Among other things, we:

- help health care decision-makers keep pace with technological change
- review and make recommendations on new and existing health technologies
- undertake comprehensive Health Technology Assessments that leverage the full depth and power of today's evidence-rich environment
- examine practices, processes, and protocols to provide a better understanding of the current landscape in health care
- scan the horizon to give health care leaders a glimpse of what the future might bring. (CADTH, 2017)

The Agency's methodology involves literature reviews, surveys, and key informant interviews, and includes cooperation with international horizon-scanning organizations.

► Early Sources of Misunderstanding

When experienced individuals who regularly deal with an issue are unable to resolve it, one or more of the following conditions are likely:

1. There is not a shared understanding of the nature of the problem.
2. There is a shared understanding, but it is not appropriate to the situation.
3. There is a realistic and relevant understanding, but it is not in some people's interest to resolve it.
4. There is an appropriate understanding and a shared desire for a solution, but there are not sufficient resources to implement the solution:
 - a. There are inadequate facilitation and leadership skills to reach the necessary compromises.
 - b. There are inadequate levels of skilled personnel to implement the preferred solution.
 - c. There are insufficient financial resources to implement the preferred solution.
 - d. The implementers cannot focus the political process on the problem or the solution sufficiently to move ahead.
 - e. Some combination of the four previous points.

This chapter primarily deals with the first two conditions—making sure that there is an accurate and appropriate definition of the problem that is understood by all involved. The other conditions relating to technological assessments, political feasibility, economic feasibility, implementation, and professional leadership are addressed elsewhere.

► Getting the Scenario Right

Assessing the Impact of a Health Policy

A World Health Organization (1999) report, the *Gothenburg Consensus Paper*, defined a *health impact assessment* as follows: “A combination of procedures, methods and tools by which a policy, programme or project may be judged as to its potential effects on the health of a population and the distribution of those effects within the population.” Distributional effects may take policy studies well beyond the population at immediate risk.

To reach agreement about the scope of potential and existing problems, possible alternatives, and desired outcomes, analysts have to reach agreement on several key areas:

1. The relevant definition of health
2. Identification of the target population
3. The current or likely future status of the health of a targeted population
4. The factors that determine the health status of concern within that population
5. The methods realistically available to change that health status
6. The responsibilities of the various actors in dealing with the identified issues and their interests
7. The societal values that are to govern the selection of alternatives and the acceptability of alternative outcomes (University of Birmingham, 2003)

FINDING AN ALTERNATIVE DEFINITION

The administrators and the board of trustees of a large academic medical center were at an impasse over the design of their new facility. At issue was whether to purchase a new and relatively untried monorail system for the distribution of supplies, laboratory samples, paperwork, and so forth. The investment would be large, and the risk was relatively high. Finally, one of the senior medical staff asked a consultant to meet with them. After listening to the arguments on both sides, the consultant asked, "Why are you in a hurry to make a decision now?" They replied, "The architect for the first building needs to know how big to make the passageways and utility channels in the plans which are nearly complete." After listening to the various concerns, the consultant asked, "How much additional would it cost to design the building to take either the new or the old technology?" Here was a new alternative. It turned out the additional cost was not much when compared with the uncertain gamble on the new technology. Both sides quickly agreed on that new alternative.

Health policy analysis seldom starts with a clean slate. The starting point is often an adverse event. This often leads to overcorrecting for earlier mistakes, rather than taking a fresh look.

Complex systems are full of problems in search of solutions, but they are also full of solutions in search of problems. A policy proposal is often put forward by someone with a specific solution already in mind. It is important, therefore, to ask whether a broader range of alternative solutions should be considered. One Secretary of Defense used to complain that the Joint Chiefs sent up the requisite three alternatives, two of which did not count. It is a waste of scarce resources to evaluate alternatives that do not count. Screening for additional alternatives, moreover, can be enlightening. The following box describes an actual situation in which this occurred.

Defining Health

BOX 6-1 presents the view of health and health care espoused in the constitution of the World Health Organization. Although the United States is a U.N. member state, one would be hard put to find consensus here on a number of points cited as basic principles.

Asking people in the United States if health is more than the absence of illness or infirmity would produce a range of responses. Some would come down on the side of physical and mental well-being but have a problem with including social well-being under the heading of health. Indeed, the fact that we have not been providing mental health care to a large proportion of the population would seem to indicate a lack of commitment to mental well-being. Some may even argue the other side that much of our health problems result from social problems, and investments into housing, personal safety, and food security would yield more benefits than added health care. This debate can be explored further under *social determinants of health*.

Those analyzing or deciding on a policy need to be aware of differences in the operational definitions of health around the table. In the best of all possible worlds, a team would agree on that definition and move on, but sometimes the art of politics depends, in part, on knowing whether to try to agree on principles, actions, or both, and whether to use limited political capital to try to bring them into alignment publicly.

BOX 6-1 Excerpts from the Preamble of the Constitution of the World Health Organization

... the following principles are basic ...

Health is a state of complete physical, mental, and social well-being and not merely the absence of disease or infirmity.

The enjoyment of the highest attainable standard of health is one of the fundamental rights of every human being without distinction of race, religion, political belief, or economic or social condition.

The health of all peoples is fundamental to the attainment of peace and security and is dependent on the fullest cooperation of individuals and States.

The achievement of any State in the promotion and protection of health is of value to all.

Unequal development in different countries in the promotion of health and control of disease, especially communicable disease, is a common danger.

Healthy development of the child is of basic importance, and the ability to live harmoniously in a changing total environment is essential to such development.

The extension to all peoples of the benefits of medical, psychological, and related knowledge is essential to the fullest attainment of health.

Informed opinion and active co-operation on the part of the public are of the utmost importance in the improvement of the health of the people.

Governments have a responsibility for the health of their peoples which can be fulfilled only by the provision of adequate health and social measures.

Adapted with permission from WHO: Constitution of the WHO, Basic Documents, 45th Ed. Page 1. Supplied 2006, October at www.who.int/governance/eb/who_constitution_en.pdf. Accessed December, 2017

Defining the Target Population

Just what population are we talking about? The history of community mental health centers illustrates how difficult—and critical—it can be to answer this question. A system designed to help the developmentally disabled and severely and persistently mentally ill morphed into a general mental health treatment system in which many practitioners avoided the original target group and concentrated on more rewarding (professionally and financially) cases (Torrey, 1997). As more and more states now focus more intently on the original target population, many of those previously served must rely more on private payment or insurance, or go without.

An analyst in charge of evaluating a maternal health program policy might start by looking at the health of all females of childbearing age. This analyst must still put a specific upper and a lower limit on the age range in order to gather data on the target population. These limits seem to be shifting over time as women are waiting until much later to have children. So, the analyst must decide whether to adjust historical data or collect new current data.

Identifying the Health Status of the Target Population

The next step after defining the target population is to assess its health status. Many data sources are available for this task, but sometimes they do not match up with the identified target population, in which case adjustments must be made.

Identifying the Factors Determining the Health Status of Concern Within That Population

Causation is the bane of the policy world. Politicians and polemicists would have us think that the right policy is certainly this or definitely that. If it were that simple, however, there would be little need for analysis. The conclusions of studies seeking causation are seldom as clear as obvious results of taking the handle off the local water pump and watching the cholera epidemic stop. Most policy problems support the characterization by the Danish mathematician and poet Piet Hein, who wrote, “Problems worthy of attack prove their worth by hitting back.” Inference is one thing, and causation is another.

Identifying the Methods Realistically Available to Change Health Status

When numerous alternative solutions are offered, the analyst needs to identify the few that are most realistic economically and politically. By politically realistic, we mean acceptable to those likely to fund and use the analysis. Some potential actors may express a preference for specific alternatives a priori. The analyst must not ignore these preferences, while keeping the process both intellectually honest and simple enough that decision makers are not likely to ignore the study or be confused by it.

Measures of Health Status

Health status can be measured in many ways. Healthcare impacts include the following:

- Mortality
- Morbidity
- Disability
- Costs
 - Direct
 - Indirect

How those impacts are relevant is highly dependent on the point of view of the user of the analysis. A labor union seeking to assess the impact of a work-related environmental hazard might be interested in all of these impacts. The corresponding employer trade association might be concerned mostly with costs in terms of health insurance premiums, mitigation costs, possible regulatory fines, worker morale, and lost productivity. Individual workers might add job security and lost wages to the list of measures.

Quality-adjusted life years (QALYs) are an outcome measure frequently mentioned in the literature. Yet the Affordable Care Act (ACA) specifically constrains the use of that measure in most government programs. International comparisons of the burden of disease often rely on disability-adjusted life years (DALYs). DALYs are the sum of two components:

Years of life lost (YLLs) for mortality

Years lived with disability (YLDs) for morbidity and disability

In some studies, the disabilities are age adjusted, and in some cases, the years further out are discounted back to a present value. A 3% discount rate is often used. Such adjustments are often controversial. YLLs are computed by comparing the longevity of the study population with that of the longest living cohort available; therefore, international comparisons have tended to use Japanese life tables.

Other aggregate measures have also been considered including life expectancy at birth and health-adjusted life expectancy at birth.

Sometimes country statistics are adjusted to remove the impact of major events that can be identified to a time period, such as an epidemic or armed conflict.

Analyses involving individual health issues tend to use disease-specific measures.

Segmentation

Looking at the health status of the target population as a whole can obscure differences between subgroups. One frequently hears about the millions of people in the United States without health insurance. Are they in trouble without it? Some are; some are not. Historically, many of the uninsured have been young people who have calculated the trade-off between the cost of health insurance and the low risk of a health crisis because they are young and healthy. Yes, they are more likely to have severe auto or motorcycle accidents than an older population, but until one happens, they are not part of the 20% of the population who account for 80% of healthcare costs. They are transferring the risk of low-probability events to the public at large because they would probably receive care anyway. Others may want insurance and need it, but are simply unable to afford it.

It is easy to talk past each other rather than solving problems. One can talk about the issue by discussing the underinsured as a group or examining the needs of specific segments. The important thing is that analysts clearly define whom they are talking about.

Defining the Methods Operationally and Optimally

In an industry like health care with a recognized high degree of waste, one has to add the step of defining the alternatives operationally by answering the following questions:

1. Has the alternative been in use?
 - a. If so, determine how it could be improved prior to applying it in this context.
 - b. If not, define it in more detail to establish operational feasibility.
2. For the more promising, feasible, and relevant alternatives, determine the optimal methods and procedures for delivery.
3. Use these optimal processes to determine costs and effectiveness.

Defining What Is a Medical Problem

Of concern to some is the medicalization of so much of human experience. How much is this tendency improving the quality of life, and is it worth paying for

BOX 6-2 Strategies Attributed to Disease-Mongering Campaigns

Develop a drug effective with a small segment of the population that is heavily impacted by the symptoms.

Redefine the disease in terms of the symptoms that the drug treats.

Inflate disease prevalence rates.

Encourage academic specialists to promote new disease definitions in seminars and articles.

Advertise to create anxiety about the symptoms, which may be quite normal.

Promote the drug as an aggressive, first-line treatment for the symptoms.

Promote the treatment of risk factors, especially if their status is measurable.

Promote the drug widely to all physicians rather than specialists handling problematic cases.

individually or collectively? We are expanding the conditions that can be treated, especially with biochemical treatments, yet these treatments have negative impacts beyond costs. They introduce side effects, some hazardous, especially when combined with other treatments. Because their manifestation varies from individual to individual, one might want to consider the dividing lines between:

- Those who would benefit from treatment and those who would not
- Those who need treatment and those who do not
- Those whose treatment clearly should be covered by a society and those for whom the presenting problem is due to a lifestyle choice

Increasingly, debates also involve permissible marketing, advertising, and commercialization, and charges of outright disease mongering. Much of this debate centers on the role of direct-to-consumer advertising and other marketing efforts, particularly by the pharmaceutical industry. **BOX 6-2** outlines promotional steps that seem to lead to the development of new, highly advertised treatments or screening policies.

► Hidden Assumptions

Other assumptions, often dealing with values, can impinge on an analysis. They may get addressed, or they may be left implicit or tacit to maintain organizational civility or political compromise. They include professional perspectives and personal conceptions of equity, due process, decision-making methods, and rights. This is not an exhaustive list. It does not include many value issues, such as the value of a human life.

If the group doing the analysis seems to be agreeing on most things but cannot reach closure, look for hidden assumptions that might be holding up the process. If the problem persists, it may be necessary to bring in a skilled process observer who will listen carefully to what people are saying and identify the stumbling blocks. It is unlikely that the team's leadership can push successfully toward closure until such hidden assumptions have been addressed.

Professional Perspectives

Social science disciplines have built-in assumptions about how societal and personal decisions are made, and these underlie differences between each discipline's jargon, research methods, and notions about things such as cognitive processes, equity, and appropriate governance. These assumptions also support aggregate assumptions (sometimes called visions) of institutional roles and how effective change takes place in a society.

Each discipline appears to redefine issues in its own terms and research approaches. MacRae (1976, pp. 109–110) used his background in public policy research to characterize how social science disciplines approach decision-making. He noted that disciplines talk to themselves, try to emulate the physical sciences, like to believe that they engage in “value-free” activities, and reward research that conforms to existing theory. He characterized the following four policy-related disciplines:

Economics deals with the satisfaction of existing individual preferences.

Psychology—especially in its relations with education and psychotherapy—is concerned with the changes that may be produced in preferences and their structures in individual personalities.

Sociology is concerned with social norms, and a related emphasis is on joint action undertaken to change them.

Political science, insofar as it escapes from the economic perspective, deals with those roles and institutions in which responsible citizens and public officials may be expected to consider the general welfare.

MacRae (1976, pp. 109–110)

In business education, most decisions are assumed to consider individual rather than social outcomes. Marketers usually see preferences as malleable. Decision theorists usually assume that preferences are given. The assumptions cited by MacRae, although subject to challenge, still seem to dominate today, and the disciplines differ in their approaches to common issues of our society involved with ethics, markets, social change programs, political processes, and social norms.

Implications for Problem Solving

Think about a meeting called to consider chronic local underemployment and homelessness. One participant cites educational differences, whereas another mentions disparities in educational opportunities. Then another speaks of imperfections in the local labor market. Oops! By now, the meeting is ready to derail. Some consider the term *labor market* to be dehumanizing. Others see it as jargon. What is a concrete, defined concept to one discipline may have a strong negative valence for another.

Multidisciplinary groups will be better able to work together cooperatively if they discuss the following early on in their work:

1. How can we express our personal assumptions and vision and incorporate them into intellectual discourse that respects others' points of view? This includes acceptance of the sensitivities of those with other approaches.
2. What notions of social equity and social change processes do we hold?

3. What is a reasonable objective for social change?
4. If we cannot agree on those assumptions and visions, how can we best cooperate on limited objectives that are compatible with our disparate viewpoints?

Is it realistic to expect any ad hoc group of busy professionals to spend the time necessary to achieve this level of trust and understanding? Yes. Without that trust and understanding, participants are unlikely to respond effectively. At a minimum, thinking about one's own assumptions and visions and their topic-specific and temporal inconsistencies is a prerequisite for a personal commitment and contribution to interdisciplinary work. Sowell (2002, p. 254) suggested that "an analysis of the implications and dynamics of visions can clarify issues without reducing dedication to one's own vision, even when it is understood to be a vision, not an incontrovertible fact, an iron law, or an opaque moral imperative." All too often visions are unexamined after years of immersion in one's profession, and becoming aware of them in oneself and in others is probably half of the uphill battle toward successful participation in interdisciplinary problem solving.

Professional Conflicts

If you have a mixture of health professionals in the group, you also have to deal with animosities between professions that have existed for years, especially those relating to either status differentials (such as nurses' anger at their treatment by physicians) or conflicting economic interests (such as between academic and community-based physicians). People do bring those experiences and attitudes into the meeting room.

Equity

Discussion aimed at defining a situation may stall because individuals have not reached consensus about the definition of equity to be applied. People's assumptions about equity are seldom on the table unless the group is very homogeneous in terms of value structures. If it is homogeneous, then the group faces the problem of the members all seeing things the same way, an outcome sometimes called *group think*. In almost any healthcare policy analysis, the issue arises of how the costs and benefits are distributed. The issue of what is fair follows. Individuals in a policy group can define equity in at least six ways (McLaughlin, 1984):

1. Equal expenditure per person
2. Equal inputs per person
3. Equal risk to each person
4. Equal satisfaction of demand (or need)
5. Equal (due) process participation
6. Equal outcomes

Decision-Making

A team must also address hidden assumptions about how the group will reach decisions. It is unlikely in policy analysis that decisions would be reached by majority vote; however, the group does need to think about the concept of consensus and how to determine when and whether it has been reached. The group also has to decide how to handle dissent. In some settings, a dissenting report or note is appropriate.

Collective Versus Individual Responsibility for Health

Social welfare versus individual welfare is an elephant in the health policy room. Garrison (2009) makes this point in looking at the ways the United Kingdom and the United States look at utility in allocating public resources. The United Kingdom favors having experts determine utility using QALY and cost data, whereas the United States has pretty much limited the concept to discussion in academic papers.

Garrison (2009) noted that these choices are determined by the following economic approaches:

- Monopsony (U.K. position): Experts use QALY and cost-utility analyses to maximize social benefit.
- Administered system competition: Experts determine a benefit package and use comparative effectiveness and cost for decisions at the margin.
- Consumer-driven (free market): Buyers (patients and payers) are provided with effectiveness and cost data and are then left to let them trade off utilities.
- Oligopolistic competition: Vendors prove effectiveness and let payers determine utilities.
- Monopoly: Providers and insurers maximize profitability.

Garrison suggested that defining a minimum acceptable benefit package is a compromise between two contending approaches—meeting a socially acceptable minimum for care and leaving marginal choices up to the individual. Garrison (2009) and Nord, Daniels, and Kamlet (2009) also pointed out a subtler difference in these analyses: experts tend to use average *ex ante* utilities, whereas patients tend to use experiential ones based on their current health status.

► Where in the World?

In the United States, the Agency for Healthcare Research and Quality (AHRQ) provides a Healthcare Horizon Scanning System. Its purpose is

to conduct horizon scanning of emerging health care technologies and innovations to better inform patient-centered outcomes research investments at AHRQ through the Effective Health Care Program. The Healthcare Horizon Scanning System provides AHRQ a systematic process to identify and monitor emerging technologies and innovations in health care and to create an inventory of emerging technologies that have the highest potential for impact on clinical care, the health care system, patient outcomes, and costs. It will also be a tool for the public to identify and find information on new health care technologies and interventions.

AHRQ (2013)

A database of scans is maintained at the International Information Network on New and Emerging Medical Technology (EuroScan International Network). It includes environmental scans by the Canadian Agency on Drugs and Technologies in Health and the National Institute of Health Research Horizon Scanning Center in the United Kingdom.

► Conclusion

Defining the right problem and the right decision-making process is critical to effective analysis. Both may be modified along the way, however. Policy analysis is a learning process, and there must be sufficient cognitive flexibility among the actors to allow for learning. At the same time, definition is a step in a larger political process. The policy process must be open to inputs expressing a variety of assumptions and viewpoints.

Key issues include reaching the operational definition of health to be used and the definition of the problem, addressing the hidden professional and personal values and assumptions of those participating in the policy process. Other key decisions relate to how much time, effort, and political capital to expend in attempting to align visions and values.

CASE 6: Small Area Variations

Background

One of the ways to examine the efficiency and efficacy of different approaches to medical care is to study variations in the types of care delivered in different areas and then compare the outcomes. The Dartmouth Atlas Working Group at Dartmouth Medical School uses Medicare data to conduct this type of “small area analysis.”

In 2006, the group reported that residents of Elyria, Ohio, received angioplasties at four times the national average. Angioplasty is an invasive, nonsurgical procedure widely utilized for treating heart attacks and alleviating symptoms of heart disease. It is also used in cases of severe heart disease in hopes of possibly preventing future heart attacks. The procedure involves pushing a collapsed balloon into the coronary artery and then expanding the balloon to press plaque against the arterial wall. Often a stent is left behind to keep the artery open. Other approaches to heart disease include drug therapy, lifestyle changes, and coronary artery bypass grafts. The latter procedure requires open heart surgery.

Elyria has a population 54,533 (2010 census) and is the county seat of Lorain County. In 2003, the rate of angioplasties in Elyria was 42 procedures per 1000 Medicare enrollees. By comparison, the rate for all of Ohio that year was 13.5 and the national rate was 11.3. All but 2 of the 35 cardiologists in Elyria at the time belonged to the North Ohio Heart Center, which relied heavily on angioplasties. The center performed 3400 angioplasties in 2004 (Abelson, 2006b).

There is considerable controversy about different treatment options for blocked coronary arteries. Some experts, according to an August 2006 *New York Times* article on the Dartmouth findings, “say that they are concerned that Elyria is an example, albeit an extreme one, of how medical decisions in this country can be influenced by financial incentives and professional training more than solid evidence of what works best for a particular person” (Abelson, 2006b).

According to medical historian Dr. David S. Jones, neither angioplasty nor coronary bypass surgery has been shown to prolong life except in cases of severe disease. Risks associated with bypass surgery include infections and brain damage resulting in memory loss and cognitive impairment. One of the concerns with angioplasty is that most heart attacks stem from tiny, often invisible lesions, and angioplasties tend to target the larger lesions that show up on angiograms. Dr. Jones argues for a greater focus on prevention through medicines and lifestyle changes (Park, 2013).

Angioplasty and coronary bypass surgery are highly profitable, and together they make up a \$100-billion-a-year industry in the United States. At the time of the Dartmouth study, Medicare was paying Elyria's community hospital \$11,000 for angioplasty with a coated stent, and the cardiologist performing the procedure received about \$800. Bypasses, however, were performed by surgeons from the Cleveland Clinic who had privileges at the community hospital. Those surgeons received up to \$2,200 per operation, and the hospital would receive up to \$25,000.

Outcomes

The founder and president of the North Ohio Heart Center responded to the Dartmouth findings by telling the *New York Times* that the center had good results with its patients and attributed the high use of angioplasty to early diagnostic interventions and aggressive treatment of coronary heart disease and to concerns about patient safety. Because of safety concerns, the center treats many of its patients in stages, doing more than one admission and procedure. Other cardiologists might perform multiple procedures at the same time. Thirty-one percent of the Elyria center's patients underwent multiple admissions and procedures, about three times the rate in Cleveland. Insurers report that the hospital's results are good, and UnitedHealth has named it a center of excellence for heart care.

► Discussion Questions

1. What do you think about using small area studies based on large Medicare databases (such as the one presented here) to identify outliers?
2. Salaried cardiologists at Kaiser Permanente in northern Ohio used drugs more often and performed cardiac procedures at slightly below the national rate. What role might different financial incentives play here?
3. If you were Anthem Blue Cross and Blue Shield in Ohio, what studies would you conduct to attempt to explain and/or deal with these striking local differences in treatments and costs?



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CHAPTER 7

The Policy Analysis Process: Health Technology Assessment

Advances in technology are a mixed blessing to policy planners. They present new possibilities for intervention and improvement, but come with new issues and uncertainties about cost, efficacy, financing, and ethical decision-making.

Reliable technological forecasts are important in health care for the following reasons:

- Long lead times for testing and approval of new technology
- Slowness of adoption of new technology due to bureaucracy, decentralized decision-making, and diffusion of power
- Impact of new technology on delivery and costs of health care (McLaughlin & Sheldon, 1974)

The accelerating rate of technological change in health care will make forecasts even more important in the future, because of:

- Increasing investments in medical research at the molecular level, leading toward breakthroughs in molecular biology and genetics leading to:
 - Increasing personalization of treatments
 - New models for clinical trials and the amortization of development costs for smaller cohorts
- Applications of information technology, especially artificial intelligence, that make observational research and epidemiology more relevant and authoritative
- Increased will to enhance dissemination and application of new knowledge through evidence-based recommendations and translational efforts

► Terminology

When the Washington State legislature established the Health Technology Clinical Committee in 2006 to review the evidence basis for up to eight medical technologies a year, the authorizing legislation, House Bill 2575, defined *health technology* as “medical and surgical devices and procedures, medical equipment, and diagnostic tests.” That definition specifically excluded pharmaceuticals, but the state already had established a separate process for evaluating pharmaceuticals and establishing a preferred drug list. In this chapter, we include pharmaceuticals and medical procedures within the definition of health and medical technologies, regardless of whether they are considered high tech.

► Technological Forecasting

Some new healthcare technologies are immediate successes. Others fail at first and then succeed, or they succeed and then fail. There are even cases of technologies that succeeded, failed, and then succeeded again. Our ability to forecast adoption rates and outcomes with new technology is limited at best. Yet we have no choice but to try. We cannot be prepared, given the scale of many events, unless we have predictive scenarios to work with.

Aiming at a Moving Target

With technological forecasting, we are always moving ahead with limited knowledge, despite our admittedly cumbersome systems to avoid unnecessary uncertainty, such as the Food and Drug Administration’s (FDA) new drug clinical trial and licensing procedures. Competing forecasts differ based on the forecaster’s particular vision of how society works. Other differences in forecasts may be driven by tunnel vision, self-interest, or one’s general outlook on life (e.g., optimistic or pessimistic). The forecasting process also must accept the context of the sponsor of the analysis, yet avoid being biased by it.

Forecasting Costs

In the United States Centers for Medicare & Medicaid Services (CMS), actuaries are responsible for making long-range forecasts about the costs of Medicare and Medicaid. Some trends, such as inflation rates, are relatively predictable. Many critical cost estimates, however, require assumptions about where technology is headed. Around 1990, when the Oregon Health Services Commission was attempting to set up a utility ranking for medical conditions and treatments, it gave a very low utility to the treatment for AIDS, because at that time there was a very low chance of survival. Today, the utility ranking would be very different given far more effective treatments for AIDS. The drug costs of treating those living with AIDS have now become a significant component of current and future health costs and have to be addressed in planning.

Forecasting Efficacy

One important component of most healthcare forecasts is how well an alternative will treat a given medical problem. Is the alternative safe? Is it effective? The effectiveness

of unregulated surgical procedures seems to be improving rapidly as less invasive methods have been developed and become part of the standard of care. Contemporary experience with infectious diseases is less consistent. Antibiotics have worked wonders, but we face a continuing battle over whether we will stay ahead of microbes' ability to develop resistance. Serious diseases such as HIV, hepatitis, and influenza evolve, and unknown ones may cross species to create new diseases, like severe acute respiratory syndrome and bird flu. Mobility can generate epidemics among immunologically naïve populations like the East African Ebola outbreak of 2014–2015 and Zika in the New World in 2016. Limited Ebola outbreaks had occurred in central Africa for decades but had been contained by traditional public health methods.

► Levels of Technological Forecasting

Sometimes it is important to forecast the effect of a given technology; other times it is adequate to forecast a specific system variable. Gilfillan (1952) visualized six systematic levels of “future causality”:

1. A specific invention
2. Alternative inventions for the same purpose
3. Technical accomplishment
4. Social and economic effects possible with a set of technical accomplishments
5. Social and economic effects predicted to flow from a set of accomplishments
6. Secondary and indirect effects of the predicted technology

All too often we focus on the invention or the technical accomplishment and not on the latter three levels of prediction. Then we face surprises and unintended consequences.

Two examples of forecasting at levels 3–5 are Moore's Law and the Polaris missile program. In 1965, Gordon Moore, cofounder of Intel, observed that the number of transistors per square inch on integrated circuits had doubled every year since the integrated circuit was invented. He predicted that this trend would continue for the foreseeable future. Although the technology has changed greatly, data density has doubled approximately every 18 months. This has been a useful predictor at the third level (technical accomplishment), which is not tied to any specific invention.

In 1957, the Soviet Union launched Sputnik, the world's first artificial satellite. In response, President Eisenhower called on the armed services to develop the capability for a deterrent nuclear ballistic missile. Admiral William Radford analyzed the trend in the size and weight of atomic warheads developed by Atomic Energy Commission contractors. He decided that over the period of time given for the task, the Navy could not develop a missile to lift existing warheads, but it could develop one that would lift a warhead of the size that he forecast the contractors could make available by the end of that period. He proposed the Polaris program on this basis, and the gamble paid off. He did not have to know which technical breakthroughs would be achieved by the contractors. He looked at the overall trends in the specific design parameters that affected his forecast and his planning. This approach can be conceived of as an envelope encompassing the results of a succession of innovations over time, as illustrated in **FIGURE 7-1**.

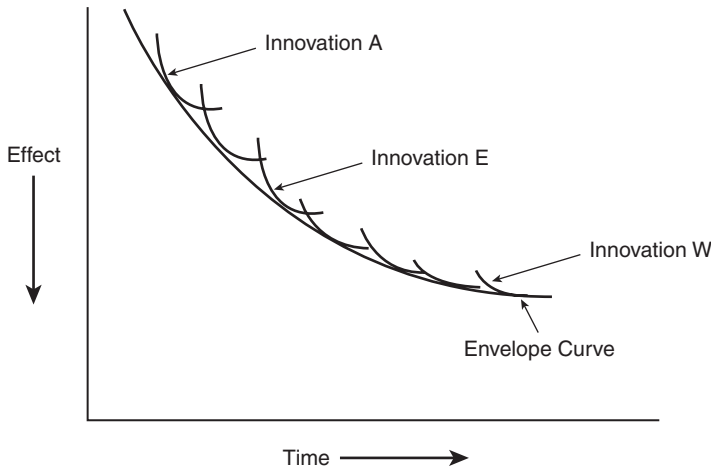


FIGURE 7-1 Using an envelope of innovations' effects to estimate a technology trend over time.

Not Just *What*, But Also *When*

To forecast the flow rates implied at the fifth level of Gilfillan's typology (predicted social and economic effects), one has to understand the influence of several interacting processes that may affect the timing of the predicted outcome, including the following:

- Regulation (in many cases)
- Dissemination
- Coding and payment
- Adoption/compliance

It is not enough for the technology to be effective or less costly. It must also be pushed and guided through these additional processes, and policy analysts must be aware of their importance and their likely impact on the timing of adoption.

Regulation

Health care is one of the more heavily regulated sectors of our society. Proponents of a new technology must often prove to regulators and potential users that it is safe and cost-effective. The FDA process for dealing with potential new prescription drugs is a well-known example. It is outlined in **TABLE 7-1**. There are really two submission processes: one for an investigational new drug, and then another for new drug approval after clinical trials have been completed. Both of these processes can be very slow and costly but are in place to protect the populace against unsafe drugs. The 21st Century Cures Act of 2016 had the goal of speeding up the licensure process through improved funding and acceptance of a wider base of evidence. So Table 7-1 is a simplified version of the process. Recently a number of complex and iterative processes have been developed to make the approval of drugs and medical equipment more rapid and more effective.

Health care is also governed by many voluntary regulatory bodies that set standards and then administer professional licensure and certification, inspect

TABLE 7-1 The Drug Discovery, Development, and Approval Process

It takes 12–15 years, on average, for an experimental drug to travel from the lab to U.S. patients. Only 5 in 5000 compounds that enter preclinical testing make it to human testing. One of those five tested in people is approved.

	Discovery/ Preclinical Testing	Phase I	Phase II	Phase III	FDA		Phase IV
Years	6.5	1.5	2	3.5	1.5	15 Total	
Test population	Laboratory and animal studies	20–100 healthy volunteers	100–500 patient volunteers	1000–5000 patient volunteers			
Purpose	Access, safety, biological activity and formulations	Determine safety and dosage	Evaluate effectiveness; look for side effects	Confirm effectiveness, monitor adverse reactions from long-term use	Review and approval process		Additional post market testing required by FDA
Success rate	5000 compounds evaluated	5 enter trials			1 approved		
		File IND at FDA		File IND at FDA			

Modified from Pharmaceutical Research and Manufacturers of America. Phrma.org

and certify facilities, and test and certify products and equipment. These are both collaborative and competitive settings where interest groups are represented and able to present their points of view. Most of the time the solutions are compromises.

Coding and Payment

In the United States, money is transferred from payers to providers based on systems of billing codes and authorizations. Each code has a specific payment attached to it. For example, payments to physicians under Medicare are set by CMS after receiving a recommendation from a voluntary organization made up primarily of representatives of the various medical specialties, the Medicare Payment Advisory Commission. CMS usually follows the commission's recommendations. Business plans can rise or fall based on these decisions. Payments' characteristics may influence process choices in healthcare delivery, such as whether infusions of an expensive biological takes place in the hospital or in an outpatient setting immediately after discharge.

Dissemination and Adoption/Compliance

After a product or process is deemed effective by regulators and evaluators, it is available for use, but that does not mean it will be used. News about the safety and effectiveness of the new technology has to reach potential adopters. Many studies have been conducted to determine how to introduce change into clinical practice. Clinical experts asked to forecast adoption rates will often overestimate the speed with which the change will take place. Adoption depends on the impact of experiences that build up to shared perceptions of technology's utility and ease of use (Davis, 1989; Tornatzky & Klein, 1982). Even when the criteria for use are met, it still takes a great deal of time to work through the various groups of potential adopters, starting with the risk takers, then the early adopters, and then the late adopters (Rogers, 1983).

Where patients and providers are concerned, one adoption issue is *compliance*—the use of a prescription or a routine as directed. Compliance often starts high but then slacks off, particularly with chronically ill patients. An example is providers' low rates of compliance with rules about hand washing and use of antibacterial aids. A hospital can bring the rate of central line infections to close to zero, yet hospitals and payers have had only mixed success, even with strong sanctions.

Disruptive Technology

Disruptive technology is difficult to forecast because straight-line extrapolations of current trends do not work. One just has to stay on top of what is happening. Christensen, Bohmer, and Kenagy (2000) have described a number of observations about successful adoption of disruptive technology in health care, including:

- Less investment in high-end technology and simplification of complex problems
- Creation of new organizations to do the disrupting
- Overcoming regulatory barriers
- Leadership favorable to change

One example of disruptive technology has been the development of mobile artificial pancreases for Type 1 diabetes assembled by patients and their families using hacked old insulin pumps. They had these devices in use well in advance of the development and regulatory approval of commercial versions.

► Forecasting Methods

Methods of forecasting include the following:

- Gathering expert opinions
- Time-series analysis
- Surveying and sampling
- Correlational and causal modeling
- Simulation and system modeling

Gathering Expert Opinion

When a technology is new, few individuals have sufficient experience to estimate trends or identify causal relationships. Often a panel of experts is assembled to offer their best estimates of likely events and outcomes. If there are concerns about the dominance of one or two individuals in the forecasting process, techniques such as nominal group or Delphi can be used to gather feedback and information while avoiding the undue influence of any one person. One major Delphi study of the future of health care consisted of three rounds of questionnaires. The first round asked the experts to suggest significant medical events that were likely to occur in a specific time period. These suggestions were then edited by the researchers to define 62 specific events. For example, one event was defined as follows: “Solution of the histocompatibility problem in transplantation, or the means around it.” Another was, “We now have a legal precedent for the right of the terminally ill patient to determine his (or her) own time and method of death.” A second questionnaire asked respondents to estimate a date of occurrence for these 62 events (in 5-year periods), a priority classification (high, low, or none), a current level of action category (much, little, or none), and the respondent’s own level of expertise on this item (some or none). The third round reported back on 33 items where there was lack of agreement and asked the experts to refine their date estimates in light of the feedback. The third questionnaire also included scenarios that indicated interactions among them and their overall impact. The respondents were asked to give their feelings about the acceptability of such an overall possible outcome (McLaughlin & Sheldon, 1974).

Time-Series Analysis

If data are available, one can plot a trend line and extrapolate it to estimate a value in the future. This can be done visually or by using regression analysis. The model can weigh the more recent data more heavily, using one or another form of exponential smoothing. If there are multiple variables in the estimating equation, these can be adjusted for time lags where applicable (Gardner & McLaughlin, 1980). The examples of Moore’s Law and the Polaris system both involved identifying and extrapolating technology trends

over time. Software is readily available for applying alternative models to time-series analysis and selecting the one that fits the historical data most closely. For example, forecasts of health expenditures must adjust existing trends for factors such as inflation, changed morbidity rates, and technological innovation (Martin, Hartman, Washington, Catlin, & National Health Expenditure Accounts Team, 2016; OCED, 2017).

Surveying and Sampling

One can also ask the affected individuals about their anticipated behavior in response to a forecast scenario or analyze their past and current behavior by gathering survey data. Unfortunately, such surveys are useful only if the participants already have experience with the same or similar technology. A clinical trial is one form of surveying and sampling. A set of patients receives the new technology, and the results are observed and compared to the results of those who did not experience the change. This approach is very expensive to develop and administer, and surveys are employed only if the stakes are very high.

Correlation and Causal Modeling

Life is simplest when we know how a given action causes a reaction. To start to figure that out, we often turn to regression analysis with time-series data, which tells us over time what is likely to happen to the dependent variable as independent variables change. One example is the assessment of survival rates for surgical procedures at an institution over time as experience mounts. For open heart surgery, survival rates increase with reduced pump time. Pump time decreases with experience, and experience usually increases more rapidly with greater volumes of procedures. In industry, this is called the *learning curve* or *progress function*. It has been shown to apply in health care as well.

Always be aware that most statistical models only show relationships, but do not prove causation. Sometimes the relationships are temporally related, so cause and effect are evident, but most of the time one can only say the degree to which two effects are correlated, leaving the issue of causation open to argument.

Simulation and System Modeling

As the picture of relationships becomes clearer, it does not necessarily become less complex. The complexity often calls for computer modeling of the system to see how the variables interact with each other. Four approaches to modeling seem to be in general use for prospective studies of technology (Homer & Hirsch, 2006; Jones et al., 2006):

- **Spreadsheet models (including those derived from regression models).** These involve the simple mathematical manipulation of estimates to provide projected results into the future.
- **Probabilistic (Monte Carlo) simulations.** These add uncertainty to the model by including the distributions of random variables in the process and then projecting out the results of a large set of runs of the model that may include both deterministic and probabilistic elements. This approach has been used to support sensitivity analysis in a wide variety of studies.

- **Feedback models (also known as *systems dynamics models*).** These are based on causal analysis and use an interlocking set of differential and algebraic equations to model complex systems which may include both physical and societal elements.
- **Markov models.** These add time-dependent transition probabilities from one state (e.g., getting better, staying the same, or dying) to a model. The steady-state outcomes of patient disease progression can be determined analytically by some Markov models; otherwise computer simulation is used to provide a solution.
- **Scenarios.** These involve taking individual predictions and putting them into a story that assumes their coexistence and their interactions, the kinds of interactions that so often lead to unintended negative consequences from otherwise desirable plans.

Integrated Approaches

A major study can combine a number of these techniques. For example, a RAND Corporation study of the health status and medical treatment of the future elderly for CMS used the following:

1. Expert panels were used to identify key technologies, their time of availability, and their impact (Shekelle et al., 2005).
2. A representative sample of 100,000 Medicare beneficiaries' histories was drawn from the Medicare Current Benefit Surveys and a forecast was made of the health of future entrants based on the National Health Information Survey.
3. A demographic and economic probabilistic simulation model was performed that moved participants ahead through time, changed their health status, estimated their survival patterns, and added a new cohort each year. This digital simulation was used to ask "what if" questions about the cost of Medicare under varying conditions of health status, medical innovation, and changes in chronic disease patterns (Goldman et al., 2004, 2005).

The RAND study showed that future costs would be quite sensitive to the implementation of certain new technologies and to the health status of the incoming cohorts, especially if obesity trends continued.

Appropriate Skill Sets

Technology assessment for health policy analysis relies on a combination of clinical assessment skills, data analysis skills, and technology evaluation skills, including the following:

1. Identifying populations affected by the technology
2. Identifying the impact of the technology on the treatment of those populations
3. Projecting clinical outcomes under a representative cross section of payment and reward systems
4. Projecting degree of adoption and/or compliance with the proposed innovation

Segmentation

Healthcare practices and institutions provide a wide variety of products. Current practice using diagnosis-related group categories identifies almost 500 products, and that differentiation is often too coarse for effective technological planning. This leads to two requirements for technology management: (1) methods to identify the affected population segments and (2) simplicity in approach to allow application to multiple product lines. Not only does one need to have large clinical databases to identify enough cases affected by a specific technology, but one must also deal with ever-shifting definitions of disease states as medical knowledge accumulates and the technology itself affects the nature and distribution of what is to be treated. An example is the treatment of AIDS. Initial classification schemes differentiated between patients with T-cell counts of 200 or more CD4 cells per mm³ and those with 100 or fewer; however, Portela (1995) found the economic impact of AIDS to be greatest in the population with T-cell counts below 50 and recommended a different segmentation for treatment planning and costing. Since then, the success of the drug “cocktails” involving expensive sustained treatment for a much larger population—HIV-infected people who are symptomatic, have T-cell counts below 350, or have viral load greater than 55,000 copies/mL—has changed the approach to fighting the disease from one of avoiding overwhelming infections to one of fighting drug toxicity and drug resistance. Each new approach brings much different technical requirements, morbidity patterns, and survival rates.

Technology Assessment and Staffing Requirements

Knowing where technology is going could also help us in developing the right workforce for the future. Although the United States lacks a central policy on the supply of professional personnel, there is continued concern about potential shortages and surpluses of trained personnel in the job market and in the educational pipeline. The most widely publicized effort to address this was the Graduate Medical Education National Advisory Committee report of 1980, which predicted a major physician surplus by 2000. It forecast a surplus of 70,000 physicians and recommended reducing the number of trainees in residency programs in several areas. That forecast seems to have been very wide of the mark. Other reports suggested a shortage of physicians as the population has aged and new technologies have been added. They have been more accurate, but the predicted shortage has not materialized because of immigration of large numbers of foreign-born physicians, the return of U.S. citizens educated in foreign medical schools, and the expanded supply of trained nurse practitioners and physician assistants to carry much of the primary care caseload. In 2005, the Association of American Medical Colleges (AAMC) announced that it was moving from a neutral position to one favoring expanding medical school enrollment based on the Council on Graduate Medical Education's estimates of a shortage of 90,000 physicians by 2020; however, the AAMC website noted that “there are many unknowns that make forecasting future supply and demand very difficult such as medical advances and changes in organization and financing” (AAMC, 2005, p. 2). A more recent study predicted a similar shortage, but highlighted shortages of specialists.

► Organizations Devoted to Healthcare Technology Assessment

Federal agencies, such as the Centers for Disease Control and Prevention (CDC), the Agency for Healthcare Research and Quality (AHRQ), the FDA, and the National Institutes of Health (NIH), play an important, albeit fragmented, role in technology assessment. However, they are not the only actors involved. Numerous other organizations are engaged in healthcare technology analysis and assessment at the state and local levels, and, of course, the private sector is heavily involved too.

The Role of States in Technology Assessment

From 1972 to 1995, the focus for technology assessment in the United States was the Office of Technology Assessment (OTA), an arm of the U.S. Congress that was overseen by six senators and six representatives equally representing both political parties. It was a small agency with a professional staff of about 140, and it provided reports on a number of scientific areas, including health care, global warming, telecommunications policy, nuclear defense, and transportation. It published a 1980 study (OTA, 1980) of how benefit/cost analysis should be applied to health that is still widely cited:

Between 1975 and 1980, another OTA group set the stage for today's booming industry in the technology assessment of health care by demonstrating the inadequacy of information on which decisions about technology were made; laying out the strengths and weaknesses of methods to evaluate technology; and crystallizing the process by which economic trade-offs could be incorporated into decisions.

OTA (1980)

The booming industry that Representative Houghton described includes what is referred to as *pharmacoconomics*. The OTA had a reputation for being professional and balanced in its reports, even though that meant many of its reports were not conclusive and were cited in debate by both sides addressing an issue (Morgan, 1995). It was defunded in 1995. For a while, there was a hope that state agencies, such as the Oregon Health Resource Commission, would take over much of that effort (Mendelson, Abramoff, & Rubin, 1995).

The ACA significantly expanded the role of the federal government in technology assessment, albeit with some severe constraints on the use of the information. The result of all of this is a patchwork of federal agencies (e.g., CDC, AHRQ, PROCI, FDA, NIH, and others), state agencies (such as Oregon's Health Evidence Review Commission and Washington's Health Technology Clinical Committee), and private industry (to justify inclusion in formularies and health plan reimbursement) focusing on new technology. AHRQ and PROCI have, however, supported considerable evaluative effort on treatments for high-volume, high-cost problems, such as acute lower back pain, where new technology may not be involved. This approach is likely to be downplayed during the Trump administration, but it will continue to expand at the state, institutional, and private sector levels. However, given the current anti-government attitudes, these efforts are unlikely to achieve widespread acceptance at the state level either.

Technology Assessment in the Private Sector

Developers of new technologies understand that they cannot get their products approved and sold until the professionals consider them safe and effective and the payers deem them cost effective. This trend accelerated as HMOs began entering into disease management and hospitals became more aggressive in limiting drug formularies. Private sector activity has grown to the point where most pharmacy schools now have departments, professorships, or centers with the word *pharmacoeconomics* in their titles, often together with the terms *disease management* and *outcome studies*. Houghton's forecast of a boom came true in the private sector, but was delayed in government until the 2009 stimulus legislation and then the ACA pushed it forward.

► Where in the World?

A number of countries with more centralized health systems have their own government units responsible for technology assessment. The United Kingdom requires rigorous cost-benefit analyses for new technologies and drugs prepared by its National Institute for Health and Care Excellence (NICE). It also offers advice about use and safety of existing technology. It is supported by the National Coordinating Centre for Health Technology Assessment at the University of Southampton, which commissions and supervises needed studies at independent academic centers.

The Canadian Coordinating Office for Health Technology Assessment publishes brief reports on new drugs and technologies that include references about them and comments on their use, cost, and evidence of effectiveness. Its *Health Technology Update* also identifies recent assessments by other agencies in Canada and in other countries. The health technology assessment (HTA) field has at least one international journal, and there is also an international association of HTA centers.

The World Health Organization (WHO) also provides technical support to help member states build capacities for technology assessment. The WHO definition of “health intervention and technology assessment” is broader than we use here. It covers the whole assessment and implementation process. This effort is also supported by the International Network of Agencies for Health Technology Assessment which is charged with:

- Mapping global HTA capacity
- Raising HTA awareness and capacity building
- Education and training
- Information sharing

The European Union has its own organization, the European Network for Health Technology Assessment (EUNetHTA). Its mission has been to:

- Facilitate efficient use of resources available for HTA
- Create a sustainable system of HTA knowledge sharing
- Promote good practice in HTA methods and processes

This organization also supports the journal, *International Journal of Health Technology Assessment*, which presents methodologies, field experience, and study outputs.

► Conclusion

Yes, it is difficult, but we must continue to forecast the impact of medical advances and resulting changes on the organization and financing of health care. Calculating the benefits of a technology starts with the science, with the measurement of its effect on the health of a population. Usually that involves a comparison between an existing technology and a new approach about which much less is known. Rules and regulations mandate trials of some new technologies to determine their safety and efficacy. Although such trials are often conducted under ideal conditions and on limited populations, they are usually sufficient to estimate the benefits in nonmonetary terms; that is, their impact on quality of life and patient survival. Even with the results of the trials in hand, it is necessary to confirm the impact of the new technology on the healthcare system when used under field conditions.

Several techniques are available to forecast the progress of technology and the overall effects of technology for the purpose of health planning. Some, such as expert opinion, are necessary when the general public has no experience with the technology. As experience develops, users and consumers can be consulted and surveyed. Where experience is extensive, time-series and causal-modeling methods can be applied. It is important to recognize, however, that in the real world there is a great deal of interaction among factors. Therefore, more complex modeling techniques are necessary to estimate the adaptation of the system to multiple factors and to assess whether the resulting adaptations are acceptable. That is why relatively sophisticated organizations are needed to conduct healthcare technology assessment and advise decision-makers and the public on which measures to choose.

CASE 7: Oregon's Health Evidence Review Commission

When the Oregon Legislature established the Health Evidence Review Commission (HERC) during the 2011 legislative session, it combined two previously existing commissions, the Health Services Commission and Health Resources Commission, that had functioned for two decades. Legislation creating the Oregon Health Plan and the Health Services Commission, Senate Bill 27, passed in 1989. The legislature established the Health Resources Commission in 1991. The HSC was charged with prioritizing healthcare services, while the HRC was a public forum charged with developing health technology assessments and forming a consensus about emerging medical technology issues. The commissions continued to function until HERC became operational in 2012.

Background

The Oregon Basic Health Services Act of 1989 tried to guarantee health care for all while still controlling health care costs through an open, publicly accountable rationing process. Rather than rationing by excluding people from Medicaid coverage or by under-reimbursing providers. "Instead, when budget limits required reductions in budget costs, health services would be eliminated according to explicit priorities

established by an independent commission through an accountable, public process" (DiPrete & Coffman, 2007).

At that time, one had to be at or below 58% of the federal poverty level to receive Medicaid benefits in Oregon, and that bill was intended to open that up to those below the 100% level. The state created a Health Services Commission to recommend the prioritization of health services. The 11-member body was to report its priorities to the governor and the Joint Legislative Committee on Health Care. An actuarial contractor would then estimate the cost of each of the prioritized services for the Joint Committee. After the Medicaid budget for services was set, the package of services offered to the Medicaid population would be developed using the priority lists. Essentially, the process would be to go down the list taking the next highest priority coverage until the estimated budget package was exhausted. Other services then would be outside this basic coverage. If the demand exceeded the funding provided, Oregon was prepared to reduce the benefit package rather than ration through eligibility thresholds or Medicaid reimbursement levels as so many other states were doing (Calkthan, 1991).

The commission used three techniques to prioritize some 800 services. It held 11 open public hearings around the state. It also had Oregon Health Decisions, a respected advocacy group, conduct 47 community forums, including one in every county. Each forum featured a slide presentation, group discussions, and a questionnaire on the participants' opinions about the relative importance of specific health situations and categories. About 70% of the more than 1,000 individuals attending these forums were healthcare workers. The commission also supported a statewide random-digit dialing survey asking 1,000 individuals to rate 31 healthcare situations on a modified Quality-of-Well-Being Scale that had been validated elsewhere. It used information to build a cost/utility scale called a "net benefit value" scale. That study produced a listing of 1,600 medical condition/treatment pairs.

The orderings produced by the telephone survey were highly debatable. Fox and Leichter (1991) reported that "crooked teeth received a higher ranking than early treatment for Hodgkin's disease, and dealing with thumb sucking was ranked higher than hospitalization of a child for starvation" (p. 22). Because it was clear that the derived utilities were not workable in terms of overall values, commission members negotiated among themselves for an acceptable ordering. For example, the four consumer representatives on the panel argued for higher rankings for preventive services because that is what the public seemed to have been saying during this process. Rather than a continuous ranking, they came up with 709 pairs grouped according to three basic categories: essential, very important, and valuable to some individuals (Kaplan, 1995) and then under 17 subcategories of outcomes, using descriptors such as survival, degree of recovery, and degree of improvement on the Quality-of-Well-Being Scale (Fox & Leichter, 1991).

Implementing this process required a federal Medicaid waiver, which the federal government denied on the grounds that the assessments of quality of life by healthy individuals tended to stereotype and discount the value of quality for the disabled, and the process, therefore, violated the Americans with Disabilities Act. Kaplan (1995) argued that this finding was in error and offered counterevidence.

Instead of debating these issues, Oregon chose to resubmit its application with the utility portion of the model excluded. Its revised waiver application considered probability of death and probability of moving up from a symptomatic to an asymptomatic state. By giving up the utility component of the model, Oregon ignored the fact that health states are valued (Kaplan, 1995). Priester (1992) argued

that the plan had been flawed because it denied access to a reasonable level of care to some, but not to others. In 1993, the Clinton administration approved the revised Medicaid waiver.

According the DiPrete and Coffman (2007) this increased the importance of the commissioner's opinions, which reflected the public process.

While the Commissioners did not feel that their original methodology was biased against any group, they reluctantly agreed to modify it in order to achieve the goal of greater health care access. Because most objective measures representing health outcomes were not allowed, the subjective collective judgment of the Commissioners became more of a factor. As a result, many of the public values on health that had been expressed through the community meetings, the telephone survey, and in public testimony were reflected through the application of Commissioner judgment in the final prioritization process.

DiPrete and Coffman (2007, p. 4)

Prioritization Methodology

The first prioritized list, issued in 1993, underwent minor revisions every two years. A major methodological revision in 2006 that emphasized prevention produced a new list in 2008. HERC's 2015 Prioritization of Health Services report describes the revised method.

The prioritization methodology first defines a rank ordered list of nine broad categories of health care to establish a basic framework for the list (see **FIGURE 1-1**). Next the methodology calls for each of the line items on the Prioritized List to be assigned to one of these healthcare categories. Once the line items have been assigned to one of the nine categories, a list of criteria is used to sort the line items within the categories (see **FIGURE 1-2**). These measures are felt to best capture the impacts on both the individual's health and the population health that the Commission believes is essential in determining the relative importance of a condition-treatment pair.

Since not every service in Category 1 is more important than every service in Category 2 and so on, a weight is applied to each category that is then multiplied by the total criteria score for each condition-treatment pair. The category weights are shown in parentheses after the title for each category in Figure 1.1. A total score is then calculated for each line using the following formula to sort all line items, with the lowest net cost used to break any ties:

Category Weight	X	Impact on Healthy Life + Impact on Suffering + Population Effects + Vulnerability of Population Affected + Tertiary Prevention (categories 6 & 7 only)	X Effectiveness	X Need for Service
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FIGURE 1-1
RANK ORDER OF HEALTHCARE CATEGORIES.

1. **Maternity & Newborn Care** (100)—Obstetrical care for pregnancy. Prenatal care; delivery services; postpartum care; newborn care for conditions intrinsic to the pregnancy.
2. **Primary Prevention and Secondary Prevention** (95)—Effective preventive services used prior to the presence of disease and screenings for the detection of diseases at an early stage. *Immunizations; fluoride treatment in children; mammograms; pap smears; blood pressure screening; well child visits; routine dental exams.*
3. **Chronic Disease Management** (75)—Predominant role of treatment in the presence of an established disease is to prevent an exacerbation or a secondary illness. *Medical therapy for diabetes mellitus, asthma, and hypertension; medical/psychotherapy for schizophrenia.*
4. **Reproductive Services** (70)—Excludes maternity and infertility services. *Contraceptive management; vasectomy; tubal occlusion; tubal ligation.*
5. **Comfort Care** (65)—Palliative therapy for conditions in which death is imminent. *Hospice care; pain management.*
6. **Fatal Conditions, Where Treatment is Aimed at Disease Modification or Cure** (40)—Appendectomy for appendicitis; medical & surgical treatment for treatable cancers; dialysis for end-stage renal disease; medical therapy for stroke; medical/psychotherapy for single episode major depression.
7. **Nonfatal Conditions, Where Treatment is Aimed at Disease Modification or Cure** (20)—Treatment of closed fractures; medical/psychotherapy for obsessive-compulsive disorders; medical therapy for chronic sinusitis.
8. **Self-limiting Conditions** (5)—Treatment expedites recovery for conditions that will resolve on their own whether treated or not. *Medical therapy for diaper rash, acute conjunctivitis, and acute pharyngitis.*
9. **Inconsequential Care** (1)—Services that have little or no impact on health status due to the nature of the condition or the ineffectiveness of the treatment. *Repair fingertip avulsion that does not include fingernail; medical therapy for gallstones without cholecystitis, medical therapy for viral warts.*

www.oregon.gov/oha/HPA/CSI-HERC/Documents/2015-Biennial-Report-to-Governor-and-Legislature.pdf

Hand adjustments were applied by the Commission where the application of this methodology did not result in a ranking that reflected the importance of the service, which was the case in fewer than 5% of the line items (compared to over 70% of cases using the previous methodology).

The following example illustrates the score given to a line item as a result of this methodology.

FIGURE 1-2
POPULATION AND INDIVIDUAL IMPACT MEASURES.

Impact on Healthy Life—What is the magnitude of the benefit to the patient from the treatment as compared to no treatment for the condition, after factoring in harms associated with the treatment. *Range of 0 (no impact) to 10 (high impact).*

Impact on Suffering—To what degree does the condition result in pain and suffering? Effect on family members (e.g., dealing with a loved one with Alzheimer’s disease or needing to care for a person with a life-long disability) is also factored in here. *Range of 0 (no impact) to 5 (high impact).*

Population Effects—The degree to which individuals other than the person with the illness will be affected. Examples include public health concerns due to the spread of untreated tuberculosis or public safety concerns resulting from untreated severe mental illness. *Range of 0 (no effects) to 5 (widespread effects).*

Vulnerability of Population Affected—To what degree does the condition affect vulnerable populations such as those of certain racial/ethnic descent or those afflicted by certain debilitating illnesses such as HIV disease or alcohol & drug dependence? *Range of 0 (no vulnerability) to 5 (high vulnerability).*

Tertiary Prevention—In considering the ranking of services within categories 6 and 7, to what degree does early treatment prevent complications of the disease (not including death)? *Range of 0 (doesn’t prevent complications) to 5 (prevents severe complications).*

Effectiveness—To what degree does the treatment achieve its intended purpose? *Range of 0 (no effectiveness) to 5 (high effectiveness).*

Need for Medical Services—The percentage of time in which medical services would be required after the diagnosis has been established. *Percentage from 0 (services never required) to 1 (services always required).*

Net Cost—The cost of treatment for the typical case (including lifetime costs associated with chronic diseases) minus the expected costs if treatment is not provided—including costs incurred through safety net providers (e.g., emergency departments) for urgent or emergent care related to the injury/illness or resulting complications. *Range of 0 (high net cost) to 5 (cost saving).*

www.oregon.gov/oha/HPA/CSI-HERC/Documents/2015-Biennial-Report-to-Governor-and-Legislature.pdf

Schizophrenic Disorders	
Category 3 Weight: 75	Effectiveness: 3
Impact on Healthy Life Years: 8	Need for Service: 1
Impact on Suffering: 4	Net Cost: 5
Effects on Population: 4	Total Score: 3600
Vulnerability of Population Affected: 0	
$75 \times [(8 + 4 + 4 + 0) \times 3 \times 1] = 3600$	

Services near the top of the list as a result of this methodology include maternity care and newborn services, preventive services found to be effective by the U.S. Preventive Services Task Force, and treatments for chronic diseases such as diabetes, major depression, asthma, and hypertension, where ongoing maintenance therapy can prevent exacerbations of the disease that lead to avoidable high-intensity service utilization, morbidity, and death. (pp. 3–6)

The 2011 Law Governing HERC

The Health Resources Commission's statutorily mandated functions include:

- Develop and maintain a list of health services ranked by priority, from the most important to the least important, representing the comparative benefits of each service to the population to be served ((ORS) 414.690(3))
- Develop or identify and shall disseminate evidence-based healthcare guidelines for use by providers, consumers and purchasers of health care in Oregon (ORS 694.698(2))
- Conduct comparative effectiveness research of health technologies. (Oregon Revised Statutes (ORS 694.698(2)).

Below are selected statutes governing the commission:

414.690 Prioritized list of health services

1. The Health Evidence Review Commission shall regularly solicit testimony and information from stakeholders representing consumers, advocates, providers, carriers and employers in conducting the work of the commission.
2. The commission shall actively solicit public involvement through a public meeting process to guide health resource allocation decisions.
3. The commission shall develop and maintain a list of health services ranked by priority, from the most important to the least important, representing the comparative benefits of each service to the population to be served. The list must be submitted by the commission pursuant to subsection (5) of this section and is not subject to alteration by any other state agency.
4. In order to encourage effective and efficient medical evaluation and treatment, the commission:
 - a. May include clinical practice guidelines in its prioritized list of services. The commission shall actively solicit testimony and information from the medical community and the public to build a consensus on clinical practice guidelines developed by the commission.
 - b. May include statements of intent in its prioritized list of services. Statements of intent should give direction on coverage decisions where medical codes and clinical practice guidelines cannot convey the intent of the commission.
 - c. Shall consider both the clinical effectiveness and cost-effectiveness of health services, including drug therapies, in determining their relative importance using peer-reviewed medical literature as defined in ORS 743A.060.
5. The commission shall report the prioritized list of services to the Oregon Health Authority for budget determinations by July 1 of each even-numbered year.

414.698 Comparative effectiveness of medical technologies

1. The Health Evidence Review Commission shall conduct comparative effectiveness research of medical technologies selected in accordance with ORS 414.695.
2. The commission may conduct the research by comprehensive review of the comparative effectiveness research undertaken by recognized state, national or international entities. The commission may consider evidence relating to prescription drugs that is relevant to a medical technology assessment but may not conduct a drug class evidence review or medical technology assessment solely of a prescription drug. The commission shall disseminate the research findings to healthcare consumers, providers and third-party payers and to other interested stakeholders. The commission shall develop or identify and shall disseminate evidence-based healthcare guidelines for use by providers, consumers and purchasers of healthcare in Oregon.
3. The Oregon Health Authority shall vigorously pursue healthcare purchasing strategies that adopt the research findings described in subsection (1) of this section and the evidence-based healthcare guidelines described in subsection (2) of this section. [2011 c.720 § 26]

414.701 Commission may not rely solely on comparative effectiveness research.

The Health Evidence Review Commission, in ranking health services or developing guidelines under ORS 414.690 or in assessing medical technologies under ORS 414.698, and the Pharmacy and Therapeutics Committee, in considering a recommendation for a drug to be included on any preferred drug list or on the Practitioner-Managed Prescription Drug Plan, may not rely solely on the results of comparative effectiveness research.

[2011 c.720 §26a]

► Discussion Questions

1. How has Oregon's effort evolved over time?
2. Evaluate the scoring model adopted in 2008?
3. What advice would you give other states as they tried to deal with cost and scarcity issues?
4. Which side would you like to be on a court case challenging a HERC recommendation not to cover a treatment based on its prioritization model?
5. How would you expect HERC to evolve in the future?



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CHAPTER 8

The Policy Analysis Process: Evidence-Based Medicine

The concepts of evidence-based medicine have many labels, including *evidence-based practice* and a more recent offshoot, *translational medicine*. The term has been attributed to David Sackett and his clinical epidemiology colleagues at McGill University, who presented it in a series of 1980 articles in a Canadian journal. In a 1996 article, they presented a revised definition, stating that evidence-based medicine is “the conscientious, explicit, and judicious use of current best evidence in making decisions about the care of the individual patient. It means integrating individual clinical expertise with the best available external clinical evidence from systematic research” (Sackett, Rosenberg, Gray, Haynes, & Richardson, 1996).

Evidence-based medicine can be implemented top-down or bottom-up, which can be confusing. With the top-down approach, experts form a consensus based on strong empirical evidence, and that consensus is then disseminated as standard guidelines and protocols. With the bottom-up approach, a provider directly deals with the patient, identifying the problem and searching for the best evidence about what to do next. Even when strong, evidence-based protocols are available, only the bottom-up process accounts for patient variability, patient preferences, situational factors, and provider expertise. The provider’s expertise may be based on evidence or experience, but a combination of both usually leads to better outcomes.

Resistance to evidence-based medicine often stems from providers’ fears that the top-down approach will deprive them of their autonomy. A further concern is that the top-down approach may be based on statistical analyses involving trials with constrained, homogeneous populations and will not represent an individual patient’s situation. For example, a randomized clinical trial may select a population without common secondary diagnoses, whereas many patients presenting in a practice will have multiple conditions. Sackett et al. (1996) addressed these concerns by stating:

Evidence-based medicine is not “cookbook” medicine.... External evidence can inform, but can never replace, individual clinical expertise, and it is this

expertise that decides whether the external evidence applies to the individual patient at all, and if so, how it should be integrated into a clinical decision.
(p. 317)

Some providers also are concerned about requiring evidence-based practices for diagnoses, procedures, and technologies when there is not an adequate body of research from which to draw conclusions. Policy makers are reluctant to pay for unproven interventions and may mandate that payment be for evidence-based practices only. In medicine, research is often funded by drug and device manufacturers, but it is much more difficult to find funds for research in areas such as mental health, public health, and social services. Public health research is poorly funded. Public health experts, for example, have long believed in the importance of inspecting food establishments, but because foodborne illnesses largely go unreported, it is difficult to measure the overall effectiveness of restaurant inspections, much less compare the value of different regulatory practices.

Looking back over 20 years of increasing emphasis on evidence-based medicine in the United Kingdom, Greenhalgh, Howick, and Maskrev (2014) observe that the sheer volume of guidelines developed and used has become “unmanageable and unfathomable” and that “statistically significant benefits may be marginal in clinical practice” (p. 2). They also suggest that

Evidence-based medicine has drifted in recent years from investigating and managing established disease to detecting and intervening in non-diseases. Risk assessment using “evidence-based” scores and algorithms (for heart disease, diabetes, cancer and osteoporosis, for example) now occurs on an industrial scale, with scant attention to the opportunity costs or unintended human and financial consequences.

Greenhalgh, Howick, and Maskrev (2014, p. 3)

These concerns are about how we have allocated resources to and institutionalized evidence-based medicine, not about its relevance or importance.

► Reducing Variation and Saving Resources

Reducing variation in how clinicians approach frequently encountered diagnoses has benefits. Standardizing treatment protocols results in less waste of resources and more consistent outcomes. Clinical guidelines are widely available from prestigious organizations as well as local management. Provider resistance to the top-down approach is common, however. Farias et al. (2013) reported on the alternative of locally developed standardized clinical assessment and management plans (SCAMPs) that offer more flexibility when applied to varied and complex patients. With SCAMPs, providers work with clinical guidelines, but clinicians are free to deviate, provided they report their reasons for doing so. After about 200 patients are treated using a particular SCAMP, provider comments, related cost data, and the most recent literature are reviewed and the SCAMP is revised. Such a process can be managed locally or applied to a professional network. The authors report that physician compliance is greatly increased by this process.

► Crosscurrents Involved

The rise to prominence of evidence-based medicine is a result of several factors:

- Rapid generation of new and revised scientific information
- Pressures on clinicians to conserve time
- Payer and patient concerns about the rising cost of care
- Concerns of all parties about the relatively high rate of medical errors
- Long lead times needed to adopt new information and change practice behaviors
- Commercial free speech and the viral movement of information on the Web, which means more information—good and bad—in the hands of the public
- Demands for effective consumer-centric care information for patients and their families that expresses relevant choices reliably and understandably

Both biased and unbiased information is available to practitioners and the public. One sees advertisement after advertisement, especially about newly defined health problems and new treatments, almost all paid for by someone who wants to profit from changes in provider or patient behaviors. Positive results are much more likely to be promoted than negative ones. Before changing a response to a diagnosis, a clinician has to vet many sources of information and revise his or her script for dealing with that situation. Pressure to see more patients and the second-guessing of providers by payers may tempt a clinician to reduce the amount of customization attempted.

In some cases, such as central line infections, what was once an acceptable risk is now considered a medical error, and the provider organization must bear more of that risk. The individual practitioner's judgment is less and less likely to be free from scrutiny, and yet that practitioner must operate in an environment where information cannot always be taken at face value.

The plight of the policy analyst is not much different. He or she will still have to rely on medical experts for interpretation and recommendations. However, information coming in is likely to be transmitted by individuals with distinct points of view, if not distinct financial interests. Policy analysts, too, are visited by lobbyists and consumer advocates. They have to make evaluations of scientific merit and economic impact. Boden and Epstein (2006) warn about the fallacies of “policy-based evidence” where advocates (political or social) conduct or cite research that begins with a policy solution and generates only arguments supporting that alternative.

► The Process of Evidence-Based Analysis

Evidence-based medicine has its roots in clinical epidemiology. It is an analysis based on:

1. Defining a clinical question
2. Searching the available, relevant information for evidence concerning the clinical question
3. Assessing the level of the evidence and its validity, and selecting the best available answer for implementation. Behavioral change, however, requires more investigation

4. Trying the approach in one's clinical practice
5. Evaluating the performance of the new or revised clinical response, and consciously incorporating positive results into one's expertise set
6. Adapting this knowledge to the specific patient

Effective evidence-based medicine rests on continuous personal, professional, and organizational learning.

The Patient as the Unit of Analysis

For the researcher, a population has tended to be the unit of analysis, whereas the presenting patient is the clinician's focus. The clinician uses observations of the patient, the patient's description of symptoms, and test and imaging results as diagnostic tools. But we read about *precision medicine* as a game changer. It means that we now have new sources of data (usually genomic) to get to the cause of the disease *in the patient's case* and to select an intervention that is specific to that case. With that ability to identify disease at the molecular level should come more effective treatments and better outcomes with fewer side effects. One question is whether there will be an improvement; in other words, whether the risks and costs of the high-markup testing more than offset the risks and costs of accepting treatments on nonresponsive individuals. *Personalized medicine* usually means both precision treatment and careful attention to patient preferences in decision-making.

Clinical Decision-Making

When a clinician or a patient searches for the best treatment in real time, they will want to consider only those alternatives that are sufficiently effective to warrant their use. They will feel a need to do something, and often opt for the most technologically advanced and most expensive thing. Hadler (2013) notes that much of American medicine incorporates treatments that have proven efficacy, but do not have a proven significant effect on clinical outcomes. They often relate only to risk factors or to a very limited segment of the population with a specific diagnosis, but they are applied widely nonetheless.

Levels of Evidence

The evidence classifications used by analysts are often presented as a pyramid or hierarchy of evidence. If you search the Internet for images related to "levels of evidence," you will be deluged with graphical representations expressing more or less the same rankings. The top of the pyramid represents the most reliable studies, which are fewest in number. At the bottom are the least reliable ones, the anecdotes, personal opinions (expert or otherwise), and case reports, which are greatest in number. The pyramid may have anywhere from 4 to 10 levels. What is counterintuitive is that expert opinions are at or near the bottom and randomized controlled clinical trials are in the middle. At the top are systematic studies that integrate findings from multiple studies. An example would be a Cochrane Collaborative Review. Below that would be meta-analyses involving multiple studies, followed by synthesis of a limited number of studies. Then come individual randomized controlled trials. Below that would likely be cohort

BOX 8-1 Three Categories for Rating the Strength of Evidence at AHRQ

Drawing on elements of these established systems, the Innovations Exchange uses three categories to provide meaningful distinctions in assessing the strength of the link between the innovation and the observed results:

Strong: The evidence is based on one or more evaluations using experimental designs based on random allocation of individuals or groups of individuals (e.g., medical practices or hospital units) to comparison groups. The results of the evaluation(s) show consistent direct evidence of the effectiveness of the innovation in improving the targeted healthcare outcomes and/or processes, or structures in the case of healthcare policy innovations.

Moderate: While there are no randomized, controlled experiments, the evidence includes at least one systematic evaluation of the impact of the innovation using a quasi-experimental design, which could include the nonrandom assignment of individuals to comparison groups, before- and-after comparisons in one group, and/or comparisons with a historical baseline or control. The results of the evaluation(s) show consistent direct or indirect evidence of the effectiveness of the innovation in improving targeted healthcare outcomes and/or processes, or structures in the case of healthcare policy innovations. However, the strength of the evidence is limited by the size, quality, or generalizability of the evaluations, and thus alternative explanations cannot be ruled out.

Suggestive: While there are no systematic experimental or quasi-experimental evaluations, the evidence includes nonexperimental or qualitative support for an association between the innovation and targeted healthcare outcomes or processes, or structures in the case of healthcare policy innovations. This evidence may include noncomparative case studies, correlation analysis, or anecdotal reports. As with the previous category, alternative explanations for the results achieved cannot be ruled out.

If the available qualitative and quantitative information is insufficient to place the innovation in one of the three categories previously discussed, the activity fails to meet the minimum inclusion criterion for evidence, and therefore is not eligible for inclusion as an Innovation Profile in the AHRQ Health Care Innovations Exchange. It may, however, qualify for inclusion as an Innovation Attempt.

studies, followed by case-control studies. The Agency for Healthcare Research and Quality (AHRQ) has boiled this down to three categories of strength of evidence shown in **BOX 8-1**.

Some representations include clinical guidelines, which tend to be reliable but vary in their underlying levels of evidence. They have been known to come from reviews and meta-analyses, as well as from a process known as GOBSAT, which stands for “Good Old Boys Sitting Around Talking.” A useful classroom exercise would be to call up the available set of pyramidal images and agree on one for use in future discussions.

Example of the Preventive Services Task Force

The U.S. Preventive Services Task Force (USPSTF) has developed its own methodology for evaluating proposed recommendations. Preventive services have a long-term and substantive impact on the cost of care. They represent an area where guidelines

are helpful because most clinicians have to rely on them because prevention avoids clinical experience. The guidelines can be controversial, however, as demonstrated by recent debates over recommended reductions on early breast cancer screenings and the PSA test for prostate cancer. The Task Force recommendations carry a great deal of weight with payers who are expected to include those specific services in their coverage. Some argue that the USPSTF has become a regulatory agency governing insurance coverage decisions. They argue that a separate organization should handle coverage requirements, which are political as well as scientific (Johns and Bayer, 2016).

TABLE 8-1 presents the system the Task Force uses to grade recommendations. There are two categories of grades: one for “Suggestions for Practice” and the other for “Certainty of Net Benefits.” Note the introductory paragraph about the change in the definition of Grade C.

TABLE 8-1 Grade Definitions After July 2012

What the Grades Mean and Suggestions for Practice

The USPSTF updated its definition of and suggestions for practice for the grade C recommendation. This new definition applies to USPSTF recommendations voted on or after July 2012. Describing the strength of a recommendation is an important part of communicating its importance to clinicians and other users. Although most of the grade definitions have evolved since the USPSTF first began, none has changed more noticeably than the definition of a C recommendation, which has undergone three major revisions since 1998. Despite these revisions, the essence of the C recommendation has remained consistent: at the population level, the balance of benefits and harms is very close, and the magnitude of net benefit is small. Given this small net benefit, the USPSTF has either not made a recommendation “for or against routinely” providing the service (1998), recommended “against routinely” providing the service (2007), or recommended “selectively” providing the service (2012). Grade C recommendations are particularly sensitive to patient values and circumstances. Determining whether the service should be offered or provided to an individual patient will typically require an informed conversation between the clinician and the patient.

Grade	Definition	Suggestions for Practice
A	The USPSTF recommends the service. There is high certainty that the net benefit is substantial.	Offer or provide this service.
B	The USPSTF recommends the service. There is high certainty that the net benefit is moderate or there is moderate certainty that the net benefit is moderate to substantial.	Offer or provide this service.
C	The USPSTF recommends selectively offering or providing this service to individual patients based on professional judgment and patient preferences. There is at least moderate certainty that the net benefit is small.	Offer or provide this service for selected patients depending on individual circumstances.

D	The USPSTF recommends against the service. There is moderate or high certainty that the service has no net benefit or that the harms outweigh the benefits.	Discourage the use of this service.
I Statement	The USPSTF concludes that the current evidence is insufficient to assess the balance of benefits and harms of the service. Evidence is lacking, of poor quality, or conflicting, and the balance of benefits and harms cannot be determined.	Read the clinical considerations section of USPSTF Recommendation Statement. If the service is offered, patients should understand the uncertainty about the balance of benefits and harms.

Levels of Certainty Regarding Net Benefit

Level of Certainty*	Description
High	The available evidence usually includes consistent results from well-designed, well-conducted studies in representative primary care populations. These studies assess the effects of the preventive service on health outcomes. This conclusion is therefore unlikely to be strongly affected by the results of future studies.
Moderate	The available evidence is sufficient to determine the effects of the preventive service on health outcomes, but confidence in the estimate is constrained by such factors as: ■ The number, size, or quality of individual studies ■ Inconsistency of findings across individual studies ■ Limited generalizability of findings to routine primary care practice ■ Lack of coherence in the chain of evidence As more information becomes available, the magnitude or direction of the observed effect could change, and this change may be large enough to alter the conclusion.
Low	The available evidence is insufficient to assess effects on health outcomes. Evidence is insufficient because of: ■ The limited number or size of studies ■ Important flaws in study design or methods ■ Inconsistency of findings across individual studies ■ Gaps in the chain of evidence ■ Findings not generalizable to routine primary care practice ■ Lack of information on important health outcomes More information may allow estimation of effects on health outcomes.

*The USPSTF defines certainty as “likelihood that the USPSTF assessment of the net benefit of a preventive service is correct.” The net benefit is defined as benefit minus harm of the preventive service as implemented in a general, primary care population. The USPSTF assigns a certainty level based on the nature of the overall evidence available to assess the net benefit of a preventive service. Reproduced from U.S. Preventive Services Task Force Grade Definitions, (08, 2016 May–June). Grade definitions. Retrieved August 21, 2017, from www.uspreventiveservicestaskforce.org/Page/Name/grade-definitions

Biases in Evidence Gathering

The gold standard for gathering new evidence is the randomized, controlled clinical trial, which uses a specifically selected population randomly divided into a control group and a treatment group. The control group, which may receive a placebo or other sham intervention or the current usual and customary treatment, is compared to the treatment group. The trial objective parallels the economist's holy grail of "all other things being equal." Concerns about bias are likely to be related to the relevance of the sample for clinical decision-making or selective reporting of the results.

More recently, there has been support for comparative effectiveness studies. They attempt to compare treatments under field conditions. While the inputs and conditions of such studies are not as tightly controlled as in randomized, controlled clinical trials, they are considered suitable for hypothesis generation. Comparative effectiveness studies have the advantage of being able to use large, relatively available databases, lowering the cost and duration of the study. Potential sources of bias include sample variety, practice variation, and unknown, uncontrolled variables. Because they use data that can be associated with charges, payments, and costs, as well as safety outcomes, they are critical to the measurement of that currently fashionable construct—*value*.

Clinician experience enters into decision-making as well. However, a clinician's experience has likely been influenced by past training, marketing efforts, event importance, recency effects, payer contracts, and perhaps even personal economic interests. Off-label use is often based on clinician experience, and it often is influenced by both legal and illegal promotion efforts.

Patient observations are important, but are subject to some of the same biases as clinician experience. Marketing efforts can shape patient attitudes and expectations in ways unsupported by effectiveness evidence. There is also the lack of observational training and less experience (based on a sample of one). However, patients living long-term with significant chronic problems often become very good observers and may tie into observational networks that amplify their limited experience. They may even have more experience than the average clinician. They are certainly an important source for socioeconomic and psychosocial support information that is crucial to effective community-based care.

Expanding Worlds of Evidence

While the evidence pyramid is a good starting point for evaluating evidence, a much wider world of evidence is opening up with the availability of different sources of evidence from mining databases and registries to adding molecular (usually genomic) data and ultimately to using artificial intelligence to identify further probable causes of diseases and cures.

Frieden (2017) evaluates the strengths and weaknesses of each and provides examples of how they have yielded improvements in health treatments and policy. These are often grouped under the title, *observational studies*. Frieden's examples included the following:

- Cross-sectional studies
- Ecological studies
- Pragmatic trials and large observational studies

- Program-based evidence
- Case reports and series
- Registries

Irving, Eramudugolla, Cherbuin, and Anstey (2017) concluded that current grading systems for evidence “have a number of shortcomings” and that “grading systems need to matched to both the research question being asked and the type of evidence being used.”

► Constraints on Variables Used in Analysis of Evidence

Congress has constrained the use of certain economic and outcome valuations. However, the USPSTF continues to specify the following outcome measures as analysis inputs:

- Deaths, where relevant
- Important health outcomes, such as strokes avoided or cancers caused
- Quality-adjusted life-years, if possible
- Harms (adverse events/states)

The task force’s experiences show the ambivalence and contentiousness that surround the rigorous use of evidence in the health sector.

► The Example of NICE

The National Health Service (NHS) in the United Kingdom has long assessed evidence and developed guidelines through its independent National Institute for Health and Clinical Excellence (NICE). Assessments and recommendations take many forms. They include NICE guidelines aimed at healthcare and social services professionals.

NICE guidelines make evidence-based recommendations on a wide range of topics, from preventing and managing specific conditions, improving health and managing medicines in different settings, to providing social care to adults and children, and planning broader services and interventions to improve the health of communities. These aim to promote integrated care where appropriate, for example, by covering transitions between children’s and adult services and between health and social care.

NICE (2017b)

NICE also offers technological appraisals of new and existing treatments. This guidance can focus on surgical procedures, diagnostic techniques, health promotion, medical devices, and medicines (pharmaceuticals and biologics). Guidance on medical technologies covers any technology intended to deliver treatment, increase patient independence, or detect or monitor health conditions. Diagnostic guidance addresses technologies that rule out diseases, help identify the cause of symptoms, provide

information about staging or severity, monitor progress of a condition, or screen undiagnosed and asymptomatic patients. Intervention procedures guidance can cover any procedure that involves entering the body by cutting or through a cavity, as well as radiation treatments.

Many other developed countries have access to similar organizations.

► Decision Aids

Evidence-based medicine underpins guidelines and protocols, but it also strives to ensure that patients are part of the decision-making process. Current information technology can be used to support the dissemination of decision aids for both patients and caregivers. Sooner or later these applications will become part of the U.S. national information technology standard for meaningful use, which is being defined and implemented in stages.

Section 3506 of the Affordable Care Act (ACA) authorized a program under a new Center for Medicare & Medicaid Innovation to develop decision aids that will "... facilitate collaborative processes between patients, caregivers or authorized representatives, and clinicians that engage the patient, caregiver or authorized representative in decision-making, provides patients, caregivers or authorized representatives with information about trade-offs among treatment options, and facilitates the incorporation of patient preferences and values into the medical plan."

The term *preference sensitive care* means medical care for which the clinical evidence does not clearly support one treatment option. The appropriate course of treatment then depends on the values of the patient or the preferences of the patient, caregivers, or authorized representatives regarding the benefits, harms, and scientific evidence for each treatment option.

States can support this movement through mandates or incentives for the implementation of decision aids (King & Moulton, 2013).

This movement is driven, in part, by discomfort with the authoritarian nature of the clinician's typical role and, in part, by the knowledge that when patients are given information and choices, the observed outcomes are better and the costs of care are often lower (Hibbard, Greene, & Overton, 2013; Veroff, Marr, & Wennberg, 2013).

The February 2013 issue of *Health Affairs* was partly devoted to the themes of patient engagement and patient activation. The previous quote from the ACA highlights patients' twin roles as active and informed consumers and as involved clinical decision-makers. The patient may be involved as an individual, in collaboration with the payer, or in collaboration with the physician. **TABLE 8-2** provides examples of activities associated with each of the roles.

Patients acting alone, or others acting on their behalf, can engage in information searches and decision-making by consulting any number of sources of information. Comparative databases on quality and cost for hospitals and other providers are available. Patients can talk to their neighbors, professionals, or local experts. However, it is important to note that the curiosity patients exhibit about treatments and quality of outcomes has not yet seemed to extend to costs. This is partly because cost information is much harder to access and because insurance often makes the data irrelevant from the patient's point of view. It is also true that they are reluctant to

TABLE 8-2 Examples of Engaged Patient Activities

Patient Roles	Informed Consumer	Clinical Decision-Maker
Independent	Use Hospital compare Use consumer satisfaction databases Do comparison shopping Ask local friends and experts	Access: PatientsLikeMe.com Guideline databases Advocacy sites General information sites Ask others about experiences
With payer	Get data on in-network providers Discuss with case managers Compare on exchanges	Access Web portals Discuss with case managers Review literature and brochures Make inquiries about coverage
With provider	Agree on appropriate entry mode Discuss rates and fees Observe and/or discuss philosophy and attitudes on cost and aggressiveness of treatment	Study journal literature Review guidelines together Use joint decision aids Talk through behaviors and preferences

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change providers just because of cost (Desai et al., 2017; Mehrotra, Dean, Sinaiko, & Sood, 2017).

Many websites are devoted to specific symptoms and diagnoses, such as the American Diabetes Association (www.diabetes.org) and PatientsLikeMe (www.patientslikeme.com). They can inform the patient together with the medical literature and databases of governmental and professional guidelines (domestic and international). These can provide access to reports backed by evidence that falls all along the evidence hierarchy, including reports from sources with major potential biases, such as television ads and websites.

There are information providers fulfilling the role of honest broker. Many payers maintain Web portals where enrollees can find tips on certain diagnoses and conditions, and they may also provide case managers for patients with certain chronic diseases or catastrophic illnesses. So do nonprofit organizations that support research and advocate for specific diagnoses. Again, validity and level of evidence can vary a great deal.

A collaborative relationship between provider and patient isn't for everybody (Carman et al., 2016). So far, demonstration efforts have identified major barriers to widespread adoption of evidence-based approaches, especially the drain on provider time, lack of payment for that time, physician perceptions about patients' ability to

understand evidence, lack of relevant information, and patients' preferences for a provider who acts as an authority figure (Lin et al., 2013; Yergian, Dardess, Shannon, & Carman, 2013). A study of the use of web-based decision aids in the NHS indicated that "clinicians did not feel the need to refer patients to use decision support tools, web-based or not, and, as a result, felt no requirement to change existing practice routines" (Elwyn, Rix, Holt, & Jones, 2012). An interesting finding in the research on this change in the healthcare culture is that although physicians have the greatest influence over patient behavior, other clinic staff and off-site personnel can contribute successfully to the support of shared decision-making (Courneya, Palattao, & Gallagher, 2013; Veroff et al., 2013).

To support collaboration, patient and physician must understand how, when, and where the patient will receive or transmit information, such as by telephone, by email, in the physician's office, or at an alternative service site. They must have a discussion of their philosophies and attitudes toward issues such as aggressiveness of treatment and costs of care. Some patients will want to control costs, whereas others will be uncomfortable when clinicians focus on costs in clinical decision-making (Gawande, 2014; Sommers, Goold, McGlynn, Pearson, & Danis, 2013). However, cost will play an increasingly important role in informing consumer behavior.

► Determining Value

Most randomized, controlled clinical trials are conducted on proposed prescription drugs for which patent protections provide a potential monopoly. In such trials, U.S. researchers only need to establish a pharmaceutical's safety and efficacy. By comparison, medical devices that can piggyback on the testing of similar devices and procedures do not require licensing. Manufacturers usually support studies in which a placebo is the control. This implies that the product being tested just has to be better than doing nothing at all. Determining whether a new product has value greater than existing products requires a comparative effectiveness study that includes cost comparisons. Because of the reluctance of manufacturers to conduct such studies, the government has had to step in. This is a relatively recent development. In 1989–1990, the predecessor to AHRQ issued a series of contracts for Patient Outcomes Review Team (PORT) studies that were quite controversial. These studies tended to evaluate high-volume and/or high-cost interventions, and the concept was not popular with the provider community. Some of the issues addressed by these studies are still open to debate. In fact, after a publication from a PORT study showing that it did not matter what type of provider treated acute (without sciatica) low back pain, neurosurgeons almost succeeded in getting Congress to defund the agency. By 2008, however, concerns about value had become so great that the American Recovery and Reinvestment Act (ARRA) contained significant funding for comparative research, and the ACA established the Patient-Centered Outcomes Research Institute (PCORI), an independent nonprofit, nongovernmental organization authorized by Congress in 2010: "Our mandate is to improve the quality and relevance of evidence available to help patients, caregivers, clinicians, employers, insurers, and policy makers make informed health decisions. Specifically, we fund comparative clinical effectiveness research, or CER, as well as support work that will improve the methods used to conduct such studies" (PCORI, 2016).

Translational Medicine: Adoption, Adaptation, and Compliance

Many are concerned with how slowly the results of basic research make their way into practice and the amount of time it takes for practice innovations to be evaluated and adopted. In addition, some experts suggest that clinician compliance with suggested guidelines is only about 50%. No one knows what the right figure is, because many patients do not duplicate the conditions envisioned by guidelines. Adoption rates affect both the quality and cost of health care, so there has been increasing interest in linking basic science and clinical practice more effectively. The attempt to build a bridge between the silos of research and practice is called *translational research*, and medical researchers involved in such multidisciplinary work have defined their new field as *translational medicine*. Interest in this interface increased sharply after the National Institutes of Health began funding such efforts through the National Center for Advanced Translational Sciences.

The field of translational medicine is still loosely defined. A consensus report from the Evaluation Committee of the Association for Clinical Research Training proposed a definition of translational research. According to the report, translational research seeks to improve the public's health by supporting the multidirectional integration of three types of research: basic, patient oriented, and population based. The following definition exemplifies the three types:

T1 research expedites the movement between basic research and patient-oriented research that leads to new or improved scientific understanding or standards of care. T2 research facilitates the movement between patient-oriented research and population-based research that leads to better patient outcomes, the implementation of best practices, and improved health status in communities. T3 research promotes interaction between laboratory-based research and population-based research to stimulate a robust scientific understanding of human health and disease.

Rubio et al. (2010)

Current efforts in translational medicine range from analyzing gene sequences associated with treatment outcomes to methods of improving patient compliance. We will have to wait and see how this type of research will be accepted in academic medicine because it often involves observational studies and critical inputs from many social science disciplines. Observational studies involving expanding registries and clinical databases show great promise but face a number of barriers (Fleurence, Naci, & Jansen, 2010; Lauer & D'Agostino, 2013).

Pros of Observational Studies

The digitization and sharing of medical records makes huge amounts of experiential data readily accessible at much lower cost than the clinical trials alternative. They are a regular component of most quality improvement initiatives. They have had an effect on the treatment of wet age-related macular degeneration and cystic fibrosis in children.

Think about the possibility of capturing all successful off-label uses of drugs. Researchers could then test the more promising ones for causality and implementation. Clinical trials are expensive and unjustifiable cost-wise for many diseases, especially as newly differentiated diagnoses based on genetic data refine and reduce relevant disease cohorts.

Cons of Observational Studies

The weaknesses of such approaches include:

- Lack of causal information because many variables are uncontrolled.
- Risk of confirmatory bias (seeing what is there and designing an analysis to confirm it).
- Risks of misinterpreting results when observations have small sample sizes.
- Fixations of medical communities on the gold standard of randomized, controlled, double-blind studies leading to rejection of useful insights.

The effect of sample size may be responsible for reports of high rates of disease in a specific county and community. Kahneman (2011) reports an illustrative example in *Thinking, Fast and Slow*, a book that should be required reading for all professional policy analysts: “A study of new diagnoses of kidney cancer in the 3141 counties of the United States reveals a remarkable pattern. The counties in which the incidence of kidney cancer is lowest are mostly rural, sparsely populated and located in traditionally Republican states in the Midwest, the South and the West. What do you make of this?” (p. 109).

Stop reading now and consider his question. What is likely to be the cause of this observation?

Then proceed with his example: “Now consider the counties in which the incidence of kidney cancer is highest. Those ailing counties tend to be rural, sparsely populated and located in traditionally Republican states in the South, the Midwest and the West.”

What do you make of this?

Incidence is a ratio, so the problem can be in either the numerators or the denominators. There were many samples, but the extreme observations (highest and lowest ratios) tended to occur in the numerous counties with low populations (small denominators) and small integer nominators. Observational studies need the same intellectual rigor as clinical trials. Recognizing this, the FDA and PCORI are underwriting research programs in *regulatory science* aimed in part at the issues of smaller populations and better application of observational data.

Understanding Choice Processes

Policy analysts will need to stay abreast of behavioral research on how decisions are made and how presentation of evidence, experience, and values affects choice process. For example, how information is framed influences the way individuals perceive the relationship between price and quality (Phillips, Schleifer, & Hagelskamp, 2016).

Carman et al. (2016) reported on a study of “informed public views of the role of evidence.” They concluded: “Although participants perceived evidence as being essential to high quality care, they also believed that personal choice or

clinical judgment could trump evidence. ... They found evidence of harm to individuals or the community to be more compelling than evidence of competitiveness” (p. 566).

There is much more to be learned here and policy analysts need to stay tuned in.

► Where in the World?

There is no shortage of sites concerned with evidence-based medicine. The AHRQ, PCORI, and the Cochrane Collaborative are prominent sources in the United States. In the United Kingdom, the Centre for Evidence-based Medicine at Oxford University is a reliable starting point. The journal *Evidence-based Medicine* offers many technical examples, while *British Medical Journal*, *Implementation Science*, and *PloS Medicine* have occasional thoughtful policy articles on this subject. You can enhance your understanding of the content of translational medicine by browsing the journal, *Science Translational Medicine*, which has more of a policy bent than some other journals on the subject.

► Conclusion

The use of evidence-based medicine is increasing, and policy analysts must know about its concepts and terms. It follows the logical paradigm adopted by systems analysts, industrial engineers, and quality and safety improvement managers—that is, the scientific method. Much of the actual research and analysis will be carried out by health professionals in the burgeoning cottage industry of providing meta-analyses, summary reviews, protocols, and guidelines. It is important to know and understand the hierarchy of evidence and stay current with the rapidly expanding development of decision aids and with studies about how to activate and engage patients and providers in collaborative decision-making. Political and economic interests will try to seize the process of policy analysis and warp it in their favor, but the antidote is maintaining high standards for validity and quality of evidence. Policy makers in the United States are increasingly demanding such professionalism behind the scenes, if not in public discourse.

CASE 8: Comparative Effectiveness: Avastin Versus Lucentis

Imagine that you are going to clean the windows of your home and you discover that your spray bottle of commercial glass cleaner is empty. You are about to go to the store to buy more when your neighbor points out that the main ingredient is vinegar and that it is far, far cheaper to buy vinegar and dilute it with water. You try your new homemade cleaner and find that it seems to work just as well as what you were using before. What would you do in the future?

This scenario resembles that faced by many people with age-related macular degeneration (AMD), as well as their providers and insurers, but with one wrinkle: What if you discovered that your homemade cleaner could be more toxic than the commercial product if mixed incorrectly?

AMD is a debilitating eye disease that is the leading cause of blindness for older Americans, with 200,000 new cases diagnosed annually. In the more severe “wet” or “exudative” form (wAMD), blood and other fluids leak from abnormal blood vessels into the macula, a yellow oval at the center of the retina.

A common, effective treatment for wAMD is injections with ranibizumab, a drug that Genentech, a division of Roche, sells under the trade name Lucentis. Chemically, Lucentis is almost identical to bevacizumab, a cancer drug that Genentech calls Avastin. The same active component found in both drugs inhibits the function of vascular endothelial growth factor (VEGF), which stimulates the growth of blood vessels. For cancer patients, it means less blood flow to support cancer growth; for wAMD patients, it means fewer vessels to leak fluids into the macula.

To turn Avastin into Lucentis, Genentech made the molecule smaller; in theory, the molecular fragment would be better able to penetrate the macula than the full-sized molecule. Genentech also raised the price dramatically. The initial cost to Medicare of a Lucentis injection was about \$2000. Injections can be given on an as-needed basis or on a monthly schedule. With a monthly schedule, the annual cost per patient per year came to \$24,000. Genentech maintains that it specifically formulated and tested Lucentis for administration in the eye, and that the higher cost is necessary to recover the company’s investment in research and development.

While waiting for the FDA to approve Lucentis, doctors discovered they could take Avastin, have compounding pharmacists repackaging it into smaller vials, and then treat their patients for around \$50 an injection. Not surprisingly, Avastin is used in more than half the injections to treat wAMD in the United States, and ophthalmologists do not observe any difference in patient outcomes. This has huge financial implications for Medicare, which funds care for 95% of patients with wAMD. If all Medicare patients were treated with Lucentis, the annual cost would be \$1–\$3 billion, but if Avastin were used instead, the cost would be less than \$1 million (Martin, Maguire, & Fine, 2010).

Genentech has maintained that repackaging by compounding pharmacies raises safety concerns. It announced in October 2007 that it would no longer sell Avastin directly to compounding pharmacies. Angry doctors and professional associations argued the move would limit supply and create affordability problems for low-income patients. Genentech relented somewhat; it said it would sell directly to physicians who could then send it to compounding pharmacies if they wished. The International Academy of Compounding Pharmacists claimed Genentech was motivated by money, not reasonable safety concerns.

The Lucentis versus Avastin controversy would appear to be low-hanging fruit for supporters of comparative effectiveness research (CER). If one treatment costs \$600 a year and another costs \$24,000, and they produce the same results, why would the CMS or any other fiscally responsible payer facilitate the use of the more expensive option? But as part of CER, someone would have to conduct a robust, head-to-head clinical trial to compare the effectiveness of the two drugs. The problem is that such trials are usually funded by manufacturers trying to prove their product’s effectiveness to win FDA approval.

Avastin’s history provides a useful example of how clinical trials, which provide the foundation for CER and evidence-based medicine, typically occur in the United States. Avastin is approved for treating colorectal, lung, kidney, ovarian, and other metastatic cancers. It is approved for treatment of breast cancer in other countries, but not in the United States. The FDA had provisionally approved it for treating advanced breast cancer in 2008, pending further study. Genentech funded the additional studies. The findings showed the drug slowed progression of the disease, but because the studies failed to demonstrate that Avastin extended patients’ lives or improved their quality of life, the FDA withdrew its approval.

But off-label use of Avastin to treat wAMD is a quite different and highly unusual case. Genentech has no interest in testing a cancer drug for treating wAMD when it already has an effective (and more profitable) wAMD-specific drug on the market. It does not want Avastin approved for wAMD, and the FDA had no standing to demand Genentech conduct a trial.

Instead, funding for a head-to-head trial, called the Comparison of Age-Related Macular Degeneration Treatment Trials (CATT), came from the National Eye Institute (NEI), which is part of the NIH. In 2006, NEI agreed to provide \$1 million to purchase, repackage, and distribute the Avastin. Researchers initially assumed Medicare would pay for Lucentis as routine care for clinical trial participants, but it turned out that existing CMS policy did not allow for that. This changed with the adoption of new policies in 2007.

This left the question of how to handle copayments for the 15% of Medicare recipients who did not have supplemental insurance and were on the hook for 20% copays. Ultimately, NEI was able to cover copays under the special circumstances provisions in the NIH guidelines. The complicated system of obtaining and billing for commercially purchased drugs, however, made it hard to devise a system for masking the identity of study participants. Resolving this issue required amendments to the Medicare Improvements for Patients and Providers Act of 2008.

Martin, Maguire, and Fine (2010) noted that the ARRA provided \$1.1 billion for CER, but after describing all roadblocks encountered by the CATT Research Group, they concluded that the infrastructure for implementing federally sponsored CER trials is inadequate. They called on the insurance industry to facilitate CER trials and on the government to develop a comprehensive policy to cover the drug-related costs of trials without relying on existing billing and payment mechanisms. The ACA requires most insurers to pay “comparative effectiveness research fees” to fund a new Patient-Centered Outcomes Research Institute to conduct research into comparative effectiveness of medical treatments, along with their risks and benefits.

The CATT Research Group published its 2-year results in 2012 (Martin et al., 2012). The group found no statistical difference between the gains in visual acuity produced by the two drugs. There was no evidence that Genentech’s “smaller molecule” approach made Lucentis a more effective treatment. The study did find that the rate of serious systemic adverse events was higher for participants treated with Avastin (40%) compared to those treated with Lucentis (32%). “The interpretation of the persistence of higher rates of serious adverse events with bevacizumab is uncertain because of the lack of specificity to conditions associated with inhibition of VEGF” (p. 1388), the authors wrote. Monthly treatment had better outcomes than treatment as needed, although this must be weighed against costs and the risks associated with each injection. The authors concluded, “The choice of drug and dosing regimen for patients must balance the comparable effects on vision, the possibility of true differences in adverse events, and the 40-fold difference in cost per dose” (p. 1397).

The next year saw the publication of data from three randomized trials conducted independent of Genentech, all of which produced similar results despite different methodologies. IVAN, funded by the United Kingdom’s National Institute for Health Research Health Technology Assessment Programme, published 2-year findings (Chakravarthy et al., 2013). The MANTA Research Group in Vienna, Austria, published 1-year results (Krebs et al., 2013). The study was funded by the Austrian Society of Ophthalmology. The GEFAL trial, which also published 1-year results, was funded by the French Health Ministry and the French Health Insurance System (Kodjikian et al., 2013). In 2016, the Netherlands weighed in with the BRAMD trial funded by the Organisation for Health Research and Development, with support from Dutch insurance companies. All found that neither drug was statistically worse than or better than the other in terms of efficacy or adverse events (Schauwvlieghe et al., 2016).

Concerns about possible contamination when compounding pharmacies repackaged Avastin have proven to have some basis. In 2012, U.S. Representative Edward J. Markey (D-Massachusetts) introduced legislation to clarify and strengthen the FDA's role in regulating compounding pharmacies. The bill was a response to a fungal meningitis outbreak that sickened hundreds of people and killed 64. The outbreak was tied to contaminated injected steroids produced by New England Compounding Center in Markey's district. That same year, the FDA reported 12 cases of blindness in Miami following injections with bacterially contaminated Avastin from a single compounding pharmacy. Other cases were reported in Nashville, TN, and Los Angeles, CA. The American Society of Retina Specialists subsequently published a set of guidelines to help providers identify high-quality compounding pharmacies. In March 2013, the FDA announced it was recalling 40 lots of syringes containing Avastin. Five reports of eye infections had reached Clinical Specialties of Martinez, GA, a compounding pharmacy that had supplied the syringes to doctors in four states. A revised version of Markey's bill was enacted in November 2013.

In late 2011, the FDA approved aflibercept, a third anti-VEGF drug using a different molecule. It is sold by Regeneron as Eylea. At \$1850 per month, it costs about the same as Lucentis. In 2012, the sales of the three drugs for AMD were estimated at \$1.66 billion based on Medicare claims data and distributed as follows (Whoriskey & Keating, 2013):

	Market (%)	Total Spent on AMD Drugs (%)
Avastin	55	4
Lucentis	34	73
Eylea	11	33

Data from Whoriskey, P., & Keating, D. (2013, December 26). Medicare rules create a booming business in hospice care for people who aren't dying. *The Washington Post*. Retrieved from www.washingtonpost.com/business/economy/medicare-rules-create-a-booming-business-in-hospice-care-for-people-who-arent-dying/2013/12/26/4ff75bbe-68c9-11e3-ae56-22de072140a2_story.html

Shaikh et al. (2015) used a retrospective chart review study to compare the costs of treatment of wAMD with the three drugs using various dosing frequencies. Eylea could be injected less frequently; still, the cost of treating one eye over 6 months was estimated at \$326 for Avastin, \$11,400 for Lucentis, and \$9,720 for Eylea. The use of more expensive drugs did not result in improved vision outcomes.

The NEI conducted a 2-year study of the effectiveness of the three anti-VEGF drugs against diabetic macular edema. All three produced similar results in cases of mild vision loss. In the first year, Eylea outperformed Lucentis and Avastin in people whose vision was worse than 20/50, but the performance difference disappeared after 2 years (Wells et al., 2016). Medicare allowable costs per injection in 2016 were approximately \$1850 for Eylea, \$1200 for Lucentis, and \$60 for Avastin.

The U.K. NICE has been working on guidance for the treatment of AMD, and final guidelines were expected to be issued in January 2018. The draft guidelines published in July 2017 (NICE, 2017a) state that "no clinically significant differences in effectiveness or safety between aflibercept, ranibizumab and bevacizumab have been seen in the trials considered by the guideline committees" (p. 169). It also specified that bevacizumab (Avastin) is not licensed for intraocular treatment of AMD and cannot be prescribed as an alternative just because it is cheaper and more cost-effective.

Clinical trials are under way for additional treatments, including one involving drops rather than injections.

► Discussion Questions

1. If you were a retina specialist, how would the data in this case affect your drug procurement processes, your clinical practices, and your recommendations to patients? (You are currently allowed a 6% markup on outpatient injectable drugs.)
2. If you had AMD, what treatment choices would you make after reading this case? Would your choice depend on whether you had significant copays or other out-of-pocket expenses related to eye injections?
3. If you were a CMS official, what coverage-related policies would you promote?
4. If you were a legislator with oversight of a state Medicaid program, how would you respond to this case?
5. If you were a policy analyst, what would you recommend and to whom?
6. What policy changes might be needed to support government-sponsored initiatives related to CER and evidence-based medicine?
7. If you were an insurer, how would you respond to the recommendation that insurers facilitate CER research? What forms might such facilitation take?



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CHAPTER 9

The Policy Analysis Process: Evaluation of Political Feasibility

On Feb. 27, 2017, President Donald Trump told the National Governor's Association, "Nobody knew that health care could be so complicated." But for every experienced governor in the room, that was akin to saying, "Nobody knew that Antarctica could be so cold." There are many reasons health policy is so complicated, but politics is certainly one of the major ones.

In any realm, but especially in health care, evaluating political feasibility can be one of the most challenging stages of policy development. Policy choices are proposed, considered, adopted, and implemented in complex, subjective, and dynamic settings. This makes them difficult to analyze systematically and rationally. Irrational factors such as emotions, ambitions, egos, disputes, personalities, allegiances, relationships, and deeply held value systems come into play. This helps explain why discussions about political feasibility are often absent from policy debates and why literature specific to political feasibility is sparse. Oberlander (2003) observed that "political calculations are often a footnote in health care reform proposals" (p. 392).

In 1994, Walt & Gilson wrote that too few attempts at health policy analysis go beyond examining the content of a policy. They proposed a *policy triangle* framework that also considered:

- The actors who make and implement policy
- The processes by which policies are developed and implemented
- The context—social, political, economic—in which the actors work

These three factors should be at the core of any political feasibility analysis.

There has been an increase in the use of structured political feasibility analysis since then, particularly employing a methodology called *stakeholder analysis*. But as a paper coauthored by Gilson almost two decades later states, "there remains surprisingly few published accounts of the use of stakeholder analysis in health policy development generally, and health financing specifically, and even fewer that draw lessons from experience about how to do *and* how to use such analysis" (Gilson et al., 2012, p. 164).

Yet no policy analysis is complete if it avoids the issue of political feasibility, and systematic techniques that help predict political feasibility do exist.

► Terminology

Brown (2006) described *political feasibility* as “the right fit between bright ideas and the values and interests that animate stakeholders with crucial pieces of power” (p. W163). Oberlander (2003) noted, “Feasibility analysis deals not with policy ideals but with what is more or less adoptable given policy constraints” (p. W3-392).

Dror (1969) identified three characterizations of political feasibility:

1. For a political actor—an individual or organization—political feasibility is a measure of the actor’s ability, within a given period, to influence various activities, specifically policy adoption and implementation. The term *political leverage* describes the actor’s influence and *political leverage domain* describes the setting or range of activities within which an actor can exercise leverage effectively.
2. For a specific policy alternative, political feasibility is a measure of the likelihood that an alternative will be adopted and implemented within a specific time period.
3. For an issue area, political feasibility is a measure of the range of policy alternatives that could potentially be adopted and implemented within a specific time period. The range of politically viable alternatives is called the *political feasibility domain*.

Dror also described four sets of “variables” for thinking about political feasibility:

1. The main actors—who they are and what they intend.
2. Other inputs into the policy arena—the political climate, the state of the economy, public opinion, technological capabilities, and so forth.
3. The interplay of the first two—how actors come together and interact, taking the other inputs into account. Some actors may ally to form a *required coalition* with enough combined political leverage to move a policy forward.
4. The threshold for adoption. In the House of Representatives, for example, the threshold for passage is a simple majority, but overriding a presidential veto requires a two-thirds majority. An actor or a coalition may have considerable leverage, but whether it has enough will depend on the threshold that must be met.

In Dror’s model, the actors, influenced and informed by various inputs, interact with other actors and seek to combine their political leverage to achieve a mass of critical leverage.

Using his lexicon, health care has traditionally been a *political feasibility domain* with a narrow set of viable options. Taking other *inputs* into account, the *actors* supporting change have rarely been able to create enough *aggregate political leverage* to meet the *threshold for adoption*. Therefore, it has been a domain marked by incrementalism, compromise, and failure. Starr (2011) has attributed this to a “policy trap”—the healthcare system is unwieldy and dysfunctional, yet enough people are

satisfied by the current arrangement that they are unwilling to support disruptive changes. Even the Affordable Care Act (ACA), as sweeping as it may have seemed, was a product of this dynamic.

Another set of variables influences whether actors become concerned about an issue or attracted to a policy option. If an issue is not on anybody's radar, there are no actors, and thus no actor interactions. In political science and public affairs, the process by which political parties and government entities decide what issues to tackle is called *agenda setting*. A prerequisite of policy change is that a set of circumstances be identified as a problem whose time has come. Many factors influence when a problem is identified, how it is defined, and whether politicians choose to address it. These include public opinion, the national political mood, media coverage, emerging social movements, interest group mobilization, voter attitudes, arising risks and threats, new research findings, critical evaluations of program performance, economic changes, the level of sympathy for the affected populations, and the availability of potential policy solutions (Hacker, 1997; Oliver, 2006).

► Overview

To assess political feasibility, one needs to draw on the fields of political science, public administration, economics, and business management. The first part of this chapter focuses on the *authorizing environments* in which policy is made. By mapping an authorizing environment (a process also referred to as *stakeholder mapping*), we identify the major actors that populate the political stage and their roles. The second part of this chapter focuses on methods for analyzing political feasibility. Finally, we examine some of the ways political feasibility analysis can be used, for good or ill, to drive political decisions and policy adoption.

Although this chapter focuses primarily on government policy making, many other channels for implementing policy exist. Organized community-based activities that are designed to change attitudes, beliefs, and individual norms also can be considered policies (Lamson & Colman, 2005). Individual private organizations are also political environments in which policy alternatives must be evaluated for political feasibility (Hansmann, 1996).

► Authorizing Environments

The term *authorizing environment* comes from the field of public administration and refers to a list of the actors from whom a public manager must receive authorization to be effective. In a pluralistic society, it applies equally well to the development, adoption, and successful implementation of a policy.

Consider the authorizing environment experienced by one of the authors while the director of the Washington State Board of Health. The board's authorities include the ability to make rules for the environmental health and safety of schools. When the board decided to update its school environmental health rules, it had to operate in a crowded and contentious authorizing environment.

The primary authority for schools rests with local school districts, and school board members are accountable to voters. Washington laws grant local jurisdictions considerable authority. The state, however, pays 69% of public schools' operating

costs and most construction costs. The legislature imposes numerous mandates on the districts.

School health and safety inspection programs, where they exist, reside with local public health jurisdictions governed by local boards of health comprised wholly or in part of county commissioners. They work closely with the state Department of Health. Local governments and local boards of health can enact their own ordinances, which may not be less stringent than state statutes and rules. They have limited authority over school districts; however, local governments can adopt and enforce building codes.

The governor appoints 9 of the state Board of Health's 10 members to 3-year terms and is the supervisor of the *ex-officio* 10th member, the secretary of health.

The legislature granted the board its regulatory powers and could repeal them or cut the board's budget. It also could overrule board action through preemptive legislation, and a special committee can investigate whether any agency's rules exceed its authority.

Other state agencies have authorities and programs related to health and safety:

- The Office of the Superintendent of Public Instruction operates a “coordinated school health” program and has sustainability standards for school construction.
- The Department of Health is a partner on the coordinated school health project, provides technical assistance to schools and local public health agencies, regulates drinking water systems, and administers money for schools to test for lead.
- The Department of Labor and Industries protects employees' workplace safety.
- The Department of Ecology addresses contaminated soils on school grounds.
- The Department of Agriculture regulates pesticide use on and near school grounds.
- The Building Code Council establishes the statewide building code.

At the federal level, the U.S. Environmental Protection Agency regulates drinking water. It also runs a voluntary program to help schools ensure a healthy environment. The Centers for Disease Control and Prevention administers a coordinated school health program.

In Seattle and other communities, parents are active and organized. They have concerns about lead and other drinking water contaminants, indoor air quality, and mold. These issues have been the subject of extensive media coverage. The Washington State PTA is an influential political player in the state.

Often aligned with parents are teachers and the Washington Education Association (WEA). School employees attribute many ailments to school environmental conditions. The WEA is a major actor in state politics.

The State Board of Health is committed to basing decisions on science, but the science is incomplete, inconclusive, and evolving.

The state, the schools, and the WEA all consulted with lawyers during rule making, and school districts have frequently challenged state policies in court.

On this one rule alone, therefore, the board's authorizing environment included schools, school boards, local government, local public health jurisdictions, the governor, the legislature, several state agencies, two federal agencies, parents, students, teachers, research scientists, the courts, labor, media, contractors, architects, and the general public (see **FIGURE 9-1**).

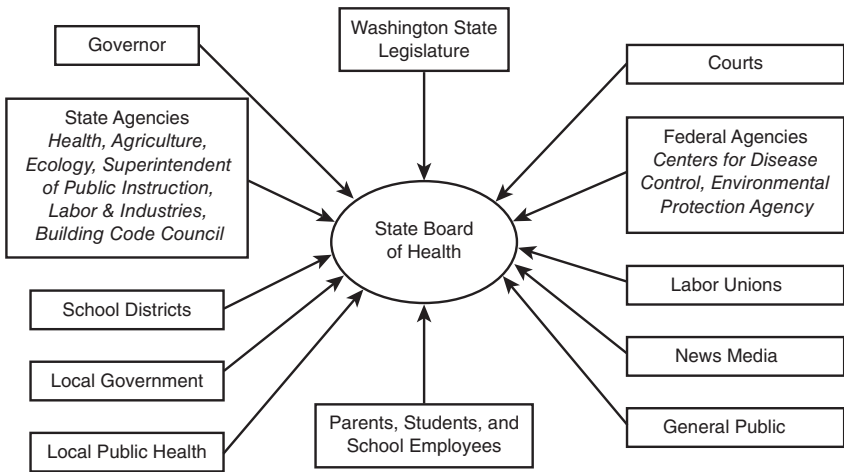


FIGURE 9-1 Washington State Board of Health authorizing environment for school health and safety rules.

This sounds (and is) incredibly complex—who knew?—but nearly all health policy making takes place within similarly complicated authorizing environments.

► Key Government Actors

In this section, we look at federal actors and consider briefly some factors that determine how they behave within their niches. This will be followed by a discussion of key actors at the state and local level and some of the ways they may differ from their federal counterparts.

Federal Government Actors

The drafters of the U.S. Constitution were wary of a government that put unrestrained power in the hands of one individual or institution, and they established a federal government with three branches—legislative, executive, and judicial—and a system of checks and balances to ensure that no one branch would dominate.

Congress: The Legislative Body

Assessments of the relative power of the three branches of government in the modern era tend to conclude that Congress is the most powerful. This is largely because only Congress has the power to make laws and create budgets. The relative power of Congress has grown over the past few decades as larger staffs have freed Congress from reliance on the executive branch to generate major policy proposals. According to Oberlander (2003), “Congress, measured in terms of its political independence, administrative capacity, and ability to pursue policies that diverge from the executive, may be the most powerful legislature in the world” (p. W3-391–W3-404).

Senators are elected by a statewide vote, which means that the least populous states have the same representation as the most populous. House members represent congressional districts that are roughly equal in population and are redrawn after every census.

When legislation is introduced, it is referred to one or more policy committees. This is where the bulk of the work gets done. Committees may hold hearings, mark up a bill, and refer it to the floor for action. Committees have subcommittees with policy expertise in certain areas. Major policy subcommittees for healthcare issues are the Subcommittee on Public Health of the Senate Committee on Health, Education, Labor, and Pensions and the Subcommittee on Health of the House Energy and Commerce Committee.

If the policy has budgetary implications, it will also go through one or more of the six powerful fiscal committees—Appropriations and Budget committees in both houses, House Ways and Means Committee, and the Senate Finance Committee. The latter two have tended to be bottlenecks for major healthcare legislation. In the House, another powerful committee, the Rules Committee, directs the progress of legislation and establishes the rules of debate.

If related but different legislation passes both houses, one house may ask the other to reconsider, or leadership from both houses may appoint a joint conference committee that will develop compromise legislation and send the new iteration back to both houses for adoption. Legislation can perish from neglect in a conference committee. Committee members can also add major policy changes with little fanfare. Just 10 days before the 1974 passage of the Employment Retirement and Income Security Act (ERISA), conferees slipped in language that restricted states from regulating the health benefit plans of self-insured companies. This “ERISA exemption” severely limited the ability of states to enact significant healthcare reforms such as play-or-pay laws.

Congress will sometimes authorize a new program but not provide an appropriation. In 1998, for example, Congress passed the Ricky Ray Hemophilia Relief Fund Act, which authorized payments of up to \$100,000 to hemophiliacs and their partners (or their survivors) infected with the human immunodeficiency virus. Congress, however, did not begin to appropriate money for the payments until 2001.

Another part of the budget process that can undermine policy committees is the reconciliation bill. Congress can adopt a budget, and then committees can go back and recommend changes to existing laws to conform to the budget instructions. These recommendations are compiled into a reconciliation bill that requires no hearing and often passes with little scrutiny. Most major health legislation during the 1980s took this route (Starr, 2011). The ACA’s passage required use of the reconciliation process to negotiate changes between House and Senate versions when the Senate lost its filibuster-proof majority after Democratic Senator Ted Kennedy of Massachusetts died in office and Republican Scott Brown won his seat in a special election. In 2017, the Republican-led Congress tried to repeal or “repeal and replace” the ACA before September 30 so it could use the reconciliation process and avoid a Senate filibuster. It failed to meet the deadline. However, the Senate could insert a new reconciliation clause tailored to repeal and replace into its next budget bill at any time.

Before major legislation can take effect, therefore, it may have to wind its way through a maze of committees and subcommittees, be scheduled for a floor vote, receive a majority vote in both houses (and perhaps 60 votes in the Senate), possibly survive a conference committee, have a budget appropriation, and be signed by the

president or have enough votes for a veto override. It is hardly surprising that major reform initiatives are easy to derail. Oberlander (2003) identified the fragmented structure of Congress as a significant barrier to healthcare reform.

The President: The Chief Executive Officer

The U.S. president is often referred to as “the leader of the free world.” The phrase perhaps overstates the importance of the office, given the relative strength of Congress. The day after the 2006 midterm elections, Representative John P. Murtha told National Public Radio, “The president has no power. The president has the perception of power.” While a president has limited authority to *make* policy, he or she has tremendous ability to *influence* policy. A president is far more likely to become a policy maker because of leadership than through the exercise of any direct authority.

The president can and does craft major legislative initiatives, but only Congress can introduce them. The president must find a sponsor, and after a bill is in the hands of Congress there is no guarantee it will pass or if it does pass that it will resemble the original proposal. In health care, perhaps the greatest example of the limits on a president’s ability to ensure a bill’s passage is the 1993 Health Security Act proposed unsuccessfully by Bill Clinton.

The president initiates the annual budget process, but the budget that emerges from Congress may be significantly different. In 2006, for example, George W. Bush proposed a fiscal year 2007 budget that would have eliminated the Urban Indian Health Program. Legislative committees marking up the Department of the Interior, Environment, and Related Agencies Appropriation Bill restored the funding and even proposed an increase in the program’s budget.

A president’s ability to push a measure through Congress depends in large part on his or her *political capital*. For presidents, political capital primarily comes down to two things—popularity and their party’s strength in Congress.

The 1965 passage of Medicaid and Medicare has been attributed to Lyndon Johnson’s phenomenal political capital. He clearly had a mandate, as he was elected with more than 61% of the popular vote, a feat unsurpassed since. The first Gallup poll of his term showed an 80% approval rating. He was a Democrat, and his party had a two-thirds majority in both houses. This gave him authority to push the agenda that had gotten him elected and a Congress unified enough to tackle even the most divisive issues.

The importance of political capital for achieving policy goals may partially explain Donald Trump’s frequent insistence that he won the 2016 election by a landslide, despite losing the popular vote, winning the Electoral College vote on the basis of razor thin margins in a few swing states, and having a lower Electoral College margin of victory than most of his contemporaries.

Presidential words can carry considerable moral authority. Presidential messages reach the public through numerous vehicles—the State of the Union address, press conferences, speeches, fireside chats, photo opportunities, public appearances, social media such as Twitter, and more. This gives the president unequaled ability to elevate issues onto the national agenda, propose strategies, and bring public pressure to bear. The president’s leadership role involves working to build coalitions, broker negotiations, and develop compromises. When Congress balks at a presidential proposal, the president can engage in education, persuasion, and horse trading. Extreme arm-twisting by a president, however, is rare.

Presidents do have authorities they can use to leverage policy. One is veto power. Another is the ability to issue executive memoranda, which provide direction to agencies. If they are published in the *Federal Register* and not challenged by Congress, they become executive orders, which have the force of law. For example, a 1999 executive memorandum required the Federal Employee Health Benefits Board to provide mental health parity (coverage for mental illnesses equivalent to that for physical ailments) by 2001.

The Bureaucracy

In theory, most bureaucracies are under the control of the executive, but the truth, as usual, is more complicated. For one thing, the president's reach is limited. Although a president fills some 2400 positions, appointees occupy only the top layers of a bureaucracy's management ranks. It is not unusual for career bureaucrats to try to subvert a president's agenda, as illustrated by leaks and other forms of resistance during the early days of the Trump administration.

Bureaucracies serve two masters—the president and Congress. Senior appointments may require congressional confirmation. Congress establishes the agencies' budgets, authorizes their programs, and enacts the laws under which they must act. Congressional committees and subcommittees oversee agencies under their purview.

Bureaucrats can draft legislation, and their expertise may be called on to craft amendments. Weissert and Weissert (2002) described the role Wilbur Cohen, then assistant secretary of Health, Education, and Welfare and the leading expert on Social Security, played in the 1965 passage of Medicare and Medicaid:

Cohen helped draft the administration bill on Medicare, consulted with members on proposed bills, and was asked to summarize various proposals to the key committees. When Wilbur Mills decided to combine several proposals into a “three-layer cake” (the layers were later known as Medicare Parts A and B and Medicaid), he asked Cohen to draw up legislative language to pull the pieces together, along with an analysis of the costs, within 12 hours.

Weissert and Weissert (2002, p. 154)

Many bureaucracies generate data, information, and research as part of their regular activities, and these work products can inform the policy development process. Examples in health care include research produced by the National Institutes of Health and data generated by the Agency for Healthcare Research and Quality.

Bureaucracies have their own networks and constituencies. They can mobilize outside groups to support or oppose legislation and work with them to convince them of the merits of a proposal (or to at least minimize resistance). They can also pull together stakeholders to craft a policy alternative that has broad support, and they can use outside contacts to try to assess political feasibility.

Finally, bureaucracies with regulatory authority are policy makers in their own right. Although Congress makes law, regulatory agencies make administrative law in the form of rules and regulations.

At the federal level, most agencies with health-related portfolios are part of the Department of Health and Human Services (DHHS). **BOX 9-1** offers a list of DHHS agencies and **BOX 9-2** lists some of the agencies that operate health-related programs but are not part of DHHS.

BOX 9-1 U.S. Department of Health and Human Services Offices and Divisions

Administration for Children and Families
 Administration for Community Living
 Agency for Healthcare Research and Quality
 Agency for Toxic Substances and Disease Registry
 Centers for Disease Control and Prevention
 Centers for Medicare and Medicaid Services
 Food and Drug Administration
 Health Resources and Services Administration
 Indian Health Services
 National Institutes of Health
 Office for Civil Rights
 Office of Global Affairs
 Office of the National Coordinator for Disease Control and Prevention
 Substance Abuse and Mental Health Services Administration

Data from U.S. Department of Health and Human Services (2017). HHS organizational chart. Accessed on October 9, 2017, from www.hhs.gov/about/agencies/orgchart/index.html

BOX 9-2 Federal Agencies with Health-Related Duties (Excludes DHHS Divisions)**Department of Agriculture**

- Animal and Plant Health Inspection Service
- Cooperative State Research, Education, and Extension Service

Department of Defense

- TRICARE
- Army Medicine Department
- Navy Medicine
- Air Force Medical Service

Department of Homeland Security**Department of Labor**

- Mine Safety and Health Administration
- Occupational Safety and Health Administration
- Employee Benefits Security Administration
- Benefits Review Board

Department of Veterans Affairs, Veterans Health Administration**Environmental Protection Agency****Federal Bureau of Prisons, Health Services Division****Federal Mine Safety and Health Review Commission****Nuclear Regulatory Commission****Occupational Safety and Health Review Commission****Uniformed Services University of the Health Sciences**

Data from the Office of the Federal Register, National Archives and Records Administration, & U.S. Government Printing Office. (2017). United States Government Manual. Retrieved on October 9, 2017, from www.archives.gov/federal-register/publications/government-manual.html

The Judiciary: The Federal Courts

The federal courts often play a critical role when it comes to issues of federal preemption, constitutional protections, and interstate commerce. When Maryland enacted a play-or-pay law requiring that any company with more than 10,000 employees spend at least 8% of its payroll on employee health care or pay into a state fund for covering the uninsured, the Retail Industry Leaders Association filed suit in federal court. U.S. District Court Judge Frederick Motz overturned the law on July 19, 2006, finding that ERISA preempted state authority.

The U.S. Supreme Court picks which cases to hear from among those brought before it. It seeks out cases involving important issues related to the Constitution or federal laws and hears cases where the rulings of the district courts appear to be in conflict. The court ultimately decided the lawsuit filed by Republican attorneys general that argued the ACA was unconstitutional. The justices upheld the constitutionality of the individual mandate but ruled that a requirement that states expand Medicaid was unconstitutional because it was overly coercive.

State Governments

States play a leading-edge role in formulating health policy and are major purchasers of health insurance and health care. Health care is the most critical cost driver for state government. Medicaid spending alone accounts for 29% of spending across all states (National Association of State Budget Officers, 2016). States are major local employers and manage and fund public health and mental health services, authorize public hospital districts, regulate health insurance, license and discipline health-care providers, and oversee healthcare facilities. They also serve as test markets for reform efforts.

States are organized on the surface as smaller versions of the federal government. The governor plays a role akin to the president. All have legislative bodies that correspond to Congress, and all but Nebraska have a bicameral legislature. States apportion legislative seats in both houses on the basis of population—the result of a Supreme Court decision in the early 1960s affirming the “one man, one vote” doctrine. States also have their own courts system.

One of the most profound policy changes in terms of health outcomes for this country came solely through the courts and started at the state level. Attorneys general from several states sued tobacco companies to recover the costs of caring for publicly insured individuals with smoking-related illnesses. In 1998, the 5 largest tobacco companies and 46 states, plus 6 commonwealths and protectorates, signed the Master Settlement Agreement (4 states settled previously). The agreement called for payments of \$206 billion over 5 years. Several of the states used the money to fund anti-tobacco programs that have contributed to lower smoking rates. Although the states filed the suits in state courts, the cases might have been consolidated in federal court if the parties had not settled.

Bureaucracies are organized differently from state to state. Some states have combined public health, social services, and healthcare purchasing into a single huge agency. Some have split off public health into a separate department so that it does not get overshadowed. Some states have healthcare purchasing agencies that may or may not include federal entitlement programs.

Local Government

Local government bodies can be even more diverse than states. Generally speaking, each state has two levels of local governments: counties (parishes in Louisiana and boroughs in Alaska) and municipalities (cities, towns, villages, and boroughs). Typically, these exist side by side, but there are examples of combined cities and counties (San Francisco is one), and, in Virginia, some large cities operate outside of the county structure. Hawaii has four counties and no municipalities, except that Honolulu is a consolidated city-county.

Just as states have despaired at looking to the federal government to address access-to-care issues and other problems linked to the nation's system of healthcare finance and delivery and have tried to find solutions on their own, so, too, have local governments and local communities. Some local community efforts have experienced significant government involvement, whereas others have relied on grassroots efforts. One example of an effort driven by local government began in the early 1990s in Hillsborough County, FL. In response to the strain placed on the local budget due to the rising costs of providing services to the disadvantaged at the local public hospital, voters approved an increase to the county property tax to purchase health coverage for low-income residents ineligible for other public health insurance programs. Part of what made the plan politically feasible was a commitment that it would in time lead to a reduction in property tax levies.

► Political Inputs

Being able to identify the actors and describe their interactions is only part of the challenge of assessing political feasibility. It is also important to understand another set of variables—inputs that influence actor interactions. There are many such inputs, but key ones include elections, constituent relations, campaign fund-raising, party agendas, the economy and its impact on government budgets, political trading, and idiosyncratic personal factors. This section discusses these types of inputs and provides examples of how these inputs have influenced public health and healthcare financing. Many of the examples are from Washington State because one of the authors spent several years as an agency executive for the state.

The Election Cycle

Elected officials, unless they are planning to retire or face term limits (or may not live until the next election, as was the case in 2017 when Senator John McCain, an Arizona Republican contending with an aggressive form of brain cancer, was the deciding vote against ACA repeal and replace legislation) must be concerned with reelection. The timing of elections can influence politicians will to take on controversial issues. If an issue is likely to interest the electorate favorably, it might rise to the top of a politician's or party's agenda preceding an election. If an issue or policy is likely to alienate or confuse voters, politicians might put the issue off until after the election. Presidents typically focus on reelection throughout most of their first term, but tend to demonstrate more independence during their second term.

Constituent Relations

A top concern for elected officials is likely to be constituent relations. The nature of constituent relations activities will differ depending on the geographic area and population represented. Generally, the smaller the direct constituency, the greater the importance given to grassroots activities such as addressing service clubs.

One of the jobs of a politician is to bring home benefits, such as state and federal funding for capital improvement projects. A member's focus on these activities could make it difficult to get other issues onto his or her agenda, or a policy proposal that makes sense on a national level may conflict with the interests of constituents back home.

Fund-Raising

The second major concern for politicians seeking reelection is campaign fund-raising. Like constituent relations, fund-raising can vary in scope depending on the office. The bigger the constituency, the more it costs to campaign. Discussions of the role of money in U.S. politics tend to focus on campaign fund-raising and often include the axiom that money buys access, not votes. That can be said of various gifts and junkets, as well as cash campaign contributions. Jesse "Big Daddy" Unruh, the powerful speaker of the California Assembly in the 1960s, once said of lobbyists, "If you can't take their money, drink their booze, eat their food, screw their women, and still look them in the eye and vote against them, you don't belong here."

There are, however, numerous examples of contributions and gifts, as well as outright bribes, influencing government decisions, or at least creating the appearance of a conflict of interest. North Carolina's former Speaker of the House, Jim Black, helped push through a bill in 2005 requiring comprehensive eye exams for children during the first 6 months of kindergarten. These exams would earn optometrists \$75–100 per exam. Black was an optometrist and accepted \$59,750 in campaign money from his professional peers that year.

Party Agendas

Candidates' ability to bring their own money to the table or to raise funds independent of the political party they represent has weakened party control in U.S. politics. Oberlander (2003) argued that the weakness of political parties in the United States is one of the political obstacles to health reform in this country. Of late, though, party politics has become more polarized and policy positions have more closely followed ideological lines. Moderate Republicans who have broken ranks to work with Democrats on health care and other issues have been defeated in their primaries by more conservative candidates. Historically, members have crossed the aisle to form bipartisan coalitions in support of health legislation, but the votes on the ACA, and on efforts to roll it back, followed party lines with few exceptions.

The Economy and the Budget

The state of the economy and the revenues flowing into government coffers can determine a politician's willingness to create new programs or retain or expand existing ones. When Washington's Governor Gregoire took office in 2005, one of

her first acts was to roll back healthcare-related changes instituted under her predecessor that had reduced the number of children enrolled in government health insurance programs. After the September 11, 2001 attacks and the economic downturn that followed, the previous administration and the legislature had needed to cut \$1–\$2 billion a year from the state budget for a few years running. By 2005, the state economy had begun to rebound, which left Gregoire with a projected surplus and allowed her to allocate an additional \$140 million to provide coverage to an estimated 70,000 more children.

Political Trading

Lawmakers will not infrequently leverage their votes, pledging to vote a certain way in return for obtaining something else they are interested in. Leadership and committee chairs can also hold a bill hostage until some kind of quid pro quo is negotiated. In Washington State, one Senate committee chair refused to release a bill to reduce commercial sources of environmental mercury, including methylmercury, which is found in fish, unless an agency withdrew its opposition to her bill banning vaccines for children and pregnant women that have more than trace amounts of thimerosal, a preservative that contains ethylmercury. Methylmercury is a persistent toxin that accumulates in the environment and is extremely harmful to humans, whereas ethylmercury appears to be metabolized quickly and cleared by the body. Yet the committee chair saw the two mercury bills as linked. Public health advocates had to weigh their concerns that the thimerosal bill, which was largely symbolic, would reinforce unsubstantiated fears about vaccine safety against the real threat to human health posed by environmental releases of elemental mercury.

Personal Issues

In the fields of public administration and public policy, rational decision-making is held out as the ideal. Decision-makers are expected to weigh the costs and benefits of various policy options and make reasoned choices that maximize the gains to themselves, their constituents, or society as a whole, but personal preferences, relationships, irrational behaviors, and unanticipated events often influence political feasibility. In the case of the thimerosal legislation, for example, the committee chair who sponsored the bill was the grandmother of a child with autism, and family members believed that the autism was linked to an immunization the mother received when pregnant.

Unexpected Events

An unanticipated event that many believe influenced the outcome of health legislation occurred in Washington State during the 2004 legislative session. Senator Rosa Franklin, an African American Democrat, sponsored a bill to establish the Joint Select Committee on Health Disparities. Many observers doubted the resolution would pass because of a lack of enthusiasm among Republicans who held a majority in the state senate. As part of an unrelated discussion of health policy, however, the chair of the senate health committee called the minority leader of the house health committee a “nigger in the woodpile.” Both men were white Republicans. Franklin defended the senate committee chair (although not his choice of words) and tearfully

accepted his apology on the floor of the state senate. There was never any public discussion of a deal, but few political observers were surprised when Franklin's resolution passed unanimously.

► Nongovernmental Actors

The Public

Politicians are constantly attempting to read the winds of public sentiment as they chart their policy courses. This is commendable up to a point. In a representative democracy such as the United States, elected officials presumably have some autonomy to pursue the policies and advance the values that got them elected without polling the electorate before every vote, but there is also an expectation that their actions will reflect the views of the majority of their constituents. At a baser level, focusing on public opinion can be about winning the next election, and subservience to public opinion polls can work against elected officials providing leadership, consistently pursuing long-term strategies, or taking risks. Politicians often focus on determining and responding to the opinions of likely voters more than those of their entire constituency. This is relevant in the context of healthcare reform, where a significant number of the uninsured are socially disenfranchised and generally not a powerful voting bloc. Additionally, older people are more likely to vote and are less likely to support significant changes to the healthcare system (Starr, 2011).

Public opinion polls consistently show that at most points in time a majority of people in the United States have favored healthcare reform. Historically, however, public support for major reforms has been "soft." The public can easily be talked out of supporting a reform through the use of carefully crafted messages. In 1993 and 1994, polls repeatedly showed that Americans supported universal access to health care. Public support continued, and even seemed to be growing as the Clinton plan moved forward. That trend began to reverse itself in late 1993, about the same time that the Health Insurance Association of America began running its famous Harry and Louise advertising campaign (Goldstein, Goldstein, Swan, & Clemeña, 2001). These ads were targeted at informed members of the public who could serve as opinion leaders (Goddard, 1998). By April 1994, only 43% of the country supported the proposal (Blendon, Brodie, & Benson, 1995).

One useful construct for understanding the vulnerability of public support for health reform is the notion of frames. The FrameWorks Institute defines a *frame* as, "The way a story is told—its selective use of particular values, symbols, metaphors, and messengers—which, in turn, triggers the shared and durable cultural models that people use to make sense of their world" (Gilliam, 2005). Essentially, people use predictable story elements to help them sort information; they respond to metaphors, plots, values, and other familiar aspects of the story. When communication is inadequate, says Gilliam, people rely on the pictures and stories already in their heads, but effective communication can allow them to see things from new and different perspectives. Healthcare reform efforts, if not understood, are subject to attack by arguing within certain frames and appealing to certain closely held values. For example, many Americans are skeptical about government and opposed to too much direct government involvement in their lives. Designers of the Harry and Louise ads conducted research that told them the public did not

BOX 9-3 Sample Frame for Promoting a Health Coverage Infrastructure

In the last 50 years, the United States has built a series of modern networks that are essential to our economy and our quality of life—our power grid, phone systems, water systems, interstate highways, and the Internet; however, with health coverage we are stuck in the 1940s because we never built a modern health coverage infrastructure. Instead, we still have job-based insurance, which has become an increasingly hit-or-miss, inefficient, and unreliable approach. What we have is the equivalent of scattered wells, individual generators, and county roads but no health coverage infrastructure we can rely on, no system for making sure that people have health coverage.

Modified from Gilliam, F. D. (2005, December 6). *Framing health care reform for public understanding and support*. Presentation delivered at Health Legislative Conference, Seattle, WA. © 2006 FrameWorks Institute; and Making the Public Case for Improving Health Care: A Frame Works Message Memo. 2009. Washington, DC: FrameWorks Institute

want a massive government program and tailored their campaign accordingly (Goddard, 1998).

According to Gilliam (2005), frames, or value-based messages, that move people toward supporting healthcare reform (i.e., increase political feasibility) include interdependence, the need for practical management, and the importance of prevention. They suggest that advocates of health reform promote the need for a health coverage infrastructure akin to a utility or transportation infrastructure.

BOX 9-3 provides an example of the kind of language that might have framed the issue effectively.

Interest Groups

When individuals and corporations with common social and political goals form alliances and influence policy in support of those goals, they comprise an interest group. Weissert and Weissert (2002) argued, “Next to Congress, interest groups may well be the most important actors in health policy” (p. 117). The use of the term *the third house* to refer collectively to the lobbyists that represent interest groups before the legislative branch is no misnomer.

Interest groups do more than just attempt to sway votes; they can be critical partners with government actors in policy development from inception to implementation. They generate proposals, provide expertise that can improve others’ proposals, offer channels for testing the feasibility of proposals with their members, broker deals, draft language, and help mobilize public opinion. After a policy is adopted, their support, apathy, or opposition can determine whether it succeeds or fails. Interest groups also lobby the executive branch and may resort to lawsuits to assert their policies.

Hundreds of interest groups are involved in healthcare issues. A majority represent occupations or health-related companies. Major, longstanding examples of these, respectively, would include the American Medical Association (AMA) representing physicians and the American Hospital Association (AHA) representing healthcare facilities. The last few decades have witnessed a splintering and proliferation of these groups. Oliver (2006) stated that the proliferation of groups “has led scholars to conclude that political influence is more generally dispersed across loosely organized ‘issue networks’ or ‘policy communities’” (p. 209).

Interest groups have multiple tools at their disposal for influencing policy, including the following:

- Political action committees: Fund-raising committees that can make contributions to candidates, parties, or issue campaigns
- Direct democracy: Sponsoring voter initiatives and referendums, hiring signature gatherers, and funding campaigns
- Grassroots lobbying: Organizing the general public (or voters) to appeal directly to lawmakers or to make their positions known at the ballot box
- Cross lobbying: Convincing other groups to support a particular policy or change an existing position that runs counter to that policy
- Coalition building: Increasing political leverage by forming coalitions to work on a specific issue or policy, often for a limited time
- Research: Conducting studies, compiling data, and producing reports that influence policy choices, often focusing on the evidence basis and economic and technological feasibility of policy options
- Framing: Using public opinion research, message development, and other communications techniques to “tell the story” about a policy in a way that changes the public discourse

The Media

The news media play a critical role in shaping policy debates and outcomes in this country. Media outlets exert their influence in three principle ways:

- Setting the frame: The news media tell stories to interest their readers and viewers, and the way those stories are organized helps establish the frames people use to interpret both the coverage and the underlying social issue. Media provide the context people use to help form opinions on an issue.
- Sorting and delivering data and information: Decision-makers rely on data and information; however, their time is limited, and we live in an era of information overload. They are not able to sort and process all the relevant information that emerges in a constant stream of studies, reports, and journal articles, and thus they tend to rely on media to sort information and identify what is relevant.
- Setting the agenda: Shortly before the Clinton administration’s took up health reform, *New York Times* articles and an editorial promoting a managed competition model developed by the Jackson Hole Group—a gathering of health policy experts, government officials, and reform-minded executives from healthcare organizations and large insurance companies that met in Jackson Hole, WY, in February 1990—helped drive the public discussion toward managed competition, which became the cornerstone of Clinton’s Health Security Plan legislation.

Here is how Goddard (1998) explained the role the media plays in U.S. public opinion formation and agenda setting:

The media communicates with opinion leaders ... about an issue. “Hey! Here’s something that’s important” ... What’s really critical is that the opinions that “informed” Americans form after exposure to them are then

communicated to policy makers. And all of a sudden the policy makers say, “Wait a minute! All these people out there who I count on for support are concerned about this issue.”

(p. 1)

Scientists and Other Experts

The notion of rationality in decision-making presupposes sufficient information and resources to understand fully the implications of each policy alternative. Although this rarely occurs in practice, most policy makers try to understand what the research tells them about the value of various policy proposals. This gives researchers who generate data and informed analysis considerable influence over political feasibility.

Research and data can come from a variety of sources. Generally, the most credible sources are academic researchers who publish their findings in peer-reviewed journals. Policy makers also turn to raw data from primary sources (especially from government agencies) and to reports and studies from think tanks, advocacy groups, pharmaceutical firms, university centers with narrowly defined missions, and a variety of other sources whose work may reflect an ideological or financial bias.

The push toward evidence-based medicine has heightened the focus on making policy based on data. It is becoming increasingly common for state governments to fund research centers that evaluate the comparative evidence basis for certain drugs and other medical technologies. Oliver and Singer (2006) found that research conducted by the California Health Benefits Review Program, which relies on university-based researchers to analyze the evidence basis for requiring that health insurance plans cover certain types of care, played a significant role in shaping legislative decision-making around mandated benefits in that state.

► Methods for Analyzing Political Feasibility

Political feasibility prediction tends to rely on seeking out expert informants and asking them how key players are likely to respond to the policy or policy alternatives.

Identifying the actors by mapping the authorizing environment, as we did in the example earlier in this chapter, is often a key early step. A subsequent step would likely involve determining the amount of support or opposition each actor is likely to generate, either by soliciting information from the actors themselves or from some other set of experts. It is also possible to solicit experts’ opinions on how much influence each actor is likely to have over the outcome. It is important to identify other inputs likely to influence the outcome of a specific policy debate. This would involve addressing a variety of questions, such as:

- Is the economy expanding or contracting? What is that likely to mean for government revenues? How expensive is the policy alternative?
- What other focusing events are likely to demand the attention of the public and decision-makers?

It may be useful to tease apart the different factors that contribute to political feasibility. Kingdon (1984) suggested that there are three streams to the policy process, and that change will be incremental unless these three streams converge

favorably, typically as a result of a crisis that redefines a problem or a change in who controls the government that creates a window of opportunity. The three streams are:

- The problem stream: People need to agree there is a problem, their understanding needs to be supported by basic science, and they need to share a common definition of what the problem is.
- The policy stream: Credible scientific, technological, and/or policy solutions must be available.
- The political stream: The problem needs to be on the agenda, and there needs to be political will to address it.

Another useful concept for thinking about political feasibility is force field analysis, a decision-making tool developed by psychologist Kurt Lewin that provides a way of examining the forces working for or against a particular decision. Lewin (1951) argued that issues were held in balance by driving forces (those promoting change) and restraining forces (those favoring the status quo). These included both strong and weak forces, as represented in the following figure (**FIGURE 9-2**). As a decision-making tool, force field analysis is often represented with a chart. In a box in the middle is the decision, or in our case, the policy. On the left side is a set of arrows representing the forces for change, and on the right side is a set of arrows representing forces against change. In political feasibility analysis, those arrows would represent actors and inputs.

After identifying the forces bearing on the policy, the analyst can assign them a score from 1 (weak) to 5 (strong) and tally to create a combined score for the forces for and against.

In their analysis of the political feasibility of mandated health insurance benefits, Oliver and Singer (2006) used a framework that examines whether the costs and the benefits are concentrated among a few or diffused among many (see **FIGURE 9-3**).

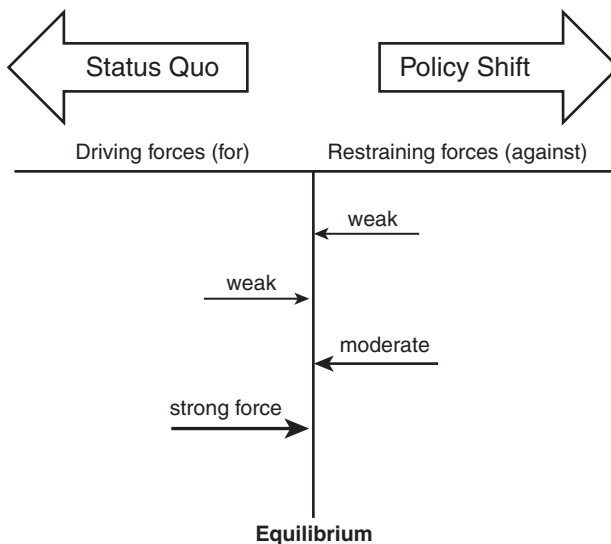


FIGURE 9-2 Force field analysis diagram.

Modified from Lewin, K. (1951). *Field theory in social science*. New York: Harper and Row.

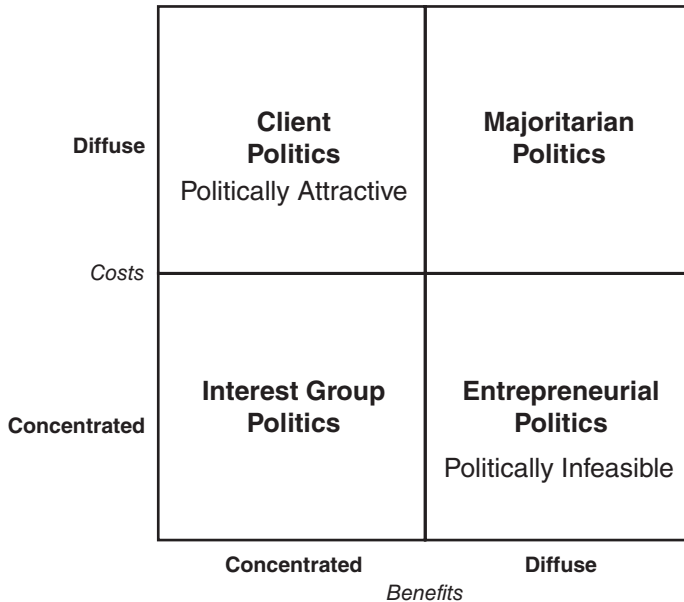


FIGURE 9-3 Framework for analysis of policy design and political feasibility.

Reproduced from Oliver, T. R. (2006). The politics of public health policy. *Annual Review of Public Health*, 27: 195–233. Reproduced with permission of ANNUAL REVIEWS in the format Republish in a book via Copyright Clearance Center.

The most politically feasible solutions are those involving “client politics” where there are strong advocates who would reap the concentrated benefits but little opposition because the costs are minor or are spread broadly among many interests or individuals. The least politically feasible are those involving “entrepreneurial politics” where the benefits are shared so broadly that there are no impassioned, self-interested advocates and yet there is great opposition from interests that would bear the concentrated costs. Incrementalism, according to Oliver (2006), is a natural byproduct of “interest group politics” that occurs when both benefits and costs are concentrated. “With clear winners and losers, the level of conflict is high and the outcome of any single proposal is highly unpredictable” (p. 211).

Dror (1969) proposed a framework for political feasibility analysis that relies on the Delphi method. He recommended that prediction panels consist of senior executives, politicians, and political observers: “While politicians are the ideal panel members for political feasibility prediction studies, this itself may be politically and personally nonfeasible, especially if such studies become widespread. Therefore, main reliance may have to be put on politics-observing persons, such as political aides, political correspondents, political science scholars, senior civil servants, etc.” (pp. 12–13)

Dror suggested that panelists be asked to complete three different types of questionnaires. One form would ask the panelists to rate the probability that specific policy alternatives might be adopted across different time frames. Another would ask whether various alternatives were feasible within a specific time frame and solicit information about barriers to adoption and what conditions would have to change to make adoption possible. These two forms are displayed in **TABLES 9-1** and **9-2**. The third one would solicit opinions about the

TABLE 9-1 Scheme 1: Direct Political Feasibility Estimation

	Political Feasibility Estimate			
Policy Alternatives	Next X Years	Next Y Years		Next N Years
Alternative 1				
Alternative 2				
....				
Alternative N				

Each cell to be filled out with a probability, or probability distribution, or alternative probabilities with explicit assumptions—depending on the capacities of the predictor.

Reproduced from Dror (1969). *The prediction of political feasibility*. Santa Monica, CA: The Rand Corporation. Permission conveyed through Copyright Clearance Center, Inc

TABLE 9-2 Scheme 2: Political Feasibility Conditions

Policy Alternatives	Is politically feasible during next X years?	If not, what changes in conditions are required to make it politically feasible?
Alternative 1		
Alternative 2		
....		
Alternative N		

Depending on interest and on capacities of the predictor, the scheme can deal with various time spans, different feasibility probabilities and probability changes, and various combinations of conditions and assumptions.

Reproduced from Dror (1969). *The prediction of political feasibility*. Santa Monica, CA: The Rand Corporation. Permission conveyed through Copyright Clearance Center, Inc

various actors—how much leverage they have, what their intentions are, and what actions they are likely to take alone or in combination with other actors. He also recommended that the informants be divided into three panels that get the surveys at different times. This would allow the coordinator to test for consistency. Researchers and analysts over the years have proposed or employed numerous variations on both the Delphi technique in general and the framework proposed by Dror. Turoff (1970), for example, has proposed a variant

technique called the Policy Delphi that can be used to identify the universe of policy options, estimate the impact and consequences of those options, and assess their acceptability.

In the past couple of decades, stakeholder analysis has emerged as the most common methodology for analyzing political feasibility. It has been employed across many disciplines, including political philosophy (Gilbert & Lawford-Smith, 2012), public health (AusAID, UNAIDS, & National AIDS Council, Papua New Guinea, 2004), business management, development, environmental and natural resource management, and healthcare finance (Brugha & Varvasovszky, 2000). Spreadsheets, manuals, software systems, and consulting services can be purchased online to help people conduct such analyses, and several governments have published guides.

There are many ways to structure stakeholder analysis, but the process generally involves three steps: (1) stakeholder identification; (2) stakeholder mapping or prioritization; and (3) gathering information from stakeholders. Each of these we have already discussed, just not within the context of a standard stakeholder analysis framework. Though anyone conducting a stakeholder analysis will want to start with stakeholder identification, these steps are iterative and not necessarily sequential. Key informant interviews, say, might identify stakeholders that researchers would not have included otherwise, or they might cast new light on the relative power of organizations or interest groups.

Stakeholder identification is basically the same process as developing a map of an authorizing environment. This can be the product of brainstorming or asking experts and readily identified stakeholders to identify the players.

Stakeholder mapping or stakeholder prioritization typically involves placing the identified stakeholders on a grid, such as the one developed for expert interviews by Oliver (2006), or the “stakeholder matrix” proposed by the Victoria state government in Australia (**TABLE 9-3**) to identify how influential they will be and how much effort should be put into encouraging their support.

A stakeholder analysis for a project to improve universal healthcare coverage in South America and Tanzania expanded on this concept by producing a “force field analysis map” that was similar to the one depicted earlier in this chapter, but it included more categories for measuring both the political power and the degree of support of each stakeholder (Gilson et al., 2012, p. 1174). This is comparable to determining whether the arrows in a force field diagram should be represented as strong, weak, or moderate.

The third step, gathering information from stakeholders, can take several forms, including use of a Delphi technique or key informant interviews. A political feasibility analysis for introducing universal health care in Afghanistan (Zeng et al., 2017), for instance, identified 11 stakeholder groups in collaboration with the Health Economics and Financing Directorate (HEFD) of the Ministry of Public Health. The researchers then identified individuals within those groups with special knowledge or authority to serve as key informants. The key informants included officials from government ministries and parliament, executives at two private insurance companies, and representatives of international donors and organizations such as the World Bank and the World Health Organization. They also conducted focus groups comprising representatives of community organizations, public and private hospital directors, managers of the country’s Basic Package of Health Services, and HEFD

TABLE 9-3 Stakeholder Analysis (Stakeholder Matrix)

Description

Stakeholder analysis is an essential part of developing a useful Engagement Plan. A common method of stakeholder analysis is a Stakeholder Matrix. This is where stakeholders are plotted against two variables. These variables might be plotting the level of 'stake' in the outcomes of the project against 'resources' of the stakeholder. Another is the 'importance' of the stakeholder against the 'influence' of the stakeholder. The concept is the same, though the emphasis is slightly different.

		Importance of Stakeholder			
		Unknown	Little/No importance	Some importance	Significant importance
Influence of Stakeholder	Significant influence	C		A	
	Somewhat influential				
	Little/No influence	D		B	
	Unknown				

Boxes A, B, and C are the key stakeholders of the project. The implications of each box is summarized as follows:

Box A

These are stakeholders appearing to have a high degree of influence on the project, who are also of high importance for its success. This implies that the implementing organization will need to construct good working relationships with these stakeholders, to ensure an effective coalition of support for the project. Examples might be the senior officials and politicians or trade unions.

Box B

These are stakeholders of high importance to the success of the project, but with low influence. This implies that they will require special initiatives if their interests are to be protected. An example may be traditionally marginalized groups (e.g., indigenous people, youth, seniors), who might be beneficiaries of a new service, but who have little ‘voice’ in its development.

Box C

These are stakeholders with high influence, who can therefore affect the project outcomes, but whose interests are not necessarily aligned with the overall goals of the project. They might be financial administrators, who can exercise considerable discretion over funding disbursements. This conclusion implies that these stakeholders may be a source of significant risk, and they will need careful monitoring and management.

Box D

The stakeholders in this box, with low influence on, or importance to the project objectives, may require limited monitoring or evaluation, but are of low priority.

Method

1. Make a list of all stakeholders.
2. Write the name of each stakeholder on a sticky note or index card.
3. Rank the stakeholders on a scale of one to five, according to one of the criteria on the matrix, such as 'interest in the project outcomes' or 'interest in the subject'.
4. Keeping this ranking for one of the criteria, plot the stakeholders against the other criteria of the matrix. This is where using sticky notes or removable cards are useful.
5. Ask the following questions:
 - Are there any surprises?
 - Which stakeholders do we have the most/least contact with?
 - Which stakeholders might we have to make special efforts to ensure engagement?

Reproduced from State of Victoria (2013). *Effective Engagement: Stakeholder Analysis (Stakeholder Matrix)*. Victoria, Australia: State of Victoria. Last updated on April 26, 2013. Retrieved from www.dse.vic.gov.au/effective-engagement/toolkit/tool-stakeholder-analysis-stakeholder-matrix. Courtesy of Department of Environment and Primary Industries, State Government of Victoria

staff. The researchers conducted a total of 22 key informant interviews and focus groups, each lasting about an hour. The goal was to gather opinions about seven parameters:

1. The perceived need for insurance
2. The level of political commitment
3. The regulatory and legal environment
4. Quality of care
5. Public awareness
6. Ability to pay
7. Technical and operational capacity

The study identified barriers that will have to be addressed to pave the way for a nationwide system of health insurance.

► Critiques of Political Feasibility Analysis

The highly qualitative nature of such techniques makes their results subject to human error and bias. Dror (1988) cautioned that “experts may easily be emotionally biased when hot political issues are touched upon and may be over-conservative in underestimating possibilities of changes and jumps in political situations” (p. 273).

Oberlander (2003) pointed out that the headlines about health reform in 2003 were reminiscent of those that appeared in 1993 and 1973. The disturbing statistics and the comprehensive reform proposals designed to address them were eerily similar. The problem, he argued, is not a lack of good ideas but a consistent failure to realistically assess political feasibility. People confuse feasibility with desirability and tend to argue that their favorite solution is the most feasible, he stated, or they can be predisposed to accept a flawed feasibility analysis that supports their position. He added that a “beltway mentality” leads people to assume that whatever appears in the *Washington Post* or has caught the imagination of Washington insiders is most politically feasible.

Both Dror and Oberlander raised concerns about the use of political feasibility predictions. Dror pointed out that such predictions assume that no unexpected occurrences would take place—and they often do. Bill Clinton ran for president on a campaign built around a central premise: “It’s the economy, stupid.” The economic downturn that hit at the end of George H. W. Bush’s term not only doomed his reelection bid, but it also produced a great deal of public anxiety about people losing their health insurance. That was in 1992, however. By the time Congress was seriously considering the Clinton Health Security Plan in late 1993 and early 1994, the economy had greatly improved, and the public was less inclined to see health insurance as an immediate crisis.

It should be noted that frameworks like stakeholder analysis can lead to an almost exclusive focus on the actors, which may contribute to inadequate analysis of inputs such as economic changes. To effectively analyze political feasibility, it is important not to be satisfied solely with identifying and consulting with stakeholders. It may also be important to consult with experts on various inputs—economists, historians, futurists, political pollsters, organizational psychologists, and other experts.

Something akin to the Heisenberg uncertainty principle of quantum physics also operates in the realm of feasibility prediction. Efforts to measure something can influence the thing being measured and change the measured result. Feasibility analyses can actively influence the outcome of a political debate—they can be merely self-fulfilling or conflicting analyses can cancel each other out. Oberlander (2003) stated that “feasibility analysis does not merely play a passive role in the political process as an objective judge of what will or can happen. Rather, policy analysis itself can influence the course of events and is often deployed as a political weapon” (p. W3–393). As an example, he pointed to the number of policy analysts that opponents of the Clinton health plan brought forward to explain why the plan was irredeemably flawed.

Dror (1969) cautioned that “care must be taken to avoid a mistake widespread in practice and sometimes supported by theory, that ‘feasibility’ becomes a dominant criterion of a preferable alternative, in the sense of ‘the more feasible the better’”

(p. 2). He suggested that political feasibility analyses should be seen as challenges, not constraints. He noted that:

...every political feasibility prediction tends to ignore the capacities of human devotion and human efforts to overcome apparently insurmountable barriers and to achieve not only the improbable but the apparently impossible. A good policy may be worth fighting for, even if its political feasibility seems to be nill [sic], as devotion and skillful efforts may well overcome political barriers and snatch victory out of the mouth of political infeasibility.

(pp. 18–19)

► Where in the World?

Software packages and various tools conducting various iterations of stakeholder analysis are readily available. Simply search the Internet. Consultants are also easy to locate and the companies offering software often are willing to do the work for you. International development organizations will fund stakeholder analyses. For example, the Afghanistan example previously cited was funded by the by U.S. Agency for International Development. Such studies have been employed to address the likely success of introducing various health insurance models in less-developed countries. There are many studies like the Afghan studies. The extent to which countries employ such studies, and how explicitly, varies. Australia, for example, offers clear guidance, while in the United States, community meetings and public comment solicitation may provide similar inputs, but in a less structured, vigorous, and acknowledged manner. To follow health-related feasibility studies, look to journals such as *Health Policy and Planning* and *Health Policy*, and sometimes the *American Journal of Public Health*. Many articles about stakeholder analysis can be found in fields outside of health care—management and public administration are prime examples, along with natural resources. When searching for political feasibility studies, use related terms such as *stakeholder management*, *social network analysis*, *social mapping*, *stakeholder analysis*, *key informant interviews*, and *political landscape*.

► Conclusion

Policy analysts may tend to avoid political feasibility assessments because they are highly qualitative in nature and subject to a great deal of uncertainty. We have discussed how medicine has become increasingly industrialized, but politics will likely remain more of an art than an industry. To address political feasibility at all, an analyst must understand the authorizing environment, the key political actors, the various inputs into the political process, and how they are likely to interact. Techniques such as force field analysis, independent looks at the three streams of policy that can create windows of opportunity, and assessments of whether the benefits or costs of a policy are concentrated or diffuse can be useful. Systematic approaches to assessing political feasibility typically rely on surveys of stakeholders but can also include

surveys of experts who are not direct stakeholders. The Delphi technique provides one structure for such inquiries. It is important, however, to select informants with care. This includes reaching out to a broad, diverse range of stakeholders, or in the case of academic or technical experts, being aware of the experts' biases and limitations. Finally, political feasibility analyses can become self-fulfilling prophecies if not used wisely. Analysts should be wary of abandoning good policy ideas because they seem unfeasible or settling on a poor idea because it would be an easy sell.

CASE 9: Green Mountain Care

Peter Shumlin assumed office as Vermont's governor in January 2011. Like all four of his Democratic primary contenders, he had run as a supporter of a single-payer health plan. The outlook was good, given the state's small, homogenous, and largely liberal population; its history of leadership in health policy; the governor's electoral mandate; the recent passage of the ACA at the national level; and the policy development foundation that had already been laid in the state.

The previous year, the legislature had passed Act 128, "An act relating to health care financing and universal access to health care in Vermont." Republican Governor Jim Douglas, who had announced he would not seek reelection, allowed it to become law without his signature on May 27, 2010.

The goals of Act 128 included: (1) providing universal health insurance to all Vermonters under a plan that would give them equal access to a standard benefits package; (2) controlling medical costs; and (3) creating a healthcare system that emphasized primary care and focused on prevention and wellness. The Act also established a commission and directed the commission to hire one or more consultants to proposal three design options—a single-payer system, a public option that would give state residents an alternative to private insurance plans, and "a third and any additional option [that] shall be designed by the consultant."

The Consultants' Report

On February 17, 2011, roughly a month after Shumlin's inauguration, the state received a 203-page report—"Health System Reform Design: Achieving Affordable Universal Health Care in Vermont"—that provided three options as required by Act 128 (Hsiao, Kappel, & Gruber, 2011).

The consultants who authored the report were William C. Hsiao and Jonathan Gruber, economics professors at Harvard and MIT, respectively, and Steven Kappel, founder and principal consultant at Policy Integrity, LLC. Hsiao had experience with the design and implementation of single-payer systems in other countries such as Taiwan. Gruber was an adviser to Rep. Nancy Pelosi and consultant to the White House during the development of the ACA, and Policy Integrity develops and evaluates policy alternatives in areas that include health care.

The report contained several pages devoted to a political feasibility study that the authors referred to as both a stakeholder analysis and a "political landscape analysis" (p. 17). (Researchers employing the term *political landscape* typically focus on identifying which actors have what levels of power and the relationships between them.) The first step was a literature review to understand the state's history around health policy and identify its major health-related institutions. The second phase primarily consisted of key informant interviews.

In a series of 64 confidential interviews, researchers spoke with almost 120 people. Interview subjects included legislators (15), members of the executive branch (6), hospital administrators (31), healthcare providers (23), representatives of large businesses (10) and small and medium businesses (13), union officials (11), health reform advocates (10), advocates from other issue areas (7), and insurance company executives (2). Most interviews were conducted in person with a pair of interviewers. The researchers also considered information gathered in “less formal stakeholder engagements.”

“Following our interviews, we categorized our finding according to key themes, recorded primary concerns across stakeholder groups, and compared current findings to those from our historical analysis” (p. 18).

They then determined the key interests and concerns of different sectors and identified policy constraints—design options that would be politically infeasible because of strong opposition from highly interested and engaged groups with significant economic and political power. Hospitals would vehemently oppose reductions in their reimbursement rates, for example. They wanted predictable and sustainable funding, but were not attached to the source of the funding. Businesses were wary of direct government control of the system and favored a third-party administrator protected from political influence. There was widespread resistance to reducing benefits.

The analysis did not simply test the feasibility of the various options; the results were used to shape the single-payer plan recommended in the report.

In contrast to previous reports on state-based single-payer plans, in our design process we explicitly considered the political landscape and the fiscal, legal, and institutional constraints on the reform.... We therefore proposed a public-private single-payer system that was financed through payroll taxes and governed by an independent board, and that offered a generous benefit package—while at the same time transforming the payment system and reforming the medical malpractice system.

Hsiao et al. (2011, p. 1233)

Vermont Enacts Single-Payer

In response, the legislature passed House Bill 202 (H 202), which established the basic design of Green Mountain Care (GMC), a system of universal coverage that borrowed heavily from the recommendations of the report. Governor Shumlin signed the bill into law on May 26, 2011, making Vermont the first state to enact a single-payer system.

Hsiao, Kappel, and Gruber estimated the single-payer system would reduce overall costs of the healthcare system by 25.3% through the following reforms:

1. Moving to a single-payer system would reduce administrative expenses by 7.3%.
2. A centralized claims database would make it possible to reduce fraud and abuse, saving 5%.
3. Integrated delivery systems (accountable care organizations) and payment reform (capitation payments adjusted for acuity and pay-for-performance) would save 10%.
4. A no-fault medical malpractice system with award limits would save 2%, largely by eliminating waste caused by providers practicing defensive medicine.
5. Insulating the system from politics and creating competition for claims administration would save another 1%.

These savings, according to projections, would reduce system costs in 2015 by \$580 million, combined with another \$56 million reduction if the state created a uniform payment schedule of no less than 105%–115% of Medicare rates. These savings would more than offset the additional costs: \$220 million to cover the uninsured, \$32 million to increase benefits for the underinsured, \$62 million to support community hospitals and primary care, and \$124 million to provide vision and dental benefits. Net savings to the system in 2015 would be \$198 million, and business and household spending on health care could be reduced accordingly.

H 202 specified an actuarial value for the plan of at least 87%, meaning about 13% of healthcare costs would be paid for out of pocket. This was consistent with the estimated actuarial value of the average private insurance plan in the state.

Taxes would be increased to pay for the system, but those increases would be more than offset by eliminating the need to purchase health insurance. The consultants estimated savings in 2016 of \$370 per household, despite a 3.1% increase in personal state income tax and a 9.4% increase in taxes for employers.

New Numbers Emerge

The bill did not specify a plan for financing the new system, however, requiring another round of studies instead.

A January 2013 study by the University of Massachusetts Medical School Center for Health Law and Economics and Wakely Consulting estimated a much lower level of savings—1.5% of system costs over 3 years (UMass Medical School & Wakely Consulting Group, 2013). It estimated that the state would need to raise an additional \$1.61 billion to fund the program.

In November 2013, a report by Avalere Health, which had been hired by Vermont Partners for Health Care Reforms, a group of providers, insurers, and employers, issued a report that challenged elements of the UMass study. For example, UMass had estimated that providers' reimbursement would be at 105% of Medicare on average and current reimbursement in the state was 107%. Avalere estimated the existing reimbursement rate at 122%. Green Mountain Care would therefore take a huge bite out of provider and hospital revenues. Avalere also challenged the assumption that there would be significant administrative savings. Avalere concluded the state would have to collect \$1.9–\$2.2 billion in additional annual revenue, a 20%–35% increase over the original \$1.61 billion estimate. Avalere also conducted an interview-based "stakeholder impact analysis."

On February 6, 2014, the legislature and the administration reached a consensus that the state would have to come up with between \$1.766 billion and \$2.175 billion in revenues to finance GMC.

According to McDonough (2015), some of the differences between the 2011 report's figures and the projections in the 2013 and 2014 economic analyses had to do with lower reimbursements from Medicaid to the state under the ACA. But policy decisions also added to the costs—a benefit structure that resulted in an actuarial value of 94% (higher than the platinum plan under the ACA), coverage for nonresidents working in the state, and the elimination of taxes for healthcare providers.

On December 7, 2014 Shumlin pulled the plug on GMC, declaring, "I have learned that the limitations of state-based financing, the limitations of federal law, the limitations of our tax capacity, and the sensitivity of our economy make that unwise and untenable at this time.... The risk of economic shock is too high" (McDonough, 2015, p. 1584).

In a presentation given that day, Michael Costa, deputy director for healthcare reform, said that going forward with GMC would have required payroll taxes of 11.5% for businesses and a sliding scale public premium of up to 9.5% for individuals—a

far cry from the original estimates of 3.1% increase in personal state income tax and a 9.4 increase in taxes for employers. As reasons for the higher estimates, Costa cited a major reduction in estimated state and federal revenues from other sources, expensive policy choices, slow economic recovery, and the high cost of transitioning small businesses onto GMC. He also said it wasn't practical to achieve the estimated administrative savings. (Vermont Health Reform, 2014)

Explanations for GMC'S Demise

McDonough (2015), a colleague of Hsiao's at Harvard who advocated for a single-payer system as a Massachusetts legislator in the 1990s, points to several political factors that doomed the plan. These include the erosion of public support, the absence of a clear mandate, weak support in the legislature, and concerns about public response to raising the state budget by 45%. Shumlin won reelection by only 1% in 2014, running against a Republican opposed to single-payer. The narrow margin sent the race to the legislature, which awarded Shumlin a second term. An April 2014 poll showed the populace effectively split on single-payer. And while all three studies projected net savings to taxpayers freed from the burden of health insurance premiums, McDonough points out that people often pay little attention to health insurance deductions on their paychecks, but could hardly ignore—and would likely erupt over—a massive tax increase.

Avik Roy (2014), an editor at *Forbes* magazine who advocates for market-based approaches to healthcare financing, has presented six reasons why the plan failed:

1. Platinum-plated coverage (a 94% actuarial value)
2. Adoption without a financial plan in place
3. A massive tax increase that he estimates would be 160%
4. Opposition from hospitals because of reduced reimbursement rates and from insurance companies that would be eliminated
5. Reduced estimates of state revenues from taxes because of the recession, other state budget restraints, and lower projected payments from the federal government under the Medicaid waiver
6. The inability to create a true single-payer system because some people would be covered by federal plans such as Medicare or by out-of-state insurers

► Discussion Questions

1. In their report, Hsiao, Kappel, and Gruber identify some limitations of their stakeholder analysis. What might these be and what changes would you suggest to their methodology?
2. Apply Dror's four variables for thinking about political feasibility to this case, and describe how they shifted over time.
3. What would a map of Governor Shumlin's authorizing environment for single-payer health insurance look like?
4. Describe the role of political capital in this instance.
5. What would a force field analysis for Green Mountain Care look like?
6. Roy suggests that passage of the ACA may doom states' efforts to enact single-payer. Based on the political landscape analysis and plan design in Vermont, why might that be the case?



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CHAPTER 10

The Policy Analysis Process—Evaluation of Economic Viability

Policy analysis involves the allocation of scarce resources. This chapter focuses on economic and financial aspects of the allocation process. Both affect the economic viability of a proposed policy alternative, which likely turns on the following questions:

- How much will it cost?
- What value will we be getting for the money?
- How does that value compare with other alternatives under consideration?
- If it is something we want to do, how will we pay for it?

To address these questions, analysts undertake many of the tasks outlined in

FIGURE 10-1.

The first is to define the health issue that is to be addressed, including population, diagnosis, incidence, and impact, and study the relevant intervention technologies. Then the group must agree on the effectiveness of the current and proposed interventions and present a value comparison together with other parts of their analysis to the decision-makers.

The team also needs to consider the points of view or interests of whoever commissioned the study. However, a fine line exists between pleasing the “customer” and maintaining the group’s professional integrity. One way to address this is by “changing hats.” The analysis can say, “When we put on Hat A, we get X, but we put on Hat B, we get Y.” All of us wear many hats in health care, including patient, payer, parent, spouse, professional, and taxpayer.

► Defining the Healthcare Process Involved

The team will usually undertake a detailed process analysis to assure agreement on how the intervention is delivered, especially if there is little field experience to go on. It is often well worth the effort to step back and visualize how a new or modified

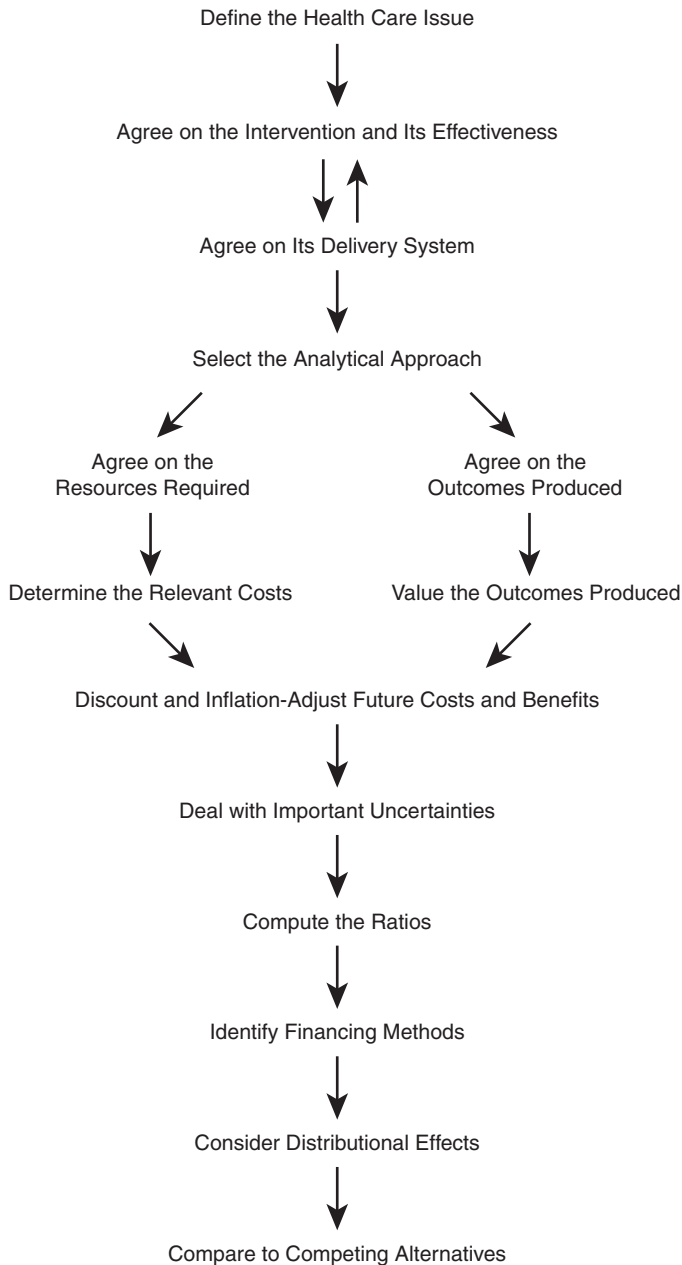


FIGURE 10-1 Steps in the cost-benefit analysis (CBA)/cost-effective analysis (CEA) process.

process will work and how the detailed implementation will go forward. Otherwise, the team will be making a stab in the dark about the resources required.

A feedback arrow indicates that sometimes the process analysis produces a revised estimate of the expected effectiveness, as the analysts learn about potential

barriers to adoption and implementation. Many more feedback loops could be added because at any point in time the team can uncover a need to revise earlier estimates.

Agreeing on Its Effectiveness

Clear evidence on the effectiveness of a proposed policy is a rarity. Most evidence is contestable even as to its science. Clinical trials may be limited, or in a few cases not even feasible. The populations involved in clinical trials and demonstrations may have been small or somewhat different from the one that will be affected by the policy. Often professional groups, having differing interests, cite studies that support their viewpoint and ignore those that do not. For example, in the case of folic acid supplementation those who favored it cited its effects on neural tube defects (NTDs), whereas those opposed cited its ability to mask other metabolic deficiencies.

Where the evidence is unclear or disputed, one way to proceed is through sensitivity analysis. After an analysis is done using the efficacy estimate that the analysis team thinks most valid, the calculations are repeated with alternative efficacy values to determine the range of values over which the conclusion holds. Many times the conclusion is not affected despite the heat generated by the differing estimates, but where the solution is sensitive to the choice of high-end or low-end parameter values, it is necessary to make the decision-makers aware of the applicable range. Perhaps they will authorize a study to narrow that range further. Sensitivity analysis is relevant to other variables besides efficacy, including costs, population affected, and inflation rates.

The relevant change in effectiveness associated with an intervention is the *marginal change*. For example, in 2005 the Washington State Legislature directed the Washington State Institute for Public Policy to report on the benefits and costs of “evidence-based” approaches to alcohol, drug, and mental illness treatment. The institute performed a meta-analysis based on a review of 206 studies in the literature that met a specific set of criteria for quality of experimental design and measurement, such as use of a control group. This was an unusually complex analysis because the Legislature specifically mandated the study of the effects of treating individuals with substance abuse disorders and/or mental illness disorders in terms of their fiscal impact and “the long-run effects on statewide education, crime, child abuse and neglect, substance abuse, and economic outcomes.” There were already systems in place for dealing with these disorders; thus, the researchers calculated the benefits based on marginal changes in costs and outcomes from expanding the existing services to provide evidence-based and consensus-based services to those not yet served. Few mental illnesses are currently cured, and many who stop abusing drugs and alcohol relapse over time. The study team had to estimate the reduced incidence or severity from implementing best practices. It estimated the number in the state’s population with each disorder and subtracted out the number already receiving services. Then they assumed that about 50% of the untreated populations would accept services, if they were made available. The analysts did not try to estimate the impact of having existing services move from their current modes of operation to the evidence-based approaches. Because the available studies were all short-term studies, the institute’s staff estimated a “decay rate” for each disorder to represent the loss of participants and program impact over time, and it included a factor in the model for those individuals who would recover on their own without treatment.

The institute's meta-analysis of the suitable studies concluded that the expansion of services would achieve a 15%–22% reduction in incidence or severity of these disorders, resulting in a saving of \$3.77 for each additional \$1 invested. Taxpayers would see direct savings of \$2.05 per additional \$1 invested or \$416 million per year in net payer benefits, if fully implemented (Aos, Mayfield, Miller, & Yen, 2006).

Agreeing in Detail on the Delivery System Involved

Analysis team members may have differing assumptions about how the intervention is to be delivered in the field. Reaching a common understanding of that process is an important early task. Discussing the process and drawing up a detailed process map are ways to get at that reality. Doing so leads directly to a description of the resources required. Team members may choose to revise their estimated effectiveness after the process is better defined and they better understand the problems of implementing the process in the field.

► Selecting the Analytical Approach

Various types of economic analysis can be performed. One key question is “What impact will the proposal have on the supply and demand for services?” Payment, access, and quality issues affect the perceived price and demand for services as well as their costs. Rychlik (2002) suggested a hierarchy of analytical approaches for comparison and decision-making:

- Establish the burden (at least the cost) of the illness usually including quality-of-life impacts of the problem.
- If the assessment shows little difference in impact from the relevant interventions or between the intervention and the status quo ante, conduct a cost-minimization study.
- If the comparison is among similar types of outcomes but there are significant differences in benefits and costs, conduct a cost-effectiveness study.
- If there are significant differences among the programs being considered, such that a common metric is necessary for benefits and costs, do a cost-benefit study.
- If quality of life after survival is an important parameter, then consider a cost utility study comparing differing quality measures using market research techniques such as conjoint analysis.

Sometimes public agencies focus on whether a proposed outlay is cost-neutral or cost-effective. To be cost-neutral, the proposal must not increase the overall costs to the agency. To be cost-effective, the proposal must be the least costly method for reaching a predetermined level of total benefits to the public. In the medical literature, however, the term *cost-effectiveness analysis* (CEA) has acquired its own specialized meaning of an analysis in which the benefits are measured in nonmonetary units, such as lives saved or quality-adjusted life years (QALYs). Such economic approaches, however, do not allow for comparisons of proposals that express outcomes in different units.

At higher policy levels, healthcare investments must be compared with other public investments, including those outside the health sector. When a common metric must be used to value both costs and benefits of the full set of proposals being analyzed, it usually turns out to be dollars. This is known as CBA. Private sector

organizations use the same techniques but often with a different set of terminology, using terms such as *return on investment* (ROI) and *internal rate of return* (IRR) when they evaluate and compare investment opportunities.

Hacker (1997) suggested that concerns about health care as an economic marketplace emerged forcefully in the 1970s, following the introduction of Medicare and Medicaid and the resulting cost inflation. With the nation's access problems seemingly addressed, government and industry turned toward issues of efficiency and effectiveness. This concern for efficiency and effectiveness in the federal government went well beyond the health sector. The Bureau of Management and Budget issued Circular A-94, entitled *Guidelines and Discount Rates for Benefit-Cost Analysis of Federal Programs*, in 1972. It was and still is intended for use across most federal government agencies and programs.

Since then there has been an explosion of studies and methodologies coming out of the subfield of pharmacoeconomics. Managed care organizations and government agencies needed these techniques and studies to justify adoption or delaying adoption of a new drug or device. Safety is no longer the sole concern.

► Basic Tools

Because of measurement difficulties, healthcare professionals frequently bypass the basic tools of economic analysis, including supply and demand analysis and benefit and cost analysis. These measurement problems pertain primarily to demand and benefit estimation, but also apply to costs.

Supply and Demand Concepts

Much of the health policy literature is concerned with aligning incentives properly through payment mechanisms, such as co-payments, withholds, discounts, and reimbursement rates. All of these really refer to changes in perceived prices and the effects these perceptions will have on the supply and demand for services. What really complicates health care is that both consumers and the consumer's agents (the health professionals) generate demand.

The policy analysis team will have to estimate the impact of those perceived price changes on the activity levels that they can expect to see in the service system. These estimates are not easy, even where a program is budget constrained. Take, for example, the situation in which a program is budget constrained and a budget increase is proposed. **FIGURE 10-2** illustrates a simple demand analysis relating to a budget constraint.

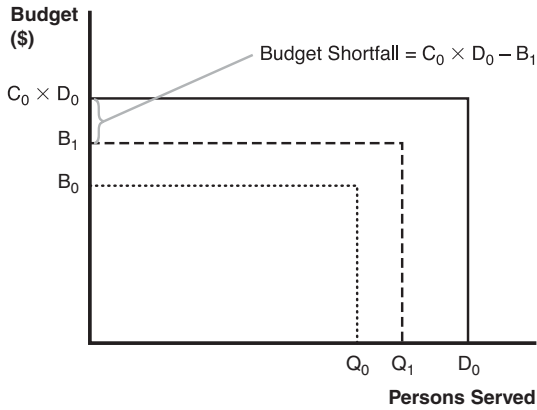
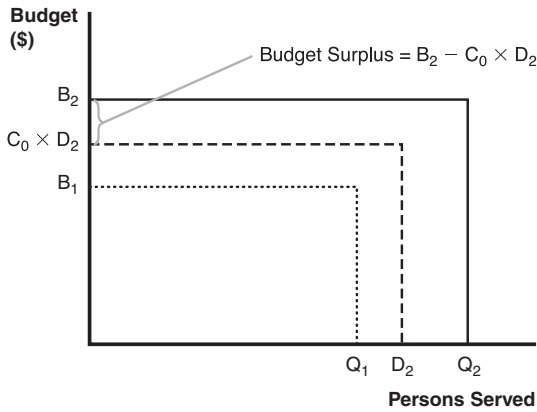
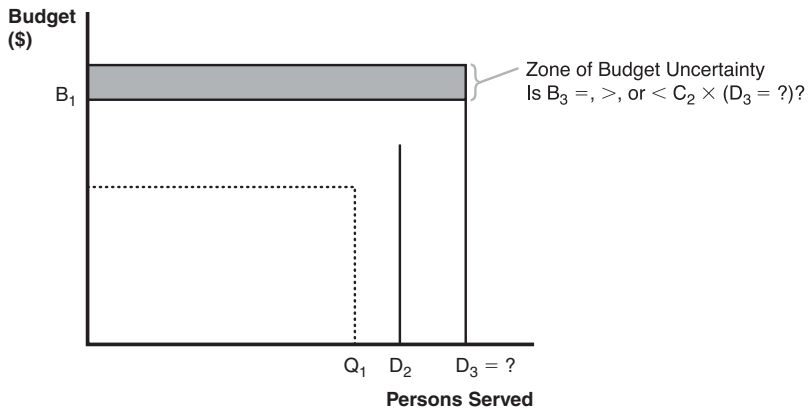
Given: Initial Budget = $B_0 = C_0 * Q_0$

where initial cost = C_0 , clients served = Q_0

Increased Budget = B_1 , where $Q_1 > Q_0$ and $B_1 > B_0$

In panel A, the new larger budget is fully consumed, but no cost change is assumed ($B_1 = C_0 * Q_1$, where Demand = D_0 still is $> Q_1$).

In panel B, the budget is increased to B_2 , but this enhanced availability of services exceeds the demand (D_2) at the current perceived cost ($Q_1 > D_1 > Q_0$) and the budget is under expended ($C_0 * D_2 < B_2$).

Panel A – Demand Greater than Budgeted**Panel B – Demand Less Than New Budgeted Quantity****Panel C – Program Changes Increase Provider Cost, But What Is New Demand?****FIGURE 10-2** Supply and demand over time in a constrained budget setting.

In panel C, the program management proposes responding by making the services more accessible (available more conveniently at more sites). This reduces the perceived cost of the service to the clients and the demand increases ($D_3 > D_2$), but at an added recruitment cost, which increases the average cost ($C_2 > C_0$). Given this situation, the program management and the policy analysts must make new estimates (C_2 , D_3) and see whether the demand will be greater than, less than, or approximately equal to the new budgeted level of activity, which is B_3 divided by C_2 . This will again determine the programmatic resources required.

If the prices charged are modified by a proposal, then the analyst must investigate supply and demand relationships further, including the following:

- The rate of change in demand with a given change in price (price elasticity)
- The rate of change in supply with a given change in price

Where data are available, these relationships can be estimated through regression analysis.

Utilities and Preferences

Health policy analysts use the terms *utility* and *preference* somewhat interchangeably. The latter sounds admittedly subjective; however, public policy has limited objective tools for measuring utility or for comparing utility across individuals. This creates an enormous challenge. Without the capacity to measure welfare, how do we maximize it? (Wheelan, 2011).

We also know that in health care one's utilities differ with one's health status. One has to be very careful in deciding whether to use utilities put forward from the general population or seek them from the subpopulation affected by the relevant diagnosis. It is also important to know which stage of disease progression they are in, if they have the disease.

Different methods are used to try to get at preferences and utilities, including:

- Direct measures:
 - Standard gamble: What probability of success would be necessary to get you to choose a proposed outcome over the status quo or some other alternative(s)?
 - Time trade-off: How many months of life would you be willing to give up to achieve the more desirable outcome?
 - Rating scale (visual analogue scale): Pick a spot of a line from 0 (worst possible outcome) and 1 (best possible outcome) that represents each alternative being considered.
- Indirect measures:
 - Generic utility instruments: Use value ratings set by the general public with off-the-shelf questionnaires. For example, in the U.K. the National Institute for Health and Clinical Excellence (NICE) has used the EQ-5D instrument; in the United States the Food and Drug Administration seems to favor the SF-6D. The Health Utilities Index, the Quality of Well-Being scale, and the 15-item 15D instrument are also used.
 - Disease specific instruments: Tailor the attributes and their weights to the disease and its observed attributes.
 - Mapping the attributes from a disease-specific instrument onto a generic instrument.

Each approach has its strengths and weaknesses. Each presents its own problems in terms of comprehension and representation (Tolley, 2009). For example, the standard gamble is easy to interpret, but it is also sensitive to a person's risk tolerance.

Valuing Costs, Benefits, and Outcomes

Analysts crossing over from other sectors will find specific problems in applying their usual evaluation methods in health care. Effective cost and benefit analysis in health care often requires understanding the nomenclature and diagnostic coding systems used in this sector. Furthermore, market failure in this industry makes it necessary to measure separately the consumer satisfaction and benefit/cost impacts of specific technological alternatives. Analysts can also expect to encounter lack of cooperation because of fear of loss of autonomy, accounting systems biased toward revenue rather than cost finding, high levels of inherent variability, compartmentalization of information systems, and poorly aligned reward systems.

The role of benefit and cost analysis in health care was investigated thoroughly in the 1960s and 1970s. During that period, the problems of benefit measurement seemed so insurmountable that most healthcare professionals doing analysis have preferred to rely on CEA. Weinstein and Stason (1977) described the difference as follows:

The key distinction is that a benefit–cost analysis must value all outcomes in economic (e.g., dollar) terms, including lives or years of life and morbidity, whereas a cost-effectiveness analysis serves to place priorities on alternative expenditures without requiring that the dollar value of life and health be presented.

The underlying premise of cost-effectiveness analysis in health problems is that, for any given level resources available, society (or the jurisdiction involved) wishes to maximize the total aggregate health benefits conferred. Alternatively, for a given health-benefit goal, the objective is to minimize the cost of achieving it. In either formulation, the analytical methodology is the same.

Weinstein and Stason (1977, p. 717)

The basic problem is not one of using dollars, however, but one of expressing all relevant factors in a single metric. The alternative approach is to express the outcomes as a vector, but because one alternative vector seldom dominates the other, one must still deal with tradeoffs among variables. The vector representation gets one into all the complexities of multidimensional scaling.

Further problems arise from the following:

- Determining the relevant costs, especially supply and demand estimation and resulting price levels
- Incorporating nonmonetary values
- Analyzing the way in which benefits and costs are distributed

Pauly (1995) suggested that CBA and CEA are used because the better normative measure, willingness to pay, is hard to assess in the real world. He defines a personal benefit as an informed individual's willingness to pay for a program, whereas

a programmatic benefit is the sum of the willingness to pay of all informed persons affected by the program, including those making altruistic contributions but not directly affected by the service process. In a few situations, willingness to pay can be imputed from what individuals are paying for insurance against an event or to mitigate the risk of an event, such as installing seat belts or highway crash barriers. As Pauly (1995) noted, “Such concepts as addition to measured gross national product associated with a health program, the additional wages to beneficiaries and providers, or addition from investment now and in the future have validity only to the extent that they proxy willingness to pay” (p. 103).

A proxy is something that stands in for the real thing. In one analysis, the costs of the early loss of a mother because of breast cancer were estimated by the value of replacement family care services plus a proxy for the emotional losses. The proxy chosen was the estimated cost of the amount of psychotherapy used by those who lost a mother early in life (Bunker, Barnes, & Mosteller, 1979).

Indicators are also used to substitute for direct measures. “Good indicators are easily measurable and highly correlated with the underlying variable of interest, which is usually impossible to measure” (Wheelan, 2011, p. 145). We may not be able to agree whether a population is healthy or not, but we often use measures to indicate success or failure, such as visits to the emergency rooms or repeat hospitalizations.

Pauly (1995) opposed two other approaches often cited in the literature: (1) the human capital approach, which emphasizes the economic cost to society, such as a worker’s daily wage times the number of work days lost, or some other measure of lost productivity (assuming full employment), or (2) the friction cost approach, which measures the loss in productivity until the system resumes full productivity with a trained and experienced new worker or the ill worker is restored to full capacity.

Benefit/Cost Concepts

Where there are multiple choices, the rational person will select one or more alternatives that maximize his or her satisfaction, or what economists would call their utility; however, our utilities are specific, if not unique, to each of us. Although individual utilities are cumbersome to capture, aggregating the utilities of a population presents even greater problems. Decisions that involve more than one person

WHOSE WILLINGNESS TO PAY?

A study to determine the need for a third London airport, as well as its location, found that a preferred location would displace a 12th century Norman church that was still in use. One group contended that the willingness to pay valuation of the church should be based on the value the current parishioners were insuring the building for against fire. A second group, however, argued that the Normans had incurred an opportunity cost for the last 8 centuries for the £100 that they invested to build it. They had foregone the opportunity to loan the money to the local usurers at a reasonable rate of interest; therefore, the church should be valued at the willingness-to-pay of the original parishioners, which would lead to a valuation of £100 plus compound interest for over 800 years. Using this calculation, it would be worth more than the construction cost of the entire new airport.

require a common measure. Most analyses are based on aggregating all of the costs and benefits to individuals regardless of what their utilities are and to whom or from whom they accrue. That is why so many studies end up choosing dollars; however, not all agree on that. Whatever the metric chosen, one ends up with a ratio of benefits to costs, and the higher that ratio the better an alternative. In health care, however, we must also consider to whom these benefits and costs accrue.

Circular No. A-94 defines *cost-effectiveness* as “a systematic quantitative method for comparing costs of alternative means of achieving the same stream of benefits or a given objective.”

In other words, the economist would say no to the request to “get me the most for the least money” because it is a mathematical impossibility. The two feasible formulations are:

- Get me the most benefit for a given sum of money (i.e., maximize my benefit-cost ratio).
- Get me a given benefit package at least cost (i.e., minimize my effectiveness-cost ratio).

Analysts often retreat to the previously prepared position of trying to produce a set of benefits defined by the politicians at the least possible cost and labeling the results as cost-effectiveness, even though it is really cost minimization. That leaves the hardest part, the valuation of benefits, up to the political process. At higher government levels where the tradeoffs are between noncomparable benefits such as health care, highways, police protection, and recreational services, the only comparable means of comparison usually turns out to be money. Pauly (1995) suggested that where money is used to measure benefits and (1) there is a fixed budget and (2) there is little variation in the preferences for outcomes, then cost-effectiveness analysis should be used, but where there is a variable budget and varying utilities of outcomes, costs effectiveness is “much less suitable, in theory than cost-benefit analysis” (p. 111).

At this point, the analysis splits into two streams. One stream estimates the costs, whereas the other values the outcome. This chapter proceeds next with the cost side, simply because that is the easier path to consider.

More recent concern about high price treatments, such as cancer drugs, that add considerably to costs for relatively small outcome improvements, have led policy makers to ask “What are we getting for the additional cost?” To address that question, analysts have been reporting on head-to-head comparisons using an *incremental cost-effectiveness index* (ICER),

$$\text{ICER} = \frac{C_1 - C_0}{E_1 - E_0}$$

where C_1 and E_1 are the respective cost and effectiveness respectively of the more expensive treatment and C_0 and E_0 are for the less expensive treatment or a control group.

► Agreeing on the Resources Required

All too often the analysis team starts by talking about monetary costs. This is the wrong place to start a cost analysis. When trying to compute the costs of a wedding

reception, few would start with a dollar amount per guest, but with estimates of the number of guests, the menu for food and drink, the portions offered, the number of helpings per person, and the staff needed. After these are defined, it is a simple matter to determine costs by multiplying these resources by their market prices and totaling them up. That gives us an estimate of the total variable cost of the reception. Then there are the fixed costs of the reception such as the chef and hiring the hall. However, the kitchen staff only has a certain capacity, and if the guest list exceeds a certain number, the staff would have to be augmented; therefore, some fixed costs apply only over a specific volume range. These are sometimes called step-variable costs or semi-fixed costs.

After we have the cost of our ideal menu and level of hospitality, it is time to figure out whether it falls within the acceptable budget range. Chances are it does not, and we would have to agree to spend more on the wedding than we had planned, cut some costs out of the reception, or cut down on some other aspect of the ceremony.

► Determining Relevant Costs

Relevant costs are those affected by the decision being considered. There are two methods of estimating costs: aggregate costs and marginal costs. One arrives at aggregated costs by taking the total costs of an organizational unit and dividing it by the number of units produced. This number is usually relatively easy to produce from existing departmental cost data, but using this method is not recommended. It does not take into account how processes, and hence costs, change with volume, nor does it include those relevant costs that occur outside of the given organizational unit. Relevant costing ignores those costs that are not affected by a decision, including those that are real but fixed. Relevant costs are also referred to as incremental or marginal costs in the economics literature. For example, the comparison of two treatments for pneumonia would not include the costs of diagnostic tests unless different tests were to be ordered as part of the new treatment regimen.

Relevant costs for the two treatments are likely to include the following:

- Changes in costs of medicines, consumable supplies, and tests caused by the introduction of the alternative method
- Changed labor costs, including physicians, nurses, and pharmacists and ancillary services
- Costs altered by the changes in length of stay or location of treatment
- Changes in costs incurred by the patient and the patient's family, including access costs and lost income, if any
- Changes in overhead costs associated with the new alternative including amortization of new specialized equipment and space requirements

Hospital costs represent an especially difficult problem because so many costs are lumped into a single overhead cost category and then allocated to the various operating departments. It is also hard to know what true costs are because published charges are so heavily discounted for federal and state governments, insurers, HMOs, and large employers.

A process-based cost study is usually a must in the hospital setting. Typical hospital cost reports are so loaded with fixed costs that it is necessary to map out the process and identify directly the resource inputs required and then price them.

Marginal/Incremental Costs

Where demand is changing, the appropriate cost is not the average cost of the service, but the cost of adding the next additional service unit (the marginal or incremental unit). Even though pricing, and hence revenue, may be related to the average cost of a unit of service, the cost of adding a unit is often not. If I am dean of a medical school and am asked to add 10 more students, I may find that the added costs during the preclinical years are virtually nil, because there are always at least 10 empty seats in the lecture halls. However, during the clinical years, I may have to divert additional faculty from seeing individual patients to rounding with the students and thereby lose revenue. Then, the anatomy professor warns that the anatomy lab is already overcrowded and I will have added new space and new equipment at considerable capital cost. Now the prospect looks less inviting. In general, expansions that do not require large fixed investments look inviting. Contractions, however, are avoided, because the fixed cost stays constant; so the corresponding cash flow requirement does not drop when volume goes down.

Handling Inherent Process Uncertainty

Earlier chapters dealt with the technological and political uncertainties of producing a desired outcome. The analysis team must decide how to deal with those uncertainties in its economic and financial analysis. Sometimes the uncertainty is handled with multiple analyses up front, a sort of branching in the analysis. However, the usual way of handling uncertainties is through sensitivity analysis in which the inputs of uncertain parameters are allowed to take on a realistic range of values to define the range over which the analysis is insensitive to change in that parameter.

For example, in the Washington State study of evidence-based treatment of substance abuse and mental illness disorders, the Washington State Institute for Public Policy conducted a sensitivity analysis with many different parameters using a Monte Carlo simulation technique. Researchers assigned a probability distribution to each range of values and ran the simulation model 10,000 times with the values of each variable sampled randomly from its distribution. This process indicated that there was only a 1% probability that the investment would provide a negative return to the taxpayers. This was very important to the credibility of the analysis because so many of the measures and variables were difficult to define and to measure (Aos et al., 2006).

A simulation is of most value if it incorporates the interaction of various factors as a policy is implemented.

FIGURE 10-3 illustrates one model that can simulate the effects of increased access. As the demand increases so do the direct costs of services and the capital costs of providing the necessary delivery infrastructure, especially supplying sufficient primary care providers, nurses, and community-based services; however, these changes will not take place in a vacuum. Policy variables can be manipulated to affect those costs, including the new covered service definitions, the management and organization of the new efforts, the amount of waste and medical error experienced, the financing and incentives of the program, and whether middlemen are used and what their margins will be. This again would seem to call for a simulation model to assess the overall impact of the planned interventions. The model is not revenue nor budget constrained, but that component could also be added. This model might start as a spreadsheet model, but be converted to a Monte Carlo simulation if one wanted to see how sensitive the model would be to uncertain values.

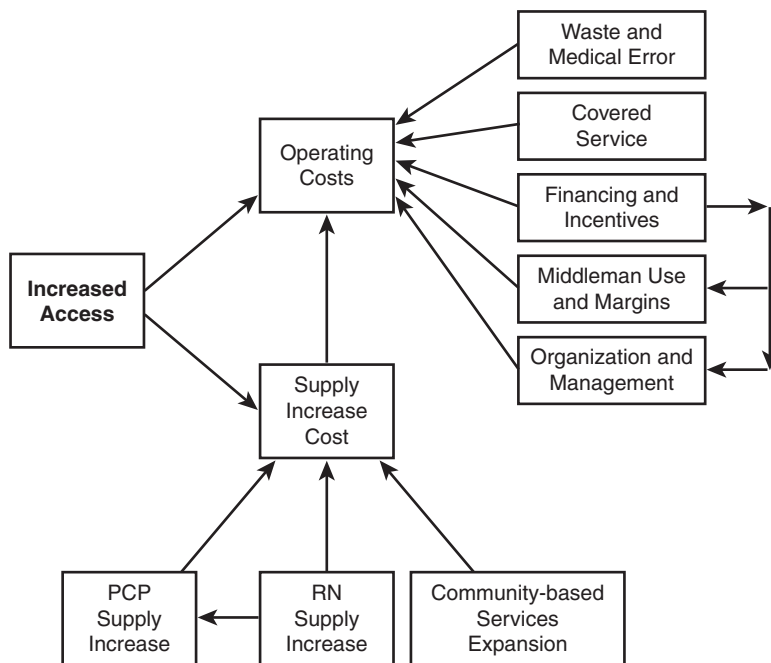


FIGURE 10-3 What happens to costs, if we resolve the general access problem.

What additional variables and/or feedback loops would you like to add? If there were many loops, a feedback type of model might be used. Certainly, the organization and management of services and the financing and incentive available would affect the capacity changes that might be needed. For example, a model of a proposal for a novel colon cancer treatment program might include an embedded Markov model of the transition rates among disease states (e.g., the rates of cure and the rates of transition from Stage 3 to Stage 4) under proposed and existing conditions.

► Valuing the Outcomes Produced

One of the hard parts of healthcare policy analysis is assigning value to the benefits. We have already addressed the willingness-to-pay argument versus the use of utility metrics such as QALYs saved or deaths avoided. Most utility comparisons are based on psychometric instruments that can be used to compare and rank alternative outcomes. QALY is one example of a utility measure. The quality of a life year is rated on a 0 to 1 scale, on which 0 is complete health and 1 is death. The attributes recognized in the comparisons included physical function limitations, social function limitations, emotional well-being, pain levels, and limitations on cognitive ability. Many intermediate states have been identified, from wearing glasses to acute pain to physical incapacitation. Any one individual may have a quirky utility curve, such as “I would rather be dead than comatose,” but the scale is based on the aggregate values of a representative population and not the individual. That is why people are urged to think about their own utilities and express their end-of-life preferences early on (Gawande, 2014).

There are many other scales for specific diseases and disease states, because the QALY may not be sensitive enough over the range experienced by that specific population or may not include a key variable. If one does try to develop a condition-specific measure, it could be useful to check out how well it maps to one of the generic quality-of-life measures and compare both against available objective measures of illness progression (Brazier et al., 2012). For example, in the case of rheumatoid arthritis, the most frequently used generic subjective utility measures seem to diverge from objective disease state measures as severity increases (Salaffi, Carotti, Ciapetti, Gasparini, & Grassi, 2011).

The estimated monetary value of a QALY varies considerably among analyses. A figure of \$50,000 per QALY is often cited in the United States. However, Braithwaite and colleagues (2008) noted that that amount had been used for a number of years and had not been adjusted for inflation. They took a willingness-to-pay approach based on U.S. investments in insured and uninsured elderly adults between 1950 and 2003 and determined a range of \$109,000–\$297,000 per QALY. They ultimately argued that the true value was closer to the World Health Organization estimate of \$109,000. NICE (2013) in the United Kingdom acknowledged that a range of £30,000–£35,000 was still its cutoff for drug evaluations.

It is difficult to reconcile cost-utilities with the ranges of cost per life saved by U.S. regulatory programs. In 2004, the Office of Management and Budget suggested a possible range of \$1 million–\$10 million, but more recently has suggested an upper limit of \$5 million (Appelbaum, 2011). Lives-saved reconciliation is difficult to achieve due to lack of a quality adjustment. For example, many Americans might prefer dying in a car accident to a long bout with terminal cancer. All this is further complicated by Americans' reluctance to accept any measures such as QALY that might be used to allocate healthcare resources (i.e., ration them); witness the constraints in the ACA.

The ACA expanded the emphasis on utilizing evidence-based medicine in the 2009 stimulus act and established the Patient-Centered Outcomes Research Institute within the Center for Medicare and Medicaid Services (CMS). However, the same legislation limited the ways the Institute's research could be used within the Department of Health and Human Services. There clearly was a concern among federal lawmakers that these research findings would find their way directly into the workings of the even more controversial Medicare Advisory Payment Commission.

Pearson and Bach (2010) noted:

Under current law and because of years of precedent, Medicare generally covers any treatment that is deemed "reasonable and necessary," regardless of the evidence on the treatment's comparative effectiveness or its cost in relation to other treatments. Likewise, with only rare exceptions, Medicare does not use comparative effectiveness information to set payment rates. Instead it links reimbursement in one way or another to the underlying cost of providing services.

(p. 1796)

This is quite different from the way comparative effectiveness research is used in other counties at various regulatory stages, such as new drug approvals and approved protocols.

Congress maintained this status quo, in part, by placing a number of constraints on the use of comparative effectiveness research. For example, Section 6301 of the

ACA amended Section 1181 of the Social Security Act to establish the Institute with the following purpose:

the Institute is to assist patients, clinicians, purchasers, and policy-makers in making informed health decisions by advancing the quality and relevance of evidence concerning the manner in which diseases, disorders, and other health conditions can effectively and appropriately be prevented, diagnosed, treated, monitored, and managed through research and evidence synthesis that considers variations in patient subpopulations, and the dissemination of research findings with respect to the relative health outcomes, clinical effectiveness, and appropriateness of the medical treatments, services, and items described in subsection (a)(2)(B).

However, the ACA went on to specify:

SEC. 1182 o42 U.S.C. 1320e–1.

- a. The Secretary may only use evidence and findings from research conducted under section 1181 to make a determination regarding coverage under title XVIII (Medicare) if such use is through an iterative and transparent process which includes public comment and considers the effect on subpopulations.
- b. Nothing in section 1181 shall be construed as—
 1. (1) superceding or modifying the coverage of items or services under title XVIII that the Secretary determines are reasonable and necessary under section 1862(l)(1); or
 2. (2) authorizing the Secretary to deny coverage of items or services under such title solely on the basis of comparative clinical effectiveness research.
- c. (1) The Secretary shall not use evidence or findings from comparative clinical effectiveness research conducted under section 1181 in determining coverage, reimbursement, or incentive programs under title XVIII in a manner that treats extending the life of an elderly, disabled, or terminally ill individual as of lower value than extending the life of an individual who is younger, nondisabled, or not terminally ill.
3. (2) Paragraph (1) shall not be construed as preventing the Secretary from using evidence or findings from such comparative clinical effectiveness research in determining coverage, reimbursement, or incentive programs under title XVIII based upon a comparison of the difference in the effectiveness of alternative treatments in extending an individual's life due to the individual's age.

Determining the Present Value of Costs and Outcomes

If we asked you for a \$10 bill and paid it back tomorrow, you probably would be okay with that. If we requested \$10 and said we will return it in 5 years, you probably would decline the honor. The value of money has a time dimension with two components: (1) whether it will have as much utility in the future or if inflation will reduce its purchasing power, and (2) the lost opportunity to create utility with it in the interim.

Analysts adjust for the latter by applying a discount rate. The discount rate may represent interest income forgone or what one would have to pay to borrow the money at interest to operate until the payoff takes place. Economists argue not over whether the time value of money should be recognized, but over the appropriate rate to use; however, there are legitimate concerns about how inflation adjusting and how discounting tends to devalue programs such as preventive care that pay off further into the future.

Discounting

The time value of money concept also applies to discounting, except that the values of future benefits and costs are reduced to a net present value (NPV). Each stream of both costs and benefits needs to be brought back to a current value using the formula:

NPV(V_t,N,r)=V_0+frac{V_1}{(1+r)}+frac{V_2}{(1+r)^2}+frac{V_3}{(1+r)^3}+dots+frac{V_N}{(1+r)^N}

where V_t = the value of the cost or benefit in time period t, N = the number of periods in the series, and r = the discount rate to be used.

Analysts can directly calculate these values using a spreadsheet model or function, or calculate it using the discount factors from an NPV table similar to **TABLE 10-1**. For example, if the annual benefit received from a program is \$10,000

TABLE 10-1 Present Value of a Dollar Received in Year N at Discount Rate r								
N	r =	0.03	0.04	0.05	0.06	0.07	0.08	0.1
1		0.97087	0.96154	0.95238	0.9434	0.93458	0.92593	0.90909
2		0.9426	0.92456	0.90703	0.89	0.87344	0.85734	0.82645
3		0.91514	0.889	0.86384	0.83962	0.8163	0.79383	0.75131
4		0.88849	0.8548	0.8227	0.79209	0.7629	0.73503	0.68301
5		0.86261	0.82193	0.78353	0.74726	0.71299	0.68058	0.62092
6		0.83748	0.79031	0.74622	0.70496	0.66634	0.63017	0.56447
7		0.81309	0.75992	0.71068	0.66506	0.62275	0.58349	0.51316
8		0.78941	0.73069	0.67684	0.62741	0.58201	0.54027	0.46651
9		0.76642	0.70259	0.64461	0.5919	0.54393	0.50025	0.4241
10		0.74409	0.67556	0.61391	0.55839	0.50835	0.46319	0.38554

(continues)

TABLE 10-1 Present Value of a Dollar Received in Year N at Discount Rate r (continued)

N	$r =$	0.03	0.04	0.05	0.06	0.07	0.08	0.09	0.1
11		0.72242	0.64958	0.58468	0.52679	0.47509	0.42888	0.38753	0.35049
12		0.70138	0.6246	0.55684	0.49697	0.44401	0.39711	0.35553	0.31863
13		0.68095	0.60057	0.53032	0.46884	0.41496	0.3677	0.32618	0.28966
14		0.66112	0.57748	0.50507	0.4423	0.38782	0.34046	0.29925	0.26333
15		0.64186	0.55526	0.48102	0.41727	0.36245	0.31524	0.27454	0.23939

Cumulative Present Value of a Dollar Received Annually for N Years Discounted at Rate r

1	0.97087	0.96154	0.95238	0.9434	0.93458	0.92593	0.91743	0.90909
2	1.91347	1.88609	1.85941	1.83339	1.80802	1.78326	1.75911	1.73554
3	2.82861	2.77509	2.72325	2.67301	2.62432	2.5771	2.53129	2.48685
4	3.7171	3.6299	3.54595	3.46511	3.38721	3.31213	3.23972	3.16987
5	4.57971	4.45182	4.32948	4.21236	4.1002	3.99271	3.88965	3.79079
6	5.41719	5.24214	5.07569	4.91732	4.76654	4.62288	4.48592	4.35526
7	6.23028	6.00205	5.78637	5.58238	5.38929	5.20637	5.03295	4.86842
8	7.01969	6.73274	6.46321	6.20979	5.9713	5.74664	5.53482	5.33493
9	7.78611	7.43533	7.10782	6.80169	6.51523	6.24689	5.99525	5.75902
10	8.5302	8.1109	7.72173	7.36009	7.02358	6.71008	6.41766	6.14457
11	9.25262	8.76048	8.30641	7.88687	7.49867	7.13896	6.80519	6.49506
12	9.954	9.38507	8.86325	8.38384	7.94269	7.53608	7.16073	6.81369
13	10.635	9.98565	9.39357	8.85268	8.35765	7.90378	7.4869	7.10336
14	11.2961	10.5631	9.89864	9.29498	8.74547	8.24424	7.78615	7.36669
15	11.9379	11.1184	10.3797	9.71225	9.10791	8.55948	8.06069	7.60608

and the discount rate chosen is 6%, then the NPV of benefits received over 5 years would be worth \$42,124 ($\text{NPV} = \$10,000 * 4.2124$).

An Example

If a proposed program were to cost \$100,000, but yield a stream of benefits annually starting at the end of 5 years for five more years, we would have the following calculations to arrive at a NPV using a 4% discount rate.

Investment at the beginning of Year 1 of \$100,000.

Benefits		
Year 6	\$40,000	$\text{NPV} = (40,000)/([1 + 0.04]^6) = \$31,612.58$
Year 7	\$40,000	$\text{NPV} = (40,000)/([1 + 0.04]^7) = \$30,396.71$
Year 8	\$40,000	$\text{NPV} = (40,000)/([1 + 0.04]^8) = \$29,227.61$
Year 9	\$40,000	$\text{NPV} = (40,000)/([1 + 0.04]^9) = \$28,103.47$
Year 10	\$40,000	$\text{NPV} = (40,000)/([1 + 0.04]^{10}) = \$27,022.57$
Total Benefit		\$146,362.97

The same result can also be calculated assuming a 10-year stream of benefits from which the initial 5 years of the stream has been subtracted. Using an NPV table, we would then calculate the following:

$$\text{NPV} = 10 \text{ year stream} - 5 \text{ year stream} = \$40,000 * (8.1109 - 4.4518) = \$146,363$$

Computing Ratios

Ratios are easy to compute. In the previous example, the NPV of the investment is \$100,000, and of the benefits it is \$146,363; thus, the benefit–cost ratio is $(146,363/100,000)$ or 1.46. This would be a comparison figure to rank order with other social or corporate investments. Each investment would then be subjected to other comparative evaluations or perhaps a minimum target IRR. IRR is the discount rate at which the benefits and the costs equal each other (i.e., the rate that brings the NPV to 0). In the previous example, the IRR is 9.155%. This rate can then be compared with the cost of capital and/or financing for the investment. It must also be evaluated in terms of the uncertainty of the values used, the distributional impact on various actors and bystanders, the impact on other programs (especially in terms of budget availability), and financial viability.

Inflation Adjusting

A constant inflation rate produces a cumulative geometric increase in costs. Over time, analyses can be very sensitive to the inflation effect, especially in health care, where overall inflation rates are high. However, the inflation rate in healthcare cost reported in the popular literature is the result of several factors, including the following:

- Input prices for labor and purchased goods (price inflation)
- Increasing technological opportunities, including pharmaceuticals and biologicals.
- Aging of the population

Because the population is defined and the rest of the technology is assumed to be fixed for comparative purposes, the appropriate inflation rate for an analysis is usually the first item, inflation in the input prices of labor and purchased goods. A reasonable lower bound on this rate is the consumer price index (CPI), but because health care is a labor-intensive professional service that does not usually exhibit the same productivity improvements as much of the rest of the economy, the appropriate rate is somewhat higher than the CPI. Available indices applicable to components of an analysis include Producer Price Indices for specific segments and medical care CPI price deflators. Newhouse (2001) estimated that historically the inflation in healthcare costs as has been about 2% above the overall CPI. He also noted that the available indices may be biased on the high side, in part because of the lack of reliable data on real transaction prices. For example, the Producer Price Index for pharmaceutical preparation manufacturing was as follows:

1995	253.9	2001	314.5
1996	259.1	2002	326.7
1997	290.1	2003	343.3
1998	298.5	2004	360.1
1999	306.6	2005	378.7

To determine a rate of change from raw data use this formula:

1. Decide on the number of years in the interval (try 5 and 10)
2. Divide the ending index by the initial one

5 years $378.7/306.6 = 1.235$
10 years $378.7/253.9 = 1.492$
3. Set the number of years equal to N, and take the Nth root of the appropriate value from the previous equation.

5 years $(1.235)^{(1/5)} = 1.0431$ meaning a 4.3% inflation rate
10 years $(1.492)^{(1/10)} = 1.0408$ meaning a 4.1% inflation rate

If one were looking at the prices charged by pharmaceutical manufacturers, the forecast of prices would grow at whatever rate you thought appropriate. If you chose 4.2% over a 5-year period, then the price of drugs now at \$200 would be calculated in the fifth year to be \$200 times 1.042⁵ or \$245.68. This is based on a mixture of products, some of which might be going up rapidly, whereas others might become generic and go down markedly.

► Dealing with Important Uncertainties

There are three possible approaches used to deal with important uncertainties in the financial analysis:

- Adding a risk premium to the discount rates used for uncertain parts of the calculations. This means that different streams of costs and benefits would be adjusted at different rates because they will have different uncertainties associated with them.
- Developing a subjective probability distribution for those values and applying that distribution to the modeling process. If there is a 60% chance that a program will cost \$100,000, a 30% chance that it will cost \$110,000, and a 10% chance that it will cost \$90,000, the expected value would be
 - $([100,000 * 0.6] + [110,000 * 0.3] + [90,000 * 0.1]) = \$102,000$ and could be substituted for the \$100,000 value to adjust for uncertainty in costs
- Investing in further research to reduce the uncertainty. For example, one could refine the cost estimate until one was relatively certain that \$100,000 was the proper mean estimate of the cost and then use that.

Evaluating Multiple Alternative Investments

Because medical care and public health are both government services in the United Kingdom, that government has wrestled with whether or not the two systems should use a common set of evaluation measures. Edwards, Charles, and Lloyd-Williams (2013) have reviewed the guidance on this from the U.K. government and international agencies. These documents seem to agree that QALYs are appropriate for medical care evaluations, but may be less so for public health investments. The latter often involve changing the behaviors of individuals and populations. These changes typically take place over considerable periods of time and affect many other aspects of personal and public life. These documents suggest that programmatic effects that are not medical be included. Some would value all costs and benefits in local currency leading to a cost-benefit analysis. Others would use multidimensional scaling to include QALYs with other effects, such as personal competency, into a single metric. Money as a common metric allows a public investment approach that can use IRR as a prioritizing metric. The 2010 Global Burden of Disease study used Disability Adjusted Life Years as a metric: a type of measure that the ACA appears to rule out for some federal programs decisions.

Other countries have set about finding the best way to use this data for three types of decisions:

1. Comparing treatments in head-to-head comparisons
2. Considering how to implement specific programs, if at all, and

3. Selecting those interventions that yield the best marginal contribution to social welfare.

The first use is the simplest. Bendavid, Duong, Sagan, and Raiker (2015) conducted a meta-study of the ICER values for international health aid programs across 20 low- and middle-income countries and 5 disease categories. They suggest that these investments were generally cost-effective, but could benefit from some reallocation among disease categories for optimal results.

The second is exemplified by the development of an optimal strategy for vaccination policies for countries in Latin America (Glassman, Cañón, & Silverman, 2016).

The simplest approach to the third situation is to set an arbitrary threshold like NICE's £30,000–£35,000 per QALY. Theoretically, that should represent the optimal opportunity cost of adding another QALY at the margin, but that figure is seldom known or available. Because social welfare has many dimensions, the investment process has to take into account a number of dimensions simultaneously, including the economic ones. This also applies to evaluating alternative clinical pathways (Antioch, Drummond, Niessen, & Vondeling, 2017).

One logical way to start prioritizing health investments is to use league tables (similar to those used to display team rankings in sports) in which treatments are listed in order of their increasing Incremental Cost per QALY or similar measure. The lowest incremental cost investments are entered into the budget first, then the second, and so forth until either a budget constraint (supply side) or an arbitrary ICER threshold (demand side) is reached. Leigh and Granby (2016) illustrated this concept with the example of the U.K. Cancer Drug Fund. At the time the fund, which is outside the regular NHS budget, was purchasing 74 drugs not approved by NICE. This demonstration was simplified by involving a single class of disease and a limited set of alternatives. Technical issues still existed because many drugs on the list had not been fully evaluated to provide the necessary data or there were no other social welfare variables considered, or no knowledge of which populations subgroups might be cut off. **FIGURE 10-4** provides an illustration of the league table approach. The treatments for diagnoses 1 through 5 would be added to the budget. Diagnosis 5 could be treated for some patients, but the budget would have to be increased to cover all the cases, or it could be rejected because it overruns the budget. Diagnosis 6 would not be treated because it violated by the budget constraint and the ICER threshold.

The ratios that we have been studying (benefit-cost and cost-effectiveness) have little meaning in and of themselves. They are useful only in comparison with either a minimum standard value or in terms of rank ordering a set of alternatives. Sometimes a set of subcategories are used, such as these:

- Absolutely necessary because of the regulation or risk to the patients
- High priority in terms of meeting overall organizational goals
- Medium priority
- Low priority

Where there are subcategories, the ratios may be used to rank order the alternative investments within each category. This approach to priorities can be a way to work in some of the political pressures that would otherwise overwhelm simple league-table rankings.

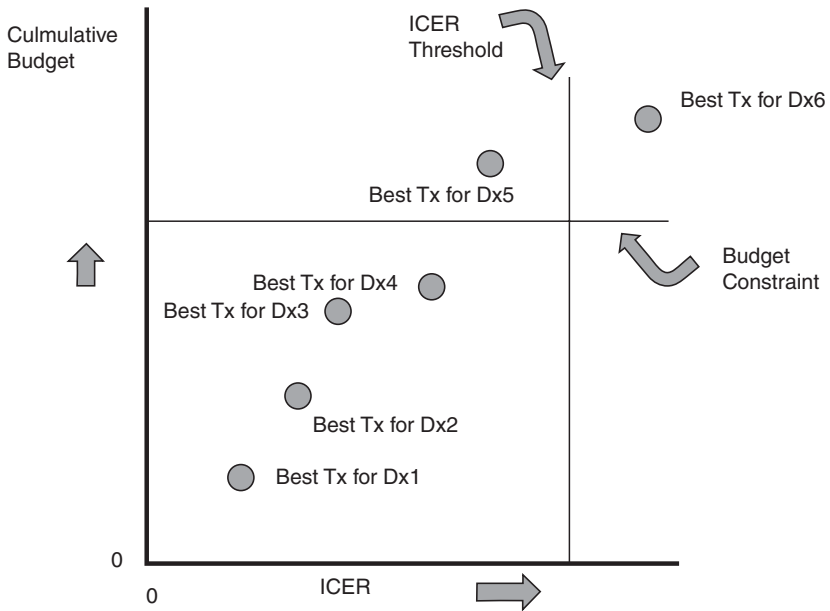


FIGURE 10-4 League Table Approach.

► Financial Feasibility

An organization's financial system must be up to the tasks of evaluating an alternative's costs and forecasting the financial implications of various funding alternatives. Cleverley and Cameron (2003) suggested three major target areas of financial strategic planning for health organizations:

1. Revenue estimation
2. Capital budgeting (more of a concern for organizations that pay income tax)
3. Financing of operations and capital investments

They expanded on these by suggesting that the underlying financial systems must

- Provide data on revenue, costs, and capital requirements according to program or product lines
- Adjust these estimates for inflation and increasing technology requirements
- Supply ROI or IRR estimates for decision
- Estimate working capital and cash reserve requirements under expected operating conditions
- Establish the desired capital structure given the debt capacity of the organization
- Provide procedures for the allocation of capital among competing internal programs and facilities

The first three steps are the same in both the public and private sectors. The last two are especially important to organizations and companies operating in the private sector. Even governmental programs that are separately funded like Medicare

have to estimate flows and reserves. It is the actuaries at the CMS who keep telling us when the Medicare trust funds will run out.

It is important to recognize that different parts of the organization may disagree on the value of an activity, such as reducing the length of stay, depending on whether they are looking at full costs or marginal costs, whether they consider the organization to be at capacity or below it, and how the reimbursement system operates (Ward, Spragens, & Smithson, 2006). In response to this concern, Voluntary Hospitals of America has developed a template for evaluating the cost, revenue, and cash flow impacts of various proposals. It is illustrated in **FIGURE 10-5**, which analyzes a proposal to improve hospital laboratory turnaround times, thus reducing the overall average length of stay by 0.1 days. It then proceeds to evaluate three types of costs: the costs of the project, the cash flow impacts of the cost reduction, and the cash flow impacts of the new patients that can be accommodated in the freed-up beds (called backfill).

► Identifying Financing Methods

Where is the money for the investment coming from, even if it is offset later by benefits? Those benefits may or may not result in offsetting cash flows into the organization. For example, money spent on smoking cessation programs by a state government would ultimately reduce the costs of its Medicaid program, but much of the benefit accrues to the federal government, to insurers, and to individual citizens.

One is unlikely to spend most of the available budget on one program with a highly favorable ratio because it would crowd out other meritorious programs and threaten the organizational and political coalitions that make a budget viable. To match the acceptable share of available funds better, any very large program investment is likely to need multiple sources of funding or be spread out over several budget cycles.

► Considering Distributional Effects

Financial decisions affect both actors and nonactors. There are issues of free riders and moral hazard associated with some alternatives. There also are impacts on provider income, payer cash flows, and patient cash flows. In a financial sense, every proposal can be considered a zero-sum game in which there are winners and losers under our current set of incentive systems. Professionals are usually paid more for doing more. Vulnerable patients often end up paying more than those strong enough to monitor their own care and bargain on price. The health policy analyst must “follow the money” without being too cynical and attributing all motivation to greed. Health care is a mixture of necessities and consumption goods. Consider that during the recent Great Recession and the great Depression of the 1930s, birth rates dropped to new lows as many families put off having more children. Health inflation rates tend to rise and fall with major shifts in the business cycle.

There are four generalized methods of rectifying maldistribution if the market fails to correct the problem over time. According to Wheelan (2011), they are:

- Regulation or deregulation
- Incentives and disincentives, including taxes and subsidies

Costs of Process Improvement			
Salaries	\$	(100,000)	
Fringe Benefits	\$	(100,000)	
Supplies & Services	\$	(20,000)	
Other	\$	-	
Costs of Process Improvement		\$	(220,000)
Cash Flow Impact of the Improvement			
Changes in Net Revenue Associated with			
Case based reimbursement	\$	-	
Per Diem reimbursement	\$	(96,000)	
Percentage of Charges reimbursement	\$	-	
Medical Device Revenue (Revenue = device cost and markup)	\$	-	
Drug Revenue (Revenue = drug cost and markup)	\$	(10,000)	\$ (106,000)
Changes in Operating Cost Associated with			
Supply costs driven by changes in patient days	\$	15,000	
Medical device changes	\$	-	
Medication substitution	\$	25,000	\$ 40,000
The Cash Flow Impact of the Improvement		\$	(66,000)
Backfill Cash Flow Estimate			
Incremental Net Revenue Associated with Backfill			
Case based reimbursement	\$	243,902	
Per Diem reimbursement	\$	96,000	
Percentage of Charges reimbursement	\$	48,780	\$ 388,683
Medical Device Revenue (Revenue = device cost and markup)	\$	-	
Drug Revenue (Revenue = drug cost and markup)	\$	10,000	\$ 398,683
Incremental Operating Costs Associated with Backfill			
Incremental supplies	\$	(15,000)	
Medications	\$	(25,000)	
Medical Devices	\$	-	\$ (40,000)
The Cash Flow Impact of Potential Case Backfill		\$	348,683
Backfill Cash Flow Estimate			
Cost of Process Improvement	\$	(220,000)	
Cash Flow Impact of the Improvement	\$	(66,000)	
Net Cash Flow Improvement Exclusive of Potential Case Backfill			\$ (286,000)
Cash Flow Impact of Potential Case Backfill			\$ 348,683
Total Cash Flow Impact from Process Improvement Initiative		\$	62,683

FIGURE 10-5 Vizient, Inc. cash flow analysis template.

Reproduced from "Faster Labs: Process Improvement Initiative Cash Flow" spreadsheet from the Financial tab of "Building the Business Case for Clinical Quality". © 2006 Vizient, Inc. All rights reserved.

- Direct government services delivery
- Contracting with the private sector to deliver service

The ACA incorporated all four methods. Everyone must have health insurance or be taxed. Subsidies are provided for those who cannot pay the full premium. The government provides insurance exchanges and also pays providers through Medicaid.

Capital Allocation Processes

Capital investment is one place where facilities and services available to the poor and the uninsured are likely to be shortchanged over time as large multisite health-care organizations seek to maintain reasonable payer mixes and returns on their investments (Hurley, Pham, & Claxton, 2005; Robinson & Dratler, 2006). The use of analytical techniques to estimate rates of return over time is unlikely to support expansion of services for those who can pay little or nothing.

► Where in the World?

Economic health process analysis assistance is widely available. There are many consulting centers in the developed world and most major universities offer courses. To fully expand your methodological skills we suggest Drummond, Sculpher, Claxton, Stoddart, and Torrance (2015) *Methods for the Economic Evaluation of Health Care Programmes*. In addition to the usual journals, *Health Economics*, *Journal of Health Economics*, *Pharmacoeconomics*, and *Health Affairs*, one should also keep an eye on *The Milbank Quarterly*, *European Journal of Health Economics*, and *Value in Health*. Astolfi, Lorenzoni, and Oderkirk (2012) provide a comparison of the models used to forecast health expenditures in 25 developed countries.

► Conclusion

Decision-makers need to know what a program will cost; what revenue, if any, it will generate; and how and by whom the balance will be financed. They also have to present a convincing case that the current and future benefits to patients, payers, and the public to justify the investment. Topics discussed included quality adjusted life years, discounting, and comparative effectiveness research.

Comparative tools such as benefit–cost analysis and cost-effectiveness analysis are basic tools of policy analysis, but they present a number of methodological, ethical, and value-based concerns. Yet they are critical to the budget approval process and to gaining top level support for proposals.

Smaller organizations also have to worry about the sources of working capital necessary to undertake and sustain both current and higher levels of activity. For them, cash flow management is also important, even if it is not usually considered part of health policy analysis. Although some may see financial viability analysis as “the dark side” of policy analysis, competency in that area is a must.

CASE 10: Increasing the Federal Cigarette Excise Tax

When the Congressional Budget Office (2012) published *Raising the Excise Tax on Cigarettes: Effects on Health and the Federal Budget*, its stated motivation was “to demonstrate the complex links between policies that aim to improve health and effects on the federal budget” (p. iii). According to this June 2012 report,

A policy that discourages smoking by raising the excise tax on cigarettes provides a good case study because a substantial body of research exists about the effect of changes in cigarette prices on smoking rates, as well as about the impact of smoking on health, health care spending, longevity, and (to a lesser extent) earnings

(p. 2)

Policy and budget analysts evaluated the impact of a hypothetical \$0.50 increase (adjusted for inflation) in the federal excise tax on cigarettes and small cigars. At the time of the report, the federal tax was \$1.01 per pack. It had been increased in 2009 from \$0.39 by the ACA. State excise taxes ranged from \$0.17 in Missouri to \$4.35 per pack in New York State, with an average \$1.48 for all states. Additional city taxes increased the combined state and local taxes per pack to \$5.85 in New York City and \$5.66 in Chicago.

The Congressional Budget Office (CBO) report noted:

A policy initiative aimed at improving the health of the population would affect the federal budget through the following links:

- The effects of the policy on people’s behaviors.
- The impact of changes in behavior on people’s health, and
- The implications for improvement in health for people’s health care spending, life expectancy, and earnings. Specifically, better health would tend to reduce health care spending per capita ... Better health might also lead to lower mortality rates and greater longevity, thus increasing size of the population and changing its age distribution. Better health could also affect total earnings, because healthier people might make different decisions about working and might be more productive at work.

To produce comprehensive estimates of how such a policy initiative would affect the federal budget, CBO must assess the magnitude of each of these complex links. That process requires a great deal of analysis by CBO and a significant amount of research by outside analysts on which CBO can draw.

(Reproduced from Congressional Budget Office Report. *Raising the excise tax on cigarettes: Effects on health and the federal budget* (Publication 44319), 19, iv–v)

The study projected the budget impacts of the status quo and the additional \$0.50 excise tax on outlays and on revenues through 2085. The net result of the analysis was a \$41 billion deficit reduction over 10 years (the usual CBO window for analysis), with smaller deficit reductions through 2085. After about 2025, federal outlays would increase slightly. Greater longevity and the costs of supporting the additional aged population would more than offset savings from health improvement,

although there would still be deficit reductions for another 15 years or so because of increased tax revenue.

Estimation Tasks

The flows of costs, payments, and revenues required estimates of the following:

- Response of smokers to the policy change
- Effects of smoking (including secondhand smoke) on health
- Effects of smoking on labor earnings, productivity, workforce participation, and disability
- Impacts of health care on Medicare, Medicaid, employment-based health insurance, subsidies through health insurance exchanges, and federal employee, military, and veteran benefits (i.e., retirements, pensions, and health care)
- Excise tax revenues
- Other federal revenues
- Inflation rates
- Longevity effects on Old Age and Survivors Insurance, Disability Insurance, and Supplemental Security Income

Estimating the Effects and Flows

One of the most difficult estimates was the underlying future rate of smoking without the intervention. Smoking rates in the United States have declined relatively steadily since 1992—at first by 2% of the population per year, and then by 1% in the most recent years. The report referred to the slower rate of decline as “hardening of the target.” The report assumed, based on the available literature, that without the increased excise tax the percentage of smokers would remain relatively constant at about 15% of the adult population.

The CBO had available a number of studies of the price elasticity of cigarettes for adults and youth, some of it from the tobacco settlement negotiations. One problem was that no price-related interventions had gone on in a vacuum. Other antismoking activities had also been taking place. The report was issued somewhat ahead of the current trend of employers and insurers penalizing smokers by charging higher healthcare premiums. The report argued that its findings are likely to hold up through a \$1 excise tax increase, but suggested that savings from even greater tax increases might be offset, in part, by increased smuggling or international Internet sales.

The CBO report also discussed three analytical models appearing in the literature: longitudinal or life-cycle, cross-sectional, and policy simulation. The report seemed to dismiss the longitudinal models as unstable and presenting present value issues. It stated that longitudinal studies seemed to give conflicting results about lifetime health spending.

The study used a regression model to analyze the healthcare cost impacts and the longevity effect of reductions in smoking. The report noted that two different approaches often are used: disease-specific analysis and regression analysis. It cited a number of sources of strong disease-specific (bottom-up) evidence, but the CBO decided to rely on regression. The researchers did not want to assume that all differences in healthcare costs, income, and longevity between smokers and nonsmokers are due to the smoking habit. Education level, gender, alcohol use, and health risk tolerance were given as examples of cohort differences unlikely to be affected by smoking status. The disease-specific approach might also miss health effects of smoking not yet documented.

CBO'S Analysis

The new work presented in the CBO report was based on cross-sectional analysis using regression followed by a set of policy simulations. The cross-sectional analysis for the effect of smoking on healthcare spending involved regressions conducted on data from the Medical Panel Expenditure Survey for five different age groupings. Smokers had higher costs than nonsmokers except in the 75+ age group. The CBO used a two-part model. A logistic regression model determined first the probability of having any expenditure in a given year, and a second general linearized model estimated the amount of expenditures with nonzero values. The regressions controlled for sex, marital status, race or ethnicity, education level, geographic location, alcohol consumption, categories of body mass index, health insurance coverage, and attitudes toward risk taking.

The conclusion from the CBO regressions was that healthcare expenditures were higher for smokers and former smokers than for nonsmokers except for the 75+ age group. This analysis gave information on the differences between the groups. Those differences were sufficiently great that the CBO decided to construct a healthcare cost comparison between a smoker and someone who was not a smoker but had the same characteristics otherwise. The percentage of healthcare cost attributable to smoking ranged from 4% to 8%, with the greatest differences in the 45–65 and 65–75 age groups. The average was 7%. This was consistent with most older studies, but higher than the 5% reported by a newer study.

To estimate the relationship between smoking and longevity, the CBO again used logistic regression with 1.0 being a death in the next year. The data came from the 1997–2004 National Health Interview Survey and death certification records from the National Death Index through 2006. Indicators (dummy variables) were added for the survey year to capture changes over time. Again the final comparison was between smokers and nonsmokers who share the same other characteristics. Their conclusion was that smoking reduced longevity by 5–6 years.

Lag Effect in Smoking Cessation Recovery

The increase in cigarette tax would affect only smokers; those who quit would gradually achieve improved health. The CBO constructed a longitudinal percentage recovery time series using disease-specific data supplied through the CDC on diseases responsible for 86% of deaths attributable to smoking. It was referred to as the “health response lag.”

Estimating Earnings Impact

The literature shows that smokers have more unemployment, lower wages, and perform less well at work, although these effects are not all due to smoking. Using data on both smoking and earning power from the Current Population Survey and its Tobacco Use Supplement, the CBO again used regression analysis and controlled for age, region, sex, race or ethnicity, education level, and marital status. The analysis indicated that apparent differences in earnings between smokers and nonsmokers with similar measured characteristics were greater than could be attributed directly to the smoking decision. Similar patterns were observed from regression analysis of the interaction between smoking and retirement based on the Health and Retirement Study. Again there seemed to be substantial unmeasured relationship as the pattern emerged that former smokers were doing better than nonsmokers who, in turn, were doing better than smokers, all with the same characteristics.

Even though the regressions indicated that smokers on average earned 11.7% less than nonsmokers and former smokers averaged 1.3% more than

nonsmokers, the CBO decided to use the following estimates of income differentials based on an evaluation of the evidence and consideration of the unmeasured factors:

- Ages 18–34: Smokers earn 4% less.
- Ages 25–44: Smokers earn 6% less.
- Ages 45–54: Smokers earn 5% less.
- Ages 55–64: Smokers earn 7% less.
- Ages 65–74: Smokers earn 5% less.

Exposure to Secondhand Smoke

The CBO report considered several ways of accounting for the health effects of secondhand smoke. There was a self-reported rate of 15% for individuals subject to secondhand smoke, and the prevalence of nicotine in blood samples of nonsmoking adults was nearly 40%. The CBO chose to calculate the number of smoke-free households with and without the tax increase, even though that ignored smoking at work and other public places. Data from the National Health and Nutrition Examination Survey, the Medical Expenditure Panel, and the Current Population Survey were combined to determine that the benefits to healthcare spending and longevity from not being subject to secondhand smoke would be about 5% of the benefits accruing to those who stopped smoking.

Modeling the Financial Flows Over Time

To model financial flows over time, first the study staff had to estimate the impact of the intervention on the size and age distribution of the national population and the relevant population segments over time, including smokers, quitters, quitters who resumed smoking, nonsmokers, and nonsmokers who started smoking, year by year. Then they could apply their estimates of the revenues, earnings, and expenditures for each subgroup year by year for each federal funding stream. They used this estimate for each year to calculate the amounts taken in and expended for each subgroup for each program. These programs were cited earlier; however, special adjustments were necessary. For example, the study took into account the revenue effects of the taxes on increased taxable incomes due to reduced health insurance premiums, changed disability program enrollments, changed cost-sharing and premium subsidies under insurance exchange provisions of the ACA, and the fact that disability programs had a higher population of mentally ill persons who were much less likely to quit smoking. The staff did not try to estimate the effect of reduced consumption due to price elasticity without stopping smoking. They cited the fact that people could and sometimes did offset their consumption of reduced numbers of cigarettes by buying stronger blends or adjusting the amount inhaled.

CBO Conclusions

FIGURES 10-6 and **10-7** show the overall findings of the study. The health improvement effects outweigh the costs of greater longevity for about 10 years, and then the increased costs of Social Security, Medicare, and other programs would mount as the survivors aged. However, the cigarette excise tax revenues would offset this and yield a net reduction in the deficit each year for the next 75 years. The deficit reductions would amount to 0.02%–0.03% of GDP, with the greatest effects in the earliest years.

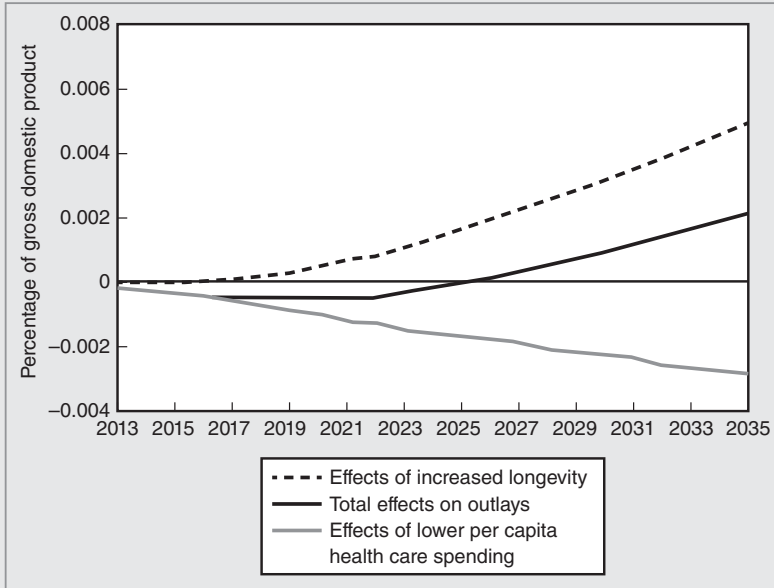


FIGURE 10-6 Effects on outlays of the illustrative increase in the cigarette tax.
 Reproduced from Congressional Budget Office Report *Raising the excise tax on cigarettes: Effects on health and the federal budget* (Publication 44319), p. 19.

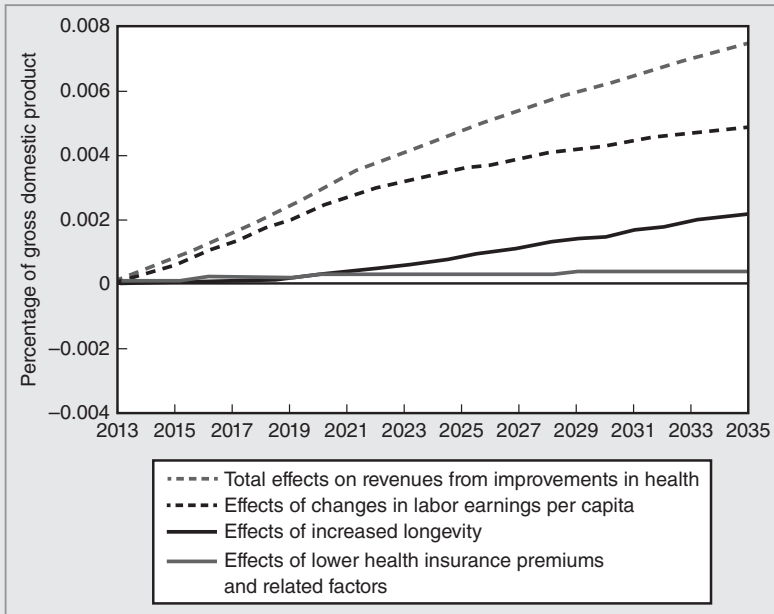


FIGURE 10-7 Health-related effects on revenues of the illustrative increase in the cigarette tax.
 Reproduced from Congressional Budget Office Report *Raising the excise tax on cigarettes: Effects on health and the federal budget* (Publication 44319), p. 19.

A Major Disclaimer

The report recognized that the budgetary analysis did not include all the factors involved in a decision to raise the federal cigarette excise tax:

If lawmakers were to consider raising the excise tax on cigarettes, or adopting other policies that would promote a healthier population, their proposals would depend on a variety of considerations besides the effects on the federal budget. Those other considerations would most likely include the impact of a proposed policy on people's health, views about the appropriate role of the government in influencing behavior, the burdens that the proposed policy might impose on people in different circumstances, and the effects of the policy on the budgets of state and local governments. Those other considerations lie beyond the scope of this analysis, which addresses only the impact of an increase in the cigarette tax on the federal budget (with related analysis of the effects on health and longevity).

(Reproduced from Congressional Budget Office Report. *Raising the excise tax on cigarettes: Effects on health and the federal budget* (Publication 44319), pp. 5, 9)

► Discussion Questions

1. How important are the budgetary effects likely to be in the decision to impose a further increase in the federal cigarette excise tax? What factors ought to have a major role in the debate? What would you add the CBO's major disclaimer about their analysis? How important should their analysis be in any real decision-making process?
2. Make sure that you understand the differences between longitudinal analysis, cross-sectional analysis (including logit), and the simulation analysis approaches. Does it appear that CBO used all three?
3. It does not appear that the CBO used a Monte Carlo simulation in the final modeling. Why do you think that they did not do so?
4. States were making large increases in their cigarette taxes during this period. What would you speculate would be the interaction between the two (state and federal), and how would that affect your analysis? Do you think some of that effect was captured by the inclusion of "region" in the regressions?



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CHAPTER 11

The Policy Analysis Process: Analysis of Values and Social Context

Final policy recommendations must also take into account values, especially where there are competing trade-offs. You might ask: Doesn't the political process take care of those value concerns? Don't the positions decision-makers take during policy debates and in their votes reflect their personal values and those of their constituencies? Yes, the political process will reflect the wider conflicts in American society over social mores, the value of human life, the role of government, the competitive marketplace, and collective versus individual rights. In this chapter, however, we consider value issues relating more specifically to health care and healthcare professions. These values may or may not be considered in the political process and should be addressed, as necessary, in any policy analysis.

Value issues discussed briefly in this chapter include:

- Equitable access
- Efficiency and value
- Patient privacy and confidentiality
- Informed consent
- Personal responsibility
- Professional ethics
- Consumer sovereignty
- Social welfare
- Rationing
- Process equity

This chapter ends with a procedure to consider the broader question of how social institutions might be influenced by interacting changes in the medical sector.

► **Equitable Access**

Priester (1992) suggested that the United States should reorder its values to give the greatest emphasis to fair equity, which he defined as giving each individual access to an “adequate level of care.” Skipping over issues of residency and eligibility, he argued, “Assuring access to health care, regardless of cause or source of need, is society’s responsibility.” He added, however, that “this does not require access to all potentially beneficial care” (p. 92). He maintained that this approach should not exceed the available resources nor deprive some segments of society of their opportunity for a reasonably full life. This would be a floor, not a ceiling; someone with more personal resources could choose to consume more health services. He saw the U.S. system as overemphasizing provider autonomy, individualism, and assumed abundance. He felt that we have too often let economic considerations outweigh ethical ones.

A similar argument could be made for removing health disparities; among the major causes of which is differential access to care. The Affordable Care Act (ACA) made a step in the direction of fair access by moving toward universal coverage and by calling for a defined essential benefit package. That in turn led to a number of issues related to cost and shared responsibility that reduced its popularity.

► **Efficiency and Value**

If resource scarcity leads society to overlook some populations or to restrict necessary services to them, efficiency becomes important. “Waste not, want not” is often not a shared value when one person’s waste is another person’s enhanced income. Because it is hard to get individuals to pull together on efficiency and cost minimization, some policy experts have emphasized “value.” Porter and Teisberg (2006) argued against a “zero-sum” mentality that attempts to maximize each individual provider’s share of the existing pie and favors reducing waste to enhance the value received by individual consumers. We now have many value-based compensation experiments that allow sharing of the savings with providers and provide incentives for higher-quality outcomes. The concept here is that value is created when you get greater quality for the same cost or the same quality for lower cost.

In a survey that asked health industry leaders to rank 11 areas of innovation for their potential impact in terms of quality and cost over the following 5 years, the highest ranking went to process improvement. The other innovation areas deemed likely to contribute to both improved quality and cost were data analytics, nonphysician delivery alternatives, disease management, and alternatives to fee-for-service. The respondents believed innovations in diagnostics, pharmaceuticals, and electronic medical records would help with costs but not quality, and they forecasted a low likelihood of help along either dimension from consumer incentives or from basic research (Chin, Hamermesh, Huckman, McNeil, & Newhouse, 2013). The good news is that process improvement is an area where individual providers can contribute.

► Patient Privacy and Confidentiality

Increasingly, electronic patient records, digitized information already collected for billing and claims, and specialized databases, such as registries, offer potential for learning more about disease processes and care outcomes. At the same time, they offer possibilities for breaches of the confidentiality we expect from the healthcare system and that is guaranteed by laws such as the Health Insurance Portability and Accountability Act (HIPAA). News stories about hacking, major data breaches, and stolen laptops crop up frequently. This is an area where trade-offs will continue to be difficult and frustrating, and privacy will continue to be important in policy analysis and decision-making.

► Informed Consent

Requirements for informed consent for human subjects in research represent a constraint on provider autonomy. They add to the staff burden, but are a regulatory requirement. **BOX 11-1** illustrates part of the federal regulations governing informed consent by research subjects in federally funded research. You might ask yourself: What values are represented here and why were they made an added requirement of all research in the first place? While the excerpt from the federal regulations cited in Box 11-1 is under review, it appears that little will be changed, and it will not add the coverage of biosamples that some individuals have requested.

BOX 11-1 Code of Federal Regulations Section 50, Subpart B, Informed Consent of Human Subjects

§50.20: General requirements for informed consent

Except as provided in 50.23 and 50.24, no investigator may involve a human being as a subject in research covered by these regulations unless the investigator has obtained the legally effective informed consent of the subject or the subject's legally authorized representative. An investigator shall seek such consent only under circumstances that provide the prospective subject or the representative sufficient opportunity to consider whether or not to participate and that minimize the possibility of coercion or undue influence. The information that is given to the subject or the representative shall be in language understandable to the subject or the representative. No informed consent, whether oral or written, may include any exculpatory language through which the subject or the representative is made to waive or appear to waive any of the subject's legal rights, or releases or appears to release the investigator, the sponsor, the institution, or its agents from liability for negligence.

Section 50.25 Basic elements of informed consent.

- a. In seeking informed consent, the following information shall be provided to each subject:
 1. A statement that the study involves research, an explanation of the purposes of the research and the expected duration of the subject's participation, a description of the procedures to be followed, and identification of any procedures which are experimental.

2. A description of any reasonably foreseeable risks or discomforts to the subject.
 3. A description of any benefits to the subject or to others which may reasonably be expected from the research.
 4. A disclosure of appropriate alternative procedures or courses of treatment, if any, that might be advantageous to the subject.
 5. A statement describing the extent, if any, to which confidentiality of records identifying the subject will be maintained and that notes the possibility that the Food and Drug Administration may inspect the records.
 6. For research involving more than minimal risk, an explanation as to whether any compensation and an explanation as to whether any medical treatments are available if injury occurs and, if so, what they consist of, or where further information may be obtained.
 7. An explanation of whom to contact for answers to pertinent questions about the research and research subjects' rights, and whom to contact in the event of a research-related injury to the subject.
 8. A statement that participation is voluntary, that refusal to participate will involve no penalty or loss of benefits to which the subject is otherwise entitled, and that the subject may discontinue participation at any time without penalty or loss of benefits to which the subject is otherwise entitled.
- b. Additional elements of informed consent. When appropriate, one or more of the following elements of information shall also be provided to each subject:
9. A statement that the particular treatment or procedure may involve risks to the subject (or to the embryo or fetus, if the subject is or may become pregnant) which are currently unforeseeable.
 10. Anticipated circumstances under which the subject's participation may be terminated by the investigator without regard to the subject's consent.
 11. Any additional costs to the subject that may result from participation in the research.
 12. The consequences of a subject's decision to withdraw from the research and procedures for orderly termination of participation by the subject.
 13. A statement that significant new findings developed during the course of the research which may relate to the subject's willingness to continue participation will be provided to the subject.
 14. The approximate number of subjects involved in the study.
- c. When seeking informed consent for applicable clinical trials, as defined in 42 U.S.C. 282(j)(1)(A), the following statement shall be provided to each clinical trial subject in informed consent documents and processes. This will notify the clinical trial subject that clinical trial information has been or will be submitted for inclusion in the clinical trial registry databank under paragraph (j) of section 402 of the Public Health Service Act. The statement is: "A description of this clinical trial will be available on <http://www.ClinicalTrials.gov>, as required by U.S. Law. This Web site will not include information that can identify you. At most, the Web site will include a summary of the results. You can search this Web site at any time."
- d. The informed consent requirements in these regulations are not intended to preempt any applicable Federal, State, or local laws which require

additional information to be disclosed for informed consent to be legally effective.

- e. Nothing in these regulations is intended to limit the authority of a physician to provide emergency medical care to the extent the physician is permitted to do so under applicable Federal, State, or local law.

Reproduced from Code of Federal Regulations (2013, April 1). *Protection of human subjects*. Washington, DC: Government Printing Office. Title 21, Volume 1, Part 50. Retrieved on December 17, 2013, at www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfcfr/CFRSearch.cfm?CFRPart=50&showFR=1&subpartNode=21:1.0.1.1.20.2

► Personal Responsibility

A significant portion of the cost of health care can be attributed to lifestyle choices, including lack of exercise; overeating or poor nutrition; not wearing seat belts or bicycle helmets; and use of tobacco, drugs, and alcohol. Many policy proposals seek to change these behavioral risk factors or shift their costs to the individuals at risk. Smokers pay higher insurance premiums. Some analysts have suggested that those involved in risky behaviors, such as not wearing motorcycle helmets, post a bond to cover their incremental medical care costs in case of an accident. Most people would agree that individuals should take more responsibility for their behavior rather than have it borne as a collective risk, but there is less agreement on the effort society should expend to make healthy choices more attractive. There is considerable evidence, for example, that the development choices we make—our *built environment*—can encourage or discourage physical activity. Yet a politician asked to support a bill to encourage “walkability” may be inclined to attribute lack of exercise solely to, as one state legislator put it, “a lack of personal fortitude.”

The question of personal responsibility comes up repeatedly in debates about health reform and the ACA in particular. Key elements of that reform can be seen as an attempt to balance a society’s responsibility to ensure equitable access with an individual’s responsibilities. The individual mandate in the ACA legislated a shared responsibility, using tax fines to ensure that young, healthy people pay for their own care if they need it and do their share to support a larger system that they will likely rely on as they age. When Governor Mitt Romney was considering the individual mandate as part of the Massachusetts reforms, he noted that the personal mandate was about personal responsibility which is a fundamental Republican value (Starr, 2011). Yet, in a video released by *Mother Jones* during the campaign, Romney said that 47% of the population feels entitled to government handouts. “I’ll never convince them that they should take personal responsibility and care for their lives,” he said (*Mother Jones*, 2012). President Obama emphasized social responsibility in his acceptance speech after winning his second term. Congressional attempts to repeal the ACA had heavily emphasized individual choice and deemphasized collective responsibility.

► Professional Ethics

Access and rationing are aspects of distributive justice. Confidentiality, truthfulness, informed consent, respect for patient and professional autonomy, and the safety of

the patient are all topics cited in discussions of professional ethics. Most professionals have had some indoctrination in biomedical ethics and are aware of the key issues. Their professions have ethical codes or statements. These codes have tended to become more general over time as patient autonomy has become more respected and monopolistic practices have come under government scrutiny. At the same time, professional societies may take strong positions without incorporating them into a code. For example, the American Nurses Association has long supported health care as a basic right delivered through a single-payer system.

What About Health Policy Experiments?

Daniels (2006) raised an interesting ethical question for health policy professionals: What if we required that health system transformations undergo an ethical review process similar to that required of medical research experiments? He noted that these social experiments can put significant populations at risk. He did not suggest what the mechanism for such reviews might be; however, he raised the issues of balancing the social value of the experimental policy changes with the risks to those affected and how we might conduct such a review. He suggested three levels of analysis:

1. Select benchmarks for the ethical analysis in terms of their effect on:
 - a. Equity
 - b. Efficiency
 - c. Accountability
2. Conduct an ethical evaluation of how well the design of the proposed reform meets its goals:
 - a. How good is the evidence used to justify the intervention?
 - b. Is the implementation planned adequate to test the results?
 - c. Will the intervention measure the effect with sufficient sensitivity to evaluate the results?
 - d. Are the key implementers involved in the planning and committed to an unbiased evaluation?
3. Determine whether there is sufficient oversight to protect the rights of those involved in the experiment.

He mostly offered examples of international situations, but, overall, he noted, “Unfortunately, there is little experience in measuring how systems establish transparency, accountability, and fair process in decisions involving resource allocation” (p. 450). Others would add preserving and enhancing patient autonomy and the way risk/benefit tradeoffs and their distributive effects are handled (Braunack-Mayer, 2006).

► Consumer Sovereignty

Those who want a highly competitive marketplace want the consumer to make decisions rather than the government. The examples in **BOX 11-2** illustrate views of health policy analysts on both sides of this economic and political ideological divide. Many health professionals do a bit of a straddle here. Most believe in patient autonomy but also recognize that consumer sovereignty often comes at the expense of professional power and influence, which some would call paternalism, but may also positively affect patient compliance and hence clinical outcomes.

BOX 11-2 Contrasts in Economic and Political Ideology

Cannon and Tanner (2005) agreed in one respect with many who support a greater government role in health: that the healthcare sector of the economy is special. But they argued that because it is special, reliance on competition is even more critical:

Unlike software, wireless communications, or banking, health care involves very emotional decisions, which often entail matters of human dignity, life, and death. However, we do not see the gravity of these matters as a reason to divert power away from individuals and toward government. Rather we see the special nature of health care as all the more reason to increase each consumer's sphere of autonomy.

(pp. 146–147)

Richmond and Fein (2005) concluded otherwise. They would prefer to use public policy to create a more equitable society:

Health and health care are vitally important in influencing life's chances and one's income and wealth should not determine the amount and quality of care one receives. We seek a system in which the financing and distribution of health services reflect our image of a just society, a society in which economic arrangements reflect a moral dimension.

(p. 4)

► Social Welfare

Interest groups at the table during policy formulation may or may not adequately represent the public's interests. In the backroom talks leading to the ACA, the proposed law that emerged protected existing high-cost providers in order to avoid opposition to expanded access for low-income individuals (Brill, 2015; Emanuel, 2014). There are many ways of evaluating outcomes in terms of social welfare. Some are economic; some are not. Economic models that would base policy decisions on benefit–cost or cost–effectiveness criteria raise issues of valuation, such as

- Social costs incurred or avoided
- Willingness to pay
- Contribution to gross domestic product
- Pain and suffering avoided

Be aware that there are counterattacks to the usual economic arguments:

Cost–benefit analysis of health and environmental policies trivializes the very values that gave rise to these policies in the first place. Moreover, through opaque and intimidating concepts like willingness to pay, quality-adjusted life-years, and discounting, economic analysts have managed to hide the moral and political questions lying just under the surface of their precise and scientific-looking numbers. It is time to blow their cover.

Ackerman and Heinzerling (2004, p. 234)

Ackerman and Heinzerling also questioned the use of quality-adjusted-life-years metrics because they discriminate against the older population, who will have fewer quality years ahead of them. This problem can be offset some by age-adjustments. The same can be said of any analysis dealing with contribution to gross domestic product, because the older population generates little output or will soon stop generating it and the output of children is so far into the future that any reasonable discount rate obliterates their benefits. This is why some suggest that economic analysis is useful for payers and for comparing treatment alternatives for a specific illness, but really does not work well when comparing alternative public investments.

Saarni et al. (2008) raised even deeper concerns about the values inherent in the use of a health technology assessment (HTA) in the international arena:

First, does the model represent and promote a “western”, individualistic perspective that fits only certain types of health-care organizations but is inappropriate to others? Further, modern health care technology (and HTA as part of it) can be seen as an “imperialistic” approach towards other health care ideologies emphasizing as it does rational, mechanical, non-intuitive and non-emotional ways of thinking.

Second, will the methods and issues of ethics be transferable between countries and cultures? It is quite clear that many moral issues and values are contextual - even more so than scientific and technological issues - so is it not counterproductive to try to “transfer” an ethical model for HTA.

Saarni et al. (2008, p. 621)

► Rationing

A society or organization may undertake explicit or tacit rationing for a number of reasons:

- Reduce overall costs to a targeted level
- Meet other resource limitations
- Maximize access
- Maximize social welfare based on the contributions of the individuals within the population
- Maximize medical utility and effectiveness
- Combinations of the previous points

For example, the attempt of the British National Health Service (NHS) to limit the costs of care has led to at least one charge of rationing in which a woman with breast cancer was denied the drug Herceptin by a local health authority on the basis of cost. An account of this has been included in **BOX 11-3**. A similar furor occurred when the National Institute for Health and Clinical Excellence (NICE) recommended limiting use of the newer, effective but expensive Hepatitis C drugs to cases where lives were threatened (NHS England, 2014).

BOX 11-3 Rationing and the Courts

Ann Marie Rogers, 54, sued the Swindon Primary Care Trust after it refused treatment for her early stage HER2 breast cancer with the drug Herceptin (trastuzumab), even after her doctor prescribed it. The Roche drug is licensed for late stage breast cancer, but some studies showed it to be effective for early stage cancer as well. Ms. Hewitt, the British Health Secretary, had praised the efforts of other women to get the treatment, and the Health Department had ordered the local health services “not to withhold it solely on the grounds of cost,” even while a government-appointed review panel was in the process of setting new guidelines for the use of the drug. Treatment with Herceptin costs \$36,000–\$47,000 a year for each patient and was supplied under very different circumstances by the local health trusts. The Swindon trust had a policy of supplying Herceptin for early stage breast cancer only in “exceptional circumstances.” Ms. Rogers had pointed to her cancer being of the type responsive to the drug and the risk factors of the death of her mother and a cousin from cancer.

When the trust refused to fund her treatment, Ms. Rogers started paying for the drug on her own, but had to stop when her own resources were used up. Thus, she sued the authority, citing the “postal code lottery,” which determined who got treated and who did not. A lower court judge ruled in February 2006 that the Swindon health service’s denial was lawful. In April, however, a three-judge appeals court overturned that verdict and said the local health service had acted unlawfully, noting that “once the Primary Care Health Trust decided, as it did, that it would fund Herceptin for some patients and that the cost was irrelevant, the only reasonable approach was to focus on the patient’s clinical needs and fund patients within the eligible group who were properly prescribed by their physician.” It observed that there had been “no rational basis for distinguishing between patients within the eligible group on the basis of exceptional clinical circumstances any more than on the basis for personal, let alone, social circumstances” (Lyall, 2006).

NICE issued a press release on April 12, 2006, in which NICE Chief Executive Andrew Dillon reaffirmed the importance of ensuring that new drugs are both safe to use, by having an effective system for licensing, and used in the right way through the work that NICE does. “Without these things, we risk exposing patients to risks and the health service to using its money unwisely,” he said (NICE, 2006a).

After a public comment period that closed in June 2006, NICE issued guidance in August 2006. “Trastuzumab, given at 3-week intervals for 1 year or until disease recurrence (whichever is the shorter period), is recommended as a treatment option for women with early stage HER2-positive breast cancer following surgery, chemotherapy (neoadjuvant or adjuvant), and radiotherapy (if applicable)” (NICE, 2006b, p. 4).

In June 2012, NICE again addressed the breast cancer issue with Technical Appraisal 257, “Breast cancer (metastatic hormone receptor)—lapatinib and trastuzumab (with aromatase inhibitor).” It stated that NICE did not recommend this as first-line treatment for post-menopausal women who have metastatic breast cancer that is hormone-receptor and HER2 positive. It explained its response as follows:

NICE looks at how well treatments work, and also at how well they work in relation to how much they cost the NHS. NICE applies special considerations to treatments that can extend the lives of people who are nearing the end of their life. Lapatinib and trastuzumab (herceptin) do not provide enough benefit to patients to justify their high cost even when the special considerations were applied, so NICE did not recommend them. (NICE, 2012)

Reduce Overall Healthcare Costs to a Target Level

An effective rationing system is not aimed at minimization. Health care cannot be driven to zero; however, when there is a budgetary limitation or a fixed-revenue situation, a system must be in place to decide which services will not be provided to which individuals. In many cases, this is done by limiting the population served and limiting the services offered.

For example, one of the largest Medicaid expenses is paying for long-term care for older people, a service not covered by Medicare. States have quietly limited the amount they pay by limiting the number of new nursing home beds they license. They know that there is a strong linear relationship between the number of beds available and the amount of Medicaid claims received. Some states have had to rely on a lottery system for enrollment of new cases when existing resources are inadequate

Meet Other Resource Limitations

Other shortages of materials, facilities, or personnel limit what can be done. A shortage of a vaccine may force public health officials to hold back on routine preventive programs so they can make their stock available in areas with active cases.

Maximize the Social Welfare

When transplanted organs first became available, it was clear that there were not enough to go around. Institutions doing transplants set up committees of individuals concerned with medical effectiveness and medical ethics to determine who would get the next available organ. This process provided an example of how social welfare considerations can influence rationing and access decisions. The committees examined a mixture of personal, family, and medical data to determine literally who should live and who would, in all probability, die. Considerations included family status, work status, medical factors such as alcoholism and comorbidities, and psychiatric factors. Outside observers, however, were suspicious that ability to pay might be creeping into the decision-making (Box 11-3).

One problem with the social welfare criterion is that it goes well beyond the issues health professionals are comfortable with, forcing them to deal with issues of economic contribution, utility of one's work, and emotional impact on others. Reaching some agreement on the importance of each of these is unlikely given current value conflicts in our society. That does not mean that we cannot reach a social consensus, but rather that it would be very difficult to justify once we have reached it.

Maximize Utility and Medical Effectiveness

"The providing or withholding of care is ideally determined by the probability that the treatment will maximize benefits and minimize harm, (i.e., by the criterion of medical effectiveness)" (Bodenheimer & Grumbach, 2005, p. 137). Previous chapters of this book have emphasized this approach. It is important to remember it is subject to constraints on available resources.

Combinations of One or More

BOX 11-4 contains a description of how the United Network for Organ Sharing (UNOS) chose to allocate available organs (not including related donors). This contractor was set up after passage of the National Organ Transplantation Act of 1984. It emphasizes the likely medical outcome and then pays attention to one's length of time on the waiting list; however, the issue of whether the transplant takes place is also contingent on the availability of private insurance or Medicare or Medicaid funds to pay for it. By now you should be aware that trying to maximize one thing while trying to minimize or maximize another is often a mathematical impossibility. It is legitimate, however, to have an analysis in which one objective is traded off against another. The problem is that most attempts to do so never specify what the trade-off ratios are because trying to arrive at a consensus ratio is likely to

BOX 11-4 How the Transplant System Works: Matching Donors and Recipients

Under the Organ Procurement and Transplantation Network (OPTN) contract with the U.S. Department of Health and Human Services' Health Resources & Services Administration, UNOS maintains a centralized computer network linking all organ procurement organizations and transplant centers. This computer network is accessible 24 hours a day, 7 days a week, with organ placement specialists in the UNOS Organ Center always available to answer questions. A person who may benefit from a transplant is referred by his or her doctor to a transplant center, which evaluates the patient. The transplant center runs a number of tests and considers the patient's mental and physical health, as well as his or her social support system. If the center decides to accept this person as a transplant candidate, it will add his or her medical profile to the national patient waiting list for organ transplant. The candidate is not placed on a ranked list at that time. Rather, his or her information is kept in a constantly updated, computerized database. When a deceased organ donor is identified, a transplant coordinator from an organ procurement organization accesses the UNOS computer. Each patient in the database is matched against the donor characteristics. The computer then generates a ranked list of candidates for each available organ in ranked order according to OPTN organ allocation policies. The match for each donor would be different and unique to the circumstances of the donor and the patients waiting. Factors affecting ranking may include tissue match, blood type, length of time on the waiting list, immune status, and the distance between the potential recipient and the donor. For the heart, liver, lung, and intestines, the potential recipient's degree of medical emergency is also considered. The organ is offered to the transplant team for the first person on the list. Often, the top patient will not get the organ for one of several reasons. When a patient is selected, he or she must be available, healthy enough to undergo major surgery, and willing to be transplanted immediately. Also, a laboratory test to measure compatibility between the donor and the recipient may be necessary. For example, patients with high antibody levels often prove incompatible to the donor organ and cannot receive the organ because the patient's immune system would reject it.

Reproduced from United Network for Organ Sharing. (n.d.). *How the transplant system works: Matching donors and recipients*. Richmond, VA: UNOS. Retrieved on December 27, 2013 from www.unos.org/donation/index.php?topic=fact_sheet_1

generate conflict over the weightings that cannot be resolved without considerable loss of momentum and goodwill. Allocation rules, therefore, tend to remain fuzzy. You can see this even in the UNOS description. Clearly, the concept of a social welfare criterion is not really operative there. Since then, UNOS has regionalized its networks and taken further steps to assure more minority representation among those treated.

► **Process Equity**

Not only must the outcome of the policy analysis and selection process—that is, the policy itself—be perceived as fair, but so should the process that produced it. For example, before settling on a policy, policy analysts must ask themselves whether all interested parties participated. If not, have the underrepresented or disenfranchised had their issues addressed by the analysts? It is not unusual for opponents of a policy to challenge the inclusiveness or integrity of the policy development process. Many governments have requirements for public hearings that are regionally representative in hopes that their process will appear fairer to those concerned and to protect against complaints of bias.

► **Influence on Society: A Broader Question**

With the medical care sector accounting for about a sixth of the economy, we have to wonder about the long-term effects of this much emphasis on health in our society. Areas of concern might include:

- A focus on illness and the possible loss of a sense of well-being
- Commercial pressure to overemphasize medical problems
- The economic and political power of the medical–industrial–university complex
- Ethical concerns about doing HTAs.

Is it healthy to be so conscious of our health? Is it healthy to be so stressed about our medical bills? Just about every sector of our society adds a health concern to its primary missions. Churches add health ministry committees and/or health-trained staff members. Schools take on more responsibility in terms of children's medications and their special needs; some have even established school-based health clinics. Tax revenues previously devoted to education and infrastructure get diverted to health care. These individual decisions may be virtuous, but they also signal a changing societal focus from toughing it out to trying to fix everything through health care.

Because health care represents an opportunity for employment growth, we see more and more commercial pressure to expand its markets—and its “top of mind” awareness for the public. In a way, this is good because it supports more innovation. However, it can create demand that siphons resources away from areas of greater need. Two issues that illustrate this are the explosive growth in direct-to-consumer advertising, especially on television, and the creation of new syndromes to justify new products.

Other countries, including New Zealand, have barred this type of commercial activity. An interesting classroom exercise would be to search out newly emphasized conditions and track down their medical and commercial origins. One place to start learning about these two issues would be to turn on your televisions and identify all the ads for drugs that one is now supposed to take to avoid previously accepted sources of stress or discomfort, including indigestion, constipation, erectile dysfunction, and insomnia.

Sterling offered a vision of a medically obsessed future in his novel *Holy Fire* (1996). In it, protagonist Mia Ziemann is 94 years old and still professionally active as a medical economist. In her world, wealth is all about how many medical procedures one can afford to extend one's longevity. The medical-industrial-research complex clearly dominates the entire economy, and young people are disenfranchised because employment opportunities rarely open through retirement or death. It is a disturbing vision, and yet one that is strangely familiar and highly plausible because Sterling drew on existing trends to shape his view of the late 21st century. Indeed, his major failing as a futurist may have been underestimating the rate of change. It also illustrates the importance of reexamining the final analysis in terms of the acceptability of tradeoffs and resulting outcomes given the social context.

► Double Checking for Interacting Policies and Contextual Change

A thoroughly analyzed policy must still be double checked against a scenario that represents what is likely to happen as the whole system responds. Alternative visions, which we call *scenarios*, need to be considered. A comparison of offsetting effects of two alternatives is a trade-off analysis. That is one step in comparing outcomes, but it involves only a very narrow picture in terms of unintended consequences and secondary and tertiary effects. Trade-offs must be considered throughout the analysis. After trade-offs have been evaluated and the alternatives narrowed down, the relevant scenarios will become apparent.

In the field of health policy, consensus is hard to come by, and even harder to sustain. Certain conclusions, however, seem inescapable:

- Without major interventions, health care will be near or at 20% of U.S. GDP over the next decade. Few governments are likely to tolerate more explosive growth in that large a portion of the economy, especially while contending with competing issues, such as international competition, immigration, terrorism, and government deficits.
- The moves of employers away from responsibility for paying health insurance premiums for workers, workers' dependents, and pensioners will continue.
- Insurance companies, if left to their own devices, will come up with less expensive products for individuals (with lesser coverage) to adapt to competitive markets. The exchange provisions of the ACA were intended to drive this process; so does the concept of "affordable access." This will be achieved by some easing of the requirements in a minimum essential benefits specification specified in the ACA.
- Consolidation will increase the possibilities for better coordination of care but risks further increases in the market power of providers.

- Communicable disease events in other countries such as the Ebola and Zika viruses will affect the United States much more because of rapid global transfers of people and agricultural products.
- The flow of physicians into the United States for training and their subsequent experiences practicing here will set up the potential for a competent workforce elsewhere, as we now see returnees performing hip and heart valve replacements in several countries at a third of the U.S. cost.
- Similarly, but more darkly, the illicit market in transplantable organs is growing rapidly, again through middlemen. There is also the international trade in prescription drugs, which is primarily attributed to Canada but is really wider and has great growth potential. Unfortunately, it opens up an entryway for counterfeit drugs as well.
- The perceived future insolvency of the Medicare “insurance” system will put pressure on the federal government, but that reckoning will likely be postponed because it will be beyond the reelection time horizons of most politicians. States will have to continue to deal with the burden of the working poor under Medicaid, which is why many Republican governors objected to the bills introduced in Congress to shift more of that burden to the states. In a recession, state revenues shrink just as potential enrollment increases.
- The states will continue to be a laboratory for implementing health policies, often through Medicaid waivers. The ACA allowed each state to set up its own coverage system within the framework of federal law. In the wake of the ACA’s passage, for example, Vermont began to develop its own system for universal coverage but found the cost burdensome. The post-ACA Republican Congress pushed for more flexibility and responsibility in the states. If the states fail, and that, in turn, threatens their fiscal integrity, Congress will be more likely to act. A number of states have experimented with turning their Medicaid programs over to private providers who accept a capitated payment. The results have been mixed. That trend will continue, especially if the federal portion of Medicaid payments are distributed as block grants.

The greatest uncertainties, therefore, are not on the financing side, but on the cost-control side. Many cost-oriented measures that seem to work over time have been described previously.

If the states find they must demand more stringent measures from Washington, or if there is another economic meltdown similar to 2008, then we are likely to see implementation of measures with more impact. They could include the following:

- Full transparency in healthcare pricing, including more competitive bidding and bundled prices for all stages of a specific medical condition
- Regionalization of care for complex or rare medical conditions, which would require greater patient travel but produce much less waste and better outcomes
- Stronger control of the referral process by primary care providers
- Other aspects of value-based competition (Porter & Teisberg, 2006) in the marketplace, including the following:
 - Integrated IT systems reporting bundled prices and quality performance for many more procedures and diagnoses
 - Shifting research budget priorities from developing and evaluating new treatments to evaluating methods already in use and innovating more in process improvement rather than in new product development (Chin et al., 2013)

► Trade-Offs

There are trade-offs everywhere one turns. **TABLE 11-1** lists a number of these, showing the two sides, the impacts of the current status quo, and some possibilities for responding to them. Dealing with them is unavoidable.

TABLE 11-1 Illustrative List of Trade-offs, Impacts, and Some of the Proposed Solutions

Trade-off	Versus	Current Impact	Proposed Solutions
First-dollar coverage	Catastrophic insurance	Many see payouts, motivating participation, but not enough money for both	Major medical option Means testing (spend-downs and Medicaid) Risk pools for high-risk patients
Selective underwriting	Community rating	Adverse selection Lifetime limits Prevention disincentives	Minimum coverage for all, with add-ons allowed
New drug safety and efficacy	Rapid deployment of effective technology	Current focus on drugs for continued use A lack of new drugs to fight acute illnesses	Sponsored development Reduced approval cycles Stronger postmarketing surveillance
Direct-to-consumer advertising	Health education campaigns	Higher drug costs Overutilization Misallocated research funds Disease mongering	Constraints on proportion of costs in advertising Redefining health Health education (counter detailing)
Liability for vaccines' effects	Vaccine development and production capacity	Inability to respond to surges in demand Failure to develop new vaccines and production methods	Limited liability Support for nonprofit producers Subsidies for standby capacity

(continues)

TABLE 11-1 Illustrative List of Trade-offs, Impacts, and Some of the Proposed Solutions (continued)

Linked electronic medical records	Consumer privacy	Medical errors Duplicate tests Slow responses to acute problems	Designing in privacy safeguards Firewalls
Need to train new health professionals	Desire to increase faculty incomes and cross-subsidize research	Overstressed learners Decreased quality of care	Better funding of supervision Reduced charges for learner-produced services
Investment driven by utilization growth	Certificate of need	Efforts to increase demand Duplication of services Reduced competition	Wider disease-specific catchment areas Capital rationing Global budgeting

► **Scenarios**

Major policy changes can send shock waves through the health system and out into the rest of society. Other actors respond and new waves come back. These moves are usually expected, but less prominent changes sometimes mount up into unexpected major problems. Sometimes a series of relatively minor changes ends up causing a perfect storm. One method for evaluating the interactive effects generated by the societal context is called *scenarios*. This involves developing a story that presents all of these changes and their impacts in context. Obviously, this is not something one does frequently, but it is worth looking when major changes are under consideration.

► **Working Out Your Own Scenarios**

One can push the policy envelope by considering extreme scenarios that suggest what the overall healthcare system might look like against the status quo. An event may be acceptable on its own, but its interaction with other events may result in an overall outcome that is totally unacceptable.

We have provided illustrative examples with five scenarios, three of which are extreme:

- A. Extrapolating current trends
- B. Extreme reliance on free market
- C. Extreme industrialization
- D. National economic crisis resulting in major tax code reform
- E. The Emanuel-Fuchs proposal

In **TABLE 11-2** we have provided, by way of illustration, event predictions for 14 changes that fit the first four of these scenarios. You and your peers certainly

TABLE 11-2 Building Some Scenarios for Cost Reduction

Change Scenario	A	B	C	D	E
Remove provider incentives to overutilize.	N	N	N	Y	
Major changes in staffing requirements.	N	Y	Y	Y	
Major reductions in provider and staff incomes and contribution margins.	?	Y	N	Y	
Reduced costs of malpractice coverage.	Y	?	Y	Y	
International competition reducing prices.	Y	Y	?	?	
Constraining treatment choices to those most effective and efficient.	N	N	Y	Y	
Encouraging labor substitution where process can be industrialized.	N	Y	Y	Y	
Empowering primary care providers to control utilization and self-referrals.	N	N	Y	Y	
Increasing supply of providers to levels in other countries.	N	Y	N	?	
Increased waiting periods and other devices to increase competition.	N	N	Y	Y	
Increasing use of IT to reduce waste and medical error without constraining competition.	Y	Y	N	?	
Global budgeting of hospitals and other institutions to ration capital, perhaps on a regional basis.	N	N	Y	Y	
Stripping away less essential services or covering outside health insurance system.	?	?	Y	Y	
Return of more voluntarism in the health sector.	N	Y	N	N	

Y = likely, N = not likely, and ? = not predicted.

Scenario A is the extrapolation of the current trends in the current system.

Scenario B is the case of an extremely strong move toward a free-market healthcare system.

Scenario C is the case of an extremely strong move in the direction of industrialization and corporate governance of health care.

Scenario D is the case of a major economic crisis that leads the country to a major overhaul of government programs, including health care, and major changes in the tax code, such as a shift from the income tax to a value-added or some other form of consumption tax.

Scenario E provides all citizens a voucher for basic health coverage, replacing current insurance and Medicaid and ultimately Medicare. It would be financed with a value-added tax and add administrative systems to oversee coverage, technology assessment, and quality measurement and to replace the current tort system for malpractice with administrative law.

come up with others, so we have also included a second table which enables you to build your own predictions, redefining the scenarios and then assessing the impact of your own set of policies and events.

Scenario A: The Status Quo Extrapolated

As you look down column A of Table 11-2, you see that not much new is happening to reduce costs. We might expect to see some responses as volume increases due to access improvements under the ACA, reductions in malpractice insurance costs, and international competition to reduce the prices of some procedures. There are also possibilities that as consumers and insurers find it harder and harder to pay their bills provider incomes will fall further, along with institutional contribution margins. Health information technology (IT) will move into place, but the motivation to share information may still be lacking. Thus, one might see some internal waste and medical error reduced, but very high investment costs with relatively little immediate impact on operating costs. Should major changes occur on the funding side, you might want to repeat this exercise with an added set of rows representing programmatic changes on the funding side.

Scenario B: Extreme Reliance on the Free Market

Here we would be likely to see little or no action to reduce costs, except to increase the competitive pressures from foreign competition and to increase the supply of providers. The primary argument for consumer-centered health care—that consumers will make choices to lower their own costs of care—is not addressed in this listing. Again, you might want to repeat the exercise with some events related to consumer-driven health care as an additional set of rows; however, with the lower resulting profit margins and professional incomes, more providers efforts to improve efficiency and compete on cost, resulting in staffing changes, better use of IT for scheduling and coordination, and more use of volunteers.

Scenario C: Extreme Industrialization

Under the corporate, industrializing scenario, the resulting oligopolistic firms will likely try to resist and/or seek protection from a number of cost-reduction pressures, such as foreign competition and increasing the supply of providers. At the same time, they will likely make a number of internal policy choices that limit provider options, restrict capital investment, constrain institutional budgets, and break jobs down into repetitive tasks doable by lower-paid personnel. They will also likely resist and lobby against measures to limit overutilization.

Scenario D: National Economic Crisis Resulting in Major Tax Code Reform

In this scenario, the nation is in severe economic difficulties, and radical change is in the air. There is near universal coverage. At the same time, there is a higher level of regulation affecting choices of treatments, capital availability, and staffing coupled with a defined minimum acceptable coverage. Resources will be very tight; thus, waiting lines will lengthen, and there will be pressures to put hospitals on fixed

budgets, reduce services that are not absolutely necessary, and concentrate specialized treatment capacities to increase throughput rates and effectiveness.

Scenario E: The Emanuel-Fuchs Proposal

In a one-page article in *Fortune*, Emanuel and Fuchs (2006) suggested “How to cure U.S. health care.” They offered a comprehensive five-part program, the centerpiece of which would be a healthcare voucher for every citizen currently younger than age 65. It would cover currently accepted levels of care with existing or new health plans. Each individual would then have a choice of 5–10 plans that would have to accept all comers.

The vouchers would be paid for by an earmarked 10% value-added tax that would be offset by replacing current employment-based insurance premiums, Medicaid expenditures, and individual and corporate tax deductions for healthcare premiums. Medicare recipients would be grandfathered (so to speak) under the current system, but those in the new program would stay with it past age 65, and Medicare would gradually be phased out.

To administer it, there would be a new system of federal and regional boards, much like the Federal Reserve System, to provide accountability, specify and modify the benefit package, and oversee technology assessment and quality evaluations. Malpractice cases would be assigned to a separate administrative system that would adjudicate and pay claims and oversee linkages with the licensure process. We have left column E of each table blank for you and your colleagues to assess and fill in.

Interpreting the Cost-Reduction Scenarios

As you can see, virtually every scenario in **TABLE 11-3** has elements to which U.S. patients would strongly object, and each is likely to generate strong opposition from one or more interest groups. That is why the most radical departures from current trends of more non-price competition and more industrialization are likely to occur only if there is a continuing economic crisis or a serious meltdown of the economic system.

► Where in the World?

A great deal has been written about biomedical ethics and the need for ethical review in HTA organization procedures. Process guidance can be found on the websites of the European Network for Health Technology Assessment and NICE. Comments on these review processes can often be found in the *International Journal of Technology Assessment in Healthcare*. However, the ethics of doing HTA in the first place and its impacts on society in general can be found separately. A number of interesting articles have appeared in the international journal *Poiesis & Praxis*.

► Conclusion

Values are the fourth leg of the analyst's desk, working alongside technology, politics, and economics to support or reject a policy recommendation.

TABLE 11-3 Your Exercise on Building Cost-Reduction Scenarios

Change Scenario	A	B	C	D
Remove provider incentives to overutilize.	N			
Major reductions in provider and staff incomes.	?			
Reduced costs of malpractice coverage.	Y			
Constraining treatment choices to those most effective and efficient.	?			
Encouraging labor substitution where process can be industrialized.				
Empowering primary care providers to control utilization and self-referrals.				
Rationing capital available to institutions.				
Increased waiting periods and other devices to increase competition.				
Increasing use of IT to reduce waste and medical error without constraining competition.				
Global budgeting of hospitals and other institutions, perhaps on a regional basis.				
No-fault health insurance, perhaps with tiered premiums				

Y = likely, N = not likely, and ? = not predicted.
Scenario A is the extrapolation of the current trends in the current system.
Scenario B is _____.
Scenario C is _____.
Scenario D is _____.

This chapter outlines various issues that a health policy analysis group must consider as they factor in concern for the general welfare of our society and our democratic traditions. This chapter also emphasizes the importance of putting event forecasts into a social context to highlight how policy choices might interact. The initial step is to outline and evaluate the trade-offs involved with each alternative. A number of examples of healthcare trade-offs are provided. Possible outcome scenarios can be developed from your own lists of trade-offs. Illustrative outcome scenarios were considered; but they were not examined exhaustively.

CASE 11: The Folic Acid Fortification Decision: Before and After

Before

At 5 P.M. on July 20, 1991, Dr. Godfrey P. Oakley, Jr., head of the Division of Birth Defects and Developmental Disabilities at the Centers for Disease Control and Prevention (CDC), took a phone call that, he says, “forever changed my life.” On the phone was a member of the British Medical Research Council (MRC) Vitamin Study Group, calling to share study results that would be published the following month in *The Lancet* (SerVaas & Perry, 1999).

The MRC study focused on women who had previously had a pregnancy in which the fetus or child had a neural tube defect (NTD), a birth defect of the spinal cord or brain. In April 1991, the MRC halted its study after almost 8 years because the data indicated that daily folic acid supplementation before pregnancy and during early pregnancy resulted in a 71% reduction in the recurrence of NTDs (CDC, 1991). It was no longer ethical not to provide folic acid to all of the women.

Dr. Oakley told the Saturday Evening Post Society in 1999:

Until that time, I thought that prevention of neural tube defects by taking folic acid supplementation was certainly no better than 50–50. If you really pressed me on what I would have thought the likelihood that a vitamin would have prevented neural tube defects, I would have said no more than 10% or 20%. But this was the first randomized, controlled trial designed and executed in a way that it proved folic acid would prevent spina bifida—not all but most of it. You could bet the farm that folic acid prevents neural tube defects.

SerVaas and Perry (1999, p. 3)

The study ignited a policy debate lasting several years. In the United States, the question of how to increase folic acid consumption in women of child-bearing years was contentious in itself, but the issue was complicated further by the controversy that surrounded implementation of the Nutrition Labeling and Education Act (NLEA) of 1990, which required the Food and Drug Administration (FDA) to regulate the health claims of food manufacturers. Among the tools policy makers used to sift through their options were three economic analyses: cost–benefit analyses produced in 1993 and 1995 and a cost-effectiveness analysis completed in 1996. All three predicted positive net economic benefits from fortifying foods with folic acid.

Background

NTDs are a class of birth defects that involve the brain and spinal cord. The most extreme form is anencephaly, in which all or part of the brain is missing. Another form is spina bifida, in which the spinal cord is not fully encased in the spine. In the early 1990s, there were about 4000 NTD-associated pregnancies per year in the United States.

Folates are a form of B vitamin that occurs naturally in leafy vegetable, legumes, nuts, and other foods. Folic acid is its synthetic form. Folates help cells replicate quickly.

Reports that increased folic acid intake could help prevent birth defects date back to 1965 (Hibbard & Smithells, 1965). Vitamin manufacturers learned how to add it to their supplements in the 1970s. The Saturday Evening Post Society launched

its campaign to promote folic acid supplementation in October 1982 in the wake of a report that women who took a vitamin supplement with 400 mcg of folic acid experienced fewer NTD-affected pregnancies. In the late 1980s, the CDC held a workshop to discuss emerging research linking folic acids to reductions in the numbers of NTDs, which renewed interest among supplement manufacturers.

Support for folic acid was limited, however. *Post* writers Cory SerVaas and Pat Perry (1999) were ridiculed for their efforts to promote folic acid. Scientific evidence was sparse, inconclusive, and based on observational epidemiological studies. Around the time of the CDC workshop, in fact, the National Research Council advised the FDA that it should lower the recommended daily allowance. A 1990 report by the Institute of Medicine's Food and Nutrition Board called taking vitamin supplements to prevent NTDs "unjustified." The National Academy of Sciences also discounted the link between NTDs and folic acid in its 1990 *Report on Nutrition and Pregnancy*.

In the midst of this debate, the NLEA skated through Congress and was signed by President George H. W. Bush. The act directed the FDA to establish standards for nutrition labels and define how certain terms such as "low fat" and "low cholesterol" could be used on food packaging. Congress also directed the FDA to investigate 10 specific health claims and develop language food manufacturers could use to convey any valid claims on their packaging. One of those claims was that folic acid reduced the risk of neural tube defects. The FDA was hard at work on its proposed rules when *The Lancet* published the MRC study (MRC Vitamin Study Research Group, 1991).

The study involved more than a thousand women in 33 centers across 17 countries. Researchers divided the subjects into four groups. Those in the first group received 400 mcg of folic acid. Those in the second group received the same amount of folic acid plus a multivitamin supplement. Those in the third group received neither the multivitamin supplement nor folic acid, and those in the final group received only the multivitamins. NTDs recurred 1% of the time when mothers received folic acid, with or without other vitamins, and 3.5% of the time when they received nothing or only the multivitamin supplement.

The MRC study did not settle the debate. It investigated women who had already had a NTD-associated pregnancy and therefore might be predisposed to having another; the results were not necessarily applicable to the general population. An FDA-contracted study released in November 1991 said it was not possible to positively conclude that folic acid prevented NTDs, but it criticized the Institute of Medicine and National Academy of Sciences reports. FDA draft rules issued later that month rejected claims that high-folate foods prevented NTDs, but it called its conclusion "tentative" and left open the possibility of further review.

The CDC, however, was less equivocal. At a 1991 conference on "Vitamins, Spina bifida, and Anencephaly," participants generally accepted the notion that women who were pregnant or might become pregnant required more folic acid. They began to wrestle with what FDA Commissioner David Kessler would call "one of the most difficult issues" of his tenure—exactly how to go about providing that folic acid.

Alternatives

There were only a few ways to ingest more folic acid at the time: consume more foods naturally high in folates, take vitamin supplements, take folic acid pills, or eat fortified foods. Each option brought its own challenges. Working against the natural nutritionists was the fact that natural folates have lower bioavailability than folic acid and break down during cooking. A typical U.S. woman ingests only 25 mcg of naturally occurring folates daily, and thus it is hard to envision women ingesting 400 mcg through dietary changes.

At the time, folic acid by itself was only available in a 100-mcg prescription formulation. The level of folic acid in most multivitamin supplements was so low that a woman trying to hit the 400-mcg target would have to ingest toxic levels of other vitamins. One option was to make folic acid more readily available in larger doses. Another was to reformulate multivitamins. Both options suffered from the same problem: Only periconceptional intake (ingestion before pregnancy and during the first few weeks after conception) is effective; yet roughly half of all pregnancies occur without early prenatal care. You would have to convince all women to take pills to supplement their folic acid intake throughout their childbearing years or risk missing more than half the pregnancies.

That left the possibility of fortification (adding a nutrient to food that does not otherwise contain it). This approach would reach everyone, including all women who are or might soon become pregnant. Food fortification and enrichment (increasing the levels of a nutrient already in a food) has a long history in the United States, beginning in 1924 with the decision to add iodine to salt to prevent goiter and other iodine-deficiency disorders. Vitamin D was added to milk in the 1930s (much later, vitamin A was added to low-fat dairy products). Flours and bread have been enriched with various nutrients—1938 through 1942 saw the addition of thiamine, niacin, riboflavin, and iron.

Four hundred micrograms of folic acid, however, struck many as a pharmacological dose. Best-selling longevity authors Durk Pearson and Sandy Shaw said fortification was equivalent to medicating competent adults without informed consent (Junod, 2006). Scientists worried about adverse reactions. One known issue was that folic acid at daily doses of a microgram or more could mask vitamin B₁₂ deficiency, particularly among the older population, prompting some scientists to worry that fortification would simply shift risks from developing embryos to adults with pernicious anemia (Gaull, Testa, Thomas, & Weinrich, 1996).

Support Builds

Fortification gained adherents after the MRC study appeared and supporters' ranks swelled as people became aware of two studies, as yet unpublished, conducted on women with no history of NTD-affected births. One, conducted in Hungary by Andrew Czeizel and his colleagues, showed benefits in the general population from consuming 800 mcg per day. Another, the "Werler study," studied women in Boston, Philadelphia, and Toronto. Although Werler and her colleagues recommended 400 mcg daily, they found that even 250 mcg offered some protection. In May 1992, Dr. Walter C. Willet argued in an *American Journal of Public Health* editorial that "fortification should be the long term goal" (Junod, 2006).

New Public Recommendation

In September 1992, the U.S. Public Health Service announced that women of child-bearing age should get 400 mcg of folic acid every day, a decision touted by the CDC but soft-pedaled by the FDA and the National Institutes of Health (Palca, 1992). The announcement did nothing to satisfy the two policy questions in the FDA's lap: whether to allow food labels to promote folic acid levels and whether to require that certain foods be fortified with folic acid.

Amid safety concerns, the FDA's folic acid advisory committee recommended against a folic acid health claim after a November 1992 meeting. In early 1993, the FDA, working under tight deadlines, adopted an NLEA rule that reflected this position; however, the committee expressed an interest in fortification, and it reconvened in April 1993 to look at a variety of unresolved issues, including

fortification. By October, the FDA had reversed its position, publishing a draft rule that would allow health claims for foods containing folic acid. There were many aspects of the rule, but its core was a provision that would allow health claims for foods that contained 40 mcg or more of folic acid per serving. On December 31, 1993, the rule became final.

The October 1993 draft rule had contained provisions about fortification, but fortification was not mandated as part of the New Year's Eve ruling. The FDA and its folic acid advisory committee continued to struggle with implementation questions.

Cost–Benefit Analysis (CBA)

Federal agencies are required to conduct regulatory impact analyses as part of rulemaking and starting in 1993 were required to assess expected costs and benefits of significant rules. FDA staff conducted a CBA and estimated that fortification with 140 mcg of folic acid per 100 grams of cereal grain products would prevent 116 NTD-affected births per year. This analysis tallied direct savings, notably medical care avoided, and estimated a saving of \$5 million for each case averted, resulting in economic benefits of \$651–\$786 million annually. The annual cost of fortification would be \$27 million, and thus the annual net economic benefit would be \$624–\$750 million (FDA, 1993).

In 1995, University of California researchers published a second CBA. They estimated 304 NTD-associated births would be avoided through fortification. Using a method that looked at lost productivity, they put the value of a case avoided at \$342,500. The economic benefit came in at \$121.5 million. From this, the researchers deducted not only the cost of fortification, which they put at \$11 million, but the cost of adverse events—namely 500 cases of neurological damage annually at a cost of \$16.4 million. Their calculations resulted in an estimated net benefit of \$93.6 million (Romano, Waltzman, & Scheffler, 1995).

These studies were before the advisory committee as it debated folic acid implementation, and they played a role in shaping the draft folic acid fortification rule published on March 5, 1996. A third analysis, published by the CDC that same year, was not influential. The CDC estimated 89 averted NTDs at a total benefit of \$16.1 million annually. That was largely offset by the \$11 million cost of fortification and an estimated \$350,000 in health costs related to 89 cases of neurological damage. That left a net benefit of \$4.7 million.

The FDA ultimately required fortification of cereal grain products with 140 mcg of folic acid for every 100 grams of grain. The rule went into effect on January 1, 1998. Between October 1998 and December 1999, the prevalence of reported cases of spina bifida declined 31%. Anencephaly declined 16%. Various studies over the years have put the total reduction of cases of spina bifida and anencephaly at 20%–30%, much more than were reflected in the three ex ante economic analyses. (Not all of the improvement can be attributed to fortification, because women of childbearing age can expect to get only about a quarter of their recommended intake of 400 mcg through fortified grains, and public education campaigns continue to promote consumption of folic acid through vitamin pills.) **FIGURE 11-1** shows the type of product labeling allowed with folate-fortified products.

The Results Come In

Studies in Chile, Canada, and South Africa also reported the effectiveness of folate fortification programs. Canada mandated 150 mcg per 100 grams in 1998; Chile required 220 mcg per 100 grams in 2000; South Africa mandated fortification in 2003 with different levels for various classifications of maize meal, wheat flour, and wheat

Nutrition Facts			Supplement Facts		
Serving Size 1 cup (228g) Servings Per Container 2			Serving Size: 1 tablet		
Amount Per Serving			Amount Per Serving		
Calories 250			Vitamin A		
Calories from Fat 110			5000IU 100		
% Daily Value*			Vitamin C		
Total Fat 12g			60mg 100		
Saturated Fat 3g			Vitamin D		
Cholesterol 30mg			400 IU 100		
Sodium 470mg			Vitamin E		
Total Carbohydrate 31g			30 IU 100		
Dietary Fiber 0g			Thiamin		
Sugar 5g			1.5mg 100		
Protein 5g			Riboflavin		
Vitamin A			1.7mg 100		
4%			Niacin		
			20mg 100		
			Vitamin B6		
			2mg 100		
			Folic Acid		
			400mcg 100		
			Vitamin B12		
			6mcg 100		
			Biotin		
			30mcg 10		
			Pantothenic Acid		
			10mg 100		
			Calcium		
			162mg 16		
			Iron		
			18mg 100		
			Iodine		
			150mcg 100		
			Magnesium		
			100mg 25		
			Zinc		
			15mg 100		
			Selenium		
			20mcg 100		
			Copper		
			2mg 100		
			Manganese		
			3.5mg 175		
			Chromium		
			65mcg 54		
			Molybdenum		
			150mcg 200		
			Chloride		
			72mg 2		
			Potassium		
			80mg 2		

FIGURE 11-1 Two labels (“nutrition facts” for foods and “supplement facts” for vitamin supplements) showing how regulations allow display of folate content.

Courtesy of the Office on Women's Health at the U.S. Department of Health and Human Services.

bread. Three Canadian population-based studies showed reductions in the incidence of NTD-related births of 50%, 54%, and 43%, respectively. A single study in Chile included measures of increased folate blood levels from fortification and reported a 43% reduction in the NTD rate within 6 months. A study in South Africa reported a 30.5% reduction. Differences among these studies included differences in fortification levels and differences in measurement and reporting regarding the inclusion or exclusion of stillbirths and terminated pregnancies.

The CDC published a before and after epidemiological study in 2004 that reported that surveillance-based population studies from 1995 and 1996 (pre-fortification) and 1999 and 2000 showed a reduction in the estimated number of NTD-affected pregnancies from 4000 to 3000. These results are summarized in **TABLE 11-4**. An editorial note in the *Morbidity and Mortality Weekly Report* noted that a 26% reduction was somewhat less than earlier studies had indicated and short of the national goal of a reduction of 50% (CDC, 2004).

In 2005, Grosse and colleagues at the CDC and several universities published an ex post economic study of fortification in the United States. It estimated 520 averted cases of spina bifida and 92 cases of averted anencephaly annually, which led to economic benefits per case of \$636,000 and \$1,020,000, respectively. That translated into \$425 million in economic benefits (\$146 million in direct costs, mostly medical) against an annual cost of fortification of \$3 million. The authors did not identify any documented adverse health effects from fortification.

TABLE 11-4 Estimated Average Annual Numbers of Spina Bifida and Anencephaly Cases Based on Prevalence per 10,000 Live Births from Surveillance Systems—United States 1995–1996 and 1999–2000

	Systems With Prenatal Ascertainment		Systems Without Prenatal Ascertainment		Fetal Deaths Elective terminations
Prefortification	Prevalence	No.	Prevalence	No.	No.
Spina bifida	6.4	2,490	5.1	1,980	
Anencephaly	4.2	1,640	2.5	970	
Total		4,130		2,950	1,180
Postfortification	Prevalence	No.	Prevalence	No.	No.
Spina bifida	4.1	1,640	3.4	1,340	
Anencephaly	3.5	1,380	2.1	840	
Total		3,020		2,180	840

For systems with prenatal ascertainment, estimated total pregnancies included live births, stillbirths, prenatally diagnosed cases, and elective terminations. For systems without prenatal ascertainment*, estimates included live births, stillbirths, and fetal deaths through 20 weeks. Fetal deaths and elective terminations were calculated as difference between systems with and without prenatal ascertainment. The numbers of NTD-affected pregnancies and births were determined as prevalence multiplied by the average total number of U.S. births during the respective periods, as derived from the U.S. National Vital Statistics System.

*Programs with prenatal ascertainment use specific case-finding techniques to identify prenatally diagnosed and electively terminated cases.

Modified from Centers for Disease Control and Prevention. (2004). Spina bifida and anencephaly before and after folic acid mandate: United States, 1995–1996 and 1999–2000. *MMWR Weekly*, 53, 362–365. www.cdc.gov/mmwr/preview/mmwrhtml/mm5317a3.htm. Accessed December 11, 2013.

This study used a 3% discount rate but noted that Office of Management and Budget guidelines called for comparisons using both 3% and 7% discount rates (at least one ex ante study had used a 5% rate). Of the \$636,000 savings per spina bifida case avoided, the study reported, \$279,000 was direct costs, mostly medical, with the rest apparently indirect costs for nonmedical caregiving. For the anencephaly cases, almost all of the \$1,020,000 in costs were indirect. In addition to the required 3% and 7% discount rate comparisons, the authors performed sensitivity analyses with only 80% of the cases avoided attributed to folate fortification and with a doubling of the fortification costs. Then they developed a worst-case scenario that assumed only

80% of the observed benefits attributed to the intervention, a doubling of the cost of fortification, and a \$25 million allowance for the potential effects of neurological damage from untreated anemia. This worst-case scenario still yielded annual net direct benefits of \$88 million after expenditures of \$6 million on fortification and an overall benefit of \$312 million. The authors observed that the benefits were exceptionally large and noted that “few public health interventions beyond immunization and injury prevention are cost saving.”

Hertkamp (2004) estimated that in Chile the fortification process cost approximately \$280,000 annually. For spina bifida, she estimated the cost of surgery and rehabilitation for each of the 110 cases avoided annually at \$100,000. She noted that bread is more of a staple of the Chilean diet. Commenting on the low priority given to folate fortification in most developing countries, she noted that prevalence data are lacking and that NTDs are not recognized as an important cause of morbidity and mortality.

A study in South Africa by Sayed, Bourne, Pattinson, Nixon, and Henderson (2008), estimated the cost of fortification at about \$176,000, a 41.6% reduction in cases resulting in 406 cases averted annually, and a cost for each case over the first 5 years of roughly \$744,000. This resulted in an estimated annual saving of approximately \$4.7 million. Dollar figures are only estimates because of conversion from rands and euros to U.S. dollars at 2017 exchange rates.

A meta-analysis of eight population-based studies found that folic acid food fortification reduced the incidence of NTDs by 46% (Blencowe, Cousens, Modell, & Lawn, 2010). The authors estimated that in low-income countries, fortification could reduce the number of neonatal deaths stemming from congenital abnormalities by 13%.

In 2015, Grosse and colleagues updated the 2005 CDC study. The results of the new cost-effectiveness analysis, published in 2016, reported significantly higher savings. One of the reasons was a higher estimate of the reduction in spina bifida births avoided—38% compared to the previous 23%—resulting in a 47% increase in the number of live-born cases avoided (stillbirths and anencephaly cases were excluded). The new estimate is based on data from five active surveillance systems with prenatal ascertainment. The second was a 59% increase in medical costs based on more complete data. And third, the newer study included—consistent with U.S. CEA guidelines that call for inclusion of societal costs—costs resulting from lost productivity for unpaid family caregivers other than the parents. The base-case analysis assumed 767 spina bifida live births avoided annually, with savings of \$791,900 per birth. That led to a saving of \$607.3 million (\$577,000 per birth for a saving of \$442 million if caregiver time costs are excluded). Deducting \$4 million in fortification costs, estimated savings came to \$603.3 million. The authors consider this a conservative number.

The authors, nonetheless, also performed a worst-case scenario, using a 5% discount in the number of cases, then further reducing the number because more than 4% of folic acid intake comes from breakfast cereals that are voluntarily fortified. This analysis reduced the number of avoided live births to 614. Similar adjustments reduced the cost per case to \$712,700 per case for total savings of \$437.5 million. A higher estimate of \$20 million for the cost of fortification produced a low-end estimate for net savings of \$417.5 million (\$298.8 million when caregiver costs are excluded).

New Concerns About Adverse Effects of High Dosages

The earlier concerns about the masking of B₁₂ deficiencies in the elderly have not been supported with reported cases. New concerns, however, surfaced by 2007 about the impact of high doses of folates on colorectal cancer (CRC). Animal studies suggested that high doses led to two conflicting effects. Doses higher than those normally

introduced by fortification protected against the onset of CRCs, but also seemed to stimulate the growth rates of existing neoplasms. These observations, reported by Mason et al. in 2007, were not followed up by randomized controlled trials, which led the British Scientific Advisory Committee on Nutrition (SACN) to report in January 2008, “The evidence for an association between folic acid and increased or reduced cancer risk is equivocal.”

A 2009 report on two Norwegian clinical trials with high doses of folic acid and vitamin B₁₂ in patients with ischemic heart disease showed significant increases in overall cancer rates (Ebbing et al., 2009). However, folic acid was considered to have a preventive effect toward heart disease and stroke (Wang et al., 2007).

Vollset and colleagues (2013) conducted a meta-analysis that examined 13 randomized trials of people taking folic acid supplements. The studies were all conducted before 2011 and combined they enrolled almost 50,000 subjects. No significant effects from folic acid supplementation on the incidence of cancer of the large intestine, prostate, lung, breast, or any significant site were found. The authors noted that folic acid dosages in the study were an order of magnitude greater than intake from fortified food. Another meta-analysis published the same year by Quin and colleagues (2013) found that folic acid supplementation has no significant effect on total cancer incidence, CRC, prostate cancer, lung cancer, breast cancer or hematological malignancy, but reduces the risk of melanoma.

A CDC-sponsored study (Crider, Bailey, & Berry, 2011) that reviewed concerns about adverse effects found no evidence that fortification masked or exacerbated neuropathies related to anemia. It found no increase in cancer; no definitive studies showing adverse health effects from exposure to unmetabolized folic acid; and no evidence that folic acid was leading to epigenetic changes in human DNA. The researchers concluded, however, that future hypotheses addressing concerns about epigenetic changes would have to be explored. Crider and colleagues noted that careful monitoring of existing and proposed programs was needed to allow the scientific community to determine the blood folate concentrations required for NTD prevention as well as to evaluate and respond appropriately to concerns that may arise.

Some studies have demonstrated a connection between folate supplementation and an increase in breathing disorders such as asthma. Such studies have produced inconsistent results, however, and were not specific to fortification. (Barua, Kuizon, & Junaid, 2014).

Fortification in Other Countries

To date, more than 50 countries have adopted mandatory fortification, but many of the programs have yet to be implemented (Crider et al., 2011). A number of English-speaking countries are considering a mandate at the recommendation of food safety agencies, but for a variety of reasons only Australia and South Africa have completed its implementation.

Australia and New Zealand

Mandatory fortification of flour for bread began in Australia in September 2009. Voluntary fortification had been allowed for more than 10 years. Food Standards Australia New Zealand (FSANZ) recommended mandatory fortification of bread in 2007 following an extensive review begun in 2004. The recommendation triggered a public comment period followed by approval of the Australia and New Zealand Food Regulation Ministerial Council, which is made up of the food and health ministers of the two countries. A 12-month phase-in period was to follow. FSANZ (2009) anticipated that a mandate would avoid about 14–48 of the 300–350 NTD-affected pregnancies

that occur annually in Australia. New Zealand, however, put the plan on hold after a media campaign against fortification by bakers increased public concern about the proposed fortification. In August 2012, the New Zealand government announced that fortification would remain voluntary, with the goal of having half of all bread products fortified eventually.

Ireland's Plans Derailed

In 1990, Ireland had one of the highest rates of NTDs in Europe, with 10–15 NTDs per 10,000 live births. Voluntary fortification and nutritional education were only marginally successful, so in 2002 the Department of Health turned to the Food Safety Authority of Ireland (FSAI). In 2003, FSAI recommended fortification, and the health minister established a National Committee on Folic Acid Food, which consulted the parties involved and in July 2006 recommended mandatory fortification. An Implementation Group on Folic Acid Fortification was formed with the expectation of final implementation. In 2008, however, the implementation group reported that folic acid intake by women of childbearing age had increased 30% as a result of voluntary fortification, and the incidence of NTDs had been reduced to 9.3 per 10,000 live births. Although Ireland's daily folic acid intake of 90 mcg is well below international targets and other countries are achieving a much lower rate of NTD-associated births (5–6 per 10,000 live births), the group concluded that mandatory fortification would be of limited public health benefit. FSAI reaffirmed this position in 2013 based on ongoing population studies of blood folate levels.

A 2016 FSAI report, however, stated that low levels of folic acid remain a reason for concern in Ireland. The report offered two options: (1) mandatory fortification of bread or flour, and (2) continued voluntary fortification along with education about supplementation. Despite advancing both options, the tone of the report strongly supported mandatory fortification. It noted that there is strong evidence to support the effectiveness of the first option and only weak evidence to support the second option. It also said mandatory fortification was consistent with European Union guidelines, and that “mandatory fortification of foods with folic acid in the USA since 1996 has shown that there is no evidence for the occurrence of adverse health effects in any population group following introduction of fortification (FSAI, 2016).

Meanwhile, Back in the United Kingdom

The U.K. Food Standards Agency (FSA) first discussed the issue of fortification in 2002 amid concerns about possible risks to the health of older people. Then SACN, an independent expert panel, reviewed the evidence and tracked the emerging science. SACN issued a November 2005 draft report (the final report was released in December 2006) that recommended mandatory fortification of flour, and in April 2006, the FSA agreed to consider four options for improving folate levels in young women:

- Do nothing.
- Increase efforts to encourage young women to change their diets and take supplements.
- Further encourage voluntary fortification of foods.
- Implement mandatory fortification of “the most appropriate food vehicle.”

The FSA consulted with consumers, stakeholders, and industry in early 2007. A SACN briefing paper issued in January 2008 recommended fortification of flour but also called for the establishment of baselines for folic acid intake and folate concentrations in the blood so future surveillance programs could identify trends. It also called for standardized testing protocols (SACN, 2008).

In 2009, SACN concluded that there was not sufficient data to support concerns about cancer, and it reconfirmed its recommendation in support of mandatory fortification. The four chief medical officers for the United Kingdom then recommended mandatory fortification to the four health ministers. A request by Food Standards Scotland for SACN to review its recommendations again—Scotland has been considering mandating fortification unilaterally—resulted in a July 2017 report once again recommending mandatory fortification.

Conclusion

Mandatory fortification with folic acid is established practice in the United States and voluntary or mandatory fortification has been introduced in more than 50 countries worldwide as a strategy to help women of childbearing age increase their intake of folate. In countries where it has been implemented, it dramatically reduced NTD-associated births. So far, no European country has adopted *mandatory* folic acid fortification. Voluntary fortification is widely practiced in the European Union under regulation 1925/2006/EC, which allows the fortification of all foods except unprocessed foods and alcoholic beverages. Foods fortified with folic acid are widely available on the European market, except in Sweden, where fortification is not practiced, and in Denmark and Norway, which require approval. The range of product categories that are fortified on a voluntary basis includes dairy products, breakfast cereals, cereal bars, fruit juices, fat spreads, bread, and beverages. Further use of fortification around the world, including in poorer nations, could dramatically reduce the global burden of NTDs.

► Discussion Questions

1. Describe the authorizing environment for folic acid fortification in the United States.
2. Why would a decision based on strong British studies influence the U.S. decision to fortify cereal products but yield a delayed response in the United Kingdom?
3. Contrast and compare the variables and their values used in the before and after CBA studies. What conclusions would you draw from them about that approach in this case? In general?



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CHAPTER 12

Implementation Strategy and Planning

Responsible health policy analysts pay attention to how their recommendations can best be implemented. No matter how good the policy analysis process, the selected policy will fail unless it has a workable process behind it and it elicits the support and full attention of local implementers. The initial failure of the Affordable Care Act (ACA) federal exchange has been attributed somewhat to its scale, but mostly to a failure by policy makers to plan for and manage its implementation. It would be easy to compile a substantial list of the basic requirements of effective program implementation that were violated during that launch, which is examined in detail in the case study that accompanies this chapter.

Dopson and Fitzgerald (2005) edited a book about the “implementation gap” for evidence-based health care in Britain’s National Health Service, where centralization of management and financing would seem to create a fertile environment for change. They pointed out that failure to achieve goals that had been carefully planned has a long history in public policy analysis. Some implementation failures can be attributed to political bargaining at a micro level; however, empirical studies suggest that implementation is a separate stage of the change process (Torenvlied & Thomson, 2003). Greater recognition of this problem has led to the development of the field of *implementation science*.

The policy process must consider how to implement a policy decision and how to promote that implementation. The Australian government considered implementation so important that it established a Cabinet Implementation Unit in late 2003 within the Department of the Prime Minister and Cabinet. The White House established the U.S. Digital Service after the problems launching the health insurance exchange.

► Levels of Implementation Failure

There are many reasons a policy might be adopted but never implemented. Sometimes plans are unrealistic. Sometimes there is a failure to execute critical elements of a plan, whereas other plans fail before subtler (all too human) resistance. Still

others are simply overtaken by events—wars, budget crises, changes of government, competing technologies, or differing ideologies (Lazar, Forest, Lavis, & Church, 2013). Many social programs started at the federal level with great effort have failed in the field over time. McLaughlin (1984), noting how often legislators develop and adopt social programs that miss their intended objectives, observed:

It is sometimes tragicomic to read the laws of entitlement, such as the one guaranteeing equality of education for the handicapped and exceptional children, and then see how they work out in group homes, classrooms and clinics. Some have blamed public parsimony. Others would cite professional narrowness, individual insensitivity, and bureaucratic myopia. But much of the blame falls on the haste with which public policy is adopted, implemented, and then displaced by new policy.

McLaughlin (1984, p. 83)

The mental health “deinstitutionalization” effort of the 1970s is a good example of failure to execute. After early antipsychotic drugs became available, the inpatient populations of psychiatric hospitals began to decline. Community mental health centers created to serve people released from mental institutions were up and running in many areas. Medicaid and Medicare provided some funding, but despite promises to transfer budgetary resources that had previously gone into psychiatric hospitals to these centers and other community organizations, the funding never arrived in sufficient amounts to provide both direct care and supportive services such as housing and employment training. A new population of homeless individuals appeared in our cities, and the terms *revolving door* and *bag lady* entered the public health vocabulary. Hospitals and community centers did not provide coordinated care; they were separate systems that competed for the same resources. Furthermore, many community mental health centers drifted away from the goal of supporting the deinstitutionalized mentally ill and began serving a much broader spectrum of patients. As Torrey (1997) noted, “Once trained, however, the vast majority of these professionals decide to provide psychotherapy for people with mental health problems rather than treat people who were mentally ill” (p. 185). It is not easy to keep well-intentioned policies from going off the tracks. As the saying goes, “The devil is in the details.”

► Implementation Planning

Implementation planning arrives early and stays late. It is important to involve all stakeholders, including the implementers. An Australian Cabinet Implementation Unit process guide, *Guide to Preparing Implementation Plans* (2006), refers to two implementation processes: (1) implementation assessment, which contributes to the documentation that accompanies the submission of a proposal to the cabinet, and (2) implementation planning, which continues that process in much greater detail once the decision is final. **FIGURE 12-1** outlines the stages of the Australian cabinet’s implementation planning process. Several topics discussed in the next section—scope, funding, risk management, and schedule (which is discussed as part of the work breakdown)—may also be inputs into the strategic decision-making process, alongside program objectives, outcomes, and governance considerations.

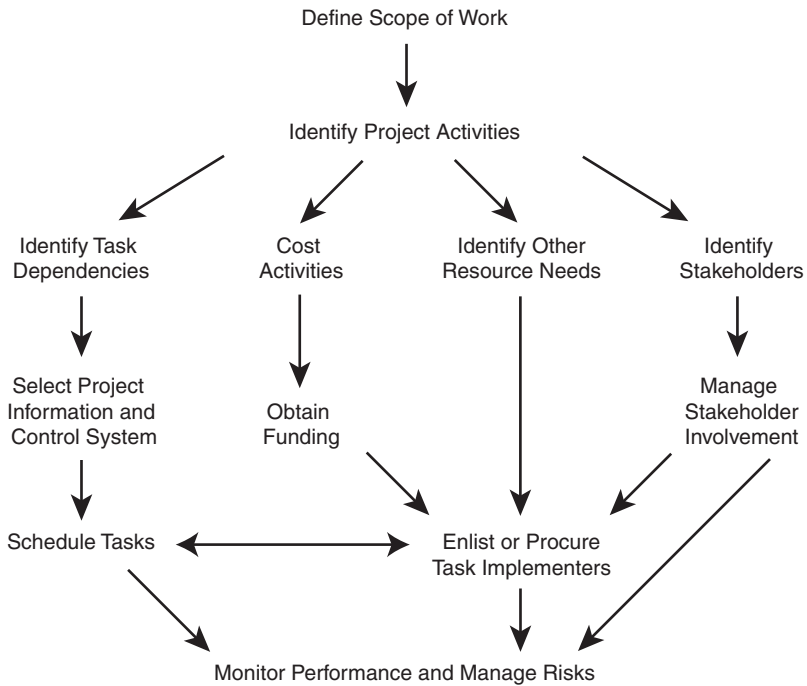


FIGURE 12-1 Stages of implementation planning.

Scope

Policy making is an art, and the “who, what, where, when, and how” may change either subtly or substantially. There is a saying that the camel was intended to be a horse, but it was designed by a committee. Note the result: the camel is highly adapted to its environment, but it is something quite different from a horse. Consequently, it needs a very different management approach to operate it and evaluate its performance.

In the late stages of implementation, it is important to establish lines of authority and accountability for implementation and to consider how the effort interacts with other initiatives and programs already under way. One senior person has to be supportive and fully in charge of any major change effort. It is also important to consider whether the new policy can be implemented effectively using existing organizational structures, management systems, and funding approaches. If major cultural shifts are required, a new organizational unit may have to be formed to implement it.

Work Breakdown

This stage is a detailed analysis of the tasks generated by the implementation requirements of the policy. Political decision-makers are likely to establish important dates (milestones) for implementation, dates that may or may not be achievable. Then the implementing organization must get to work:

- Identifying the tasks to be performed, such as submitting a detailed budget, hiring personnel, finding office space, issuing rules and regulations, establishing advisory committees, and specifying reporting requirements.

- Identifying the units and individuals responsible for each task and gaining their commitment to completing the task within a specific time period.
- Establishing reporting responsibilities for the status of each task.
- Including coordination tasks as well as intradepartmental task.
- Developing a master schedule and an estimated time of completion for the project along with mechanisms for monitoring progress.

Depending on the complexity and urgency of the project, project implementation staff may choose to use any one of a number of project management techniques and their associated software to show the project duration and whether the original target is likely to be met. If it would not be met, implementation planners may decide to “crash” the project in order to remain on schedule. This process works best when the estimate of how long an activity will take comes from the individual or team that will actually perform that activity. This encourages realistic estimates and the team’s commitment to meeting its own estimates. For example, the Obama administration shortened the testing period for the insurance exchanges, but failed to have top-level oversight of the process or to test whether this mission-critical system was up to its assigned tasks.

Funding

Few proposals get considered without a cost estimate attached; however, there are additional steps related to funding once that estimate is finalized. Congress authorizes many more initiatives than it funds. For example, in 1998, Congress passed and President Clinton signed the Ricky Ray Hemophilia Relief Act, which authorized a compassionate payment of \$100,000 to each hemophiliac infected between 1982 and 1987 from contaminated blood products, or to their families if they had died. This was to compensate for lax government control of the blood supply. The authorization bill did not include the funding required, estimated to be \$750 million. Only after considerable effort by advocacy groups was \$75 million appropriated in the fiscal year. Additional funding was approved incrementally over the next few years.

Risk Management

The Australian *Guide to Preparing Implementation Plans* states the following:

By understanding the potential risks which may affect the implementation of a policy measure, agencies can reduce the likelihood or consequence of “unpleasant surprises” that may jeopardise the achievement of policy objectives.

Cabinet Implementation Unit (2006, p. 23)

It suggests that likely risks include:

- Unclear objectives and deliverables
- Unrealistic schedules
- Shortages of key resources—funds, people, or equipment
- Lack of infrastructure and supports
- Lack of agency internal capacity

Whatever the risk, the planning process needs to assess the likelihood that it will occur, its severity and impact, how to mitigate it, and who is responsible for preventive measures. It also needs to address monitoring and how to initiate needed actions. In situations of high uncertainty and high impact, such as national security intelligence, analysts are now required to add estimates of their degree of certainty regarding their findings.

One way for implementers to ensure that they will be able to respond to changing conditions is to keep the planning flexible. How does a plan stay flexible?

- By not getting too detailed too early
- By not planning to use the available resources up to their limit
- By checking in with all implementers from time to time to see if anything has changed
- By having periodic reviews to allow other parts of the plan to adjust to the “as-built” changes that naturally occur
- By making sure the staff knows from the start that there will probably be changes

This does not mean that the planning is incomplete. However, it does mean that there are contingencies built in and that the people on the job are prepared to respond without fear to unexpected situations as they arise.

Stakeholder Engagement

Implementation must include a review of the stakeholders, including the following:

- Who needs to be kept informed?
- Who needs to participate in what detailed planning activities?
- Who can be an opinion leader or champion of the program?
- Who needs further training and motivation?
- Who can be an enabler?
- Who can be a blocker and needs to be co-opted?

Then, as **TABLE 12-1** illustrates, implementation planners must identify the type of commitment needed from each implementation stakeholder, how to secure it, key messages that need to be delivered, and who is responsible for maintaining the relationship. Decisions will need to be made about how to deliver the message and maintain the relationship. Should it be delivered personally, by email, through the media, through a representative, and so on? After those coordination and communication tasks are identified, they can be scheduled and assigned.

It often is best to approach this work with stakeholders as meaningful consultation and collaboration. This can make initial implementation easier because stakeholders who feel that their ideas have been considered and their concerns addressed are less likely to try to subvert the process. Chances of success over time are greater if the people and organizations affected by the project recognize its value and are invested in its success. There will be times, though, when an implementing agency is given a mandate to implement a policy change on a specific timeline over the objections of important stakeholders, and this may force the implementation planners to develop a policy for “stakeholder management.”

Most techniques for evaluating political feasibility involve collecting information from stakeholders. Political feasibility inquiries can yield a lot of information

TABLE 12-1 Communications Strategy Tool

Key Stakeholder	Desired Commitment and Strategy for Securing Commitment	Key Messages That Need to Be Delivered	Responsible Officer

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that is important during the implementation process. If Delphi processes, key informant interviews, or other stakeholder outreach efforts were part of a feasibility evaluation prior to policy adoption, then implementers should review that material. Even if they feel they know the material, it could be helpful to turn to it with a fresh eye. A stakeholder concern that was not enough to prevent a policy’s adoption, for example, could frustrate or even derail that same policy’s implementation

Individuals tasked to influence the policy implementation process, including lobbyists, public relations firms, and think tanks, will often go further and analyze the actors and their influence on implementation through a social network analysis (Hoppe & Reinelt, 2010; Ibarra & Hunter, 2007). Kibbe and McLaughlin (2004, 2008), for example, used network analysis to confirm how Internet approaches were systematically underrepresented in the national standard-setting processes for health information technology implementation.

Resources, Including Sourcing and Procurement

Key resource needs should have been identified by this point in any process, because the funding requirements depend on the resources needed, especially personnel, equipment, and support systems. There may also be other, less tangible resources that are mission critical, such as office space, computers and communication equipment, loaned personnel, and contractor personnel with special skills.

Assuring quality of contractor personnel throughout the project is an important management task. Contractors frequently assign their most competent personnel to preparing and marketing a proposal, but then substitute underutilized, less experienced, less skilled, or less costly personnel when the work gets under way. Implementation managers must stay alert so they get the quality people and services for which they originally bargained.

Quality Assurance

The policy proposal usually specifies the quality and quantity of outcomes anticipated, and it may refer to the process and outcome measures to be used; however, implementation planners may still have to develop, validate, and install measures and measurement systems to monitor progress and suggest improvements as the implementation proceeds. When planning quality assurance measures, analysts should also assure collection of the kinds of data, information, and documentation needed to conduct an evaluation after the policy has been implemented.

When it comes to sequencing these implementation activities, there is no magic order. After the proposal stage, many activities may have to go forward on parallel tracks, with the implementation team making adjustments as they encounter problems and opportunities. New technology or results from ongoing studies may lead to program changes after the policy makers have completed their work. If this happens, the implementation team should inform the policy makers of the change so that there will be no big surprises when the effort is being reviewed (e.g., during a site visit). The list of implementation activities is often quite detailed, and it is up to the implementation team to decide how much planning time to invest in each activity. The systems necessary to plan and monitor some of these stages may be in place already. Budgeting and scheduling functions are basic management activities everywhere.

► Setting Up to Succeed

Health care is going through a transition from a professional model that emphasizes individual responsibility, autonomy, and accountability to an organizational one, preferably one where organizational learning and transformation are norms. To succeed at implementation in this environment, several things need to happen, including the following:

1. **Shared responsibility is accepted.** Team leaders and members at all professional levels must come to share overall responsibility while still accepting individual responsibility for assigned tasks. The industrialization of health care often means that treatment processes are carried out by multiple actors whose actions must be coordinated. Involving the implementers early enhances the likelihood of successful implementation because it builds their commitment to the plan.
2. **Leadership takes place at multiple levels, but an adult must be in charge.** The team involved in developing a policy must have facts about how the system works at the operational level as well as the strategic level. Many developing countries still suffer from the system developed by the British Colonial Office for managing the Empire. This consisted of mostly locals organized into two cadres: (1) the civil service developed policy, and (2) another organization (the health service, for example) implemented it. A small group of British officials controlled the flow of information from one to the other. The policy analysts produced brilliant reports that circulated in files held together with red ribbon (called *red tape* by the English), but these analyses often proved unworkable for implementers in the field. It was an efficient way to use a small expatriate

THE ORIGIN OF A POLICY

The labor, delivery, recovery, and postpartum nursing team at a community hospital was reviewing its procedures and cost items to become more competitive in its city. One policy that puzzled team members was sending all placentas to the pathology laboratory after delivery. Team members kept asking around for the source of that rule and found that it stemmed from an incident many years earlier. At the time, the daughter of the chief of obstetrics was having a difficult delivery and encountered a problem that might have been handled more effectively if the placenta from her delivery had been saved and analyzed. The angry chief thoroughly chewed out the OB nursing staff for not saving it. To avoid such a confrontation in the future, the nursing supervisor instituted the rule that all placentas would go to pathology. Both the chief of obstetrics and the nursing supervisor had long since retired, but the rule lived on. The three obstetricians currently practicing at the hospital developed a set of criteria for sending placentas to the laboratory. This new rule decreased the number of pathology reviews of placentas by 95%, resulting in substantial cost savings to the patients and the hospital.

- staff, but it did not necessarily produce effective delivery systems. All too many policies and plans have become “shelf art,” sitting there unused because implementers did not see them as practical or relevant.
3. **People understand the technical core and the professional and public processes, values, and mission of the organization.** Policies and processes rarely bolt out of control. More typically, they silently drift away from optimal as individuals and groups make incremental local adjustments in response to local events, stimuli, and experiences. The example in the box on pathology review of placentas illustrates how one process moved away from its intended objectives. Planners and implementers must be aware of the full context of the issue being addressed.
 4. **Expectations are managed.** Some participants will be optimistic and expect too much, whereas others will be pessimistic or cynical and expect too little. Wilson and McLaughlin (1984) suggest that one needs to work out a psychological contract with the planning participants, one that recognizes the scarcity of resources and directly addresses the WIIFM (what’s in it for me?) question. **BOX 12-1** outlines some of the factors such a psychological contract might consider. In essence, the contract would specify what each party gets from the exchange beyond monetary considerations.
 5. **Planning is continuous.** The healthcare environment is continuously changing. There are many more opportunities and challenges than most organizations can take on at any one time. Through functional and cross-functional teams, managers involve providers and others in setting priorities, developing and evaluating alternatives, and meeting the many new challenges.
 6. **Orientations are prospective rather than retrospective.** Dopson and Fitzgerald (2005) and Senge and colleagues (1994) observed that behaviors and beliefs take time to shift and that changing them required both abstract reasoning and experiential reinforcement. It is often easier to do things the way they have always been done. Looking to the past for precedent is usually a rational step in assessing a situation, but there is little

BOX 12-1 The Psychological Contract

The Employee Gets

Performance standards that represent realistic trade-offs between funds, personnel, schedules, and service levels

Personal courtesy and respect

A supportive environment

Meaningful and purposeful work

Reasonable conflict and tension levels, mediated by clear standards for priority setting and evaluation of work

Opportunities for personal development

Profession and organizational recognition for good work

Security, as long as funds are available

Process for psychological contract change processes that reflect changing resource conditions

Scarce resources

The Organization Gets

An honest day's work, at least

Loyalty to the organization

Initiative, especially in resource use

Job effectiveness and efficiency in meeting overall organizational goals

Flexibility and willingness to wear multiple hats under tight staffing

Acceptance of reasonable trade-offs among professional norms and organizational needs

Participation in psychological contract change processes that reflect changing resource conditions

Reproduced from Wilson, M. P., & McLaughlin, C. P. (1984). *Leadership and management in academic medicine* (p. 310). San Francisco, CA: Jossey-Bass.

assurance that what an organization has been doing will be effective in the future. "Transformational leadership continually brings forward the vision of the future organization and indicates how the organization can get from where it is in the present to where it wants to be in the future" (Upshaw, Steffen, & McLaughlin, 2013, pp. 293–294).

7. **Performance is assessed and rewarded.** Effective policy change requires that the organization be prepared to commit real resources and provide support mechanisms for recognizing creativity and innovation. Personnel evaluation processes and procedures must be in place to encourage implementers, as well as policy makers, to seek new ways to get things done and to prepare for the changing future. The evaluation must include performance on intergroup tasks and teams as well as day-to-day tasks (Zmud & McLaughlin, 1989).

► That All-Important Start

Starting right is crucial. If a team is involved, especially a multidisciplinary one (which would be the case for just about anything clinical), plan to offer experienced leadership throughout the team formation cycle, which usually progresses through the following four stages:

1. Forming
2. Storming
3. Norming
4. Performing

Forming

The implementing team needs to understand the factors behind the policy change and how it links to the institution's strategies. The team's initial efforts should be well supported and include tasks that are carefully chosen to build a shared experience of success. At this stage, the team must also consider whether it includes representation of all the important implementers.

Storming

Because healthcare settings are complex and have many built-in tensions, team members often start out blaming, finding fault, or expressing distrust of others. The team members have to be kept focused on their task until they begin to understand it and accept each other's viewpoints.

Norming

The group can then set up norms of operation, detailing, for example, how decisions are made and how work is allocated. Do we vote? Do we have to have a consensus? What do we have to clear with higher authority? Do some professional groups have veto power?

Performing

After the barriers at each of the preceding stages are dealt with, the team can get on with its assigned tasks effectively or, having addressed them and failed to get over them, seek further guidance or facilitation.

► Providing for Periodic Reviews

Major change projects often require that multiple teams go forward in parallel. Since one or more may veer off onto its own path, it is important to stage periodic review sessions in which each team reports what it is doing, compares its progress against measures such as budget and schedule, and shares what implementation

barriers it is experiencing. This process enables management to assess overall progress and adjust the plan where necessary. It also allows individual team leaders to see where they need to interface with other teams to meet their objectives and how to adjust their efforts to reflect what they learn at the review. The larger and more complex the project, the more important these periodic reviews become. Lack of effective periodic reviews was a serious problem in the case of the ACA federal exchange startup.

► Implementing Policies That Affect Clinical Operations

A problematic area for implementing policy is getting professionals to change the way they carry out professional tasks. This has been a topic of concern to those who study quality of care, provide continuing professional education, or sell pharmaceuticals and other supplies to the health industry. The move toward greater use of evidence-based medicine is a case in point. For example, Dopson and Fitzgerald (2005) provided a meta-analysis of studies that looked at implementing evidence-based practices in the U.K. Health Service. The concluding summary by Ferlie (2005) emphasized the role of leadership, organizational support, and the differences between *knowledge* (i.e., information) and *practice*:

In the best case examples, there appeared to be a circular relationship between research evidence and experience—they reinforced each other and were woven together. At other times, there was a tension between craft knowledge and formal evidence. It should be remembered that clinical practice contains elements of judgment and tacit knowledge more reminiscent of craft skills than traditional conceptions of science. It may be unwise to force a stark choice between the two modes (experience or science), but there may be a need to balance both

Ferlie (2005, p. 188)

► The Postmortem

Assuming that the group doing the analysis and planning will do this kind of thing again, it is critical to review the group's own performance. This post-project analysis should be done by the planning group if the project is small or medium-sized or by an independent evaluator if it is large or mission critical. Evaluation should study two aspects of the project: process and outcome. Process analysis should take place relatively quickly after the project is concluded and should ask this question: "If we had to do this analysis again, what would we do differently?" Because this can be threatening interpersonally, a little humor would not hurt. The list of phases of a project in **BOX 12-2** can be used to lighten things up a little and may also point out opportunities for improvement.

BOX 12-2 A Little Humor Might Help: Six Phases of a Project

1. Enthusiasm
2. Frustration
3. Panic
4. Assigning blame
5. Punishing the innocent
6. Rewarding the noncontributors

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Clearly, one should avoid using the postmortem merely to assign blame and must make sure that the accolades and rewards go to all those who really contributed to the process.

Outcomes assessment has to wait until the outcomes are evident. It can only be justified for major policy proposals that were actually implemented; otherwise, a second type of analysis might be needed to focus on what might have been done differently to get that proposal implemented. The failure to implement might be primarily due to outside factors. Either way, one should be able to go back through the analysis and see how well the group defined the situation, assessed the technology, outlined the economics, adjusted to the political and values issues, planned the implementation, and presented the product to the decision-makers. Such a review is a key to improved organizational learning about policy analysis.

Where in the World?

Issues of implementation are so pervasive in large-scale programs that the literature is extensive but widely scattered. The journal *Implementation Science* is one place to start even though many of its articles are not specific to the health sector. Most of the relevant literature can be searched under project management and organizational change processes.

► Conclusion

There is no silver bullet that will make implementation easy; but the elements of successful implementation are well known. They involve top management consideration and support. They require an understanding of the social context in which change takes place. Would-be implementers must focus on the players, their roles, their barriers to action, and the roles of leadership and influencers in bringing about change. From an academic viewpoint, all of these should be worked out so that strategic policies are not derailed by the implementation details; however, decision-makers are usually not comfortable with that level of detail as they deal broadly with a wide range of issues. They are very uncomfortable with surprises after the fact, but they also do not want to be bogged down reading the fine print. That is just a managerial reality. Perhaps the best answer is to turn to the experienced presenters on the team who know the behavior patterns of key decision-makers and ask for their assessment of the appropriate level of detail to present at each stage.

CASE 12 : The Troubled Launch of HealthCare.gov

In January 2016, the Office of the Inspector General (OIG) of the U.S. Department of Health and Human Services (HHS) issued a “Case Study” entitled “HealthCare.gov: CMS Management of the Federal Marketplace.” The launch of HealthCare.gov, on October 1, 2013, had been a public relations disaster. *Forbes magazine* observed: “The Patient Protection and Affordable Care Act, better known as Obamacare, will probably be remembered as President Obama’s most important domestic policy initiative. However, inept federal management of the HealthCare.gov website that is central to implementing Obamacare has left many users with a negative first impression of the program” (Thompson, 2013).

That negative first impression continued long after the site was salvaged.

The Case Study

The OIG investigation covered 5 years of development and was intended to strengthen the future performance of the Center for Medicare and Medicaid Services (CMS). That case study included **BOXES 12-3** and **12-4**: one on the breakdown and the other on the recovery. The case study was confirmed by other reports as being accurate.

The Various Actors

CMS may have been the focus of the study, but it was neither the only actor nor the only one at fault. Congress, the Supreme Court, the states, the Office of the Secretary of HHS, the White House, and CMS influenced the implementation: “The ingredients are familiar: partisan hurdles throw up by the GOP, the jumpy political instincts of administration aides, administration insularity, spin that borders on deception, bureaucratic clots, and the bold and sprawling scope of the project” (Dickerson, 2013).

The Congress

Despite a general consensus that something needed to be done about the uninsured, the process behind the passage of the ACA had been full of twists and turns. Because of the Senate rule requiring 60 senate votes for cloture in the face of a filibuster, every vote was critical; so was the support (or at least acquiescence) of most key health players. As the requests of each player were accepted or compromises reached, the Senate bill grew larger and larger and more and more complex. Then the loss of Senator Kennedy’s former seat in the November 2010 election forced quick adoption of the Senate bill by both houses using the reconciliation process (McDonough, 2011). A more thorough vetting of the bill might have smoothed the way for implementation. The totally partisan passage of the bill by the Democrats left the Republican majority in the House dead set against funding its implementation. Consequently, the Office of Consumer Information and Insurance Oversight (OCIIO), which was the implementation hub at CMS, would have not been funded adequately, if at all. In response, the Secretary downgraded the office to a center (the Center of Consumer Information and Insurance Oversight or CCIIO), so that it was no longer a budget line item. As a result, however, it lost much of its clout within the bureaucracy including control over the contracting process (Brill, 2015). Later, concern about political pushback led the White House to order a slowdown in the process of writing regulations and contract specifications until after the 2012 elections. This delay negatively impacted the contracting process and in turn led to a crunch before the October 1, 2013 deadline, accompanied by inadequate

BOX 12-3 Key Contributing Factors to Breakdown

PREPARATION AND DEVELOPMENT

March 2010–December 2012

Policy Development Delays: Initial work to create the Federal Marketplace required extensive policy development that delayed HHS and CMS in planning for the technical and operational needs of the HealthCare.gov website.

Poor Transition to CMS: A poor transition of the Federal Marketplace from HHS to CMS early on caused inefficiencies that resulted in communication breakdowns and needlessly complex implementation.

Lack of Clear Leadership: HealthCare.gov lacked clear project leadership to give direction and unity of purpose, responsiveness in execution, and a comprehensive view of progress.

Mismanagement of Key Contract: CMS mismanaged the key website development contract, with frequent changes, problematic technological decisions, and limited oversight of contractor performance.

FINAL COUNTDOWN TO LAUNCH

January 2013–September 2013

Compressed Timeline for Technical Build: CMS continued to change policy and business requirements, which compressed the timeframe for completing the website's technical development.

Resistance to Bad News: CMS leaders and staff failed to recognize the magnitude of problems, became resistant to bad news about the website's development and failed to act on warnings and address problems.

Path Dependency: As problems worsened, CMS staff and contractors became path dependent, continuing to follow the same plan and schedule rather than change course as circumstances warranted.

Corrections Weak and Late: CMS attempted last-minute corrections that were weak and too late to effect change, retaining a fixed deadline for launch, despite poor progress.

Office of the Inspector General. (2016). *HealthCare.gov: CMS management of the federal marketplace: A case study* (Document OEI-06-14-00350), p. iii.

testing of the website (Brill, 2015; Emanuel, 2014). The congressional battle had also left many of the key actors in the White House, HHS, and CMS exhausted and turnover proved high (Emanuel, 2014). The OIG case study also reports that some contractors were unwilling to bid on the Federal Marketplace contracts due to uncertainty about whether the law might be repealed. The 16-day government shutdown on October 1, 2013 also slowed the recovery.

The Supreme Court

The ACA required states to expand their Medicaid programs with extensive federal support. Many states challenged that requirement, as well as the individual mandate, and those cases went to the Supreme Court. The OIG case study reports that some work on the website slowed down because of uncertainties about the Supreme Court's ruling. The Court, on June 28, 2012, upheld the individual mandate but supported the states opposed to Medicaid expansion.

BOX 12-4 Key Contributing Factors to Recovery

LAUNCH, CORRECTION, AND FIRST OPEN ENROLLMENT

October 2013–March 2014

Quick Pivot to New Strategy: CMS and its contractors began correction of website problems immediately following launch, making a quick pivot to change their strategy.

Adoption of Badgeless Culture: CMS and its contractors adopted a badgeless culture that encouraged full collaboration by CMS staff and contractors regardless of employer status and job title, fostering innovation, problem solving, and communication among teams.

Integration of All Functions: CMS integrated all functions into its organizational structure to align with project needs, enhancing CMS and contractor accountability and collaboration.

Planning for Problems: CMS planned for problems, establishing redundant (backup) systems in the event of further breakdowns, and restructuring its key development contract to ensure better performance.

TURNAROUND & SECOND OPEN ENROLLMENT

April 2014–February 2015

Ruthless Prioritization: CMS adopted a policy of ruthless prioritization to reduce planned website functionality, focusing resources on the highest priorities.

Quality over On-Time Delivery: CMS prioritized quality over on-time delivery, employing extensive testing to identify and fix problems and delaying new website functionality if unready for perfect execution.

Simplifying Processes: CMS simplified systems and processes to enable closer monitoring of progress, increased transparency and accountability, and clearer prioritization.

Continuous Learning: CMS adopted continuous learning for policy and technological tasks, balancing project plans with system and team capacity, and changing course as needed to improve operations.

Office of the Inspector General. (2016). *HealthCare.gov: CMS management of the federal marketplace: A case study* (Document OEI-06-14-00350), p. iv.

The States

One provision of the bill allowed the states to choose between participating or not in HealthCare.gov. Those decisions had to play out in 50 legislatures and 50 governors' offices. The deadline for the states to submit final plans to CMS was December 14, 2012 and the states' decisions were not reviewed and questions resolved until January 2013. That led to further delays in the specifications for the federal marketplace. Differences in the views of longtime federal employees and newer employees with state insurance backgrounds also led to delays in writing regulations and contract specifications.

Health and Human Services

Established under the Office of the Secretary, the OCIO had been an innovation-oriented unit in Bethesda, MD with direct hiring authority. It had even successfully developed an ACA browsing site called Plan Finder. Yet some thought that the

resources and experiences of CMS with programs such as Medicare Part D were relevant to HealthCare.gov, and that work going on outside Washington would be less associated with the hostility toward Obamacare.

The transition did not go well as the OCIO changed to CCIO. Its culture clashed with the very hierarchical culture of CMS, and various functions were parceled out from the new center to CMS's Office of Information Services (OIS), the Office of Acquisition and Grants Management, Office of Financial Management, and Office of Communications. But the small Center staff remained in Bethesda, while those other functions were situated in Baltimore. The OIG case study reported: "[D]ocuments also revealed major differences in understanding between the divisions regarding shared responsibilities and assessments of project progress. In interviews staff in CCIO and OIS gave very different descriptions of each other's tasks and in some cases could not identify the staff positions or subdivisions responsible for critical project tasks" (OIG, 2016, p. 9).

Brill (2015) reports interviewing key individuals about who was in charge of the implementation and getting seven different answers. He also asked one official about the existence of a command center and was informed that there were to be four or five command centers—three for CMS, one at HHS, and maybe one at the White House. The OIG report mentions two command centers operated jointly by CMS and its contractors.

According to the OIG case study, when Todd Park, the new White House Chief Technology Officer coming over from the HHS, became concerned, he convinced Secretary Sibelius to hire McKinsey & Co. to assess the CMS process. On March 28, 2013, a McKinsey team report described a process that "for all of its efforts to be polite, depicted a debacle" (Brill, 2015, p. 273). The McKinsey report referred to the fact that the program design process was not yet completed, and there had been "no end-to-end business process view" across the many government actors. It called for "[f]ast, locked-down decisions" and noted a "lack of any single, empowered decision-making authority." It also hinted at the possibility of delaying the launch. The report was not widely circulated at HHS and was dismissed by White House health policy staff as "second guessing from the sidelines." The Administration was confident CMS was already working on those problems (Brill, 2015, pp. 274–275).

The OIG report noted: "In all, CMS received 18 'documented warnings' of concerns regarding the HealthCare.gov build between July 2011 and July 2013, all containing substantial detail about the project's shortcomings and formally submitted to CMS leadership or project managers at CMS" (p. 21).

That case study also noted reluctance within CMS to ask, quoting someone from the center: "You know it has to be bad if I am requesting help" (p. 25).

The White House

Implementation had been a concern at the White House, but it was not always a high priority. It often came behind enrollment and public relations. Economist and staffer David Cutler wrote a letter recommending bringing in an executive familiar with information technology start-ups to head up the implementation. In late 2011, White House Chief Technology Officer Aneesh Chopra informed HHS and CMS officials about fragmented leadership and recommended the appointment of a project CEO. There was no response. In rollout meetings at the White House during the summer of 2013, President Obama "always ended every meeting by saying something to the effect of 'This is all good, but remember it won't matter, if we don't get the technology right'" (Brill, 2015, p. 301). However, the cost-oriented economics staff and the Chief Technology Officers, Chopra at the White House and Todd Park at HHS, were shut out of the process. Senior White House and HHS policy-oriented individuals had no

experience with ecommerce start-ups, and CMS leadership thought their experience with large transactional information systems was sufficient to this task. Senior policy staff were more interested in stimulating enrollment and thought good-looking front-end visuals developed directly with experienced contractors meant the project was on track.

As the 2013 elections had approached, the White House political staff slowed down or even stopped the critical work at CMS to write more than 10,000 pages of regulations and system specifications rather than stimulate public criticism (Emanuel, 2014).

It is fairly clear that little notice of the problems at CMS reached the President. The OIG report notes a climate of not admitting problems or asking for help at CMS. People were in denial or hoping that the next fix would solve their problems (OIG, 2015). There was also a certain reluctance to bother the president. Valerie Jarrett, the domestic policy advisor with almost daily access to the president, reportedly responded to an early recommendation about program management with, "The President likes to hear solutions, not problems. If there are problems, bring them to Nancy-Ann (de Parle); she is the one the President put in charge" (Brill, 2015, p. 203). Nancy-Ann, a CMS director during the Clinton administration, had moved up into a domestic policy role.

Brill notes that while House Republican investigative hearings uncovered many revealing documents about the problems at CMS, the memos of biweekly briefings with the president had only passed along positive reports.

CMS

The downgrading of the managerial hub of HealthCare.gov activity from an office to a center made working there less attractive. The key IT person, Henry Chao, deputy HHS chief technology officer, was assigned to the center, but soon returned to his headquarters job. CCIIO had four directors in four years. John Kingsdale, former CEO of the Massachusetts Connector, already at HHS, turned the lead job down citing insufficient authority (Emanuel, 2014).

When the deadline and the delays in writing specifications necessitated a rapid response, the center could only choose from a list of preapproved contracts for an "indefinite-delivery, indefinite-quantity" contract. The chosen one, CGI Federal, had worked for CMS but had little experience with ecommerce. It also had a checkered history of system successes and failures. At one point, there were 55 contractors and subs involved, mostly on extensions of existing contracts, but no assigned integrator nor any shared timeline. There was still, however, the arbitrary commitment to the October 1 launch.

This was a very complex project technically, Thompson (2013) wrote:

There are just too many moving pieces. A typical user might have to navigate 75 screens to get to their goal of obtaining health insurance, and the whole system contains over a thousand screens. A total of 55 contractors were hired to produce the various pieces, and in order for all the steps to work CMS had to involve five federal agencies, 36 states, and 400 private sector insurers offering well over 4,000 plans.

Thompson (2013, p. 2)

After the McKinsey report, CCIIO hired a consulting firm to do a series of "independent verification and validation reports" concerning contractor progress. It was a technically oriented analysis that identified many risks and problems and

questioned whether the system had even been designed with sufficient capacity to handle the launch. That July 17 report, however, apparently was not shared with higher levels of management.

During much of the build period, the health insurance companies were on edge about their interactions with CMS. Karen Ignani, head of Americas Health Insurance Plans, had suggested to CMS officials in Baltimore that the site would not be ready. "Why not let October be a month where people can browse, but not buy? Or perhaps you could delay everything a month? What are your contingency plans?" The response had been that there were no contingency plans and that her problems were all "hypothetical" (Brill, 2015, p. 302).

As the work fell further and further behind and the July 30 launch of a stripped-down system failed, CMS and the contractors added more people. But it was too little, too late. There were no strategic changes in the application, just more resources. The OIG case study referred to this approach negatively as "path dependent." Then certain functions, such as Spanish language capability, were stripped out to be added back after the launch. The employer mandate and the small business exchange had already been set aside.

The Launch and the Turnaround

On October 1 very few potential enrollees got through to the endpoint of a purchase, and a public uproar ensued. Things improved day by day, but only to slightly less disastrous levels. On top of that, the 16-day federal government shutdown occurred on October 1.

As a small coterie of grim-faced advisers shuffled into the Oval Office on the evening of Oct. 15, President Obama's chief domestic accomplishment was falling apart . . . "We created this problem we didn't need to create. . . . And it's of our own doing, and it's our most important initiative."

Stolberg and Shear (2013)

By October 18, the President had had enough and asked Todd Park and advisor Zions to take over. He essentially told them that they had no financial, contractual, or procurement constraints and needed to get the job done (Schlesinger, 2016). They called on contacts in the private sector and within a couple of days a new team of technical experts appeared from around the country.

Here the OIG case study can lead someone astray. It talks about the activities of CMS, the contractors, and an "ad hoc technology team." This team included individuals with experience at Google, Amazon, Facebook, and Twitter, and some who had worked on the Obama election information infrastructure. Some had worked at both. Because one could not volunteer, they were signed on as employees of Quality Software Services, Inc., considered one of the more competent existing contractors, but they took the lead on testing and the integration of functions. They first had to see whether the system could be salvaged. They found no measurement system, so the first day they built a "dashboard" and then got CMS to order in standard monitoring systems. They found the system was slowed because all queries went through the same data base, so they followed industry procedure and established a cache of reusable information, which greatly sped up the system. They established a project structure of ignoring blame, no hierarchy of rank or badges, and prioritizing risks and working through them systematically (Schlesinger, 2016). By January 1, the system was operating tolerably and an improved system was launched for the second enrollment with a new lead contractor, Accenture Federal.

► Discussion Questions

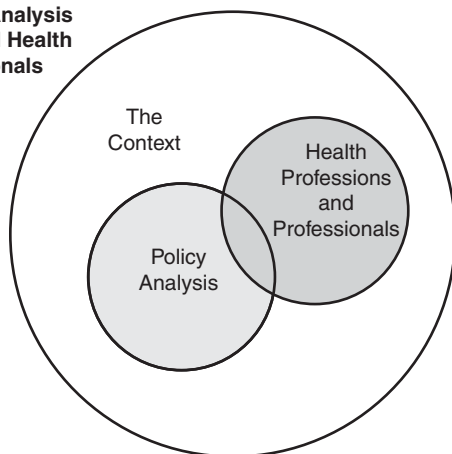
1. The two tables in this case show implementation issues in this situation. Which are a function of the task the Obama Administration faced and which are inherent in the way government agencies work?
2. What is your opinion of the prioritization of “Quality over On-Time Delivery”?
3. Thompson (2013) argued that the management of HealthCare.gov should have been handled like a military weapon system program. Does that make sense to you?
4. What might Congress have done to enhance the implementation of the Federally Facilitated Marketplace?
5. What can healthcare provider organizations learn from this story? Does the story fit with your experience?



PART III

The Professional as Participant

**The Policy Analysis
Process and Health
Professionals**



We expect that most users of this book will be current or future health professionals. They need to understand the role of the professions and professionals in the health policy process. Although not disinterested parties, they can bring to the table their commitment, their technical expertise, and their concern for the patient. However, they also bring their economic interests, just as with all of the other actors, and their economic interests will be known to the other parties. Likewise, health professionals must understand the positions of the others at the table and be prepared to evaluate how they are affected by various policy alternatives.

Chapter 13 addresses the leadership roles open to professionals and the skills that they need to develop to perform them appropriately. Chapter 14, the concluding chapter, sums up some of the issues raised in earlier chapters and sends you on your way, hopefully as an effective future participant in the policy process.



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CHAPTER 13

Health Professional Leadership

Normal is getting narrower and narrower.

Personal observation by an experienced nurse practitioner.

As a health professional, you can be an important participant in health policy processes, bringing to bear your experiences, your knowledge of both the science and the art of health care and your ability to distinguish between the two, and your commitment to the patient. Typically, professionals also bring a commitment to lifelong learning. The power of the professions, especially physicians, has been waning of late, but that has a lot to do with the height of their past dominance. In an open, market-driven, information-rich society, the old monopoly power described by Starr (1982) has not been sustainable. Health professionals now need to undertake new leadership roles, or else their status will be undermined further by those filling the managerial void.

► Disinterestedness

Much of the diminished respect for health professionals stems from the public's perception of lack of disinterestedness (i.e., the concept of freedom from special interest or the ability to set aside one's own interests to seek the best possible outcome for others). The opposite is the oft-repeated phrase, "All they care about is money." Schlesinger (2002) argued that this loss of faith had seemed to intensify with the advent of Medicare and Medicaid and that has led to a loss of political power as well. One parameter of successful professional leadership will be the ability to engender faith that the professional has the interests of other constituencies in mind.

► Informational Credibility

Disintermediation, particularly in the form of general and direct-to-consumer advertising, has affected the informational monopoly of the health professions.

This is not a one-way street. The claims and counterclaims of the various interested parties can be hard to sort out. One leadership role for the health professional is to guide the general public through that welter of information. This is not just a physician's task. It involves all health professionals. Maintaining the power of a profession in the future will require maintaining acceptance as a unique and relevant information domain. Procedural control alone is a slender reed on which to stake the future of a profession. Conceding the informational domain to others is risky.

► To Influence Globally, Start Locally

Health professionals earn their power to participate effectively in the political process through leadership in their professions, in their institutions, and in their communities. Although some leaders and spokespersons appear to have burst onto the national scene directly—Dr. Donald Berwick in government and health quality improvement, for example, Dr. Atul Gawande with his *New Yorker* articles and his books, and Dr. Paul Farmer in international health—most rise slowly through the ranks of their profession as team players. The routes to leadership positions are varied. Health professionals are in leadership roles in medical centers, community hospitals, insurance companies, and government agencies. Each presumably came by his or her position by training, intelligence, hard work, and trustworthiness. They were able to convince others to work beside them and for them because they could be trusted to take the interests of others into account.

Leadership career paths often overlooked in the health policy arena include those in corporations and in entrepreneurial ventures. A number of very influential health professionals have stopped delivering care directly and have moved into the management of health institutions, insurance companies, occupational health, medical device and supply companies, pharmaceutical companies, and government agencies. They represent those institutions but seem able to do so without negating the trust of healthcare decision-makers. Their leadership roles may have been thrust on them, or they may have sought them. In either case, they took a prepared mind and a sense of what they want to accomplish into the arena of health-care policy.

The press seems to emphasize the importance of careers in publicly held companies, as considerable wealth can be created by developing a company and taking it public. After the company goes public, however, it is beholden primarily, if not solely, to one set of stakeholders, the stockholders. Therefore, there is still a major role in health care for the nonprofit organization that does not have stockholders and balances a number of competing interests. A deeper knowledge of nonprofit organizations and their behaviors is necessary for determining their role in setting and implementing health policy. This is especially true of entrepreneurial not-for-profit organizations that can participate in the marketplace fully as stock corporations. Leaders must understand the similarities and differences in how these types of organizations function. The term *governance* is often applied to the roles of management, staff, and boards in both for-profit and nonprofit organizations. The professional leader must be able to function effectively and help govern effectively in one or the other or both.

► Process Innovation

There seems to be a consensus developing that there is great potential in the area of process innovation. This goes well beyond improving current processes and into major changes that meet the criteria outlined by the Institute of Medicine (IOM) in its Learning Healthy System initiative and its Innovation Collaborative. Those criteria include:

- A participatory, team-based transparent culture
- Patient-anchored and tested processes
- Fully active and engaged patients and the public
- Informed, facilitated, shared, and coordinated decisions
- Care that starts with best practice every time
- Transparent and constantly assessed outcomes
- Incentives aligned for value
- Knowledge that is an ongoing, seamless product of services and research
- Health information that is a reliable, secure, and reusable resource for the patient and the common good
- Leadership that is multi-focal, networked and dynamic. (IOM, 2012a, b)

There is plenty there to work on locally and nationally.

► Health Policy Analysis: A Relevant School for Leadership

Participating in policy discussions and analyses can help prepare one for leadership. By reviewing and critiquing the alternative scenarios provided by scholars—such as the consumer-oriented free-market approach of Herzlinger (1997) versus the community-based planning approach of Shortell, Gillies, Anderson, Erickson, and Mitchell (1996) versus new approaches being undertaken by the various states—one can learn a great deal.

Evaluating the alternatives calls for an understanding of the types of risks that healthcare organizations and healthcare managers may choose to handle or not handle in the design of their system. These risks are described as follows (McLaughlin, 1997; McLaughlin & Kaluzny, 1997):

- Underwriting
- Marketing
- Clinical operations
- Financial
- Regulatory
- Integrative

The would-be professional leader has to think through the following questions:

- Which of these risks am I now comfortable handling?
- Which other ones do I need and want to learn to handle?
- How can I use my work and educational experiences to learn to handle those that I want and need to handle?

This exercise can help the potential professional leader outline what he or she needs to learn about managerial skills and activities. One must learn to analyze the various organization structures used for healthcare delivery in terms of how they allocate these risks and facilitate their handling.

► Governance

Not only do healthcare professional leaders make decisions, they also provide what Karl Weick (1995) called *sense making* for others. They must be able to understand and articulate the role of the governance process in their operation. Healthcare professionals guard and maintain the technological core of their organizations. As leaders, they must understand how change processes operate and how health professionals and other actors can best work together in the policy-making process. Through understanding the risks encountered, analyzing the nature of local markets and delivery organizations, and meeting the governance needs of organizations delivering care, health profession leaders become equipped to analyze the local healthcare system and how it is best led.

► Communities

If professionals are to manage populations rather than just individuals, they must develop a sense of how to do that in a community setting. They must experience and participate in change processes undertaken by groups involving payers, providers, public health agencies, and patient organizations in their own community. Leaders must develop sufficient respect among their colleagues to be trusted with the data needed for community health improvement when it might otherwise be seen as proprietary information for competitive use.

► Enhancing the Professional's Role

It is possible to enhance professional performance in health policy roles in a number of ways, including the following:

- Preparation
- Skills development
- Training others
- Networking
- Practicing leadership

Preparing to Learn and to Lead

Professionals need opportunities to adapt to policy analysis roles beyond those associated with clinical care. Potential leaders have to walk in the shoes of those who are leading, consider multiple sides of the issues, use facts and fit them into conceptual and mathematical models used to reduce and refine the array of available alternatives, and then help select those that are likely to succeed in the field.

One very important role is that of a team member. Policy analysis teams require a wide variety of skills, including management, economics, operations, and medicine. As the owners of the technological core of health care, health professionals can always claim a place at the table; however, they must be prepared to contribute overall as well.

Developing Skills

The policy analyst must understand the financial implications of what is being discussed; think in terms of markets and competition; adjust to social, economic, and political change; analyze and optimize processes; and motivate individuals and teams. All of these move in the direction of exhibiting competence and gaining the respect of one's peers and the senior management team. Another skill of senior management is being able to interpret publicly what is going on in a way that supports positive outcomes. One learning and practice task would be explaining to colleagues and to the general public any changes in the healthcare landscape—for example, interpreting the provisions of current healthcare legislation to one's peers, patients, and the public.

Learning and Training Others

Another function of professional leadership is training the next generation of professionals. For example, if health policy is going to focus on motivating the system to reduce waste, then the next generation is going to have to think in terms of value-based patient care and focus on managing the entire medical condition from start to finish. Paul Batalden and others at Dartmouth have already started to incorporate this into their training of physicians there and elsewhere. They refer to it as employing microsystems strategies as compared with organization-centered or issue-centered strategies for process improvement (Mohr & Batalden, 2006). They suggest that there are eight dimensions of effective microsystems (Mohr, Batalden, & Barach, 2006, p. 408):

1. Constancy of purpose
2. Investment in improvement
3. Alignment of role and training for efficiency and staff satisfaction
4. Interdependence of the care team to meet patient needs
5. Integration of information and technology into workflows
6. Ongoing measurement of outcomes
7. Supportiveness of the larger organization
8. Connection to the community to enhance care delivery and extend influence

These eight dimensions align very well with the concepts of the value-based competition model (Porter & Teisberg, 2006). Adopting that approach in both clinical process improvement and in clinical training is one way to walk the talk, to learn the full implications of such an approach, and to develop the skills and insights applicable at higher levels of policy analysis. If one does not normally use something, one of the best ways to come to understand it fully is to try to teach it to others.

Organized Efforts for Leadership Development

Increasingly, leadership training is available within organizations and training programs, and from universities and professional associations. Sonnino (2016) offers a

comprehensive review of leadership programs and “their progress and pitfalls.” She notes that the biggest pitfall is lack of time to undertake the experiential learning that is essential to an effective training program. One can read a lot on leadership theory, but effectiveness lies in being able to adapt available approaches to a culture and to one’s own personality, or to slowly change that culture or undertake a program of personal growth. The literature emphasizes the importance of learning to work on leadership in teams, including understanding and implementing effective conflict management strategies.

Building Networks

Health policy analysis takes place in a network of participants, professions, and organizations. One learns how influence is exerted nationally, locally, and in one’s work group by knowing when to speak up and when to hold back, when to be the advocate and when to be the analyst, and how to support and move forward the multidisciplinary team. By doing so, one develops skill at working with other disciplines and acquires the contacts that become important assets as one attempts to exert leadership at higher and higher levels in the policy analysis process. One need not wait for a senior management opening to put those skills to use.

► Where in the World?

There are many places to look for leadership development information. The *Journal of Health Leadership* can be helpful and there are numerous training programs available from universities, contractors, and aid agencies. The problem is that management styles need to be sensitive to culture (Hofstede, Hofstede, & Minkov, 2010), so it is important to find a program that has been tailored to your part of the world and your language. One also has to be concerned that the training is not heavy on indoctrination. Cultural imperialism is not a myth.

► Conclusion

Professionals play a very important role in policy analysis; however, they need to acquire those skills necessary to achieve positions of leadership in policy making. Professionals, especially leader-physicians, must learn to take a disinterested view in many of their interactions with others, offsetting the growing public perception that they are much too concerned with the monetary aspects of care. If they fail to do so, their professional and political influence will continue to wane as their informational and procedural monopolies weaken.

To start, professionals can begin to influence health policy locally. They can gain experience and leadership skills at that level before moving up. As they move up, they will learn about the governance processes of both for-profit and nonprofit organizations and the suitability of each for specific purposes. They will gain knowledge about managing nonclinical types of risks in the healthcare setting and about how to become a member of a team that can deal with an entire medical condition rather than just their subspecialty’s aspect of it.

Learning by doing is available in all settings—in training newer health professionals, improving local care processes, and health policy analysis at the community level. There is plenty of room for professional leaders in the health policy process, if they are willing to invest the necessary time and effort. They fail to do so at their own risk.

CASE 13: The Data Sharing Proposal

In a “Perspective” article in the August 4, 2016, *New England Journal of Medicine*, “Strengthening Research through Data Sharing,” Sen. Elizabeth Warren (D-Mass.) began: “Data sharing has incredible potential to strengthen academic research, the practice of medicine, and the integrity of the clinical trials system.” (Warren, 2016, p. 401)

She expressed enthusiasm for the work of the International Committee of Medical Journal Editors (ICMJE) and its proposal to “require a commitment from authors to make available the data that underlie the reported results of their work within 6 month after we publish them.” (ICMJE, 2016, p. 384)

This was another salvo in the tug-of-war over access to the data from clinical trials that has involved the editors of medical journals, the World Health Organization (WHO), and others against those who believe that these data should remain proprietary and that such a requirement adds unnecessary cost to what is an already extremely expensive process and offers little potential benefit.

Background

The double-blind randomized controlled clinical trial (RCT) has been the gold standard of clinical regulatory science and evidence-based medicine since the 1950s. The 1962 Kefauver-Harris Amendments to the Food and Drug Administration Act called for manufacturers to prove the effectiveness of new pharmaceuticals based on “adequate and well-controlled clinical studies conducted by qualified experts.” They also required informed consent. By 1970, the FDA had mandated submission of RCTs results in most situations. There have been few objections, other than the cost (more than \$10,000 per enrollee), and the fact that the control group may miss out on a potential effective treatment.

Since 2012 there has been a clear international trend in transparency and consumer involvement in health care. Increased calls for “open science” have led clinical trial participants and consumers to push for access alongside researchers, journal editors, and funding organizations. Some people have argued that clinical trial data should be considered a public good because of the voluntary contributions by the trial participants, and in some cases, support from nonprofits and governments (IOM, 2015). They have further argued that these voluntary participants deserve to have the results they had a part in producing made available to them and to researchers regardless of the outcome.

Funders and journals on both sides of the Atlantic had been requiring that grants and articles address the planned or actual availability of data. Much of this was focused on enabling the replication of research following several high profile cases of falsification of data. By 2010, the European Medical Agency allowed data to be released under a freedom of information procedure, and in 2011, an international group that included the Wellcome Trust and the National Institutes of Health went on record supporting data sharing.

Significant external pressure came from an English nonprofit AllTrials. An offshoot of Sense About Science, the organization was founded by Dr. Ben Goldacre, who is

associated with the Centre for Evidence-Based Medicine at Oxford University, and the author of *Bad Science*, a blogger, and a columnist for *The Guardian*. In 2013, AllTrials sent an open letter to the European Medical Agency noting that the results of about half of clinical trials were not published. “This is dangerous and expensive and it holds back good medicine. It is also a betrayal of our trust in clinical trial regulation, and the trust of the families of those patients who volunteer for trials having had terminal diagnoses.” (AllTrials, 2015) It was picked up by the press. Simultaneously, Sense About Science set up a petition on the AllTrials website that stated: “All trials past and present should be registered, and the full methods and the results reported. We call on governments, regulators and research bodies to implement measures to achieve this.” (AllTrials, 2015)

Within a month, 20,000 individuals had signed up and support grew to more than 80,000 individuals and 500 organizations by the end of 2014.

Goldacre and his associates held discussions with journal editors and other interested stakeholders and a steering committee was established. The Wellcome Trust and Medical Research Council signed on. There was pushback. Some interested parties were concerned that the deidentified records would be reidentified. Others thought the effort would be extremely costly with little or no return. They denied there was any problem and threatened lawsuits if such requirements were added. Many large pharmaceutical companies already had procedures for screening and permitting researchers access to their deidentified clinical trials data.

Overt resistance diminished, however, when GlaxoSmithKline (GSK) suddenly agreed to make its clinical data available to researchers whose requests were approved by a relatively independent committee. This was surprising to many, but not to others who were aware of the historical and current linkages between the Wellcome Trust and GSK.

GSK encouraged development of a public website, clinicalstudydatarequest.com (CSDR), with an independent review panel. By December 2015, the resulting repository contained more than 3000 trials from 14 pharmaceutical companies. In 2015, the Wellcome Trust took over administration of the panel. Between May 2013 and November 2015, 177 proposals were submitted and access granted CSDR in 144 cases, which led Strom, Buyse, Hughes, and Knoppers (2016) to ask, “Data Sharing—Is the Juice Worth the Squeeze.” The title is catchy, but it may be too early to evaluate the results given long research and publication lead times. Anecdotal evidence indicated that the process was cumbersome and firewalls made integration of data into meta-analyses difficult.

Existing U.S. Requirements for Clinical Trials

The Food and Drug Administration Modernization Act of 1997 established the federal government as a player in the sharing of clinical trials information. It required the registration of interventional clinical trials for new drug indications on ClinicalTrials.gov once they were approved by the FDA. The ICMJE made that a requirement to be considered for publication. The 2007 FDA Amendment Act required that results be reported on ClinicalTrials.com for Phase II and Phase III trials. Government agencies and private donors continued the push for data availability under a variety of conditions.

In late 2012, IOM conducted a workshop on sharing clinical research data. This highlighted the benefits of sharing data and discussed ways of overcoming its negative aspects.

Acceptance of the sharing concept continued to grow. The Pharmaceutical Research and Manufacturers of America, the European Federation of Pharmaceutical Industries and Associations, and the Biotechnology Industry Organization issued “Principles” statements agreeing to share data with researchers, publish both positive and negative results, and make public full clinical studies report synopses. A number of companies committed to using ClinicalTrialsDataRequest.com and NIH announced it expected that researchers would include informed consent for sharing genomic data and commit to sharing it by the first publication of the study results. The trade associations for pharmaceutical and biological manufacturers in Europe and the United States issued statements supporting data sharing. Yet policies and procedures varied greatly. In response, a large group of drug companies and government and private donors asked IOM for a consensus study to generate guidelines for responsible sharing.

The IOM responded by setting up the 13-member expert Committee on Strategies for Responsible Sharing of Clinical Trials. Its charge was to:

- Develop principles for responsible sharing
- Outline “a selected set of data and data sharing activities” for sharing clinical trials
- Summarize the benefits and risks for key stakeholders
- Address challenges to effective sharing, including
 - Protection of human subjects and privacy
 - Intellectual property and other legal issues
 - Implementation requirements and resource constraints
 - Incentives and disincentives for researchers to participate in sharing and conducting studies with secondary data
- Suggest alternatives for the supporting infrastructure for collecting, managing and distributing the data

This committee held workshops, developed a framework, and accepted comments on its interim work.

The IOM Study

After several hearings and commissioned studies, the committee published its report in January 2015. It contains four recommendations:

- Data sharing should be the norm and stakeholders should commit to “maximizing the benefits, minimizing the risks, and overcoming the challenges.”
- Investigators should release data at specific times after reaching specified milestones in the clinical trials process, namely:
 - Trial registration: Before enrollment
 - Technical and lay summaries of results including adverse results: Within 12 months of study completion
 - “Full-data package” within 18 months of study completion
 - “Post-publication data package”: 6 months after publication
 - “Post-regulatory data package”: later of 30 days after approval or 18 months after study completion
 - “Post-regulatory data package” for abandoned trials: 18 months after complete abandonment

- Data holders should have specific data use agreements, independent review panels that include representatives of the public, and transparency about access.
- There should be an ongoing international group to address “key issues of infrastructure, technology, sustainability and staff availability” (IOM, 2015).

The report acknowledged that the clinical trials world is in a state of flux, with many trials being conducted outside the regulatory process framework by government agencies, universities, and even patient groups. It notes that trials are taking place that were gathering evidence after licensure and involving treatments and practices already in use.

Benefits

The IOM report notes that access to trials’ data and secondary analysis could lead to a deeper understanding of the effectiveness of the tested intervention. For example, trials used to justify prophylactic anti-influenza treatment in nursing homes in the United Kingdom were conducted almost entirely on younger, healthier patients.

Trial data could help investigators avoid interventions that are questionable or dangerous. Trial data could also contribute to generalized medical knowledge and specifically might lead to other uses of the tested interventions.

Senator Warren notes in her article that compliance with current promises to make data available was poor and that stronger external motivation was needed. Audits of performance in registering trials at English universities had shown poor understanding and compliance among university investigators.

AllTrials has emphasized the significant loss of information in the half of past trials that had never been published.

Risks

The IOM report cites the risks of data security and risks of reidentification and invasion of privacy as major concerns. A commissioned appendix on reidentification is included with the report.

There is concern, according to the report, about misuse and mishandling of the data by less sophisticated researchers, corporate or professional enemies, or even competitors. For example, a certain number of adverse events occur randomly and could be cited as problems even when they are not caused by the intervention. Some commenters were concerned that data could be used inappropriately in jury-based product liability or negligence cases. There was also the possibility of disclosure, and worry that the time and energy needed to deal with negative claims would keep firms, researchers, and patients from participating in clinical trials. Competitors might even use the data in regulatory filings in countries that “do not recognize data exclusivity periods or do not grant patents for certain kinds of research.”

The report also pointed out the issue of individuals being less willing to participate in trials if they knew that half of them failed to communicate their results to the participants. This became an increasing risk as patients formed information networks on the Web and even conducted and shared observational studies—and very occasionally conducted their own trials. Failure to report led to a bias in the evidence and could lead to duplication of effort and unnecessary patient risk.

The report relayed observations that concerned the reduced motivation for conducting primary research if one could avoid the work of patient recruiting, treatment, data collection, data integrity, and data reporting and get equal credit for secondary analysis. Some commenters worried about encouraging

substandard workmanship by second-tier researchers and (implicitly) having to compete with them for scarce funding.

Resources

There is no doubt that complete packaging and storage of all clinical trials data would take considerable resources. Firms and institutions were very concerned about who would bear the costs of preparing the inputs. There is also the question of who would maintain the data repositories.

Alternative Arrangements

The report notes that multinational pharmaceutical companies have begun to develop alternative approaches to data sharing. The initial leader, GSK, set up its own independent review board. Johnson & Johnson and others have used a “trusted intermediary” such as the Yale Open Data Access (YODA) Project. Investigators submit proposals to YODA and agree to abide by its conditions, which include a prohibition against use for commercial or litigation purposes (Krumholtz & Waldstreicher, 2016). The Academic Research Organization Consortium for Continuing Evaluation of Scientific Studies—Cardiovascular cited new discoveries and confirmation of results and added the “possibility of aggregation of data sets from multiple trials to facilitate high-quality systematic meta-analysis” (ACCESS-CV, 2016). This would require a major effort to make data from different studies comparable. Consortium members expressed strong concerns about fishing expeditions and argued for requirements that a full study design be posted on ClinicalTrials.gov.

The Contrary Response

A strong rejoinder to Senator Warren’s piece appeared in the same issue. Authors from the International Consortium of Investigators for Fairness in Trial Data Sharing (2016), which claims the support of 282 investigators in 33 countries, argued that the ICMJE proposal would impose risks:

- Direct costs (the authors cite a paper reporting 27,000 RCTs published worldwide in 2010)
- Opportunity costs of diverting money and staff from actual conduct of trials
- “Misleading or inaccurate analyses and analyses aimed at unfairly discrediting or undermining the original publication.”

They suggest the data sharing requirements be limited to trials “whose results are under question and those that are likely to influence care.” They also object to the stipulation that data would be released within 6 months after publication. They point out that the primary researchers often need multiple articles to present their findings and then go on to mine the data for additional articles over a period of more than a year. This is especially the case for large multisite studies where different local teams would also like to publish their own output. They argue that the IOM committee’s proposal would be a disincentive for coinvestigators and site investigators. It would also lead principle investigators to delay their initial publications until multiple articles were ready for submission.

Their counterproposal is to allow investigators 2 years of exclusive use plus 6 months for each year elapsed conducting the study up to a maximum of 5 years. Their alternative for confirming validity would be to have an independent statistician examine the data before publication. They also would require the secondary users to compensate the primary research teams for some of the cost of the research and all of the cost of transforming the data. (ACCESS-CV, 2016)

Ongoing Steps

AltTrials continues to campaign for a more open data sharing environment in Europe by calling for:

- Going back in time and registering and sharing data from trials not already published
- Bringing the registering and reporting levels of trials conducted by governments, universities, and funders up to level required of trials for regulatory approval. A study has shown poor conformance to existing standards (STAT, 2016)
- Having funders and institutions audit their members and grantees performance regarding sharing and withhold support and partial payment for poor performers

They have also worked behind the scenes with members of the UK and EU parliaments to memorialize openness in regulation and law.

What About Observational Studies?

ClinicalTrials.gov contains a significant number of observational study protocols and so do registries at WHO and in other countries (Khanoyan & Cuervo, 2016). The rationale is the same as for RCT trials. However, the pressure for registration is not as great.

The FDA Issues “Final Rule”

The FDA issued its “Final Rule” on Trials Reporting on ClinicalTrials.gov, which became effective in April 2017. One report by staff from the Library of Medicine and NIH involved in the project noted that slightly less than 10% of the 224,000 records on the site included results and stated that the rule was intended to increase compliance by providing clarification (Zarin, Tse, Williams, & Carr, 2016).

The rule also expanded reporting requirements. The original law had required reporting results only if the product had been approved for at least one use. Now study methods and results for unapproved products (other than medical devices, which the 2007 law specifically excluded) must be reported within 1 year. Two more years are allowed if product development continues. Secondary results must also be posted. However, data posting is not required. NIH decided not to require posting of narrative summaries because further research was needed to avoid summaries that were “promotional or misleading.”

ICMJE Publishes “Requirement”

On June 8, 2017, the ICMJE “requirement” was published in the *NEJM* (ICMJE, 2017). These editors cited progress but noted that “challenges” existed. They would insist that manuscripts about clinical trials submitted after July 1, 2018 include a statement about data availability and that clinical trials enrolling after January 1, 2019 include a data sampling plan with the trial registration. They wrote: “These initial requirements do not yet mandate data sharing, but investigators should be aware that editors may take into consideration data sharing statements when making editorial decisions. These minimum requirements are intended to move the research environment closer to fulfilling our ethical obligation to participants. Some ICMJE member journals already maintain or may choose to adopt more stringent requirements for data sharing.” (p. 2277)

► Discussion Questions

1. Why has the imposition of data sharing requirements been much slower in the United States?
2. What do you think will be the outcome of the debate in the United States?
3. What policy analysis process would you use to determine the appropriate national policy for sharing trials data?
4. What would you recommend as the exclusive use period for clinical trial administrators?
5. How important would it be to have narrative results summaries on Clinical-Trials.gov?
6. What would be your recommendation of a data sharing infrastructure for the United States, if any?



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CHAPTER 14

Conclusion: All Those Levers and Still No Fulcrum

► Where to Stand

A variety of levers can be used to try to move healthcare delivery in one direction or the other. Levers, however, require a strong fulcrum, a solid base against which the lever can operate when sufficient force is applied. In the United States, there is a clear absence of a reliable fulcrum. The passage of the Affordable Care Act (ACA) provided a fulcrum, albeit a sometimes shaky one, and it was a contentious issue in the 2016 election. The new Republican administration has wanted to replace it with yet another iteration of health reform, one that is a combination of returning Medicaid to early versions and reducing federal responsibility for funding it in order to offset tax cuts as part of its proposed tax overhaul. Meanwhile, some versions of the repeal-and-replace legislation before the U.S. Congress in 2017 proposed tax credits for health insurance premiums as an alternative to subsidies and the individual mandate.

Federal government bureaucrats know that the efforts of lobbyists, senior White House staffers, or chairs of congressional committees can undermine in a few days what has taken months of study and consensus building to achieve. At worst, one's program, or even one's agency, can disappear from the budget overnight. State offices are subject to the same risks, although governors sometimes stand more firmly because a state must meet its financial obligations, rather than print money. Insurers continue to take their cut and pass on any added costs. Providers continue to maximize revenue. Employers continue to opt out of defined benefit programs. More and more of the costs of providing coverage and care accrue to state and federal governments through Medicare, Medicaid, and other programs and to the individuals who actually have insurance. A 2013 survey of more than 200 key healthcare industry executives showed deep pessimism about our ability to improve both quality and inflation-adjusted costs, thus improving value. Only 1% were strongly positive, and 22% were strongly negative. To a parallel question about the current

quality of U.S. health care, 16% answered that they were strongly positive, and 22% were strongly negative (Chin, Hamermesh, Huckman, McNeil, & Newhouse, 2013).

Fitting into Our Culture of Individualism

There are practical reasons for the “on the one hand, and on the other hand” approach Harry Truman objected to when he called for a “one-handed” economist. Each of us brings a value system to any policy analysis, and those values inevitably get mixed in with more objective information. We are, therefore, understandably reluctant to declare one approach, one solution, or one system to be absolutely and unequivocally superior.

It is clear, however, that there are some things that fit well into the culture of the United States and others that do not. One of our current cultural problems is that loud sets of voices are calling for one extreme or another with almost religious zeal. Reagan (1999) pointed to the horns of one healthcare dilemma. On the one hand, some would prefer that the federal government be the single purveyor of health care, but in a country that has long valued individualism and a free-market economic system it seems that there has to be some acceptance of market forces in the process. On the other hand, it is clear that health care has been and will continue to be a highly imperfect marketplace.

No matter how much information U.S. consumers receive, it is unlikely to be sufficient for them to make good decisions about all aspects of personal health care. Some segments of U.S. society cannot generate sufficient income to participate in that marketplace, whereas others may lack the skills or attributes necessary to participate fully and effectively in an unfettered healthcare market. Advocates of a single-payer system, who note that the United States is the only developed country without a national health system, are unlikely to see the United States fall in line anytime soon. Yet the current dependence on employers as the basic source of funding for care for workers and their families, while working and especially during retirement, is collapsing in the face of increasing international competition.

But does it really matter? Gawande suggested that we don’t want the system we have, but fear losing in the process. He asks:

Do we want a single payer of care (Medicare for all), each state to have its own payer for care (Medicaid for all), a nationwide marketplace where we all choose from a selection of health plans (Health.gov for all), or personal accounts that we can use to pay directly for health care (Health Savings accounts for all)? Any of these can work. They all have been made to work universally somewhere in the world.

Gawande (2017, p. 53)

All we have to do is find the collective will to make it work.

Limits of the Free-Market Approach

Those who argue that insurance, because it insulates healthcare consumers from any economic consequences, has been responsible for waste and overutilization have a

strong point. They want to shift more costs to the consumer who then will reduce utilization. There are three issues with that approach, however, in addition to the risk that for lower income individuals it will lead to underutilization with negative long-term consequences. The first is that much of the impact of cost-shifting may already have been achieved with recent increases in deductibles and copays. The second is that information on quality and price is so opaque that the market cannot function very effectively until major changes take place to put realistic and relevant information into the hands of the consuming public. Those changes seem to be under way with respect to hospital prices and quality outcomes for specific institutions. Interpreting the data is still hard in many cases and drilling down to individual providers is difficult.

The third constraint on the free-market approach is information asymmetry—some purchasers are capable of researching a medical condition and some are not. Educated individuals may be able to make an informed market decision because they have:

- The technical information and the background to interpret it
- The market information on price and quality of providers of care for this medical condition
- Enough confidence in their knowledge to act on it
- Access to a primary care provider to test out their conclusions

Six months later, however, they may have a set of symptoms that leaves them completely baffled and having to rely on the recommendations of their primary care provider or a local specialist, or are in acute pain and sedated at that moment. Yes, that individual may be capable of independent, rational consumer decision-making behavior in the first instance. In the second situation one has to rely on one's agents: one's physicians, and, to a lesser extent, one's insurer.

► The Physician's Dilemma

What would likely changes do to the status of U.S. physicians? Many factors contribute to the industrialization of the healthcare sector and further weaken physician bargaining power. For example, if most medical centers were required to quote a single price up front for treating a medical condition and that single price bundles the professional, technical, and inpatient and aftercare components of that treatment, the physicians would have to bargain for their share, and they would likely be in a relatively weak position. Some physicians might counter this development by setting up their own specialty hospitals, but one might anticipate further constraints on referrals if that response became widespread.

More information on process, price, and quality; more bundling of pricing; more evidence-based protocols; requirements for reduced waste; and more intense case management would all, in part, constrain physician autonomy. At some point, health system policy makers will have to reconsider the patient-physician relationship and its attendant agency issues, and try to reach an appropriate balance between industrialization and professionalism. One answer is for physicians to provide more assertive leadership when it comes to defining and maintaining the physician's role, especially the role of the primary provider, without trying to get

back to the monopolistic mind-set that once tainted state and local American Medical Association efforts.

► **The ERISA Problem**

Trying to make rational sense out of the healthcare system is difficult enough, but effectively having three complete sets of regulatory requirements, one under state insurance laws, one following on prior federal legislation, and one under the Employee Retirement Income Security Act of 1974 (ERISA), does not make sense, especially because the current responsibility for health and welfare in the United States rests constitutionally with the states. At the time ERISA passed, it was seen as an enabler of nationwide collective bargaining between large employers and large industrial unions, but in an era when we have so many “model uniform codes” for so many types of commercial and personal business transactions, it would not seem to be an insurmountable obstacle to adopt a model uniform code for employer-based health insurance. The requirement for a defined minimum essential health benefit within the ACA-based exchanges was an initial step in the direction of establishing uniform national coverages.

► **Many ACA Provisions Stay in Place, But Uncertainty Continues**

Although the Republicans failed to repeal the ACA in 2017, they maintained the option to return to that effort through future years’ budget bills. Subsidies to insurers for the costs of catastrophic cases are uncertain and further administrative actions to undermine the implementation of the exchanges is likely. The unfortunate fact is that the alternatives put forward in 2017 were less about improving health care and more about scaling back federal contributions to Medicaid to finance part of the other major Republican agenda item, tax reform.

The ACA will continue to be disruptive even if the repeal attempts succeed. An overwhelming majority of the public now favor barring preexisting condition exclusions.

Some health care advocates have openly stated that they see the ACA as a stepping stone to a single-payer system. That was an argument used by Tea Party activists in the House against the initial 2017 Repeal and Replace legislation. That assessment assumes that what is in place will be a complete failure that leads to greater government intervention rather than retrenchment.

Republican proposals in the House and Senate in 2017 did not adequately address issues of access or cost. Regardless of the party in power, health care will have to compete for attention with other agenda items, such as national security, immigration reform, inflationary pressures, and lack of savings, and the rising cost of interest necessary to attract foreign capital. Along the way, we will probably experience some positive effects from prevention programs, witness increased implementation of pay-for-performance, and see some offsetting federal efforts to strip out of Medicare and Medicaid some services that are not essential to health and safety in the short run. The pressure on both low-income working families and older people to pay a higher percentage of their income for health care will continue, despite the offsets from individual or household subsidies.

► Why Not an Unraveling?

In January 2006, in advance of President Bush's State of the Union speech, *The Economist* published a scathing review of the U.S. healthcare system, including a forecast of a great "unraveling" in the future. If true, such a disaster is still years away for the following reasons:

- Measures undertaken will have had some effect. One example is more competitive bidding in federal government programs.
- Efficiency improvement in other sectors, such as energy production, will enable consumers to allocate a larger proportion of the gross domestic product to their health care with somewhat less pain.
- Increasing employment in the health sector is seen by many as economically healthy given the absence of growth in other sectors.
- The increased use of modern information technology may ultimately reduce administrative costs in some segments of the industry.
- Deregulation might actually reduce costs.
- Increased patient participation in making choices about such things as purchasing policies from the exchanges, or directly from competing insurers, and shopping among high-deductible plans has been making patients increasingly conscious of costs and interested in containing them, even though response to date has been slow.
- Additional actions may be taken to constrain the exploding costs of post-discharged care and biological drugs including those now used to treat wet macular degeneration, hepatitis C, hemophilia, and many forms of cancer and autoimmune disease.

One can, however, envision a scenario further into the future of an unraveling as the full impact of the baby boomers retiring is felt. They are coming in much larger numbers, tend to be less healthy, and have fewer defined-benefit pensions and less employment-based retiree health care. If their 401K accounts continue to yield low rates of return, there will be huge increases in demand for all safety net services.

► Conclusion

The United States is unlikely to undertake major healthcare reforms at the federal level beyond those associated with the ACA and/or its replacement in the near future. The states, however, will continue their reform efforts. Major reforms will not take place until the states are in financial trouble. The weakness of state-centered Medicaid is that states have their greatest trouble financing it during recessions, just as enrollment spikes.

Employers continue to support the notions of value-based care, especially transparency in pricing, and an end to price discrimination. We are likely to see excise taxes on specific healthcare goods and services and gainsharing incentive schemes added to state programs similar to Medicare Advantage.

Available measures and expected revenue adjustments are likely to delay the anticipated "unraveling" of the healthcare system. As the health sector grows, people may become increasingly concerned about upsetting the economy with radical

measures. The issues to be faced are not just the increasing cost to the sector, but the distributional effects of its growth, the equity concerns of transfers of income from the young on behalf of the old and from middle class toward the poor and the wealthy. These may also generate pressures to reduce waste and overall cost in the health sector as employers, patients, and taxpayers resist further premium support or taxation.

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